The Wilson Loop Functions on Two Orientations of the Lattice

1 Introduction

The paper constructs, for every ensemble $\mathcal{C}$ of admissible loops on the torus $(\mathbb{R}/T\mathbb{Z})^4$, the limit $W(\mathcal{C})$ of the renormalised expectations $W_K(\mathcal{C})$ as the lattice spacing $\varepsilon_K = uL^{-K}$ tends to zero (Paper, Theorem 1.5), and it shows that the limits are covariant under the translations by vectors of the lattice of the loops and under the reflections in coordinate hyperplanes (Paper, Corollary 1.6). One expects more. The lattice breaks the invariance of the continuum theory under rotations, and the limit in which the lattice is removed should restore it, as far as the torus permits. We ask whether this can be proved for the limits of the paper.

For most ensembles the question cannot be posed within the construction: a rotation which is not a symmetry of the lattice maps a path of the lattice to a path which is not one, and the limits are defined for paths of the lattice only. There is one family of ensembles for which it can. Let the loops lie in planes parallel to e_0 and e_1, and let the rotation turn the plane of e_2 and e_3 and fix the other two coordinates. It then maps each loop to a translate of itself, and it maps the ensemble to an ensemble of the lattice whenever it maps the positions of the planes to points of the lattice. This happens for rotations which are not symmetries of the lattice, because a circle around a point of the square lattice may pass through points which the symmetries of the lattice do not relate: $5^2 = 3^2 + 4^2$. The smallest instance is a loop C with a parallel copy, and the question is whether

$$W(C, C + 5\varepsilon_q e_2) = W(C, C + 3\varepsilon_q e_2 + 4\varepsilon_q e_3) \;, \tag{1.1}$$

where ε_q is the spacing of the lattice of the loops. The two sides are two numbers of one lattice theory, and none of the properties which we know relates them. Nor could these properties relate them: the kernel $(d_2, d_3) \mapsto e^{-|d_2|-|d_3|}$ is continuous, it is invariant under the symmetries of the square lattice, and it is reflection positive in both coordinate directions, since $e^{-(x_2+y_2)}$ is a product and $e^{-|x_3-y_3|}$ is a positive definite kernel, and it takes different values at $(5, 0)$ and $(3, 4)$.

The first difficulty is the torus. The rotation which turns $(5, 0)$ into $(3, 4)$ is not an isometry of $(\mathbb{R}/T\mathbb{Z})^4$, and on this torus the two configurations are not congruent, so that there is no reason for (1.1) to hold there. We therefore change the torus. Writing $z = x_2 + ix_3$, the reflection $\rho\colon z \mapsto \frac{3+4i}{5}\bar{z}$ of the plane maps 5 to $3+4i$, and it is an isometry of the torus whose periods in this plane are $T(2 + i)$ and $iT(2 + i)$, a torus which the lattices of the paper fit as well. On this torus ρ maps the lattice Λ_K of spacing ε_K to a second lattice $\Lambda'_K = \rho\Lambda_K$, tilted against the first, and it maps the lattice theory on the one to the lattice theory on the other. The expectation of the configuration $(5, 0)$ on Λ_K is therefore equal to the expectation of the configuration $(3, 4)$ on Λ'_K, at every K and every coupling (Proposition 4.2). The configuration $(3, 4)$ consists of paths of both lattices, and (1.1) becomes the statement that

the limit of its renormalised expectations is the same along the two families of lattices: the independence of the continuum limit of the orientation of the lattice, in the one case in which the two orientations share an observable.

A comparison of two lattice theories is what the paper does when it proves that the limit exists. It compares the construction on Λ_K with that on Λ_{K+1}, two runs of the renormalisation group which share every lattice but the finest (Paper, Section 9.1). The lattices Λ_K and Λ'_K share nothing, but if the blocking factor L is a multiple of 5, then Λ_K is a sublattice of Λ'_{K+1} (Lemma 2.2). A run which starts from the lattice theory on the tilted lattice Λ'_{K+1} and passes, by one renormalisation step, to the lattice Λ_K of the paper, and from there to $\Lambda_{K-1}, \Lambda_{K-2}, \ldots$, shares every lattice but the finest with the construction of the paper on Λ_K. Its coarse lattices, its cubes, its localisations, and its expansions are those of the paper. What differs is the fine lattice, on which every background field and every propagator of the run is defined: it is turned by ρ against everything else.

The proofs of Bałaban on which the paper rests are written for a fine lattice of which the coarser lattices are sublattices spanned by multiples of the same basis. Our aim is to record what changes in them when the fine lattice is turned, and only that: each statement which is used is assigned to one of three classes, those which hold with the same proof, those which hold after a modification, which we state and prove, and those which we have not examined. We proceed in the order of the proofs. Sections 2–4 contain the setting: the arithmetic of the two lattices, the torus, and the identity between the expectations which reduces (1.1) to a comparison of two runs. Section 5 determines the symmetries which the tilted run retains; they leave three invariant quadratic forms of the field strength where the symmetries of one lattice leave one, which affects the definition of the running coupling. Section 6 defines the averaging operations of the tilted run and proves the properties which replace those of [2, Sections A and B], and Section 7 gives their momentum representation, in which the turned lattice appears as one fixed trigonometric polynomial multiplying Bałaban's symbols and as a matrix in place of a scalar. Section 8 describes a block as a union of cells of 5^4 points, which gives contours inside the blocks and a Poincaré inequality, and Section 9 proves the L^2-bounds of [2] for the tilted run by counting paths. Section 10 goes through [2] formula by formula and Section 11 through the first two sections of [1]. Sections 12–14 then prove the bounds on the scalar operator of the whole lattice which these two papers use, the kernel with its exponential decay, the equation which relates the scales, and the bounds in the supremum and Hölder norms; Section 15 bounds the kernel of the gauge-fixing projection, and Section 16 proves Bałaban's Proposition 1.2 on the propagator of the vector field. Section 17 completes the first paper: the inverse of $Q_k G Q_k^*$, which is $\Delta_k + a$, the minimiser H_k with its bounds, the second derivatives in L^2, and the commutators of the averages with functions, whose values do not depend on the contours. Section 18 begins the second paper on the propagators: it replaces the local inverses on boxes with Neumann conditions, from which Bałaban's expansion of the scalar operator of a sequence of domains starts, by operators of two neighbouring scales on the whole lattice, and Section 19 proves Bałaban's lemma on the distance in which the decay of these operators is measured, for contours which run along the turned directions in the region of the finest scale. Section 20 then carries out Section B of that paper for the tilted run: Bałaban's Proposition 2.2 on the scalar operator of a sequence of domains, and his Proposition 2.3 on the inverse of $Q' G'^2 Q'^*$, with the bound on the kernel of the gauge-fixing projection of the sequence. Section 21 prepares Section C of that paper: it treats the operators H'_j and Δ'_j of the gauge functions which relate the gauge conditions of two neighbouring scales, with their kernels and the covariance of the corresponding Gaussian integral, and Section 22 proves, for the blocks of the turned lattice, the inequality which bounds a field in the axial gauge by its curl. Section 23 proves Bałaban's representation

of the propagator of two neighbouring scales, by minimising quadratic forms in the place of his transformations of Gaussian integrals, and his Proposition 2.5; Section 24 completes the second paper on the propagators with the propagator of a sequence of domains, the inverse of its averages, and the minimiser, his Propositions 2.6 and 2.7 and Corollary 2.8. Section 25 treats the operators with Dirichlet conditions outside a neighbourhood of the first domain, which the later papers use: since the averages are the identity in the region of the finest scale, these operators are given by those of the whole lattice through exact formulae. Section 26 turns to the fields with values in the group: it defines the average of the first step of the tilted run, proves the bounds on its plaquettes, and goes through the paper on the averaging operations. Sections 27, 28, and 29 go through the papers on the propagators in a background field, on the spaces of regular configurations, and on the variational problem. In these the turned lattice enters through the earlier sections and through three statements which we prove: further bounds on the scalar operator of a sequence at the trivial background, Bałaban's lemma on a configuration in the axial gauge of one step, and the construction of a regular configuration with given averages from which the induction on the number of the levels starts. Section 30 takes up the one place at which the first paper on the renormalisation transformations uses the symmetries of the lattice, the analysis of the vacuum polarisation: under the symmetries of the tilted run its part of the second order has three coefficients, one for each class of planes, while the corresponding part of the action has one. Section 31 shows that the three coefficients can be subtracted with the actions of the three classes of planes of the turned lattice, writes the effective action with a coupling and two further numbers, and proves those parts of the bound on these numbers which have no counterpart in the run of the paper: the logarithm for unequal weights of the three classes, by which the two numbers are damped, and the recursion which they satisfy. The parts which have a counterpart are listed there and are not written out. Section 32 sketches, without proofs, the transfer of the earlier sections to the averages of [9], which the covariance requires. Section 33 says what the equality requires beyond the bound on the two numbers, and it settles the three points which are not a transfer of the paper's comparison of two runs: the coupling of the tilted run at the stopping scale, which cannot be fixed by the choice of the bare coupling, the continuity at the momentum zero of the bounded part of the polarisation of the bare lattice, and the normalisations. Section 34 says with what degree of detail each paper has been treated and lists what remains; the first item which is not a modification of an argument of Bałaban is this bound.

Throughout, G, L, u, m_T, $T = 2L^{m_T}u$, and $\varepsilon_K = uL^{-K}$ are as in Paper, Section 1.1; in particular L is odd.

2 The Two Lattices

We identify the plane of the coordinates x_2 and x_3 with $\mathbb{C}$ through $z = x_2 + ix_3$, and we write a point of $\mathbb{R}^4$ as (x_0, x_1, z). The points of this plane with integer coordinates are the Gaussian integers $\mathbb{Z}[i] = \mathbb{Z} + i\mathbb{Z}$. We put

$$\omega := \frac{3+4i}{5} = \frac{(2+i)^2}{5} = \frac{2+i}{2-i}\,, \qquad \rho(x_0, x_1, z) := (x_0, x_1, \omega\bar{z})\,. \tag{2.1}$$

Since $|\omega| = 1$ and $\omega\,\overline{\omega\bar{z}} = z$, the map ρ is a linear isometry of $\mathbb{R}^4$ with $\rho^2 = 1$. It fixes e_0 and e_1, it fixes $2 + i$ because $\omega(2 - i) = 2 + i$, and it maps $i(2 + i)$ to $\omega(-i)(2 - i) = -i(2 + i)$; it is therefore the reflection in the hyperplane spanned by e_0, e_1, and $2e_2 + e_3$. We write $e'_\nu := \rho e_\nu$, so that $e'_0 = e_0$, $e'_1 = e_1$, and, because $\omega\bar{1} = \omega$ and $\omega\bar{i} = (4 - 3i)/5$,

$$e'_2 = \tfrac{1}{5}(3e_2 + 4e_3)\,, \quad e'_3 = \tfrac{1}{5}(4e_2 - 3e_3)\,, \qquad e_2 = \tfrac{1}{5}(3e'_2 + 4e'_3)\,, \quad e_3 = \tfrac{1}{5}(4e'_2 - 3e'_3)\,, \tag{2.2}$$

the last two by applying ρ to the first two. In particular $\rho(5e_2) = 3e_2 + 4e_3$.

Lemma 2.1 (Gaussian Integers). *(i) For $w \in \mathbb{Z}[i]$, $\omega w \in \mathbb{Z}[i]$ if and only if $w \in (2-i)\mathbb{Z}[i]$. Consequently $\mathbb{Z}[i] \cap \omega\mathbb{Z}[i] = (2+i)\mathbb{Z}[i]$.*

(ii) The group $(2+i)\mathbb{Z}[i]$ has the index 5 in $\mathbb{Z}[i]$ and in $\omega\mathbb{Z}[i]$.

(iii) $\omega\,\overline{(2+i)\mathbb{Z}[i]} = (2+i)\mathbb{Z}[i]$.

(iv) $5\omega\mathbb{Z}[i] \subset \mathbb{Z}[i]$ and $5\mathbb{Z}[i] \subset \omega\mathbb{Z}[i]$, each with the index 25.

Proof. (i) Let $w = a+bi$ with integers a, b. Then $5\omega w = (3a-4b)+(4a+3b)i$, and $\omega w \in \mathbb{Z}[i]$ if and only if $3a \equiv 4b$ and $4a \equiv -3b$ modulo 5. Multiplying the first congruence by 2 and the second by 4, each of them is equivalent to $a \equiv 3b$, that is, to $a + 2b \equiv 0$ modulo 5. On the other hand $(2-i)(m+ni) = (2m+n)+(2n-m)i$, for which $a+2b = 5n$; and if $a+2b = 5n$, then $m := 2n - b$ satisfies $2m + n = 5n - 2b = a$, so that $w = (2-i)(m+ni)$. This is the first assertion. For such $w = (2-i)w'$ we have $\omega w = (2+i)w'$, so that the elements of $\omega\mathbb{Z}[i]$ which lie in $\mathbb{Z}[i]$ are those of $(2+i)\mathbb{Z}[i]$.

(ii) The multiplication by $2+i$ is the linear map of $\mathbb{Z}^2$ with the columns $(2,1)$ and $(-1,2)$, of determinant 5, so that its image has the index 5 in $\mathbb{Z}[i]$. The multiplication by ω^{-1} maps $\omega\mathbb{Z}[i]$ onto $\mathbb{Z}[i]$ and $(2+i)\mathbb{Z}[i]$ onto $(2-i)\mathbb{Z}[i]$, which has the index 5 in $\mathbb{Z}[i]$ for the same reason.

(iii) Complex conjugation maps $\mathbb{Z}[i]$ onto itself, so that $\omega\,\overline{(2+i)\mathbb{Z}[i]} = \omega(2-i)\mathbb{Z}[i] = (2+i)\mathbb{Z}[i]$.

(iv) $5\omega = 3 + 4i \in \mathbb{Z}[i]$ gives the first inclusion, and the multiplication by $3+4i$ has the determinant 25. Since $\omega\bar{\omega} = 1$ and $5\bar{\omega} = 3 - 4i \in \mathbb{Z}[i]$, we have $5\mathbb{Z}[i] = \omega(5\bar{\omega})\mathbb{Z}[i] \subset \omega\mathbb{Z}[i]$, with the index of $(3-4i)\mathbb{Z}[i]$ in $\mathbb{Z}[i]$, which is 25. *q.e.d.*

For an integer $K \geq 0$ we consider the two lattices in $\mathbb{R}^4$

$$\Lambda_K := \varepsilon_K\mathbb{Z}^4 = \varepsilon_K(\mathbb{Z}^2 \times \mathbb{Z}[i])\,, \qquad \Lambda'_K := \rho\Lambda_K = \varepsilon_K(\mathbb{Z}^2 \times \omega\mathbb{Z}[i])\,. \tag{2.3}$$

The lattice Λ_K is the lattice of the paper before the periods are divided out, and Λ'_K is a lattice of the same spacing whose axes are the vectors e'_ν.

Lemma 2.2 (The Two Families of Lattices). *(i) $\Lambda_K \cap \Lambda'_K = \varepsilon_K(\mathbb{Z}^2 \times (2+i)\mathbb{Z}[i])$, a sublattice of the index 5 in Λ_K and in Λ'_K, and ρ maps it onto itself.*

(ii) If L is a multiple of 5, then $\Lambda_K \subset \Lambda'_{K+1}$ and $\Lambda'_K \subset \Lambda_{K+1}$, each with the index L^4.

Proof. (i) is Lemma 2.1(i)–(iii). (ii) Since $\varepsilon_K = L\varepsilon_{K+1}$ and $L\mathbb{Z}[i] \subset 5\mathbb{Z}[i] \subset \omega\mathbb{Z}[i]$ by Lemma 2.1(iv), the definition (2.3) gives $\Lambda_K = \varepsilon_{K+1}(L\mathbb{Z}^2 \times L\mathbb{Z}[i]) \subset \Lambda'_{K+1}$. Both are lattices of rank four, and ρ preserves volumes, so that the index is the quotient $\varepsilon_K^4/\varepsilon_{K+1}^4 = L^4$ of the volumes of their cells. Applying ρ gives $\Lambda'_K \subset \Lambda_{K+1}$ with the same index. *q.e.d.*

Thus, for L a multiple of 5, the two families form one chain $\cdots \supset \Lambda'_{K+1} \supset \Lambda_K \supset \Lambda'_{K-1} \supset \Lambda_{K-2} \supset \ldots$, each lattice of the index L^4 in the one before, while no lattice of one family contains a lattice of the same index of the other.

3 One Torus for Both Lattices

The torus $(\mathbb{R}/T\mathbb{Z})^4$ of the paper is of no use for our question: $\rho(Te_2) = \frac{1}{5}T(3e_2 + 4e_3)$ is not a period, so that ρ does not act on it. We need a group of periods which ρ preserves and which lies in both lattices, and Lemma 2.2(i) provides one.

Definition 3.1 (The Tilted Torus)**.** Let

$$\Gamma := T(\mathbb{Z}^2 \times (2+i)\mathbb{Z}[i]) \,, \qquad \mathbb{T}_\Gamma := \mathbb{R}^4/\Gamma \,. \tag{3.1}$$

The periods of $\mathbb{T}_\Gamma$ are Te_0, Te_1, and the two vectors $T(2e_2 + e_3)$ and $T(-e_2 + 2e_3)$ of the plane, which are orthogonal and of the length $\sqrt{5}\,T$.

Lemma 3.2 (The Torus and the Lattices)**.** *(i) $\Gamma \subset \Lambda_K \cap \Lambda'_K$ for every $K \geq 0$, and the quotients Λ_K/Γ and Λ'_K/Γ have $5(T/\varepsilon_K)^4$ elements each.*

(ii) $\rho\Gamma = \Gamma$, so that ρ induces an isometry of $\mathbb{T}_\Gamma$, which maps Λ_K/Γ onto Λ'_K/Γ.

(iii) Γ has the index 5 in $T\mathbb{Z}^4$, so that $\mathbb{T}_\Gamma$ covers the torus of the paper five times.

Proof. (i) The number $T/\varepsilon_K = 2L^{m_T+K}$ is an integer, so that $\Gamma \subset \varepsilon_K(\mathbb{Z}^2 \times (2+i)\mathbb{Z}[i])$, which is $\Lambda_K \cap \Lambda'_K$ by Lemma 2.2(i). The index of Γ in this group is $(T/\varepsilon_K)^4$, and the index of this group in each of the two lattices is 5. (ii) is Lemma 2.1(iii), and $\rho\Lambda_K = \Lambda'_K$ by definition. (iii) is Lemma 2.1(ii). *q.e.d.*

4 The Identity between the Two Lattice Theories

We now define the lattice theory on $\mathbb{T}_\Gamma$ for each of the two lattices and show that ρ carries the one to the other. Let Ξ be Λ_K or Λ'_K. A *bond* of Ξ is an ordered pair $b = (x, y)$ of points of Ξ at the distance ε_K, with the reversed bond $b^{-1} = (y, x)$; the bonds from x are the eight pairs $(x, x \pm \varepsilon_K e_\mu)$ for $\Xi = \Lambda_K$ and $(x, x \pm \varepsilon_K e'_\mu)$ for $\Xi = \Lambda'_K$. A *plaquette* of Ξ is a square of side ε_K with corners in Ξ, and ∂p is its boundary, a closed path of four bonds. The group $\Gamma \subset \Xi$ acts on the bonds and on the plaquettes by translations, and a translation by $\gamma \neq 0$ fixes no bond and no plaquette and reverses no bond. A *gauge field* on (Ξ, Γ) is a map U from the bonds of Ξ to G with $U(b^{-1}) = U(b)^{-1}$ and $U(b + \gamma) = U(b)$ for every $\gamma \in \Gamma$. The lattice theory at the coupling $g_0 > 0$ is the probability measure

$$\mathrm{d}\mu^{\Xi}_{\Gamma,g_0}(U) := \frac{1}{Z^{\Xi}_{\Gamma,g_0}} \exp\Big(-\frac{1}{g_0^2}\sum_p (1 - \operatorname{Re}\operatorname{tr} U(\partial p))\Big)\,\mathrm{d}U \,, \tag{4.1}$$

where the sum runs over one plaquette from each orbit of Γ on the unoriented plaquettes of Ξ and $\mathrm{d}U$ is the product of the normalised Haar measures over one bond from each orbit of Γ on the unoriented bonds. Neither depends on the choices, because U is periodic, $\operatorname{Re}\operatorname{tr}$ is invariant under conjugation and, on unitary matrices, under inversion, and the Haar measure is invariant under inversion. This is the measure of Paper, (1.2), with the group of periods Γ in place of $T\mathbb{Z}^4$ and, for $\Xi = \Lambda'_K$, with the lattice turned by ρ. We write $\mathbb{E}^{\Xi}_{\Gamma,g_0}$ for its expectation. For a closed path C of bonds of Ξ in $\mathbb{R}^4$ and a character χ we put $W_\chi(C; U) := \chi(U(C))$, where $U(C)$ is the ordered product of U over the bonds of C, and for a finite family $\mathcal{C} = ((C_1, \chi_1), \ldots, (C_n, \chi_n))$ of such paths with characters, $W_{\mathcal{C}} := \prod_i W_{\chi_i}(C_i)$. For a linear map O we write $O\mathcal{C}$ for the family of the images OC_i with the same characters.

Lemma 4.1 (Transport). *Let Ξ be Λ_K or Λ'_K, and let O be a linear isometry of $\mathbb{R}^4$ with $O\Gamma = \Gamma$ for which $O\Xi$ is again one of the two lattices. Then, for every family $\mathcal{C}$ of closed paths of bonds of Ξ with characters and every $g_0 > 0$,*

$$\mathbb{E}^{O\Xi}_{\Gamma,g_0}[W_{O\mathcal{C}}] = \mathbb{E}^{\Xi}_{\Gamma,g_0}[W_{\mathcal{C}}] \,. \tag{4.2}$$

Proof. The map O is a bijection from the points of Ξ onto those of $O\Xi$ which preserves distances, so that it maps bonds to bonds, reversed bonds to reversed bonds, and plaquettes to plaquettes with their boundaries; and since $O(b+\gamma) = Ob + O\gamma$ with $O\Gamma = \Gamma$, it maps the orbits of Γ on the bonds and on the plaquettes of Ξ bijectively onto those of $O\Xi$. For a gauge field U on (Ξ, Γ) put $(O_*U)(b) \coloneqq U(O^{-1}b)$ for the bonds b of $O\Xi$. Then O_*U is a gauge field on $(O\Xi, \Gamma)$, the map O_* is a bijection between the two spaces of gauge fields which carries the product of the Haar measures of the one to that of the other, and $(O_*U)(O\gamma) = U(\gamma)$ for every path γ of bonds of Ξ. In particular $(O_*U)(\partial(Op)) = U(\partial p)$ for every plaquette p of Ξ, so that the sums in (4.1) for O_*U on $O\Xi$ and for U on Ξ agree, and O_* carries μ^{Ξ}_{Γ,g_0} to $\mu^{O\Xi}_{\Gamma,g_0}$. Since $W_{O\mathcal{C}}(O_*U) = W_{\mathcal{C}}(U)$, this gives (4.2). *q.e.d.*

We apply the lemma to $O = \rho$ and to the ensembles for which the question of the introduction can be posed. Let $q \geq 0$, and let $C_1, \dots, C_n$ be closed paths of bonds of Λ_q in the plane $\{z = 0\}$, so that all their bonds are parallel to e_0 or e_1. For $d = (d_1, \dots, d_n)$ with $d_j \in \varepsilon_q\mathbb{Z}[i]$ we write

$$\mathcal{C}_d \coloneqq ((C_1 + d_1, \chi_1), \dots, (C_n + d_n, \chi_n)) \,, \qquad \rho d \coloneqq (\omega\bar{d}_1, \dots, \omega\bar{d}_n) \,, \tag{4.3}$$

where $C_j + d_j$ is the translate of C_j to the plane $\{z = d_j\}$. Since ρ fixes the plane $\{z = 0\}$ pointwise and is linear, $\rho(C_j + d_j) = C_j + \omega\bar{d}_j$, that is, $\rho\mathcal{C}_d = \mathcal{C}_{\rho d}$.

Proposition 4.2 (Reduction). *Let $q \geq 0$, let $C_1, \dots, C_n$ and $\mathcal{C}_d$ be as above, and let $d_j \in \varepsilon_q(2+i)\mathbb{Z}[i]$ for every j. Then $\rho d_j \in \varepsilon_q(2+i)\mathbb{Z}[i]$ for every j, the families $\mathcal{C}_d$ and $\mathcal{C}_{\rho d}$ consist of closed paths of bonds of both Λ_K and Λ'_K for every $K \geq q$, and for every $K \geq q$ and every $g_0 > 0$,*

$$\mathbb{E}^{\Lambda_K}_{\Gamma,g_0}[W_{\mathcal{C}_d}] = \mathbb{E}^{\Lambda'_K}_{\Gamma,g_0}[W_{\mathcal{C}_{\rho d}}] \,. \tag{4.4}$$

Consequently

$$\mathbb{E}^{\Lambda_K}_{\Gamma,g_0}[W_{\mathcal{C}_{\rho d}}] - \mathbb{E}^{\Lambda_K}_{\Gamma,g_0}[W_{\mathcal{C}_d}] = \mathbb{E}^{\Lambda_K}_{\Gamma,g_0}[W_{\mathcal{C}_{\rho d}}] - \mathbb{E}^{\Lambda'_K}_{\Gamma,g_0}[W_{\mathcal{C}_{\rho d}}] \,. \tag{4.5}$$

Proof. By Lemma 2.1(iii), $\rho d_j = \omega\bar{d}_j \in \varepsilon_q(2+i)\mathbb{Z}[i]$. The sites of $C_j + d_j$ are the points (x_0, x_1, d_j) with $(x_0, x_1, 0)$ a site of C_j; they lie in $\varepsilon_q(\mathbb{Z}^2 \times (2+i)\mathbb{Z}[i]) = \Lambda_q \cap \Lambda'_q$, and the bonds of $C_j + d_j$ are parallel to $e_0 = e'_0$ or $e_1 = e'_1$ and of the length ε_q. A straight path of the length ε_q between two points of Λ_q, or of Λ'_q, along one of its axes is a path of L^{K-q} bonds of Λ_K, or of Λ'_K; so $C_j + d_j$ is a closed path of bonds of both lattices, and the same holds for $C_j + \rho d_j$. Lemma 4.1 with $\Xi = \Lambda_K$ and $O = \rho$, for which $O\Gamma = \Gamma$ by Lemma 3.2(ii) and $O\Xi = \Lambda'_K$, gives (4.4) because $\rho\mathcal{C}_d = \mathcal{C}_{\rho d}$, and (4.5) follows by subtracting both sides of (4.4) from $\mathbb{E}^{\Lambda_K}_{\Gamma,g_0}[W_{\mathcal{C}_{\rho d}}]$. *q.e.d.*

The left side of (4.5) measures how far the lattice theory on Λ_K is from being invariant under ρ on this ensemble, and the right side how much the expectation of one ensemble differs between the two lattices. The first instance in which $\mathcal{C}_{\rho d}$ is not the image of $\mathcal{C}_d$ under a symmetry of Λ_q is $n = 2$, $d_1 = 0$, and $d_2 = 5\varepsilon_q = (2+i)(2-i)\varepsilon_q$, for which $\rho d_2 = (3+4i)\varepsilon_q$; this is (1.1). The step factors of the paper do not depend on the position of a loop (Paper, Proposition 7.13), so that $C_j + d_j$ and $C_j + \rho d_j$ have the same normalisation on Λ_K, and

(1.1) is the statement that the left side of (4.5), divided by this normalisation, tends to zero as $K \to \infty$ along the couplings of the paper. On the right side, the first term is the object of the construction of the paper on Λ_K, carried out on the torus $\mathbb{T}_\Gamma$, and the second, read at $K+1$, is the object of the tilted run of the introduction.

5 The Symmetries of the Tilted Run

The tilted run has only those symmetries which preserve the lattices of both families. We determine them, and what they leave of the argument by which a single coupling describes the leading term of an effective action.

Lemma 5.1 (The Common Symmetries)**.** *Let $\mathfrak{G}_0$ be the group of the linear isometries O of $\mathbb{R}^4$ with $O\mathbb{Z}^4 = \mathbb{Z}^4$ and $O\rho\mathbb{Z}^4 = \rho\mathbb{Z}^4$. Its elements are the maps*

$$(x_0, x_1, z) \mapsto (A(x_0, x_1), i^k z) \,, \tag{5.1}$$

with A one of the eight signed permutation matrices of the first two coordinates and $k \in \{0,1,2,3\}$. It has 32 elements, it contains the reflections in the hyperplanes $\{x_0 = 0\}$ and $\{x_1 = 0\}$, and it contains no reflection of the plane of x_2 and x_3.

Proof. An isometry O with $O\mathbb{Z}^4 = \mathbb{Z}^4$ permutes the eight vectors $\pm e_\mu$, the elements of $\mathbb{Z}^4$ of the length 1, and an isometry with $O\rho\mathbb{Z}^4 = \rho\mathbb{Z}^4$ permutes the eight vectors $\pm e'_\nu$. By (2.2) the only vectors common to the two sets are $\pm e_0$ and $\pm e_1$. An element O of $\mathfrak{G}_0$ permutes both sets, hence their intersection, so that $Oe_0, Oe_1 \in \{\pm e_0, \pm e_1\}$; the map O therefore preserves the plane of e_0 and e_1 and its orthogonal complement, and it acts on the first as a signed permutation matrix A and on the second as a signed permutation matrix of e_2 and e_3, that is, as $z \mapsto i^k z$ or as $z \mapsto i^k \bar{z}$. Every such map preserves $\mathbb{Z}^4$, and it preserves $\rho\mathbb{Z}^4 = \mathbb{Z}^2 \times \omega\mathbb{Z}[i]$ if and only if its second part preserves $\omega\mathbb{Z}[i]$. The maps $z \mapsto i^k z$ do. The maps $z \mapsto i^k \bar{z}$ send $\omega\mathbb{Z}[i]$ to $\bar{\omega}\mathbb{Z}[i]$, which is equal to $\omega\mathbb{Z}[i]$ only if $\omega^2 \in \mathbb{Z}[i]$, and $\omega^2 = (-7+24i)/25$. This gives (5.1) and the count $8 \cdot 4$. The reflections named are the cases $A = \mathrm{diag}(-1,1)$ and $A = \mathrm{diag}(1,-1)$ with $k = 0$. *q.e.d.*

The group of all symmetries of $\mathbb{Z}^4$, the signed permutation matrices, has 384 elements and exchanges any two coordinate planes. The group $\mathfrak{G}_0$ does not, and this has the following consequence. Let $F = (F_{\mu\nu})$ be an antisymmetric tensor, on which an isometry O acts by $(OF)_{\mu\nu} = \sum_{\kappa,\lambda} O_{\mu\kappa} O_{\nu\lambda} F_{\kappa\lambda}$.

Lemma 5.2 (Invariant Quadratic Forms)**.** *The real quadratic forms of the antisymmetric tensors F which are invariant under $\mathfrak{G}_0$ are the combinations*

$$\alpha F_{01}^2 + \beta F_{23}^2 + \lambda \sum_{a \in \{0,1\}} \sum_{j \in \{2,3\}} F_{aj}^2 \tag{5.2}$$

with real α, β, λ, a space of dimension three. Those which are invariant under all signed permutation matrices are the multiples of $\sum_{\mu<\nu} F_{\mu\nu}^2$.

Proof. Let q be an invariant quadratic form, a combination of the products of two of the six components F_{01}, F_{23}, F_{aj}. The three elements θ_0, θ_1, and σ of $\mathfrak{G}_0$ which change the sign of x_0, of x_1, and of z multiply each component by a sign: F_{01} by $(-,-,+)$, F_{23} by $(+,+,+)$,

F_{0j} by $(-,+,-)$, and F_{1j} by $(+,-,-)$. A product of two components is invariant under the three only if the two have the same signs, so that

$$\mathsf{q}(F) = \alpha F_{01}^2 + \beta F_{23}^2 + \sum_{j,j'} (m^0_{jj'} F_{0j} F_{0j'} + m^1_{jj'} F_{1j} F_{1j'}) \tag{5.3}$$

with symmetric matrices m^0 and m^1. The element of $\mathfrak{G}_0$ which exchanges x_0 and x_1 exchanges F_{0j} and F_{1j}, so that $m^0 = m^1 =: m$. The element $z \mapsto iz$ maps e_2 to e_3 and e_3 to $-e_2$, so that it replaces the vector (F_{a2}, F_{a3}) by $(-F_{a3}, F_{a2})$; the invariance of (5.3) then requires $m_{22} = m_{33}$ and $m_{23} = -m_{23}$, that is, $m = \lambda \mathbf{1}$. This is (5.2). Conversely each of the three forms is invariant: an element (5.1) multiplies F_{01} by $\det A = \pm 1$, fixes F_{23}, because $z \mapsto i^k z$ has the determinant 1, and replaces the matrix (F_{aj}) by its product with A on one side and with a rotation matrix on the other, which preserves the sum of the squares of its entries. Finally a signed permutation matrix which exchanges the planes of two pairs of coordinates exchanges the corresponding components up to signs, so that the invariance under all of them forces $\alpha = \beta = \lambda$. *q.e.d.*

Modification of [9, (1.20)–(1.22), p. 264]. Bałaban defines the coefficient β_{j+1} of the flow of the coupling as a second derivative of the Fourier transform of the vacuum polarisation $\Pi_{j+1,\mu\nu}$ of a step, for one pair $\mu \neq \nu$, and the definition does not depend on the pair because, by his (1.21), the vacuum polarisation is covariant under the rotations which leave the lattice invariant. In the tilted run the covariance holds for the elements of $\mathfrak{G}_0$ only, and by Lemma 5.2 the second derivatives define three numbers, one for each of the forms (5.2): those of the pair $(0,1)$, of the pair $(2,3)$, and of the four pairs (a,j). We do not know whether the three agree, and the consequences for the inductive description of the densities have not been examined.

6 The Averages of the Tilted Run

From here on L is a multiple of 5. After $k \geq 1$ steps of the tilted run, and in the unit of the lattice which it has reached, the situation is that of [2, Section B] with one change: the unit lattice is $\mathbb{Z}^4$ and the lattices of the intermediate steps are $L^{-j}\mathbb{Z}^4$, $1 \leq j \leq k-1$, as there, but the fine lattice is

$$T_\eta \coloneqq \eta\rho\mathbb{Z}^4 = \eta(\mathbb{Z}^2 \times \omega\mathbb{Z}[i]) \,, \qquad \eta = L^{-k} \,, \tag{6.1}$$

in place of $\eta\mathbb{Z}^4$. Its bonds are the pairs $\langle x, x \pm \eta e'_\nu \rangle$. We call a number a a *scale* if $a/(5\eta)$ and $1/a$ are integers; the numbers L^{-j}, $0 \leq j \leq k-1$, are scales, and so is 5η. For a scale a we have $a\mathbb{Z}^4 \subset T_\eta$, because a/η is a multiple of 5 and $5\mathbb{Z}[i] \subset \omega\mathbb{Z}[i]$ by Lemma 2.1(iv). A real field is a function A on the bonds of T_η with $A(b^{-1}) = -A(b)$, and $A(\gamma) \coloneqq \sum_{b\in\gamma} A(b)$ for a path γ of bonds. We consider infinite lattices in this section and the next; all the operations commute with the translations by $\mathbb{Z}^4$ and therefore act on the fields which are periodic with respect to a subgroup of $\mathbb{Z}^4$.

Bałaban's averages are sums, over the points x of a cube of the fine lattice, of the values of a function at x or of the sums of a field over the straight segment $[x, x(b)]$ from x to its translate by a bond b of the coarse lattice [2, (1.6), (1.11), (1.18), pp. 18–20]. In the tilted run the cubes keep their meaning, as the sets of the points of T_η in a cube of the coarse lattice, but for the directions e_2 and e_3 the segment does not consist of bonds of T_η, and we replace it by a path which follows it.

Definition 6.1 (Blocks, Staircases, and Averages)**.** Let a be a scale.

(i) The *block* of $y \in a\mathbb{Z}^4$ is $B_a(y) := \{x \in T_\eta \,|\, y_\mu \le x_\mu < y_\mu + a,\ \mu = 0, \dots, 3\}$.

(ii) For $x \in T_\eta$ the *staircases* from x are the following paths of bonds of T_η. For $\mu \in \{0, 1\}$, $\sigma_\mu^a(x)$ is the straight path of a/η bonds in the direction e_μ. The path $\sigma_2^a(x)$ repeats $a/(5\eta)$ times the seven steps $\eta e_3', \eta e_2', \eta e_3', \eta e_2', \eta e_3', \eta e_2', \eta e_3'$, and $\sigma_3^a(x)$ repeats $a/(5\eta)$ times the seven steps $\eta e_2', -\eta e_3', \eta e_2', -\eta e_3', \eta e_2', -\eta e_3', \eta e_2'$.

(iii) For a function λ on T_η and $y \in a\mathbb{Z}^4$,

$$(Q_a'\lambda)(y) := \left(\frac{\eta}{a}\right)^4 \sum_{x \in B_a(y)} \lambda(x)\,. \tag{6.2}$$

(iv) For a real field A and a bond $b = \langle y, y + ae_\mu\rangle$ of $a\mathbb{Z}^4$,

$$(Q_a A)(b) := \left(\frac{\eta}{a}\right)^5 \sum_{x \in B_a(y)} A(\sigma_\mu^a(x))\,, \tag{6.3}$$

and $(Q_a A)(b^{-1}) := -(Q_a A)(b)$.

We write $Q_k' := Q_1'$ and $Q_k := Q_1$.

For $a = 1$ these are the operators Q_k' of [2, p. 20] and Q_k of [2, (1.18), p. 20], with T_η turned and with $\sigma_\mu^1(x)$ in place of $[x, x(b)]$, and for the directions e_0 and e_1 nothing has changed. For two scales a and a' with a'/a an integer we also need Bałaban's own averages between the lattices $a\mathbb{Z}^4$ and $a'\mathbb{Z}^4$, which we denote by

$$(Q'^{\mathrm{st}}_{a,a'}\lambda)(y) := \left(\frac{a}{a'}\right)^4 \sum_z \lambda(z)\,, \qquad (Q^{\mathrm{st}}_{a,a'}B)(\langle y, y+a'e_\mu\rangle) := \left(\frac{a}{a'}\right)^5 \sum_z B([z, z+a'e_\mu])\,, \tag{6.4}$$

the sums over the points $z \in a\mathbb{Z}^4$ with $y_\mu \le z_\mu < y_\mu + a'$ for every μ, for functions λ and fields B on $a\mathbb{Z}^4$.

Lemma 6.2 (Blocks and Staircases)**.** *Let a and a' be scales with a'/a an integer.*

(i) The blocks $B_a(y)$, $y \in a\mathbb{Z}^4$, are pairwise disjoint, their union is T_η, each has $(a/\eta)^4$ points, and $B_a(y + w) = B_a(y) + w$ for $w \in a\mathbb{Z}^4$. The block $B_{a'}(y)$ is the union of the blocks $B_a(z)$ over the points $z \in a\mathbb{Z}^4$ with $y_\mu \le z_\mu < y_\mu + a'$.

(ii) The staircase $\sigma_\mu^a(x)$ ends at $x + ae_\mu$, and $\sigma_\mu^{a'}(x)$ is the concatenation of the staircases $\sigma_\mu^a(x + nae_\mu)$, $0 \le n < a'/a$. Each site of $\sigma_\mu^a(x)$ is at a distance of at most $\frac{3}{5}\eta$ from the segment $[x, x + ae_\mu]$.

Proof. (i) Since $a\mathbb{Z}^4$ is a subgroup of T_η and the cube $y + [0, a)^4$ contains exactly one point of every class of $\mathbb{R}^4$ modulo $a\mathbb{Z}^4$, the block $B_a(y)$ contains exactly one point of every class of T_η modulo $a\mathbb{Z}^4$. The number of these classes is the quotient a^4/η^4 of the volumes of the cells of the two lattices, because ρ preserves volumes. The remaining assertions follow from the corresponding ones for the cubes.

(ii) For $\mu \in \{0, 1\}$ there is nothing to prove. By (2.2), seven steps of σ_2^a add up to $\eta(3e_2' + 4e_3') = 5\eta e_2$ and seven steps of σ_3^a to $\eta(4e_2' - 3e_3') = 5\eta e_3$, which gives the end point and the concatenation. The distance of a point $\eta(ne_2' + n'e_3')$ from the line through the origin in the direction $3e_2' + 4e_3'$ is $\eta|4n - 3n'|/5$. After one to seven steps of σ_2^a from the origin the

pair (n, n') is $(0,1)$, $(1,1)$, $(1,2)$, $(2,2)$, $(2,3)$, $(3,3)$, $(3,4)$, with $|4n-3n'| = 3,1,2,2,1,3,0$, and the foot of the perpendicular lies between the origin and $5\eta e_2$ in each case; the eighth site lies on the line, so that the same holds in every period. For σ_3^a the pairs are $(1,0)$, $(1,-1)$, $(2,-1)$, $(2,-2)$, $(3,-2)$, $(3,-3)$, $(4,-3)$, and the distance from the line in the direction $4e_2' - 3e_3'$ is $\eta|3n+4n'|/5$, with the same values. *q.e.d.*

The next lemma replaces the calculation [2, (1.16)–(1.18), p. 20] by which a composition of averages is an average, and it contains a second statement which has no counterpart there: the tilted average is one of Bałaban's averages, preceded by an average over cells of 5^4 points which is the same at every step.

Lemma 6.3 (Composition)**.** *For scales a and a' with a'/a an integer,*

$$Q_{a'} = Q^{\mathrm{st}}_{a,a'} Q_a \ , \qquad Q'_{a'} = Q'^{\mathrm{st}}_{a,a'} Q'_a \ . \tag{6.5}$$

In particular $Q_k = Q^{\mathrm{st}}_{L^{-1},1} Q_{L^{-1}}$ and $Q_k = Q^{\mathrm{st}}_{5\eta,1} Q_{5\eta}$, and likewise for Q'_k.

Proof. Let $b = \langle y, y + a'e_\mu \rangle$, and let z run over the points of $a\mathbb{Z}^4$ with $y_\nu \le z_\nu < y_\nu + a'$. The segment $[z, z+a'e_\mu]$ consists of the a'/a bonds $\langle z + nae_\mu, z + (n+1)ae_\mu \rangle$ of $a\mathbb{Z}^4$, so that by (6.4) and (6.3)

$$\begin{aligned} (Q^{\mathrm{st}}_{a,a'} Q_a A)(b) &= \Big(\frac{a}{a'}\Big)^5 \Big(\frac{\eta}{a}\Big)^5 \sum_z \sum_{n=0}^{a'/a-1} \sum_{x \in B_a(z+nae_\mu)} A(\sigma^a_\mu(x)) \\ &= \Big(\frac{\eta}{a'}\Big)^5 \sum_z \sum_{x \in B_a(z)} \sum_{n=0}^{a'/a-1} A(\sigma^a_\mu(x + nae_\mu)) \ , \end{aligned} \tag{6.6}$$

where we have used $B_a(z + nae_\mu) = B_a(z) + nae_\mu$. By Lemma 6.2(ii) the inner sum is $A(\sigma^{a'}_\mu(x))$, and by Lemma 6.2(i) the points x run over $B_{a'}(y)$; this is $(Q_{a'}A)(b)$. The proof for Q' is the last assertion of Lemma 6.2(i). *q.e.d.*

Lemma 6.4 (The Identities of the Averages)**.** *Let a be a scale, and write $(\partial^\eta \lambda)(b) := \eta^{-1}(\lambda(b_+) - \lambda(b_-))$ for a function λ on T_η and a bond $b = \langle b_-, b_+ \rangle$ of T_η, and ∂^a for the same operation on $a\mathbb{Z}^4$.*

(i) $Q_a \partial^\eta = \partial^a Q'_a$.

(ii) *Q'_a maps the constant function 1 to 1, and so does its adjoint. For $v \in \mathbb{R}^4$, Q_a maps the constant field $A(\langle x, x+\eta e'_\nu \rangle) := v \cdot e'_\nu$ on T_η to the constant field $B(\langle y, y + ae_\mu \rangle) := v \cdot e_\mu$ on $a\mathbb{Z}^4$, and its adjoint maps B to A. Here the adjoints are taken with respect to the sums over the points, or the positive bonds, with the weights η^4 on T_η and a^4 on $a\mathbb{Z}^4$.*

(iii) *Q_a and Q'_a commute with the translations by the vectors of $a\mathbb{Z}^4$.*

Proof. (i) The sum of $\partial^\eta \lambda$ over a path from x to x' is $\eta^{-1}(\lambda(x') - \lambda(x))$, so that $(Q_a \partial^\eta \lambda)(\langle y, y + ae_\mu \rangle) = (\eta/a)^5 \eta^{-1} \sum_{x \in B_a(y)} (\lambda(x + ae_\mu) - \lambda(x))$ by Lemma 6.2(ii). Since $x \mapsto x + ae_\mu$ maps $B_a(y)$ onto $B_a(y + ae_\mu)$, this is $a^{-1}((Q'_a \lambda)(y + ae_\mu) - (Q'_a \lambda)(y))$.

(ii) $Q'_a 1 = 1$ is the count of Lemma 6.2(i), and $(Q'^*_a \omega)(x) = \omega(y)$ for $x \in B_a(y)$, because every point lies in one block. The sum of the constant field A over a path from x to x' is $\eta^{-1} v \cdot (x' - x)$, so that every term of (6.3) is $\eta^{-1} a v \cdot e_\mu$, and $(\eta/a)^5 (a/\eta)^4 \eta^{-1} a = 1$. For

the adjoint we count the staircases through a bond. By (6.3) and the weights, $(Q_a^* B)(b') = (\eta/a)\sum_\mu \sum_x \pm B(\langle y(x), y(x)+ae_\mu\rangle)$ for a positive bond b' of T_η, the inner sum over the points $x \in T_\eta$ whose staircase $\sigma_\mu^a(x)$ contains b' or its reverse, with the sign $-$ in the second case, and with $y(x)$ the corner of the block of x. A bond in the direction e_0 or e_1 lies on a/η staircases of that direction and on no other. Each period of σ_2^a has three steps $\eta e_2'$ and four steps $\eta e_3'$, and each period of σ_3^a has four steps $\eta e_2'$ and three steps $-\eta e_3'$; since every translate of a staircase by a vector of T_η is a staircase, a bond in the direction e_2' lies on $3a/(5\eta)$ staircases σ_2^a and on $4a/(5\eta)$ staircases σ_3^a, and a bond in the direction e_3' on $4a/(5\eta)$ staircases σ_2^a and, reversed, on $3a/(5\eta)$ staircases σ_3^a. For the constant field B this gives the values $\frac{1}{5}(3v\cdot e_2 + 4v\cdot e_3) = v\cdot e_2'$ and $\frac{1}{5}(4v\cdot e_2 - 3v\cdot e_3) = v\cdot e_3'$ by (2.2).

(iii) The blocks and the staircases are carried to each other by these translations. *q.e.d.*

Statement (i) is [2, (1.55), p. 27], and with it [2, (1.9), (1.13), (1.20)]. Statement (ii) says that the average of a constant field is the same constant vector, with its components in the basis e_ν' replaced by those in the basis e_μ. On the constant fields Q_a is therefore an isometry with $Q_a^* Q_a = 1$, where Bałaban's average is the identity, and Q_a and Q_a' map the orthogonal complements of the constants to the orthogonal complements of the constants.

7 The Momentum Representation

Bałaban's bounds on the operators built from Q_k, Q_k', and the Laplace operator of the fine lattice are read off from their momentum representation, in which Q_k' is the multiplication by the function u_k of [2, (1.31), p. 23] and Q_k the multiplication of the component μ by $u_k v_\mu$ [2, (1.61), p. 28]. Both are products of quotients of sines, and their properties are evident from this form. For the tilted averages the factorisation $Q_k = Q^{\mathrm{st}}_{5\eta,1} Q_{5\eta}$ of Lemma 6.3 gives the corresponding form: Bałaban's functions for the lattice of the spacing 5η, multiplied by one trigonometric polynomial of 25 terms for the cell, and with a matrix of polynomials of at most four terms in place of the scalar v_μ in the two turned directions.

For $p \in \mathbb{R}^4$ we write $e_p(x) \coloneqq e^{ip\cdot x}$, and for a number $a > 0$

$$\begin{aligned} \partial_\mu^a(p) &\coloneqq \frac{e^{iap_\mu} - 1}{a} , \qquad & \partial_\nu'(p) &\coloneqq \frac{e^{i\eta p\cdot e_\nu'} - 1}{\eta} , \\ \Delta(p) &\coloneqq \sum_\nu |\partial_\nu'(p)|^2 , \qquad & \Delta_0(p) &\coloneqq \sum_\mu |\partial_\mu^1(p)|^2 . \end{aligned} \tag{7.1}$$

The functions ∂_ν' and Δ are the symbols of the difference operators and of the Laplace operator of T_η, and ∂_μ^1 and Δ_0 those of the unit lattice, as in [2, (1.31), p. 23, and p. 24]. For numbers a and a' with a'/a an integer we put

$$D_\mu^{a,a'}(p) \coloneqq \frac{a}{a'} \sum_{n=0}^{a'/a-1} e^{ianp_\mu} , \tag{7.2}$$

which is $\partial_\mu^{a'}(p)/\partial_\mu^a(p)$ wherever $\partial_\mu^a(p) \neq 0$ and satisfies $D_\mu^{a,a''} = D_\mu^{a,a'} D_\mu^{a',a''}$. Bałaban's functions are $u_k = \prod_\mu D_\mu^{\eta,1}$ and $v_\mu = D_\mu^{\eta,1}$. The cell enters through the set $\mathsf{C} \coloneqq \{c \in \omega\mathbb{Z}[i] \,|\, 0 \le \operatorname{Re} c < 5,\ 0 \le \operatorname{Im} c < 5\}$, which has 25 elements by Lemma 6.2(i) for $a = 5\eta$, and through

the functions of $\theta = (\theta_2, \theta_3)$

$$\begin{aligned} \kappa(\theta) &:= \frac{1}{25} \sum_{c \in \mathsf{C}} e^{i\theta \cdot c} \,, \\ s_{22}(\theta) &:= e^{i\theta \cdot e_3'} + e^{i\theta \cdot (e_2' + 2e_3')} + e^{i\theta \cdot (2e_2' + 3e_3')} \,, \\ s_{23}(\theta) &:= 1 + e^{i\theta \cdot (e_2' + e_3')} + e^{i\theta \cdot (2e_2' + 2e_3')} + e^{i\theta \cdot (3e_2' + 3e_3')} \,, \\ s_{32}(\theta) &:= 1 + e^{i\theta \cdot (e_2' - e_3')} + e^{i\theta \cdot (2e_2' - 2e_3')} + e^{i\theta \cdot (3e_2' - 3e_3')} \,, \\ s_{33}(\theta) &:= -e^{i\theta \cdot (e_2' - e_3')} - e^{i\theta \cdot (2e_2' - 2e_3')} - e^{i\theta \cdot (3e_2' - 3e_3')} \,, \end{aligned} \tag{7.3}$$

where c is read as the vector $(\operatorname{Re} c, \operatorname{Im} c)$ and $\theta \cdot e_\nu'$ is the product with the components of e_ν' in the plane.

Lemma 7.1 (The Symbols of the Tilted Averages)**.** *Let $p \in \mathbb{R}^4$.*

(i) $Q_k' e_p = u_k(p)\, e_p$ on $\mathbb{Z}^4$, with

$$\begin{aligned} u_k(p) &:= \sum_{x \in B_1(0)} \eta^4 e^{ip \cdot x} \\ &= D_0^{\eta,1}(p)\, D_1^{\eta,1}(p)\, D_2^{5\eta,1}(p)\, D_3^{5\eta,1}(p)\, \kappa(\eta p_2, \eta p_3) \,. \end{aligned} \tag{7.4}$$

(ii) For $w = (w_\nu) \in \mathbb{C}^4$ and the field $A(\langle x, x + \eta e_\nu' \rangle) := w_\nu e^{ip \cdot x}$, we have $(Q_k A)(\langle y, y + e_\mu \rangle) = u_k(p) \sum_\nu V_{\mu\nu}(p) w_\nu\, e^{ip \cdot y}$, where $V_{\mu\mu}(p) := D_\mu^{\eta,1}(p)$ for $\mu \in \{0, 1\}$,

$$V_{\mu\nu}(p) := \tfrac{1}{5} D_\mu^{5\eta,1}(p)\, s_{\mu\nu}(\eta p_2, \eta p_3) \tag{7.5}$$

for $\mu, \nu \in \{2, 3\}$, and $V_{\mu\nu} := 0$ for the other pairs.

(iii) $\sum_\nu V_{\mu\nu}(p)\, \partial_\nu'(p) = \partial_\mu^1(p)$ for every μ.

(iv) u_k and V are periodic with respect to $2\pi\eta^{-1}\rho\mathbb{Z}^4$, which is a subgroup of $2\pi\mathbb{Z}^4$; $u_k(0) = 1$; $u_k(l) = 0$ for every $l \in 2\pi\mathbb{Z}^4$ which does not lie in $2\pi\eta^{-1}\rho\mathbb{Z}^4$; and $V_{\mu\nu}(0) = e_\mu \cdot e_\nu'$.

Proof. (i) By (6.2), $(Q_k' e_p)(y) = \sum_{x \in B_1(0)} \eta^4 e^{ip \cdot (y+x)}$, which is the first equality. By Lemma 6.2(i), $B_1(0)$ is the disjoint union of the sets $z + B_{5\eta}(0)$ over the $z \in 5\eta\mathbb{Z}^4$ with $0 \le z_\mu < 1$, and $B_{5\eta}(0)$ is the set of the points $\eta(n_0, n_1, c)$ with $n_0, n_1 \in \{0, \dots, 4\}$ and $c \in \mathsf{C}$. Hence

$$\begin{aligned} u_k(p) &= \Big(\sum_z (5\eta)^4 e^{ip \cdot z} \Big) \Big(\frac{1}{5^4} \sum_{n_0, n_1 = 0}^{4} \sum_{c \in \mathsf{C}} e^{i\eta(n_0 p_0 + n_1 p_1)} e^{i\eta(p_2, p_3) \cdot c} \Big) \\ &= \prod_{\mu=0}^{3} D_\mu^{5\eta,1}(p) \cdot D_0^{\eta,5\eta}(p)\, D_1^{\eta,5\eta}(p)\, \kappa(\eta p_2, \eta p_3) \,, \end{aligned} \tag{7.6}$$

and $D_\mu^{\eta,5\eta} D_\mu^{5\eta,1} = D_\mu^{\eta,1}$.

(ii) For $\mu \in \{0, 1\}$ the sum of A over $\sigma_\mu^1(x)$ is $w_\mu e^{ip \cdot x} \sum_{n < 1/\eta} e^{i\eta n p_\mu} = \eta^{-1} D_\mu^{\eta,1}(p) w_\mu e^{ip \cdot x}$, and (6.3) with (i) gives the assertion. For $\mu = 2$, the bonds of the first period of $\sigma_2^1(x)$ start at the points $x + \eta c$ with $c = 0$, e_3', e_2', $e_2' + e_3'$, $e_2' + 2e_3'$, $2e_2' + 2e_3'$, $2e_2' + 3e_3'$, $3e_2' + 3e_3'$, and they have the directions $e_3', e_2', e_3', e_2', e_3', e_2', e_3'$; the sum of A over them is $e^{ip \cdot x}(s_{22}(\eta p) w_2 + s_{23}(\eta p) w_3)$. The n-th period is the translate of the first by $5\eta n e_2$, so that the sum of A over $\sigma_2^1(x)$ is

this number multiplied by $\sum_{n<1/(5\eta)} e^{5i\eta n p_2} = (5\eta)^{-1} D_2^{5\eta,1}(p)$. For $\mu = 3$ the steps in the direction e'_2 start at the points $x + \eta j(e'_2 - e'_3)$, $j = 0, \dots, 3$, and a step $-\eta e'_3$ from $x + \eta c$ is the reversed bond $\langle x + \eta(c - e'_3), x + \eta c\rangle$, on which A takes the value $-w_3 e^{ip\cdot(x+\eta(c-e'_3))}$; the three such steps end at $x + \eta j(e'_2 - e'_3)$, $j = 1, 2, 3$. This gives s_{32} and s_{33}.

(iii) Multiply each term of $s_{\mu\nu}(\theta)$ by $e^{i\theta\cdot e'_\nu} - 1$ and add: each step of a period from c to c' contributes $e^{i\theta\cdot c'} - e^{i\theta\cdot c}$, a step $-\eta e'_3$ included, since $-(e^{i\theta\cdot e'_3} - 1)e^{i\theta\cdot(c-e'_3)} = e^{i\theta\cdot(c-e'_3)} - e^{i\theta\cdot c}$. The sum over a period telescopes to $e^{5i\theta_\mu} - 1$, because the period ends at $5e_\mu$. Hence $\sum_\nu V_{\mu\nu}\partial'_\nu = \frac{1}{5} D_\mu^{5\eta,1}(p)\eta^{-1}(e^{5i\eta p_\mu} - 1) = D_\mu^{5\eta,1}(p)\,\partial_\mu^{5\eta}(p) = \partial_\mu^1(p)$ for $\mu \in \{2, 3\}$, and for $\mu \in \{0, 1\}$ it is $D_\mu^{\eta,1}\partial_\mu^\eta = \partial_\mu^1$.

(iv) For $n \in \mathbb{Z}^4$ and $x \in T_\eta$ the product $2\pi\eta^{-1}\rho n \cdot x$ is a multiple of 2π, because ρ is an isometry and $\rho x \in \eta\mathbb{Z}^4$; this gives the periodicity of u_k, and that of V follows in the same way from the positions of the bonds. The inclusion holds because $\eta^{-1}\omega\mathbb{Z}[i] \subset \mathbb{Z}[i]$ by Lemma 2.1(iv). The value $u_k(0) = 1$ is the count of Lemma 6.2(i). For $l \in 2\pi\mathbb{Z}^4$ the function $x \mapsto e^{il\cdot x}$ is a character of the group T_η which is trivial on $\mathbb{Z}^4$, and $B_1(0)$ contains one point of every class of T_η modulo $\mathbb{Z}^4$; the sum of a nontrivial character over a finite group vanishes, and the character is trivial only if $l \cdot x \in 2\pi\mathbb{Z}$ for every $x \in T_\eta$, that is, $l \in 2\pi\eta^{-1}\rho\mathbb{Z}^4$. Finally $s_{22}(0) = 3$, $s_{23}(0) = 4$, $s_{32}(0) = 4$, $s_{33}(0) = -3$, which are the numbers $5e_\mu \cdot e'_\nu$ of (2.2). *q.e.d.*

Statement (iii) is the identity $Q_k\partial = \partial_1 Q'_k$ in the momentum representation, and (iv) contains the two properties of u_k which Bałaban uses after [2, (1.33), p. 23]. In his setting a momentum p of the fine lattice is written as $p = p' + l$ with p' in the dual of the unit lattice and $l \in 2\pi\mathbb{Z}^4$, the classes of l being taken modulo $2\pi\eta^{-1}\mathbb{Z}^4$. Here they are taken modulo $2\pi\eta^{-1}\rho\mathbb{Z}^4$, and the number of the classes is again η^{-4}.

The bounds on u_k are those of [2, (1.36), p. 24] with one change. For $p \in \mathbb{R}^4$ let q_μ, for $\mu \in \{0, 1\}$, be the number with $q_\mu \equiv p_\mu$ modulo $2\pi/\eta$ and $-\pi/\eta < q_\mu \le \pi/\eta$, and for $\mu \in \{2, 3\}$ the number with $q_\mu \equiv p_\mu$ modulo $2\pi/(5\eta)$ and $-\pi/(5\eta) < q_\mu \le \pi/(5\eta)$, so that $(p_2, p_3) = (q_2, q_3) + \frac{2\pi}{5\eta} m$ with $m \in \mathbb{Z}[i]$. We call p *central* if $m \in (3 + 4i)\mathbb{Z}[i]$.

Lemma 7.2 (Bounds on the Symbols). *(i)* $\kappa(0) = 1$*;* $\kappa(\theta + \frac{2\pi}{5}m) = \kappa(\theta)$ *for* $m \in (3 + 4i)\mathbb{Z}[i]$*;* $\kappa(\frac{2\pi}{5}m) = 0$ *for* $m \in \mathbb{Z}[i] \setminus (3 + 4i)\mathbb{Z}[i]$*; and* $|\kappa(\theta) - \kappa(\theta')| \le 5\sqrt{2}|\theta - \theta'|$.

(ii) For every $p \in \mathbb{R}^4$*, with* q *as above,*

$$|u_k(p)| \le \prod_{\mu=0}^{3} \min\Big(1, \frac{\pi|\partial_\mu^1(p)|}{2|q_\mu|}\Big) \cdot \begin{cases} 1 & \text{if } p \text{ is central,} \\ \min(1, 5\sqrt{2}\,\eta|(q_2, q_3)|) & \text{otherwise,} \end{cases} \tag{7.7}$$

and $|V_{\mu\nu}(p)| \le \frac{4}{5}\min(1, \pi|\partial_\mu^1(p)|/(2|q_\mu|))$ *for* $\mu, \nu \in \{2, 3\}$*; a quotient with* $q_\mu = 0$ *is read as* ∞.

(iii) For $p \in [-\pi, \pi]^4$*,*

$$|u_k(p)| \ge \Big(\frac{2}{\pi}\Big)^4 (1 - 10\pi\eta)\,, \tag{7.8}$$

and the matrix $V^{\mathrm{pl}}(p) \coloneqq (V_{\mu\nu}(p))_{\mu,\nu\in\{2,3\}}$ *satisfies* $|V^{\mathrm{pl}}(p)w| \ge \frac{2}{\pi}(1 - 32\eta)|w|$ *for every* $w \in \mathbb{C}^2$.

Proof. (i) $\kappa(0) = 1$ because C has 25 elements. For $m \in \mathbb{Z}[i]$ the function $c \mapsto e^{i\frac{2\pi}{5}m\cdot c}$ is a character of the group $\omega\mathbb{Z}[i]$, trivial on $5\mathbb{Z}[i]$, since $m \cdot c \in 5\mathbb{Z}$ for $c \in 5\mathbb{Z}[i]$; and C contains one element of every class of $\omega\mathbb{Z}[i]$ modulo $5\mathbb{Z}[i]$. The character is trivial on $\omega\mathbb{Z}[i]$ if and only if $m \cdot \omega w \in 5\mathbb{Z}$ for every $w \in \mathbb{Z}[i]$, that is, since $m \cdot \omega w = \mathrm{Re}(\bar\omega m \bar w)$, if and only if

$\bar\omega m \in 5\mathbb{Z}[i]$, which means $m \in 5\omega\mathbb{Z}[i] = (3+4i)\mathbb{Z}[i]$. If it is trivial, multiplying the terms of κ by it changes nothing, which is the periodicity; if it is not, its sum over the classes vanishes. The last assertion holds because $|c| < 5\sqrt{2}$ for $c \in \mathsf{C}$.

(ii) For a number a with $1/a$ an integer and a real t with $|at| \le \pi$ we have $|e^{iat}-1|/a = 2|\sin(at/2)|/a \ge 2|t|/\pi$, so that $|D^{a,1}_\mu(p)| = |\partial^1_\mu(p)|/|\partial^a_\mu(q)| \le \pi|\partial^1_\mu(p)|/(2|q_\mu|)$ for $a = \eta$, $\mu \in \{0,1\}$, and for $a = 5\eta$, $\mu \in \{2,3\}$, because ∂^a_μ has the period $2\pi/a$ in p_μ; and $|D^{a,1}_\mu| \le 1$ by (7.2). Further $|\kappa| \le 1$, and if p is not central, then $\kappa(\eta(p_2,p_3)) = \kappa(\eta(q_2,q_3)+\frac{2\pi}{5}m)-\kappa(\frac{2\pi}{5}m)$ by (i), which is at most $5\sqrt{2}\,\eta|(q_2,q_3)|$ in modulus. With (7.4) this is (7.7), and the bound on V follows from (7.5) and $|s_{\mu\nu}| \le 4$.

(iii) For $|t| \le \pi$ and $a \le 1$ we have $|e^{it}-1| = 2|\sin(t/2)| \ge 2|t|/\pi$ and $|e^{iat}-1|/a \le |t|$, so that $|D^{a,1}_\mu(p)| \ge 2/\pi$; and $|\kappa(\eta(p_2,p_3))-1| \le 5\sqrt{2}\,\eta|(p_2,p_3)| \le 10\pi\eta$ by (i). This gives (7.8). Let R_0 be the matrix with the entries $\frac15 s_{\mu\nu}(0) = e_\mu \cdot e'_\nu$, which is orthogonal by (2.2). Each $s_{\mu\nu}$ has at most four terms $\pm e^{i\theta\cdot c}$ with $|c| \le 3\sqrt{2}$, so that $|\frac15 s_{\mu\nu}(\theta) - \frac15 s_{\mu\nu}(0)| \le \frac{12\sqrt{2}}{5}|\theta|$, and the norm of the matrix of these differences is at most twice this number. For $\theta = \eta(p_2,p_3)$, $|\theta| \le \sqrt{2}\,\pi\eta$, this is at most $\frac{48\pi}{5}\eta \le 32\eta$. Since $V^{\mathrm{pl}}(p)$ is the product of the diagonal matrix of the numbers $D^{5\eta,1}_\mu(p)$, of modulus at least $2/\pi$, with the matrix $\frac15 s(\theta)$, the assertion follows. *q.e.d.*

The first product in (7.7) is Bałaban's bound $\prod_\mu |p'_\mu|/|p'_\mu + l_\mu|$, with the difference that in the turned directions q_μ is the representative of p_μ in an interval of the length $2\pi/(5\eta)$ rather than $2\pi/\eta$. Each value of q therefore belongs to 25 of the classes of l; one of them is central, and on the other 24 the symbol carries the additional factor $\eta|q|$, while $\Delta(p)$ is of the order of η^{-2} there.

As a first application we give the bound on the operator $\phi = Q_k \Delta^{-1} Q_k^*$ of [2, (1.58), (1.62), pp. 27–28], which in the tilted run is a matrix in the turned directions. For p' in the dual of the unit lattice let

$$\phi_{\mu\mu'}(p') \coloneqq \sum_l \frac{|u_k(p'+l)|^2}{\Delta(p'+l)} \sum_\nu V_{\mu\nu}(p'+l)\,\overline{V_{\mu'\nu}(p'+l)}\,, \tag{7.9}$$

the sum over the classes of $l \in 2\pi\mathbb{Z}^4$ modulo $2\pi\eta^{-1}\rho\mathbb{Z}^4$. By Lemma 7.1(i) and (ii) this is the momentum representation of ϕ: $\widetilde{(\phi\omega)}_\mu(p') = \sum_{\mu'} \phi_{\mu\mu'}(p')\tilde\omega_{\mu'}(p')$. It is diagonal for $\mu \in \{0,1\}$, where it is the function ϕ_μ of [2, (1.62)] with the tilted u_k, and a Hermitian matrix on the indices $\{2,3\}$.

Lemma 7.3 (The Matrix ϕ)**.** *Let $32\eta \le \frac12$. There are constants $0 < \gamma_0 \le \gamma_1$, which do not depend on k and L, such that $\gamma_0|w|^2 \le \Delta_0(p')\sum_{\mu,\mu'} \bar w_\mu \phi_{\mu\mu'}(p') w_{\mu'} \le \gamma_1|w|^2$ for every $p' \in [-\pi,\pi]^4$, $p' \ne 0$, and every $w \in \mathbb{C}^4$.*

Proof. The quadratic form is $\sum_l |u_k(p'+l)|^2 \Delta_0(p')\Delta(p'+l)^{-1}|V(p'+l)^* w|^2$, a sum of non-negative terms. For the lower bound we keep the term $l = 0$. Since $|e^{it}-1| \le |t|$, we have $\Delta(p') \le \sum_\nu |p'\cdot e'_\nu|^2 = |p'|^2$, and $\Delta_0(p') \ge (4/\pi^2)|p'|^2$ because $|e^{it}-1| \ge 2|t|/\pi$ for $|t| \le \pi$. With Lemma 7.2(iii), applied to V^{pl} and to the adjoint of this invertible matrix, and with $|D^{\eta,1}_\mu(p')| \ge 2/\pi$ for $\mu \in \{0,1\}$, the term is at least $(4/\pi^2)(2/\pi)^8(1-10\pi\eta)^2(2/\pi)^2(1-32\eta)^2|w|^2$, and $10\pi\eta \le 32\eta \le \frac12$.

For the upper bound, $|V(p)^* w| \le \frac85 |w|$ by Lemma 7.2(ii). Let $\hat p$ be the representative of $p'+l$ with $|\eta\hat p \cdot e'_\nu| \le \pi$ for every ν. Then $\Delta(p'+l) = \Delta(\hat p) \ge (4/\pi^2)|\hat p|^2$, and $\Delta_0(p') \le |p'|^2$. For $l = 0$ we have $\hat p = p'$ and the term is at most $(\pi^2/4)(8/5)^2|w|^2$. For $l \ne 0$ the difference

$\hat p - p'$ is a nonzero element of $2\pi\mathbb{Z}^4$ by Lemma 7.1(iv), so that one coordinate of $\hat p$ is at least π in modulus, and $\Delta_0(p')/\Delta(p'+l) \le (\pi^2/4)|p'|^2/\pi^2 \le \pi^2$. By (7.7), $|u_k(p'+l)| \le \prod_\mu \min(1, \pi/|q_\mu|)$, since $|\partial^1_\mu| \le 2$, and each value of q, which is of the form $q_\mu = p'_\mu + 2\pi n_\mu$ with integers n_μ, belongs to at most 25 classes of l. Hence the terms with $l \ne 0$ add up to at most $25\pi^2(8/5)^2 \prod_\mu \sum_{n\in\mathbb{Z}} \min(1, \pi/|p'_\mu + 2\pi n|)^2 |w|^2 \le 25\pi^2(8/5)^2(1+\pi^2/4)^4|w|^2$, because $|p'_\mu + 2\pi n| \ge \pi(2|n|-1)$ for $n \ne 0$ and $\sum_{n\ge1}(2n-1)^{-2} = \pi^2/8$. *q.e.d.*

8 The Cells

Several of Bałaban's arguments are local. They use paths of fine bonds inside a block, and they use the lower bound for the Laplace operator of a block with Neumann conditions on the functions of mean zero, which for a cube of the fine lattice is explicit [1, (2.27), p. 580]. In the tilted run a block is a cube of the coarse lattice filled with points of the turned lattice, and neither the paths nor the bound are evident. Both follow from the description of a block as a union of *cells*, the blocks $B_{5\eta}(z)$, $z \in 5\eta\mathbb{Z}^4$, of the scale 5η: by Lemma 6.2(i) the cells of a block form a cubic array with the axes e_μ, and each of them is a translate of the same set of 5^4 points, which we now describe.

Lemma 8.1 (The Cell)**.** *Let $\mathsf{R} \subset \mathbb{Z}^2$ be the set of the pairs (m,n) with $m \in \{3,4\}$ for $n = -3$, $2 \le m \le 5$ for $n = -2$, $1 \le m \le 6$ for $n = -1$, $0 \le m \le 6$ for $n = 0$, $2 \le m \le 5$ for $n = 1$, and $m \in \{3,4\}$ for $n = 2$.*

(i) $\mathsf{C} = \{me'_2 - ne'_3 \,|\, (m,n) \in \mathsf{R}\}$, the vectors e'_2 and e'_3 being read in the plane. Hence the cell of $z \in 5\eta\mathbb{Z}^4$ consists of the points $z + \eta(n_0e_0 + n_1e_1 + me'_2 - ne'_3)$ with $n_0, n_1 \in \{0,\dots,4\}$ and $(m,n) \in \mathsf{R}$.

(ii) Every row $\{m \,|\, (m,n) \in \mathsf{R}\}$ is an interval which contains 3 and 4.

(iii) The points $z + \eta(3e'_2 + 4e'_3)$ and $z + \eta(4e'_2 - 3e'_3)$ are the corners $z + 5\eta e_2$ and $z + 5\eta e_3$ of the neighbouring cells.

Proof. (i) An element of $\omega\mathbb{Z}[i]$ is $c = \omega(m+ni) = m\omega + n\omega i$ with integers m, n, and ω and $\omega i = (-4+3i)/5$ are the vectors e'_2 and $-e'_3$ of the plane by (2.2). Its real and imaginary parts are $(3m-4n)/5$ and $(4m+3n)/5$, so that $c \in \mathsf{C}$ if and only if $0 \le 3m-4n < 25$ and $0 \le 4m+3n < 25$, that is, $\max(\frac{4}{3}n, -\frac{3}{4}n) \le m < \min(\frac{25+4n}{3}, \frac{25-3n}{4})$. For $n = -3,\dots,2$ these bounds are $\frac{9}{4} \le m < \frac{13}{3}$, $\frac{3}{2} \le m < \frac{17}{3}$, $\frac{3}{4} \le m < 7$, $0 \le m < \frac{25}{4}$, $\frac{4}{3} \le m < \frac{11}{2}$, and $\frac{8}{3} \le m < \frac{19}{4}$, which give the rows of R. For $n \ge 3$ they require $m \ge 4$ and $m < 4$, and for $n \le -4$ they require $m \ge 3$ and $m < 3$. (ii) is read off from the list, and (iii) is (2.2). *q.e.d.*

By (ii) and (iii) two neighbouring corners of cells are joined by short paths which stay in the cell of the first: for the direction e_2 three steps $\eta e'_2$ followed by four steps $\eta e'_3$, through the points $(m,n) = (1,0),(2,0),(3,0),(3,-1),(3,-2),(3,-3)$ of R, and for the direction e_3 four steps $\eta e'_2$ followed by three steps $-\eta e'_3$, through $(1,0),\dots,(4,0),(4,1),(4,2)$. We call these paths, and the straight paths of five bonds in the directions e_0 and e_1, the *connectors* from z.

Definition 8.2 (Contours in a Block)**.** Let a be a scale, $y \in a\mathbb{Z}^4$, and $x \in B_a(y)$, let $z \in 5\eta\mathbb{Z}^4$ be the corner of the cell of x, and write $x = z + \eta(n_0e_0 + n_1e_1 + me'_2 - ne'_3)$ as in Lemma 8.1(i). The contour $\Gamma^a_{y,x}$ is the following path of bonds of T_η from y to x. It first goes from y to z through corners of cells, changing the coordinates in the order $\mu = 3,2,1,0$, each step from

a corner to the next along the connector. It then makes n_0 steps ηe_0 and n_1 steps ηe_1, and finally it goes through the points of the cell with the plane coordinates $(1,0)$, $(2,0)$, $(3,0)$, from there along the column $m = 3$ to $(3,n)$, and along the row n to (m,n).

By Lemma 8.1 every site of $\Gamma^a_{y,x}$ lies in $B_a(y)$, the part of the contour after z lies in the cell of x and has at most $4+4+9$ bonds, and the contour has at most $\frac{7}{5}\cdot 4a/\eta + 17$ bonds. These contours take the place of [2, (1.7), p. 18] and of the contours $\Gamma^{(k)}_{y,x}$ of [1, (1.4), p. 572], which are required to lie in the block.

Lemma 8.3 (A Poincaré Inequality)**.** *For every scale a, every $y \in a\mathbb{Z}^4$, and every function λ on $B_a(y)$ with $\sum_{x\in B_a(y)} \lambda(x) = 0$,*

$$\sum_{x\in B_a(y)} |\lambda(x)|^2 \le C_P \Big(\frac{a}{\eta}\Big)^2 \sum_b |\lambda(b_+) - \lambda(b_-)|^2 \,, \qquad C_P \coloneqq 1550 \,, \tag{8.1}$$

the sum on the right over the bonds $b = \langle b_-, b_+\rangle$ of T_η with both ends in $B_a(y)$.

Proof. Write $\delta\lambda(b) \coloneqq \lambda(b_+) - \lambda(b_-)$, and let $N \coloneqq a/(5\eta)$, so that $B_a(y)$ is the union of N^4 cells, whose corners form a cube of N^4 points of $5\eta\mathbb{Z}^4$. For a point x of a cell with the corner z, the last part of the contour of Definition 8.2 is a path of at most 17 bonds inside the cell, so that $|\lambda(x)-\lambda(z)|^2$ is at most 17 times the sum of $|\delta\lambda|^2$ over the bonds of the cell; summing over the 625 points,

$$\sum_x |\lambda(x) - \lambda(z)|^2 \le 625\cdot 17 \sum_b |\delta\lambda(b)|^2 \,, \tag{8.2}$$

the sums over the points and the bonds of the cell. Let Λ be the restriction of λ to the corners, and $\bar\Lambda$ its mean. For two neighbouring corners z and $z+5\eta e_\mu$ in $B_a(y)$ the connector has at most 7 bonds, all with both ends in $B_a(y)$, so that $|\Lambda(z+5\eta e_\mu) - \Lambda(z)|^2$ is at most 7 times the sum of $|\delta\lambda|^2$ over its bonds, and a bond of T_η lies on at most four connectors, those from the corner of its cell. For a function f on $\{0,\dots,N-1\}$ with the mean $\bar f$ we have $\sum_i |f_i - \bar f|^2 = \frac{1}{2N}\sum_{i,j}|f_i - f_j|^2$ and $|f_i - f_j|^2 \le N\sum_m |f_{m+1} - f_m|^2$, hence $\sum_i |f_i - \bar f|^2 \le \frac12 N^2 \sum_m |f_{m+1} - f_m|^2$. On the cube of the corners, let E_μ be the average over the coordinate μ; then $\Lambda - \bar\Lambda = \sum_{\mu=0}^3 E_0\cdots E_{\mu-1}(1-E_\mu)\Lambda$ is a sum of mutually orthogonal functions, the operators E_μ are contractions, and the inequality in one variable applies to $(1-E_\mu)\Lambda$ on every line. Therefore

$$\sum_z |\Lambda(z) - \bar\Lambda|^2 \le \frac{N^2}{2}\sum_\mu \sum_z |\Lambda(z+5\eta e_\mu) - \Lambda(z)|^2 \le \frac{N^2}{2}\cdot 7\cdot 4 \sum_b |\delta\lambda(b)|^2 \,, \tag{8.3}$$

the last sum over the bonds with both ends in $B_a(y)$. Since λ has the mean zero, $\sum_x |\lambda(x)|^2 \le \sum_x |\lambda(x) - \bar\Lambda|^2$, and $|\lambda(x) - \bar\Lambda|^2 \le 2|\lambda(x) - \lambda(z)|^2 + 2|\Lambda(z) - \bar\Lambda|^2$ for x in the cell of z. With (8.2) and (8.3) the left side of (8.1) is at most $(2\cdot 625\cdot 17 + 2\cdot 625\cdot 14N^2)\sum_b |\delta\lambda(b)|^2 \le 38750\, N^2 \sum_b |\delta\lambda(b)|^2$, and $38750/25 = 1550$. *q.e.d.*

In the notation of [1, (2.27)], the lemma for $a = 1$ says that the Laplace operator of a unit block with Neumann conditions is at least C_P^{-1} on the functions of mean zero, where for a cube of $\eta\mathbb{Z}^4$ the bound is π^2.

9 Bounds in Position Space

Bałaban obtains the bounds of his Proposition 1.1 and the inequality [2, (1.67)] by inspection of the momentum representation, in which every operator is a scalar in the index of the field. With the matrices of Section 7 this inspection becomes laborious, and we prove the same statements in another way: four inequalities for the tilted average, proved by counting paths, the Poincaré inequality of Lemma 8.3, and the momentum representation of two scalar operators only. In this section the fine lattice is the torus T_η/Γ' with a group of periods $\Gamma' \subset \mathbb{Z}^4$, the scalar products are the sums over the points, the positive bonds, or the plaquettes, with the weight η^4 on the fine lattice and 1 on the unit lattice, and $\|\cdot\|$ are the corresponding norms. For a field A and a positive direction ν we write $(\partial'_\nu A)(b) \coloneqq \eta^{-1}(A(b+\eta e'_\nu) - A(b))$, so that $\langle A, \Delta A\rangle = \sum_\nu \|\partial'_\nu A\|^2 = \|\partial A\|^2 + \|\partial^* A\|^2$ by [2, (1.21)], and ∂_1, ∂_1^* are the operations of the unit lattice.

Lemma 9.1 (Four Inequalities for the Average)**.** *For every real field A on T_η/Γ',*

(i) $\|Q_k A\|^2 \le 2\|A\|^2$;

(ii) $\|\partial_1^* Q_k A\|^2 \le 62\,\langle A, \Delta A\rangle$;

(iii) $\|\partial_1 Q_k A\|^2 \le 24\|\partial A\|^2$;

(iv) $\|A\|^2 \le (2C_P + 64)\langle A, \Delta A\rangle + 4\|Q_k A\|^2$.

Proof. Let n_μ be the number of the bonds of a staircase σ^1_μ, so that $n_\mu \le \frac{7}{5\eta}$, and recall from the proof of Lemma 6.4(ii) that a bond of T_η lies on at most $\frac{7}{5\eta}$ of the staircases $\sigma^1_\mu(x)$, $x \in T_\eta$, $\mu = 0, \dots, 3$. We use the Cauchy–Schwarz inequality in the form $|\sum_{t\in\mathcal{T}} c_t|^2 \le |\mathcal{T}| \sum_t |c_t|^2$.

(i) By (6.3), $(Q_k A)(\langle y, y + e_\mu\rangle)$ is η^5 times a sum of the values of A over the $\eta^{-4} n_\mu$ pairs of a point $x \in B_1(y)$ and a bond of $\sigma^1_\mu(x)$. Its square is therefore at most $\eta^{10}\eta^{-4} n_\mu$ times the sum of $|A(b)|^2$ over these pairs, and in the sum over y and μ every bond occurs at most $\frac{7}{5\eta}$ times. Hence $\|Q_k A\|^2 \le \frac{49}{25}\eta^4 \sum_b |A(b)|^2$.

(ii) We have $(\partial_1^* B)(y) = \sum_\mu (B(\langle y - e_\mu, y\rangle) - B(\langle y, y + e_\mu\rangle))$, and $(Q_k A)(\langle y, y + e_\mu\rangle) - (Q_k A)(\langle y - e_\mu, y\rangle) = \eta^5 \sum_{x\in B_1(y)} (A(\sigma^1_\mu(x)) - A(\sigma^1_\mu(x - e_\mu)))$. The staircase $\sigma^1_\mu(x)$ is the translate of $\sigma^1_\mu(x - e_\mu)$ by e_μ. For a bond b the difference $A(b + e_\mu) - A(b)$ is the sum of the n_μ differences of A between the translates of b by two consecutive sites of $\sigma^1_\mu(0)$, each of which is $\pm\eta(\partial'_\nu A)(b'')$ for a translate b'' of b and a direction ν. The difference of the two averages is therefore $\eta^5 \cdot \eta$ times a sum of $\eta^{-4} n_\mu^2$ values $\pm(\partial'_\nu A)(b'')$, and in the sum over y every pair (b'', ν) occurs at most n_μ^2 times. This gives $\sum_y |\cdot|^2 \le \eta^{12}\eta^{-4} n_\mu^4 \sum_{b'',\nu} |(\partial'_\nu A)(b'')|^2 \le (\frac{7}{5})^4 \langle A, \Delta A\rangle$ for every μ, and $|(\partial_1^* Q_k A)(y)|^2$ is at most four times the sum of the squares of the four differences; $16(\frac{7}{5})^4 \le 62$.

(iii) Let P be the plaquette of the unit lattice with the corners y, $y + e_\mu$, $y + e_\mu + e_\nu$, $y + e_\nu$. Since $B_1(y + e_\mu) = B_1(y) + e_\mu$, we have $(\partial_1 Q_k A)(P) = \eta^5 \sum_{x\in B_1(y)} A(\ell(x))$ with the closed path $\ell(x)$ which runs along $\sigma^1_\mu(x)$ and $\sigma^1_\nu(x + e_\mu)$ and back along $\sigma^1_\mu(x + e_\nu)$ and $\sigma^1_\nu(x)$. Let $s_0 = 0, s_1, \dots, s_{n_\mu} = e_\mu$ be the sites of $\sigma^1_\mu(0)$ and $s'_0, \dots, s'_{n_\nu}$ those of $\sigma^1_\nu(0)$. The points $x + s_i + s'_j$ form a mesh whose boundary is $\ell(x)$, and the sum of A over $\ell(x)$ is the sum over i and j of the sums of A over the closed paths through $x + s_i + s'_j$, $x + s_{i+1} + s'_j$, $x + s_{i+1} + s'_{j+1}$, $x + s_i + s'_{j+1}$. If the two steps $s_{i+1} - s_i$ and $s'_{j+1} - s'_j$ are parallel, such a path runs through each of its bonds once in each sense and contributes nothing; otherwise it is the boundary of a plaquette p of T_η and contributes $\pm\eta(\partial A)(p)$. Hence $(\partial_1 Q_k A)(P)$ is $\eta^5 \cdot \eta$ times a sum

of at most $\eta^{-4}n_\mu n_\nu$ values $\pm(\partial A)(p)$, and in the sum over y every plaquette occurs at most $n_\mu n_\nu$ times. This gives $\sum_y|(\partial_1 Q_k A)(P)|^2 \le (\frac{7}{5})^4\|\partial A\|^2$ for each of the six pairs $\mu<\nu$, and $6(\frac{7}{5})^4 \le 24$.

(iv) Let $A_\nu(x) \coloneqq A(\langle x, x+\eta e'_\nu\rangle)$, and let $\bar A_\nu(y)$ be the mean of A_ν over $B_1(y)$. Lemma 8.3 with $a=1$, applied to $A_\nu - \bar A_\nu(y)$ on every block, gives $\sum_\nu\sum_x \eta^4|A_\nu(x)-\bar A_\nu(y(x))|^2 \le C_P\langle A,\Delta A\rangle$, where $y(x)$ is the corner of the block of x, because the differences of A_ν along the bonds of a block are among the numbers $\eta(\partial'_{\nu'}A)(b)$. Since a block has η^{-4} points, $\|A\|^2 \le 2C_P\langle A,\Delta A\rangle + 2\sum_y\sum_\nu|\bar A_\nu(y)|^2$. By Lemma 6.4(ii) the average of the constant field with the values $\bar A_\nu(y)$ is $\sum_\nu(e_\mu\cdot e'_\nu)\bar A_\nu(y)$, so that $(Q_kA)(\langle y,y+e_\mu\rangle) - \sum_\nu(e_\mu\cdot e'_\nu)\bar A_\nu(y)$ is η^5 times the sum, over $x\in B_1(y)$ and over the bonds of $\sigma^1_\mu(x)$, of the differences between the value of A on the bond and the value of the same component at x. Each of these differences is a sum of at most n_μ terms $\pm\eta(\partial'_{\nu'}A)(b'')$ along the staircase, and the counting of (ii) bounds the sum over y of the squares by $(\frac{7}{5})^4\langle A,\Delta A\rangle$ for every μ. The matrix $(e_\mu\cdot e'_\nu)$ is orthogonal, so that $\sum_\nu|\bar A_\nu(y)|^2 \le 2\sum_\mu|(Q_kA)(\langle y,y+e_\mu\rangle)|^2 + 2\sum_\mu|\cdot|^2$ with the differences just bounded, and $\sum_y\sum_\nu|\bar A_\nu(y)|^2 \le 2\|Q_kA\|^2 + 8(\frac{7}{5})^4\langle A,\Delta A\rangle$. Since $16(\frac{7}{5})^4 \le 64$, this is (iv). *q.e.d.*

The two scalar operators are those of [2, (1.38), (1.58)] for functions. As there, Δ^{-1} denotes the inverse of the Laplace operator $\Delta=\partial^*\partial$ of T_η/Γ' on the functions orthogonal to the constants, and $\Delta_0 \coloneqq \partial_1^*\partial_1$ is the Laplace operator of the unit lattice.

Lemma 9.2 (Two Scalar Operators)**.** *Let $10\pi\eta \le \frac12$. On the functions of the unit lattice which are orthogonal to the constants, the operators $\mathsf{S} \coloneqq Q'_k\Delta^{-2}Q'^*_k$ and $\mathsf{T} \coloneqq Q'_k\Delta^{-1}Q'^*_k$ satisfy*

$$\mathsf{S} \ge c_{\mathsf{S}}\Delta_0^{-2}\,, \qquad \mathsf{T} \le C_{\mathsf{T}}\Delta_0^{-1}\,, \qquad c_{\mathsf{S}} \coloneqq \tfrac14(\tfrac{2}{\pi})^{12}\,, \quad C_{\mathsf{T}} \coloneqq \tfrac{\pi^2}{4} + 100(1+\tfrac{\pi^2}{4})^4\,. \tag{9.1}$$

Proof. By Lemma 7.1(i) and its adjoint, both operators are multiplications in the momentum representation of the unit lattice, by $\mathsf{S}(p') = \sum_l|u_k(p'+l)|^2\Delta(p'+l)^{-2}$ and $\mathsf{T}(p') = \sum_l|u_k(p'+l)|^2\Delta(p'+l)^{-1}$, for $p'\in[-\pi,\pi]^4$, $p'\neq0$. As in the proof of Lemma 7.3, $(4/\pi^2)|p'|^2 \le \Delta_0(p') \le |p'|^2$ and $(4/\pi^2)|p'|^2 \le \Delta(p') \le |p'|^2$. The term $l=0$ of S is at least $(2/\pi)^8(1-10\pi\eta)^2|p'|^{-4}$ by (7.8), and $|p'|^{-4} \ge (4/\pi^2)^2\Delta_0(p')^{-2}$. The term $l=0$ of T is at most $(\pi^2/4)|p'|^{-2} \le (\pi^2/4)\Delta_0(p')^{-1}$. For $l\neq0$ we have $\Delta(p'+l)\ge4$ and $\sum_{l\neq0}|u_k(p'+l)|^2 \le 25(1+\pi^2/4)^4$ by the proof of Lemma 7.3, and $1 \le 16\Delta_0(p')^{-1}$. *q.e.d.*

Proposition 9.3 (Proposition 1.1 and the Inequality (1.67) for the Tilted Run)**.** *Let $32\eta\le\frac12$ and $a>0$. There are constants $\gamma_0>0$ and C, which depend on a and do not depend on k, L, and the torus, such that the operator $\Delta_a = \Delta - \partial P\partial^* + aQ_k^*Q_k$ of [2, (1.69)] and its inverse G satisfy*

$$\Delta_a \ge \gamma_0(\Delta+I)\,, \tag{9.2}$$

$$\|GJ\|,\ \|\nabla GJ\|,\ \|G\nabla^*J\|,\ \|\nabla G\nabla^*J\| \le \gamma_0^{-1}\|J\|\,, \qquad \|\nabla\nabla GJ\|,\ \|G\nabla^*\nabla^*J\| \le C\|J\|\,, \tag{9.3}$$

where $\nabla A \coloneqq (\partial'_\nu A)_\nu$, and such that the action Δ_k of [2, (1.65)] satisfies

$$\tfrac{1}{24}\|\partial_1 B\|^2 \le \langle B,\Delta_k B\rangle \le \gamma_\phi^{-1}\|\partial_1 B\|^2\,, \tag{9.4}$$

where γ_ϕ denotes the lower constant γ_0 of Lemma 7.3.

Proof. We use the operator identities of [2], which hold in the tilted run by Section 10: $R = I - P$ is the orthogonal projection onto $\Delta N(Q'_k)$, where $N(Q'_k)$ is the null space of

Q'_k; $P = \Delta^{-1}Q'^*_k \mathsf{S}^{-1} Q'_k \Delta^{-1}$ on the functions orthogonal to the constants; and $\langle A, \Delta_a A\rangle = \|\partial A\|^2 + \|R\partial^* A\|^2 + a\|Q_k A\|^2$.

A gauge transformation. Let A be a field. There is $\lambda \in N(Q'_k)$ with $\Delta\lambda = R\partial^* A$, and $A_s := A - \partial\lambda$ satisfies $\partial A_s = \partial A$, $Q_k A_s = Q_k A - \partial_1 Q'_k \lambda = Q_k A$ by Lemma 6.4(i), and $\partial^* A_s = \partial^* A - \Delta\lambda = P\partial^* A$. Hence

$$\langle A, \Delta_a A\rangle = \|\partial A_s\|^2 + a\|Q_k A_s\|^2 + \|\Delta\lambda\|^2 \,. \tag{9.5}$$

The gauge part. Since $\partial\partial\lambda = 0$, we have $\langle \partial\lambda, \Delta\partial\lambda\rangle = \|\partial^*\partial\lambda\|^2 = \|\Delta\lambda\|^2$. The function λ has the mean zero on every unit block, so that $\|\lambda\|^2 \le C_P\|\partial\lambda\|^2$ by Lemma 8.3 with $a = 1$, and $\|\partial\lambda\|^2 = \langle\lambda, \Delta\lambda\rangle \le \|\lambda\|\|\Delta\lambda\|$; together, $\|\partial\lambda\|^2 \le C_P\|\Delta\lambda\|^2$.

The divergence on the slice. Put $f := \partial^* A_s$, so that $f = Pf$ is orthogonal to the constants, and $\mu := Q'_k\Delta^{-1} f$. Then $\|f\|^2 = \langle f, Pf\rangle = \langle \mu, \mathsf{S}^{-1}\mu\rangle \le c_{\mathsf{S}}^{-1}\|\Delta_0\mu\|^2$ by Lemma 9.2. The field $A^T := A_s - \partial\Delta^{-1} f$ satisfies $\partial^* A^T = 0$ and $\partial A^T = \partial A_s$, and $\partial_1\mu = Q_k\partial\Delta^{-1} f = Q_k A_s - Q_k A^T$ by Lemma 6.4(i). Hence $\Delta_0\mu = \partial_1^* Q_k A_s - \partial_1^* Q_k A^T$, where $\|\partial_1^* Q_k A_s\|^2 \le 16\|Q_k A_s\|^2$, each bond of the unit lattice entering ∂_1^* at its two ends, and $\|\partial_1^* Q_k A^T\|^2 \le 62\langle A^T, \Delta A^T\rangle = 62\|\partial A_s\|^2$ by Lemma 9.1(ii). Therefore

$$\|\partial^* A_s\|^2 \le C_4(\|\partial A_s\|^2 + \|Q_k A_s\|^2) \,, \qquad C_4 := 124\, c_{\mathsf{S}}^{-1} \,. \tag{9.6}$$

The lower bound. By (9.6), $\langle A_s, \Delta A_s\rangle \le (1 + C_4)(\|\partial A_s\|^2 + \|Q_k A_s\|^2)$, and by Lemma 9.1(iv), $\|A_s\|^2 \le (2C_P + 64)\langle A_s, \Delta A_s\rangle + 4\|Q_k A_s\|^2$. With the bounds on the gauge part and $\langle A, (\Delta + I)A\rangle \le 2\langle A_s, (\Delta + I)A_s\rangle + 2\langle\partial\lambda, (\Delta + I)\partial\lambda\rangle$, the identity (9.5) gives (9.2) with

$$\gamma_0^{-1} := 2\max\Big(\frac{(2C_P + 65)(1 + C_4) + 4}{\min(1, a)},\ 1 + C_P\Big) \,. \tag{9.7}$$

The bounds on G. By (9.2), Δ_a is invertible and $(\Delta + I)^{1/2} G (\Delta + I)^{1/2}$ has the norm at most γ_0^{-1}; since $\|\nabla A\|^2 = \langle A, \Delta A\rangle$, this is the first half of (9.3). On the torus the operators ∂'_ν commute with each other and with their adjoints, so that $\|\nabla\nabla A\|^2 = \|\Delta A\|^2$, and $\Delta G = I + \partial P\partial^* G - aQ_k^* Q_k G$. Here $\|Q_k^* Q_k G J\| \le 2\gamma_0^{-1}\|J\|$ by Lemma 9.1(i). For a function g orthogonal to the constants, $Pg = \Delta^{-1}Q'^*_k\omega$ with $\omega = \mathsf{S}^{-1}Q'_k\Delta^{-1} g$, so that $\|\partial Pg\|^2 = \langle Pg, \Delta Pg\rangle = \langle\omega, \mathsf{T}\omega\rangle \le C_{\mathsf{T}}\langle\omega, \Delta_0^{-1}\omega\rangle$, while $\|g\|^2 \ge \|Pg\|^2 = \langle\omega, \mathsf{S}\omega\rangle \ge c_{\mathsf{S}}\langle\omega, \Delta_0^{-2}\omega\rangle \ge \frac{1}{16}c_{\mathsf{S}}\langle\omega, \Delta_0^{-1}\omega\rangle$, because $\Delta_0 \le 16$. With $g = \partial^* GJ$, $\|g\| \le \|\nabla GJ\|$, this gives $\|\Delta GJ\| \le (1 + 2a\gamma_0^{-1} + (16C_{\mathsf{T}}/c_{\mathsf{S}})^{1/2}\gamma_0^{-1})\|J\|$, and the last bound of (9.3) follows by taking adjoints.

The action. By [2, (1.47)–(1.65)], $\langle B, \Delta_k B\rangle = \|\partial H_k B\|^2$ is the minimum of $\|\partial A\|^2$ over the fields with $Q_k A = B$ and $R\partial^* A = 0$, and by the gauge transformation above the second condition may be dropped. Lemma 9.1(iii) for $A = H_k B$ is the lower bound of (9.4). For the upper bound write $B = B_0 + B'$ with B_0 constant and B' orthogonal to the constants, and $B' = B^T + \partial_1\nu$ with $\nu := \Delta_0^{-1}\partial_1^* B'$, so that $\partial_1^* B^T = 0$ and $\langle B^T, \Delta_0 B^T\rangle = \|\partial_1 B^T\|^2 = \|\partial_1 B\|^2$. Let $\phi = Q_k\Delta^{-1}Q_k^*$, whose momentum representation is (7.9); by Lemma 7.3, $\phi^{-1} \le \gamma_\phi^{-1}\Delta_0$ on the fields orthogonal to the constants. The field $A := Q_k^* B_0 + \Delta^{-1}Q_k^*\phi^{-1}B^T + \partial Q'^*_k\nu$ satisfies $Q_k A = B_0 + B^T + \partial_1 Q'_k Q'^*_k\nu = B$, by Lemma 6.4 and because $Q'_k Q'^*_k = I$, and $\partial A = \partial\Delta^{-1}Q_k^*\phi^{-1}B^T$. Hence $\langle B, \Delta_k B\rangle \le \|\partial A\|^2 \le \langle\Delta^{-1}Q_k^*\phi^{-1}B^T, Q_k^*\phi^{-1}B^T\rangle = \langle B^T, \phi^{-1}B^T\rangle \le \gamma_\phi^{-1}\|\partial_1 B\|^2$. *q.e.d.*

The proposition has the content of [2, Proposition 1.1, (1.89), (1.90), and (1.67)]; the two bounds on second derivatives hold with a constant other than γ_0^{-1}.

10 The Modifications in the First Paper on the Propagators

We go through [2] in its order. The paper treats the Gaussian theory of one real field, with the action $\frac{1}{2}\langle \partial A, \partial A\rangle$ on the fine lattice, and it derives the propagators of the later papers from the averages. In the tilted run its fine lattice is T_η of (6.1), on a torus T_η/Γ' with a group of periods $\Gamma' \subset \mathbb{Z}^4$ in place of the box of [2, p. 17]; the operators ∂, ∂^*, and Δ are those of this lattice, whose bonds have the directions e'_ν, and a field has the components A_ν in these directions, while the fields B and the multipliers ω on the unit lattice have the components in the directions e_μ.

Section A, (1.1)–(1.15). The action [2, (1.3), (1.5)] and its gauge invariance [2, (1.4)] concern the fine lattice alone and are unchanged. The blocks [2, (1.6)] are those of Definition 6.1(i), and the average [2, (1.8), (1.11)] is replaced by Definition 6.1(iv); the transformation laws [2, (1.9), (1.13)] are Lemma 6.4(i). The contours [2, (1.7)] enter the axial gauge condition [2, (1.10)], which requires for every point x of a block a path of fine bonds from the corner y to x; we take the path which changes the coordinates of $x - y$ in the basis $\eta e'_\nu$ in the order $\nu = 3, 2, 1, 0$. Such a path may leave the block, by less than its side, which is of no consequence in [2, (1.10)–(1.15)]: these use only that the conditions $A(\Gamma_{y,x}) = 0$ fix the gauge transformations which vanish at the corners, one condition for each point $x \neq y$.

Section B, (1.16)–(1.20). The computation [2, (1.16)] of a composition is Lemma 6.3: the average of the first step of the tilted run is $Q_{L^{-1}}$ in the unit of the lattice reached after k steps, the later ones are Bałaban's, and the composition is Q_k. The formula [2, (1.20)] is Lemma 6.4(i).

Sections C to E, the operator identities. The derivations of [2, (1.21)–(1.28), (1.38)–(1.44), (1.46)–(1.60), (1.64), (1.65), (1.68)–(1.82), (1.91)–(1.107)] are calculations with the operators ∂, ∂^*, Δ of the fine lattice, with Q_k, Q'_k, their adjoints, and with ∂_1 on the unit lattice. They use the following properties and no others: the decomposition [2, (1.21)] of the action, which holds on every lattice with four orthogonal directions of bonds and hence on T_η; the identity [2, (1.55)], which is Lemma 6.4(i); that Q'_k maps constants to constants and their orthogonal complement to the orthogonal complement; and, in [2, (1.57)–(1.59), (1.74), (1.82)], that Q_k and Q_k^* map the constant fields to the constant fields with $Q_k^* Q_k = 1$ on them. The last two properties are Lemma 6.4(ii). These derivations therefore hold without change, and so do their results in operator form: the projection R [2, (1.38), (1.44)], the operator H_k [2, (1.60), (1.103)], the action Δ_k [2, (1.65)], the operator G [2, (1.81), (1.82)], and the identities [2, (1.95), (1.97), (1.99), (1.107)].

The momentum representation, (1.29)–(1.37) and (1.45). The Fourier transform [2, (1.29)] is that of the finite group T_η/Γ', whose dual consists of the classes modulo $2\pi\eta^{-1}\rho\mathbb{Z}^4$ of the p with $p \cdot \gamma \in 2\pi\mathbb{Z}$ for $\gamma \in \Gamma'$, and the decomposition $p = p' + l$ is as described after Lemma 7.1. The symbols [2, (1.31)] are replaced by (7.1) and (7.4). The solution [2, (1.32)–(1.35), (1.37)] uses of u_k only that $u_k(l) = 0$ for the classes $l \neq 0$ and $u_k(0) = 1$, which is Lemma 7.1(iv), and it holds without change. The bound $\Delta_0(p')/\Delta(p' + l) \leq O(1)(1 + |l|^2)^{-1}$ holds with l replaced by $\hat{l} \coloneqq \hat{p} - p'$ for the representative $\hat{p}$ of the proof of Lemma 7.3: by the estimates given there the quotient is at most $\pi^4/|\hat{p}|^2$, and $|\hat{p}| \geq \max(\pi, |\hat{l}| - 2\pi)$ for $\hat{l} \neq 0$. The bound [2, (1.36)] is replaced by (7.7). The formula [2, (1.45)] and the bounds on it are taken from [1], which we have not examined.

The momentum representation, (1.61)–(1.63), (1.66), (1.67), (1.83)–(1.88), and (1.101). The representation [2, (1.61)] of Q_k is replaced by Lemma 7.1(ii): the scalar v_μ becomes the matrix V. The function ϕ_μ of [2, (1.62)] becomes the matrix (7.9), and the bounds $\gamma_0 \leq$

$\Delta_0\phi_\mu \le \gamma_1$ stated there are Lemma 7.3. The formulae [2, (1.63), (1.66), (1.83)–(1.88), (1.101)] are the operator formulae [2, (1.60), (1.65), (1.81), (1.99)] written in components under the assumption that ϕ is diagonal. In the tilted run they have to be written with the matrices V and ϕ on the indices $\{2,3\}$, in which products such as $\overline{v_\mu}\phi_\mu^{-1}$ become $V^*\phi^{-1}$ and the function ϕ_μ of [2, (1.84)] becomes $\mathbf{1} + a\phi$. Bałaban reads three bounds off from these formulae. Two of them, the inequality [2, (1.67)] and Proposition 1.1 [2, (1.89), (1.90)], are Proposition 9.3, which we have proved without the formulae in components. The third, the bound on the sum over l stated after [2, (1.63)], from which the Hölder continuity of the derivatives of H_kB follows, we obtain by another route. By [2, (1.103)], $H_k = GQ_k^*(Q_kGQ_k^*)^{-1}$, so that the properties of H_k reduce to those of G, which are Proposition 16.2, and to the decay of the kernel of $(Q_kGQ_k^*)^{-1}$; this operator is $\Delta_k + a$, and its kernel decays by Lemma 17.1. The resulting bounds on H_k are Corollary 17.2.

Section F, Proposition 1.2. The bounds [2, (1.110)–(1.113)] are Proposition 16.2, and the bounds [2, (1.114)] are Lemma 16.1 for the first derivatives and Lemma 17.3 for the second. The proof is organised differently from Bałaban's. His expansion [2, (1.119)–(1.123)] of G over a partition of unity, with the commutators [2, (1.120), (1.121)] of the averages with functions, is not used; the decay comes from a weighted form of the lower bound (9.2). The bounds [2, (1.126), (1.127)] on the kernel of $\partial P\partial^*$ are Lemma 15.1, the identities [2, (1.132), (1.133)] are used as they stand, and the bounds [2, (1.135)–(1.137)] on the scalar operator are Propositions 14.3 and 14.4, which rest on Sections 12 and 13. The expansion over the partition of unity is, however, the model of the expansions of the later papers. Its commutators with the averages along the staircases are the subject of Lemma 17.4, which gives [2, (1.128)], so that the expansion is available in the tilted run as well.

11 The Modifications in the Paper on the Scalar Green's Functions

The paper [1] treats the operators $G_k(\Omega, A) = (-\Delta^\eta_{A,\Omega} + m^2 + aP_k(A))^{-1}$ for functions with values in $\mathbb{R}^N$, on a union Ω of big blocks, with Neumann conditions on its boundary and with an external field A. We have read its Sections 1 and 2, with the following result for the tilted run.

The setting, (1.1)–(1.8). The blocks [1, (1.1)] are those of Definition 6.1(i), and the big blocks are cubes of the unit lattice as before. The quadratic form [1, (1.3)] is the sum over the bonds of T_η with both ends in Ω. The covariant average [1, (1.4)] needs contours inside the blocks, which are those of Definition 8.2.

The random walk representation, (2.2)–(2.13). The partition of unity consists of functions of the coordinates x_μ/M, whose differences along a bond of T_η are $O(\eta/M)$ as before. The formulae [1, (2.3)–(2.5)] concern one bond or the bonds at one point and do not involve the blocks; the commutator [1, (2.8)] of $P_k(A)$ with a function uses the contours of the block only. The derivation of [1, (2.9)–(2.13)] is therefore unchanged.

Lemma 2.1 and Corollary 2.3. The proof of the lemma reduces the L^2-bounds, by the expansion [1, (2.24)–(2.26)] in the regular part of the field, to the lower bound [1, (2.27)] for one unit block at $A = 0$, where the Laplace operator of the block with Neumann conditions is bounded below by π^2 on the functions orthogonal to the constants. In the tilted run this bound is Lemma 8.3, with C_P^{-1} in place of π^2, so that [1, (2.28)] holds with $c_0^{-1} = \min\{C_P^{-1}, a_k\}$, and [1, (2.29)] is unchanged. We have not checked the expansion [1, (2.24)] itself, which is quoted from another paper. Corollary 2.3 follows from the lemma and the random walk representation as in the text.

Lemma 2.2 and the last assertion of the Theorem. These concern a rectangular parallelepiped $\square$ with Neumann conditions, and their proof passes from the operator of the whole lattice to that of $\square$ by multiple reflections [1, (2.42)]. The reflections in the faces of $\square$ which are orthogonal to e_2 or e_3 do not map T_η to itself: by Lemma 5.1 no reflection of the plane of these two directions preserves both lattices. The formula [1, (2.42)] is therefore not available in the tilted run, and neither are, by this proof, the bounds of Lemma 2.2 in the supremum, Hölder, and L^p norms up to the boundary of a parallelepiped. In the proof of the Theorem the lemma is applied to the cubes $\square_j$ in the interior of Ω, on which the functions h_j vanish near the boundary; a replacement of the operators of these cubes by operators without a boundary has not been carried out.

Lemma 2.4. For the whole lattice the representation [1, (2.43)–(2.48)] is the solution of a linear equation in the momentum representation, and it holds with the function u_j of (7.4). The bounds [1, (2.35), (2.36)], with the analytic continuation in p' on which the exponential decay rests, are Proposition 12.2; of the lower bound on u_j stated in [1, p. 585] for all l, which we have not proved for the tilted u_j, only the case $l = 0$ is used, and this case is (7.8). The bound [1, (2.37)] is Lemma 14.2, and the identity [1, (2.34)], which is quoted there from a paper that is not among our sources, is Lemma 13.1. The resulting bounds for the whole lattice at $A = 0$ are Proposition 14.3.

Section 3, (3.1)–(3.10). The representation [1, (3.6)] expresses the covariance $C_\Lambda^{(k)}$ of the unit lattice, the inverse of $\Delta^{(k)} + aL^{-2}P$ on a set Λ, through the operator [1, (3.5)], which has the averages of the next scale on $B^k(\Lambda)$ and those of the scale k elsewhere. It is derived by integrating over the blocks with the formula [1, (3.3)], which uses of the next average only that it is the mean over the L^d points of a block and of Q_k only that it is an average; it therefore holds for the tilted run. At $A = 0$ and on the whole lattice the operator [1, (3.5)] is $\mathcal{D}_{\Lambda'}^{-1}$ of Section 18, the relation between the two operators is Lemma 18.1(ii), read in the other direction, and the bounds on $C_\Lambda^{(k)}$ which the later papers use are Lemma 18.1(iii). The comparison [1, (3.9), (3.10)] of the operators of two sets is the resolvent identity, as in Lemma 20.5.

Section 4, (4.1)–(4.22). This is the lower bound of the quadratic form $\Delta^{(k)}$ by the squared differences on the unit lattice, Proposition 3.1′. Its proof separates pairs of neighbouring blocks by Neumann conditions and computes, for two blocks at a constant field, a Gaussian integral over the fields on the two blocks [1, (4.14)–(4.22)]. At $A = 0$ and on the whole lattice the statement is the inequality $a_j - a_j^2 Q'G'Q'^* \geq c_\Delta \Delta_0$ in the proof of Lemma 14.2, which we obtain from the momentum representation. For $A \neq 0$ we have not examined it.

Section 5, (5.1)–(5.27). The Theorem of this section bounds the inverse of an operator on a subset of $\mathbb{Z}^d$ which is bounded below and has an exponentially decaying kernel, together with its change under a change of the subset. It concerns a unit lattice of the paper. Its first statement [1, (5.7)] is Lemma 14.1, which we prove by a conjugation with weights and which holds on every countable set with a metric; the statement [1, (5.8)] on the change of the subset follows from the first and the formula (25.4), as in Proposition 25.1(i).

In the theory without matter fields the operators of this paper occur at $A = 0$ only: the scalar operators in a background field of [5] are those of the adjoint representation, and they are treated there by an expansion around $U = 1$ (Section 27). The statements of [1] for an external field $A \neq 0$, and its boxes with Neumann conditions, are therefore not used by the later sections.

12 The Kernel of the Scalar Operator on the Whole Lattice

We now carry out, for the tilted run, the part of the proof of [1, Lemma 2.4] which concerns the whole lattice: the bounds [1, (2.35), (2.36)] on the kernel of $G'Q_k'^*$, with their exponential decay. Bałaban obtains the decay by moving the integration over p' into the complex domain, and says of the required bounds on the integrand in a complex neighbourhood that they are "more troublesome, but equally elementary" [1, p. 586]. For the tilted symbols they follow from two observations: u_k is the Fourier transform of a probability measure on the unit cube, which bounds it and its differences in every strip, and its decay in l is read off from the product (7.4) for complex arguments as for real ones.

In this section the fine lattice is the infinite lattice T_η, $a > 0$, and $G' \coloneqq (-\Delta + aQ_k'^*Q_k')^{-1}$ on the square-summable functions on T_η, where $-\Delta = \partial^*\partial$ and the adjoint is taken with the weights η^4 on T_η and 1 on $\mathbb{Z}^4$, so that $(Q_k'^* g)(x) = g(y)$ for $x \in B_1(y)$ and $Q_k'^*Q_k'$ is the orthogonal projection onto the functions which are constant on the blocks. The inverse exists: by Lemma 8.3 with $a = 1$, applied on every block to the difference of a function and its mean, $-\Delta + aQ_k'^*Q_k' \geq \min(C_P^{-1}, a)$. We write $\mathcal{K}(x, y)$, $x \in T_\eta$, $y \in \mathbb{Z}^4$, for the kernel of $G'Q_k'^*$, that is, $(G'Q_k'^* g)(x) = \sum_y \mathcal{K}(x, y)g(y)$.

The symbols extend to entire functions of $p \in \mathbb{C}^4$: $u_k(p) = \sum_{x \in B_1(0)} \eta^4 e^{ip\cdot x}$, and

$$\Delta(p) \coloneqq \eta^{-2} \sum_\nu (2 - 2\cos(\eta p \cdot e_\nu')) \,, \tag{12.1}$$

which for real p is the function of (7.1); for real p, $\overline{u_k(p)} = u_k(-p)$. Both are periodic with respect to $2\pi\eta^{-1}\rho\mathbb{Z}^4$. For $p \in \mathbb{C}^4$ we define q from the real part of p as before Lemma 7.2, and $\hat{p} \in \mathbb{R}^4$ is the representative of the real part of p modulo $2\pi\eta^{-1}\rho\mathbb{Z}^4$ with $|\eta\hat{p} \cdot e_\nu'| \leq \pi$ for every ν.

Lemma 12.1 (The Symbols in a Strip)**.** *Let $10\eta \leq 1$, and let $p \in \mathbb{C}^4$ with the imaginary part τ, $|\tau_\mu| \leq \frac{1}{4}$ for every μ.*

(i) $|u_k(p)| \leq e^{9/4} \prod_\mu \min(1, \pi/|q_\mu|)$.

(ii) If s is the real part of p, then $|u_k(p) - u_k(s)| \leq 2\sum_\mu |\tau_\mu|$.

(iii) $\operatorname{Re}\Delta(p) \geq (4/\pi^2)|\hat{p}|^2 - 2|\tau|^2$ *and* $|\Delta(p)| \leq |\hat{p}|^2 + 2|\tau|^2$*; and if the real part s of p satisfies $|\eta s \cdot e_\nu'| \leq \pi$ for every ν, then* $|\Delta(p) - \Delta(s)| \leq 2|\tau|^2 + 4|s||\tau|$.

Proof. (i) Let $t \coloneqq \frac{1}{4}$. By (7.2), $|D_\mu^{a,1}(p)| \leq a\sum_{n<1/a} e^{an|\tau_\mu|} \leq e^t$. Further $|e^{ip_\mu} - 1| \leq 1 + e^t \leq 2e^t$, and for $z = ap_\mu$ with the real part ς and the imaginary part ϑ, $|e^{iz} - 1|^2 = (1 - e^{-\vartheta})^2 + 4e^{-\vartheta}\sin^2(\varsigma/2) \geq e^{-t}(4/\pi^2)(aq_\mu)^2$, because $\varsigma \equiv aq_\mu$ modulo 2π and $|aq_\mu| \leq \pi$. Hence $|D_\mu^{a,1}(p)| = |e^{ip_\mu} - 1|a/|e^{iap_\mu} - 1| \leq 2e^t e^{t/2}\pi/(2|q_\mu|)$, and altogether $|D_\mu^{a,1}(p)| \leq e^{2t}\min(1, \pi/|q_\mu|)$, for $a = \eta$ and for $a = 5\eta$. Since $|c| < 5\sqrt{2}$ for $c \in \mathsf{C}$ and the imaginary part of $\eta(p_2, p_3)$ has the length at most $\sqrt{2}\,\eta t$, $|\kappa(\eta p_2, \eta p_3)| \leq e^{10\eta t} \leq e^t$. The product (7.4) of four factors D and of κ gives (i).

(ii) For $x \in B_1(0)$ we have $0 \leq x_\mu < 1$, so that $|\tau \cdot x| \leq \sum_\mu |\tau_\mu| \leq 1$ and $|e^{ip\cdot x} - e^{is\cdot x}| = |e^{-\tau\cdot x} - 1| \leq 2\sum_\mu |\tau_\mu|$, because $e^v - 1 \leq 2v$ for $0 \leq v \leq 1$; and u_k is an average over these x.

(iii) By periodicity we may replace the real part of p by $\hat{p}$. Let $\varsigma \coloneqq \eta\hat{p}\cdot e_\nu'$ and $\vartheta \coloneqq \eta\tau \cdot e_\nu'$, so that $|\varsigma| \leq \pi$ and $|\vartheta| \leq \eta|\tau| \leq \frac{1}{20}$. Then $2 - 2\cos(\varsigma + i\vartheta) = 2(1 - \cos\varsigma) - 2\cos\varsigma(\cosh\vartheta - 1) + 2i\sin\varsigma\sinh\vartheta$, where $(2/\pi^2)\varsigma^2 \leq 1 - \cos\varsigma \leq \frac{1}{2}\varsigma^2$, $\cosh\vartheta - 1 \leq \vartheta^2$, and $|\sinh\vartheta| \leq 2|\vartheta|$. The real

part is therefore at least $(4/\pi^2)\varsigma^2 - 2\vartheta^2$, the modulus is $4\sin^2(\varsigma/2) + 4\sinh^2(\vartheta/2) \le \varsigma^2 + 2\vartheta^2$, and the difference from the value at $\vartheta = 0$ is at most $2\vartheta^2 + 4|\varsigma||\vartheta|$ in modulus. Summing over ν, with $\sum_\nu \varsigma^2 = \eta^2|\hat p|^2$, $\sum_\nu \vartheta^2 = \eta^2|\tau|^2$, and $\sum_\nu |\varsigma||\vartheta| \le \eta^2|\hat p||\tau|$, and dividing by η^2 gives (iii). *q.e.d.*

Proposition 12.2 (The Kernel of $G'Q_k'^*$)**.** *Let $32\eta \le \frac12$ and $0 < a_- \le a \le a_+$. There are constants $\delta_0 > 0$ and c_0, and for every $\alpha \in [0,1)$ a constant c_α, which depend on a_- and a_+ and not on k and L, such that for all $x, x' \in T_\eta$ with $|x - x'| \le 1$, all $y \in \mathbb{Z}^4$, and all ν,*

$$\begin{aligned} |\mathcal{K}(x,y)|\ ,\ |(\partial'_\nu \mathcal{K})(x,y)| &\le c_0 e^{-\delta_0|x-y|}\ , \\ |(\partial'_\nu \mathcal{K})(x,y) - (\partial'_\nu \mathcal{K})(x',y)| &\le c_\alpha |x-x'|^\alpha e^{-\delta_0 \min(|x-y|,|x'-y|)}\ , \end{aligned} \tag{12.2}$$

where ∂'_ν acts on the first argument.

Proof. The representation. We write a momentum of the fine lattice as $p'+l$ with $p' \in [-\pi,\pi]^4$ and l in the set $\mathfrak{L}$ of the η^{-4} classes of $2\pi\mathbb{Z}^4$ modulo $2\pi\eta^{-1}\rho\mathbb{Z}^4$. By Lemma 7.1(i), for f on T_η and g on $\mathbb{Z}^4$, $(\widetilde{Q'_k f})(p') = \sum_l u_k(p'+l)\tilde f(p'+l)$ and $(\widetilde{Q'^*_k g})(p'+l) = u_k(-p'-l)\tilde g(p')$. The equation $(-\Delta + aQ'^*_k Q'_k)\phi = Q'^*_k g$ therefore reads $\Delta(p'+l)\tilde\phi(p'+l) + a u_k(-p'-l)w(p') = u_k(-p'-l)\tilde g(p')$ with $w(p') \coloneqq \sum_l u_k(p'+l)\tilde\phi(p'+l)$; multiplying by $u_k(p'+l)/\Delta(p'+l)$ and summing over l gives $w = \mathsf{T}\tilde g/(1+a\mathsf{T})$ with $\mathsf{T}(p') = \sum_l u_k(p'+l)u_k(-p'-l)/\Delta(p'+l)$, and then, as in [1, (2.47), (2.48)],

$$\mathcal{K}(x,y) = (2\pi)^{-4}\int_{[-\pi,\pi]^4} \Phi(p')\,\mathrm{d}p'\ , \qquad \Phi(p') \coloneqq \sum_{l\in\mathfrak{L}} \frac{e^{i(p'+l)\cdot(x-y)}\,u_k(-p'-l)}{\Delta(p'+l)(1+a\mathsf{T}(p'))}\ . \tag{12.3}$$

The function Φ is periodic with respect to $2\pi\mathbb{Z}^4$, a translation of p' by an element of $2\pi\mathbb{Z}^4$ permuting the classes. We single out the class $l = 0$ and put, for p' in the set U of the $p' \in \mathbb{C}^4$ whose real parts have the coordinates at most $\pi + \frac14$ and whose imaginary parts have the coordinates at most $\frac14$ in modulus,

$$\mathsf{T}_*(p') \coloneqq \sum_{l\neq 0} \frac{u_k(p'+l)u_k(-p'-l)}{\Delta(p'+l)}\ , \qquad \mathcal{D}(p') \coloneqq \Delta(p')(1+a\mathsf{T}_*(p')) + a u_k(p')u_k(-p')\ , \tag{12.4}$$

so that $\Delta(p')(1+a\mathsf{T}(p')) = \mathcal{D}(p')$ and

$$\Phi(p') = \frac{1}{\mathcal{D}(p')}\Big(e^{ip'\cdot(x-y)}u_k(-p') + \sum_{l\neq 0} e^{i(p'+l)\cdot(x-y)}u_k(-p'-l)\frac{\Delta(p')}{\Delta(p'+l)}\Big)\ . \tag{12.5}$$

The terms $l \neq 0$ on U. Let $p' \in U$ and $l \neq 0$, and let $\hat p$ be the representative of the real part of $p'+l$. The difference of $\hat p$ and the real part of p' is a nonzero element of $2\pi\mathbb{Z}^4$, by Lemma 7.1(iv), so that one coordinate of $\hat p$ is at least $2\pi - \pi - \frac14$ in modulus, $|\hat p| \ge 2.8$, and by Lemma 12.1(iii), with $|\tau|^2 \le \frac14$,

$$\operatorname{Re}\Delta(p'+l) \ge \frac{4}{\pi^2}|\hat p|^2 - \frac12 \ge \frac{2}{\pi^2}|\hat p|^2\ . \tag{12.6}$$

In particular T_* and $\mathcal{D}$ are analytic on U. We need three sums over the classes $l \neq 0$, for $\beta \in \{0, 1, 1+\alpha\}$:

$$\Sigma_\beta(p') \coloneqq \sum_{l\neq 0} |\hat p|^{\beta-2} \prod_\mu \min(1, \pi/|q_\mu|) \le C_\beta\ , \tag{12.7}$$

with constants C_0, C_1, and $C_{1+\alpha}$ which depend on nothing, respectively on α. To see this, recall that q is of the form $q_\mu = r_\mu + 2\pi n_\mu$ with r the real part of p' and integers n_μ, and that every such q belongs to 25 classes, one of them central. For the central class $\hat p = q$, the class is not zero only if $n \neq 0$, and then $|q_\mu| \geq 2\pi|n_\mu| - \pi - \frac14 \geq |n_\mu|$ and $|\hat p| \geq \max_\mu |n_\mu|$, so that the central classes contribute at most $\pi^4 \sum_{n\neq 0} (\max_\mu |n_\mu|)^{\beta-2} \prod_\mu \max(1,|n_\mu|)^{-1} \leq \pi^4 (\sum_{n\in\mathbb{Z}} \max(1,|n|)^{-1-(2-\beta)/4})^4$, which is finite for $\beta < 2$. For the other 24 classes the plane part of $\hat p$ is $(q_2, q_3) + \frac{2\pi}{5\eta} m$ with a nonzero $m \in \mathbb{Z}[i]$, and $|(q_2,q_3)| \leq \sqrt2\,\pi/(5\eta)$, so that $|\hat p| \geq (2-\sqrt2)\pi/(5\eta) \geq 1/(3\eta)$; the number of the values of n_μ with $|q_\mu| \leq \pi/\eta$ is at most $1/\eta + 2$, and since $\pi/|q_\mu| \leq 2/|n_\mu|$ for $n_\mu \neq 0$, $\sum_{n_\mu} \min(1, \pi/|q_\mu|) \leq 5 + 4\log(2/\eta)$. These classes therefore contribute at most $24(3\eta)^{2-\beta}(5+4\log(2/\eta))^4$, which is bounded on $0 < \eta \leq 1$ for every $\beta < 2$. With (12.6), Lemma 12.1(i), and $\min(1,\pi/|q_\mu|)^2 \leq \min(1,\pi/|q_\mu|)$,

$$|\mathsf{T}_*(p')| \leq \frac{\pi^2}{2} e^{9/2}\, \Sigma_0(p') \leq C_* \coloneqq \frac{\pi^2}{2} e^{9/2} C_0 \tag{12.8}$$

for every $p' \in U$.

The denominator. Let $s \in [-\pi,\pi]^4$. Then $\Delta(s) \geq 0$, $\mathsf{T}_*(s) \geq 0$, and $u_k(s)u_k(-s) = |u_k(s)|^2 \geq c_u \coloneqq \frac14 (2/\pi)^8$ by (7.8), so that $\mathcal{D}(s) \geq a c_u$. Let $\tau \in \mathbb{R}^4$ with $|\tau_\mu| \leq \delta \leq \frac18$. The function $z \mapsto \mathsf{T}_*(s + z\tau/\delta)$ is analytic and bounded by C_* on the disc $|z| < \frac14$, so that its derivative is at most $8C_*$ on the disc $|z| \leq \frac18$, and $|\mathsf{T}_*(s+i\tau) - \mathsf{T}_*(s)| \leq 8C_*\delta$. By Lemma 12.1(ii) and (i), $|u_k(s+i\tau)u_k(-s-i\tau) - |u_k(s)|^2| \leq 8\delta(e^{9/4}+1)$, and by Lemma 12.1(iii), with $|s| \leq 2\pi$ and $|\tau| \leq 2\delta$, $|\Delta(s+i\tau) - \Delta(s)| \leq 8\delta^2 + 16\pi\delta \leq 52\delta$ and $\Delta(s) \leq 4\pi^2$. Hence

$$|\mathcal{D}(s+i\tau) - \mathcal{D}(s)| \leq \delta(52(1+aC_*) + 32\pi^2 a C_* + 84a)\,, \tag{12.9}$$

and we let $\delta_0 \leq \frac18$ be so small that the right side with $\delta = \delta_0$ and $a = a_+$ is at most $\frac12 a_- c_u$. Then $|\mathcal{D}(s+i\tau)| \geq \frac12 a c_u$ whenever $|\tau_\mu| \leq \delta_0$, the function Φ is analytic in a neighbourhood of the set of these $s + i\tau$, by (12.5), and by its periodicity and Cauchy's theorem in each variable, $\int \Phi(s)\,\mathrm{d}s = \int \Phi(s+i\tau)\,\mathrm{d}s$ over $[-\pi,\pi]^4$.

The bounds. Let $\tau_\mu \coloneqq \delta_0$ if $x_\mu \geq y_\mu$ and $\tau_\mu \coloneqq -\delta_0$ otherwise, so that $|e^{i(s+i\tau+l)\cdot(x-y)}| = e^{-\delta_0 \sum_\mu |x_\mu - y_\mu|} \leq e^{-\delta_0|x-y|}$. For $p' = s + i\tau$ we have $|u_k(-p')| \leq e^{9/4}$, $|\Delta(p')| \leq 4\pi^2 + 1$, and, by (12.6) and Lemma 12.1(i), the sum over $l \neq 0$ in (12.5) is at most $(4\pi^2+1)\frac{\pi^2}{2} e^{9/4} \Sigma_0 e^{-\delta_0|x-y|}$. With (12.7) and the lower bound on $\mathcal{D}$ this is the first bound of (12.2). The difference operator ∂'_ν multiplies the term l by $\eta^{-1}(e^{i\eta(p'+l)\cdot e'_\nu} - 1)$, which is at most $2(|\hat p|+1)$ in modulus because $|e^{iz} - 1| \leq |z| e^{|\mathrm{Im}\, z|}$; the term $l = 0$ stays bounded, and the terms $l \neq 0$ are bounded through Σ_1, which gives the second bound. For the third we may assume $|x' - y| \leq |x - y|$ and choose τ for the pair (x', y). The difference of the two kernels has the terms of $(\partial'_\nu \mathcal{K})(x', y)$ multiplied by $e^{i(p'+l)\cdot(x-x')} - 1$, and for $|x - x'| \leq 1$ this factor is at most $2\min(2, (|\hat p|+1)|x-x'|) \leq 4(|\hat p|+1)^\alpha |x-x'|^\alpha$ in modulus. The terms $l \neq 0$ are then bounded through $\Sigma_{1+\alpha}$, with a constant which depends on α. *q.e.d.*

The proposition replaces [1, (2.35), (2.36)] for the whole lattice. On a torus T_η/Γ' the kernel is the sum of $\mathcal{K}(x, y+\gamma)$ over $\gamma \in \Gamma'$, which converges by (12.2).

13 The Equation of the Scales

The bounds on the scalar operator in the supremum and Hölder norms are obtained in [1] from the identity (2.34) there, which expresses the operator of k steps through those of fewer steps and through operators on the intermediate lattices, and which is quoted from a paper

that is not among our sources. We derive it for the tilted run. Let $\ell_j := L^{j-k}$, $1 \le j \le k$, be the spacings of the lattices of the run in the unit of the last one, let $a > 0$, and put

$$\frac{1}{\alpha_j} := \frac{1}{a}\sum_{i=1}^{j} \ell_i^2 \,, \qquad P_{(j)} := Q'^{*}_{\ell_j} Q'_{\ell_j} \,, \qquad G_{(j)} := (-\Delta + \alpha_j P_{(j)})^{-1} \,, \tag{13.1}$$

on the functions on T_η, or on a torus T_η/Γ'. The adjoints are taken with the weights η^4 on T_η and ℓ_j^4 on $\ell_j\mathbb{Z}^4$, so that $Q'_{\ell_j} Q'^{*}_{\ell_j} = I$ and $P_{(j)}$ is the orthogonal projection onto the functions which are constant on the blocks $B_{\ell_j}(y)$; the inverse exists by Lemma 8.3 for the scale ℓ_j, as in Section 12. In the unit of the lattice $\ell_j\mathbb{Z}^4$ the constant is $a_j := \alpha_j \ell_j^2$, with $1/a_j = a^{-1}\sum_{i=0}^{j-1} L^{-2i}$, so that $a(1-L^{-2}) \le a_j \le a$.

Lemma 13.1 (The Equation of the Scales). *For $1 \le j \le k-1$ let $Q'^{\mathrm{st}} := Q'^{\mathrm{st}}_{\ell_j,\ell_{j+1}}$ be Bałaban's average* (6.4) *from $\ell_j\mathbb{Z}^4$ to $\ell_{j+1}\mathbb{Z}^4$, and let, on the functions on $\ell_j\mathbb{Z}^4$,*

$$\Delta^{(j)} := \alpha_j - \alpha_j^2\, Q'_{\ell_j} G_{(j)} Q'^{*}_{\ell_j} \,, \qquad C^{(j)} := (\Delta^{(j)} + a\ell_{j+1}^{-2} Q'^{\mathrm{st}*} Q'^{\mathrm{st}})^{-1} \,. \tag{13.2}$$

Then $\Delta^{(j)} \ge 0$, the inverse $C^{(j)}$ exists, and

$$G_{(j+1)} = G_{(j)} + \alpha_j^2\, G_{(j)} Q'^{*}_{\ell_j} C^{(j)} Q'_{\ell_j} G_{(j)} \,, \qquad G_{(k)} = G_{(1)} + \sum_{j=1}^{k-1} \alpha_j^2\, G_{(j)} Q'^{*}_{\ell_j} C^{(j)} Q'_{\ell_j} G_{(j)} \,. \tag{13.3}$$

Proof. Write $\mathsf{q} := Q'_{\ell_j}$, $\Pi := Q'^{\mathrm{st}*} Q'^{\mathrm{st}}$, and $\beta := a\ell_{j+1}^{-2}$. By Lemma 6.3, $Q'_{\ell_{j+1}} = Q'^{\mathrm{st}}\mathsf{q}$, so that $P_{(j+1)} = \mathsf{q}^*\Pi\mathsf{q}$, and Π is an orthogonal projection. By (13.1), $1/\alpha_{j+1} = 1/\alpha_j + 1/\beta$, that is, $\alpha_{j+1} = \alpha_j\beta/(\alpha_j+\beta)$. The operator $\alpha_j - \alpha_j^2(\alpha_j+\beta\Pi)^{-1}$ vanishes on the null space of Π and is $\alpha_j - \alpha_j^2/(\alpha_j+\beta) = \alpha_{j+1}$ on its range, so that it is $\alpha_{j+1}\Pi$, and

$$G_{(j+1)}^{-1} = -\Delta + \alpha_{j+1}\mathsf{q}^*\Pi\mathsf{q} = G_{(j)}^{-1} - \alpha_j^2\, \mathsf{q}^*(\alpha_j + \beta\Pi)^{-1}\mathsf{q} \,. \tag{13.4}$$

For bounded operators M, K with bounded inverses and a bounded operator U such that $\mathsf{M} - \mathsf{U}\mathsf{K}\mathsf{U}^*$ has a bounded inverse, $\mathsf{K}^{-1} - \mathsf{U}^*\mathsf{M}^{-1}\mathsf{U}$ has the bounded inverse $\mathsf{K} + \mathsf{K}\mathsf{U}^*(\mathsf{M} - \mathsf{U}\mathsf{K}\mathsf{U}^*)^{-1}\mathsf{U}\mathsf{K}$, and $(\mathsf{M} - \mathsf{U}\mathsf{K}\mathsf{U}^*)^{-1} = \mathsf{M}^{-1} + \mathsf{M}^{-1}\mathsf{U}(\mathsf{K}^{-1} - \mathsf{U}^*\mathsf{M}^{-1}\mathsf{U})^{-1}\mathsf{U}^*\mathsf{M}^{-1}$, as one checks by multiplication. We apply this with $\mathsf{M} = G_{(j)}^{-1}$, $\mathsf{U} = \alpha_j\mathsf{q}^*$, and $\mathsf{K} = (\alpha_j + \beta\Pi)^{-1}$, for which $\mathsf{M} - \mathsf{U}\mathsf{K}\mathsf{U}^* = G_{(j+1)}^{-1}$ by (13.4) and $\mathsf{K}^{-1} - \mathsf{U}^*\mathsf{M}^{-1}\mathsf{U} = \alpha_j + \beta\Pi - \alpha_j^2\mathsf{q}G_{(j)}\mathsf{q}^* = \Delta^{(j)} + \beta\Pi$. This gives the existence of $C^{(j)}$ and the first identity of (13.3), and the second follows by adding the first over j. Finally $G_{(j)}^{-1} \ge \alpha_j\mathsf{q}^*\mathsf{q}$ gives $\langle \mathsf{q}^*w, G_{(j)}\mathsf{q}^*w\rangle = \sup_\phi(2\langle w, \mathsf{q}\phi\rangle - \langle\phi, G_{(j)}^{-1}\phi\rangle) \le \sup_\phi(2\langle w,\mathsf{q}\phi\rangle - \alpha_j\|\mathsf{q}\phi\|^2) \le \|w\|^2/\alpha_j$, hence $\Delta^{(j)} \ge 0$. *q.e.d.*

The second identity of (13.3) is [1, (2.34)] for the whole lattice or a torus, with $G_{(1)}$ for the operator $C^{(0),\eta}$ there; the factor $(L^j\eta)^{-4}$ of that formula is contained in our adjoints. The operators $G_{(j)}$, Q'_{ℓ_j} are those of the tilted run at the scale ℓ_j, to which Proposition 12.2 applies after a change of the unit of length, and the operators $C^{(j)}$ act on the lattices $\ell_j\mathbb{Z}^4$ of the paper.

14 The Scalar Operator in the Supremum and Hölder Norms

With the equation of the scales and the kernel of Section 12 we obtain the bounds of [1, (1.9), (1.10)] for the whole lattice at the field $A = 0$, by the summation [1, (2.38), (2.39)]. This replaces Lemma 2.2 of that paper, which concerns a box, by a statement without a boundary;

the lattice in this section is T_η, and the statements hold on a torus T_η/Γ' with the kernels summed over the periods. Two further inputs are needed, the decay of the kernels of $G_{(1)}$ and of $C^{(j)}$. Bałaban derives the second from his Proposition 2.3; we use that both are inverses of operators which are bounded below and have kernels of rapid decay.

Lemma 14.1 (Inverses of Operators with Decaying Kernels)**.** *Let Y be a countable set with a metric d, and let M be a bounded self-adjoint operator on the square-summable functions on Y, with the kernel $\mathsf{M}(y,y')$, such that $\mathsf{M} \geq \gamma > 0$ and, for some $\theta > 0$ and every y, $\sum_{y'} |\mathsf{M}(y,y')|(e^{\theta d(y,y')} - 1) \leq \frac{1}{2}\gamma$. Then $|\mathsf{M}^{-1}(y,y')| \leq (2/\gamma) e^{-\theta d(y,y')}$ for all $y, y' \in Y$.*

Proof. Fix $y_0 \in Y$ and $R > 0$, and let $w(y) \coloneqq \min(\theta d(y,y_0), R)$, a bounded function with $|w(y) - w(y')| \leq \theta d(y,y')$. The operator $\mathsf{M}_w \coloneqq e^w \mathsf{M} e^{-w}$ has the kernel $e^{w(y)-w(y')}\mathsf{M}(y,y')$, and the kernel of $\mathsf{M}_w - \mathsf{M}$ is at most $|\mathsf{M}(y,y')|(e^{\theta d(y,y')} - 1)$ in modulus, a bound which is symmetric in y and y'; by the hypothesis and Schur's test $\|\mathsf{M}_w - \mathsf{M}\| \leq \frac{1}{2}\gamma$. Hence $\|\mathsf{M}_w \phi\| \geq \frac{1}{2}\gamma \|\phi\|$ for every ϕ, and the same holds for the adjoint, so that M_w is invertible with $\|\mathsf{M}_w^{-1}\| \leq 2/\gamma$. Since $e^{\pm w}$ are bounded, $\mathsf{M}_w^{-1} = e^w \mathsf{M}^{-1} e^{-w}$, and $|\mathsf{M}^{-1}(y,y_0)| = e^{-w(y)+w(y_0)} |\mathsf{M}_w^{-1}(y,y_0)| \leq (2/\gamma) e^{-w(y)}$. Letting $R \to \infty$ gives the claim. *q.e.d.*

Lemma 14.2 (The Operators of One Scale)**.** *Let $L \geq 64$ and $a > 0$. There are positive constants c_1, θ_1, c_C, θ_C, which depend on a and L and not on k, such that the kernel g_1 of $G_{(1)}$ and the kernels C_j of $C^{(j)}$, $1 \leq j \leq k-1$, with respect to the sums over the points of T_η and of $\ell_j \mathbb{Z}^4$, satisfy*

$$|g_1(x,x')| \leq c_1 \eta^2 e^{-\theta_1 |x-x'|/\eta} , \qquad |\mathsf{C}_j(y,y')| \leq c_C \ell_j^2 e^{-\theta_C |y-y'|/\ell_j} . \tag{14.1}$$

Proof. For g_1, let $\mathsf{M} \coloneqq \eta^2(-\Delta + \alpha_1 P_{(1)})$ on the functions on T_η, with the metric $d(x,x') \coloneqq |x-x'|/\eta$. Since $\eta^2 \alpha_1 = a/L^2$, its kernel with respect to the sum over the points is 8 on the diagonal and -1 for the eight neighbours of a point, plus $(a/L^2)L^{-4}$ for two points of one block $B_{\ell_1}(y)$, whose distance d is at most $2L$. By Lemma 8.3 for the scale $\ell_1 = L\eta$, $\mathsf{M} \geq L^{-2} \min(C_P^{-1}, a)$, and $\sum_{x'} |\mathsf{M}(x,x')|(e^{\theta d(x,x')} - 1) \leq 8(e^\theta - 1) + (a/L^2)(e^{2L\theta} - 1)$, which is at most half this lower bound for $\theta = \theta_1$ small. Lemma 14.1 and $G_{(1)} = \eta^2 \mathsf{M}^{-1}$ give the first bound.

For C_j we take ℓ_j as the unit of length. Then T_η becomes the fine lattice of the spacing L^{-j}, $G_{(j)}$ becomes $\ell_j^2 G'$ with the operator G' of Section 12 for the constant $a_j \in [a(1-L^{-2}), a]$, and $C^{(j)} = \ell_j^2 \mathsf{M}^{-1}$ with $\mathsf{M} \coloneqq a_j - a_j^2 Q' G' Q'^* + (a/L^2)\Pi$ on the functions on $\mathbb{Z}^4$, where Π is the orthogonal projection onto the functions which are constant on the cubes of side L. The kernel of $Q'G'Q'^*$ is the average of $\mathcal{K}(x,y')$ over $x \in B_1(y)$, which by (12.2) is at most $c_0 e^{2\delta_0} e^{-\delta_0 |y-y'|}$ in modulus, and the kernel of Π is L^{-4} for two points of one cube. Hence $\sum_{y'} |\mathsf{M}(y,y')|(e^{\theta|y-y'|} - 1)$ tends to zero with θ, uniformly in k and j. For the lower bound we use the momentum representation: $Q'G'Q'^*$ is the multiplication by $\mathsf{T}(p')/(1 + a_j \mathsf{T}(p'))$, by the proof of Proposition 12.2, so that $a_j - a_j^2 Q'G'Q'^*$ is the multiplication by $a_j/(1 + a_j \mathsf{T}(p'))$, and $\mathsf{T} \leq C_\mathsf{T} \Delta_0^{-1}$ by Lemma 9.2, whose proof applies to the whole lattice. With $\Delta_0 \leq 16$ this gives $a_j - a_j^2 Q'G'Q'^* \geq c_\Delta \Delta_0$, $c_\Delta \coloneqq a_j/(16 + a_j C_\mathsf{T})$. The Laplace operator Δ_0 of $\mathbb{Z}^4$ is at least the sum of those of the cubes of side L with Neumann conditions, each of which is at least $2/L^2$ on the functions of mean zero by the inequality for the cube of the corners in the proof of Lemma 8.3. Therefore $\mathsf{M} \geq L^{-2} \min(2c_\Delta, a)$, and Lemma 14.1 gives the second bound after the return to the original unit. *q.e.d.*

Proposition 14.3 (The Scalar Operator on the Whole Lattice)**.** *Let $L \geq 64$, so that $32L^{-j} \leq \frac{1}{2}$ for every $j \geq 1$, and let $a > 0$. There are constants $\delta > 0$ and c, and for every $\alpha \in [0,1)$ a*

constant c'_α, which depend on a and L and not on k, such that for every bounded function f on T_η, all $x, x' \in T_\eta$ with $|x - x'| \le 1$, and every ν,

$$\begin{aligned} |(G_{(k)}f)(x)|\ ,\ |(\partial'_\nu G_{(k)}f)(x)| &\le c\, e^{-\delta \operatorname{dist}(x,\operatorname{supp} f)} \sup|f|\ , \\ |(\partial'_\nu G_{(k)}f)(x) - (\partial'_\nu G_{(k)}f)(x')| &\le c'_\alpha |x-x'|^\alpha e^{-\delta \operatorname{dist}(\{x,x'\},\operatorname{supp} f)} \sup|f|\ . \end{aligned} \tag{14.2}$$

Proof. By Lemma 13.1, $G_{(k)}f = G_{(1)}f + \sum_{j=1}^{k-1} \mathsf{T}_j f$ with $\mathsf{T}_j \coloneqq \alpha_j^2 G_{(j)} Q'^*_{\ell_j} C^{(j)} Q'_{\ell_j} G_{(j)}$. Let $\mathcal{K}_j(x,y)$, $x \in T_\eta$, $y \in \ell_j \mathbb{Z}^4$, be the kernel of $G_{(j)} Q'^*_{\ell_j}$. In the unit ℓ_j it is ℓ_j^2 times the kernel of Proposition 12.2 for the constant a_j and the fine spacing L^{-j}, so that, with the constants of that proposition for $a_- = a(1-L^{-2})$ and $a_+ = a$,

$$|\mathcal{K}_j(x,y)| \le c_0 \ell_j^2 e^{-\delta_0 |x-y|/\ell_j}\ , \qquad |(\partial'_\nu \mathcal{K}_j)(x,y)| \le c_0 \ell_j e^{-\delta_0 |x-y|/\ell_j}\ , \tag{14.3}$$

and $|(\partial'_\nu \mathcal{K}_j)(x,y) - (\partial'_\nu \mathcal{K}_j)(x',y)| \le c_\alpha \ell_j^{1-\alpha} |x-x'|^\alpha e^{-\delta_0 \min(|x-y|,|x'-y|)/\ell_j}$ for $|x-x'| \le \ell_j$; for $\ell_j < |x-x'| \le 1$ the same bound holds with $2c_0$ in place of c_α, by the second bound of (14.3). With the weights of the adjoints, $(Q'_{\ell_j} G_{(j)} f)(y') = \ell_j^{-4} \sum_{x'} \eta^4 \mathcal{K}_j(x',y') f(x')$, and therefore

$$(\mathsf{T}_j f)(x) = \alpha_j^2 \sum_{y,y'} \mathcal{K}_j(x,y)\, \mathsf{C}_j(y,y')\, \ell_j^{-4} \sum_{x'} \eta^4 \mathcal{K}_j(x',y') f(x')\ . \tag{14.4}$$

The sums $\sum_{x'} \eta^4 e^{-\delta|x'-y'|/\ell_j}$ over T_η and $\sum_y e^{-\delta|y-y'|/\ell_j}$ over $\ell_j \mathbb{Z}^4$ are at most a constant, depending on δ, times ℓ_j^4 and times 1, because $\eta \le \ell_j$. Taking half of each exponent for the summations and the other half for the decay, (14.3), (14.1), and $\alpha_j^2 = a_j^2 \ell_j^{-4}$ give

$$|(\mathsf{T}_j f)(x)| \le c'' \ell_j^{-4} \cdot \ell_j^2 \cdot \ell_j^2 \cdot \ell_j^{-4} \ell_j^4 \ell_j^2\, e^{-\delta \operatorname{dist}(x,\operatorname{supp} f)/\ell_j} \sup|f| = c'' \ell_j^2\, e^{-\delta \operatorname{dist}(x,\operatorname{supp} f)/\ell_j} \sup|f| \tag{14.5}$$

with $\delta \coloneqq \frac{1}{2}\min(\delta_0, \theta_C)$, and in the same way $|(\partial'_\nu \mathsf{T}_j f)(x)| \le c'' \ell_j e^{-\delta \operatorname{dist}(x,\operatorname{supp} f)/\ell_j} \sup|f|$ and the Hölder bound with the factor $\ell_j^{1-\alpha}|x-x'|^\alpha$. Since $\ell_j = L^{j-k} \le 1$, the sums over j of ℓ_j^2, ℓ_j, and $\ell_j^{1-\alpha}$ are bounded uniformly in k, the last by a constant which depends on α. For the first term, (14.1) gives $|(G_{(1)}f)(x)| \le c\eta^2 e^{-\theta_1 \operatorname{dist}(x,\operatorname{supp} f)/(2\eta)} \sup|f|$, the bound with η in place of η^2 for its difference quotients, and, because two different points of T_η are at least η apart, the Hölder bound with the factor $2\eta^{1-\alpha}|x-x'|^\alpha$. *q.e.d.*

The proposition is [1, (1.9), (1.10)] for the whole lattice and $A = 0$, proved along [1, (2.38), (2.39)]. The constants depend on L, through Lemma 14.2, as Bałaban expects for the operators $C^{(j)}$ [1, p. 588].

The first paper on the propagators also needs the scalar operator applied to a derivative of the source, and, for sources which are Hölder continuous, two derivatives [2, (1.135)–(1.137), pp. 39–40]. We write $(\partial'^*_\nu f)(x) \coloneqq \eta^{-1}(f(x-\eta e'_\nu) - f(x))$ for the adjoint of ∂'_ν, and for $0 < \epsilon \le 1$ we let $\|f\|_\epsilon$ be the supremum of $|f(x) - f(x')|/|x-x'|^\epsilon$ over the pairs of different points with $|x-x'| \le 1$, as in [2, (1.109)].

Proposition 14.4 (Derivatives of the Source)**.** *Let $L \ge 64$ and $a > 0$, and let δ be as in Proposition 14.3. For $\alpha \in [0,1)$ and $\epsilon > 0$ with $\alpha + \epsilon < 1$ there are constants c, which depend on a, L, α, and ϵ and not on k, such that for every bounded function f on T_η, all $x, x' \in T_\eta$ with $|x-x'| \le 1$, and all ν, ν', with $D \coloneqq \operatorname{dist}(\{x,x'\}, \operatorname{supp} f)$,*

(i) $|(G_{(k)}\partial'^*_\nu f)(x)| \le c\, e^{-\delta D} \sup|f|$ *and* $|(G_{(k)}\partial'^*_\nu f)(x) - (G_{(k)}\partial'^*_\nu f)(x')| \le c|x - x'|^\alpha e^{-\delta D} \sup|f|$*;*

(ii) $|(\partial'_{\nu'} G_{(k)} \partial'^*_{\nu} f)(x)| \le c\, e^{-\delta D/2} (\|f\|_\epsilon + \sup|f|)$;

(iii) $|(\partial'_{\nu'} G_{(k)} \partial'^*_{\nu} f)(x) - (\partial'_{\nu'} G_{(k)} \partial'^*_{\nu} f)(x')| \le c|x-x'|^\alpha e^{-\delta D/2} (\|f\|_{\alpha+\epsilon} + \sup|f|)$.

Proof. We use the decomposition of the proof of Proposition 14.3. By a summation by parts, $\sum_{x'} \eta^4 \mathcal{K}_j(x',y')(\partial'^*_\nu f)(x') = \sum_{x'} \eta^4 (\partial'_\nu \mathcal{K}_j)(x',y') f(x')$, so that $\mathsf{T}_j \partial'^*_\nu f$ is (14.4) with $\partial'_\nu \mathcal{K}_j$ in place of the last kernel. By (14.3) this replaces one factor ℓ_j^2 of (14.5) by ℓ_j: $|(\mathsf{T}_j \partial'^*_\nu f)(x)| \le c'' \ell_j e^{-\delta D/\ell_j} \sup|f|$, and the difference of the values at x and x' is at most $c'' \min(2\ell_j, |x-x'|) e^{-\delta D/\ell_j} \sup|f| \le 2c'' \ell_j^{1-\alpha} |x-x'|^\alpha e^{-\delta D/\ell_j} \sup|f|$, by the bound on $\partial' \mathcal{K}_j$. The sums over j of ℓ_j and $\ell_j^{1-\alpha}$ are bounded, which gives (i) for these terms.

For (ii) and (iii), $(\partial'_{\nu'} \mathsf{T}_j \partial'^*_\nu f)(x) = \sum_{x''} \eta^4 k_j(x,x'') f(x'')$ with the kernel

$$k_j(x,x'') := \alpha_j^2 \ell_j^{-4} \sum_{y,y'} (\partial'_{\nu'} \mathcal{K}_j)(x,y)\, \mathsf{C}_j(y,y')\, (\partial'_\nu \mathcal{K}_j)(x'',y') \;, \tag{14.6}$$

for which (14.3), its Hölder counterpart, and (14.1) give, as in (14.5),

$$\begin{aligned} |k_j(x,x'')| &\le c'' \ell_j^{-4} e^{-\delta|x-x''|/\ell_j} \;, \\ |k_j(x,x'') - k_j(x',x'')| &\le c'' \ell_j^{-4} \Big(\frac{|x-x'|}{\ell_j}\Big)^\alpha e^{-\delta \min(|x-x''|, |x'-x''|)/\ell_j} \;, \end{aligned} \tag{14.7}$$

the second for $|x-x'| \le \ell_j$. Since ∂'^*_ν annihilates the constants and the sums converge absolutely, $\sum_{x''} \eta^4 k_j(x,x'') = 0$ for every x. Hence $(\partial'_{\nu'} \mathsf{T}_j \partial'^*_\nu f)(x) = \sum_{x''} \eta^4 k_j(x,x'') (f(x'') - f(x))$. Here $|f(x'') - f(x)| \le \|f\|_\epsilon |x-x''|^\epsilon$ for $|x-x''| \le 1$ and $|f(x'') - f(x)| \le 2\sup|f|$ otherwise, and $\sum_{x''} \eta^4 \ell_j^{-4} e^{-\delta|x-x''|/\ell_j} |x-x''|^\epsilon$ is at most a constant times ℓ_j^ϵ, as is $e^{-\delta/(2\ell_j)}$. This gives $|(\partial'_{\nu'} \mathsf{T}_j \partial'^*_\nu f)(x)| \le c\, \ell_j^\epsilon (\|f\|_\epsilon + \sup|f|)$, with a sum over j which is bounded. If $D \ge 1$, then f vanishes at x and the first bound of (14.7) gives directly $|(\partial'_{\nu'} \mathsf{T}_j \partial'^*_\nu f)(x)| \le c\, e^{-\delta D/(2\ell_j)} \sup|f|$, and $\sum_j e^{-\delta D/(2\ell_j)} \le e^{-\delta D/2} \sum_{i \ge 0} e^{-\delta(L^i-1)/2}$. The two cases give (ii). For (iii), the vanishing of the sums of $k_j(x,\cdot)$ and of $k_j(x',\cdot)$ gives

$$(\partial'_{\nu'} \mathsf{T}_j \partial'^*_\nu f)(x) - (\partial'_{\nu'} \mathsf{T}_j \partial'^*_\nu f)(x') = \sum_{x''} \eta^4 (k_j(x,x'') - k_j(x',x''))(f(x'') - f(x)) \;. \tag{14.8}$$

For $|x-x'| \le \ell_j$ the second bound of (14.7) and the estimate of the sum, with $\alpha+\epsilon$ in place of ϵ, bound this by $c(|x-x'|/\ell_j)^\alpha \ell_j^{\alpha+\epsilon} (\|f\|_{\alpha+\epsilon} + \sup|f|)$; for $|x-x'| > \ell_j$ the two terms are bounded separately by $c\, \ell_j^{\alpha+\epsilon} \le c|x-x'|^\alpha \ell_j^\epsilon$ times the same norms. For $D \ge 1$ the decay is obtained as in (ii), with $\sum_i L^{i\alpha} e^{-\delta(L^i-1)/2} < \infty$.

The term $G_{(1)}$ is treated in the same way with η in place of ℓ_j: by (14.1) the kernels of $G_{(1)} \partial'^*_\nu$ and of $\partial'_{\nu'} G_{(1)} \partial'^*_\nu$ with respect to the weights η^4 are at most a constant times $\eta^{-3} e^{-\theta_1|x-x''|/(2\eta)}$ and $\eta^{-4} e^{-\theta_1|x-x''|/(2\eta)}$, the second has vanishing sums over x'', and two different points are at least η apart. *q.e.d.*

Statement (ii) is [2, (1.137)], and (i) and (iii) are the corresponding bounds for the scalar operator in [2, (1.111), (1.113)].

15 The Kernel of the Gauge-Fixing Projection

The operator Δ_a of [2, (1.69)] contains the nonlocal term $\partial P \partial^*$, and the passage from L^2-bounds to bounds in the supremum norm needs the bounds [2, (1.126), (1.127)] on its kernel.

Bałaban refers for them to the representation [2, (1.44)], $P = G'Q_k'^*(Q_k'G'^2Q_k'^*)^{-1}Q_k'G'$, to [1, Lemma 2.4], and to the analyticity of the momentum representation [2, (1.45)]. In the tilted run the first two are available by Proposition 12.2, and for the inverse in the middle we use Lemma 14.1.

Lemma 15.1 (The Kernel of $\partial P\partial^*$)**.** *Let $32\eta \le \frac{1}{2}$, let $a > 0$, and let G' and $\mathcal{K}$ be as in Section 12. The operator $\mathsf{N} := Q_k'G'^2Q_k'^*$ on the square-summable functions on $\mathbb{Z}^4$ has a bounded inverse, $P := G'Q_k'^*\mathsf{N}^{-1}Q_k'G'$ is the orthogonal projection onto the orthogonal complement of $\Delta N(Q_k')$, and for every field A of finite support*

$$\begin{aligned}(\partial P\partial^* A)_\nu(x) &= \sum_{\nu'}\sum_{x'}\eta^4\varpi_{\nu\nu'}(x,x')A_{\nu'}(x')\;,\\ \varpi_{\nu\nu'}(x,x') &:= \sum_{y,y'}(\partial'_\nu\mathcal{K})(x,y)\,\mathsf{N}^{-1}(y,y')\,(\partial'_{\nu'}\mathcal{K})(x',y')\;.\end{aligned} \tag{15.1}$$

There are constants c_ϖ and $\delta_\varpi > 0$, and for $\alpha \in [0,1)$ a constant $c_{\varpi,\alpha}$, which depend on a and not on k and L, such that for $|x-x'| \le 1$

$$\begin{aligned}|\varpi_{\nu\nu'}(x,x'')| &\le c_\varpi e^{-\delta_\varpi|x-x''|}\;,\\ |\varpi_{\nu\nu'}(x,x'') - \varpi_{\nu\nu'}(x',x'')| &\le c_{\varpi,\alpha}|x-x'|^\alpha e^{-\delta_\varpi\min(|x-x''|,|x'-x''|)}\;.\end{aligned} \tag{15.2}$$

Proof. With the weights of Section 12, $(Q_k'G'f)(y) = \sum_x \eta^4\mathcal{K}(x,y)f(x)$, so that N has the kernel $\mathsf{N}(y,y') = \sum_x \eta^4\mathcal{K}(x,y)\mathcal{K}(x,y')$, which by (12.2) is at most a constant times $e^{-\delta_0|y-y'|/2}$ in modulus. In the momentum representation N is the multiplication by $\sum_l |u_k(p'+l)|^2\Delta(p'+l)^{-2}(1+a\mathsf{T}(p'))^{-2}$, by the proof of Proposition 12.2, and the term $l=0$ of this sum is $|u_k(p')|^2/\mathcal{D}(p')^2$ with $\mathcal{D}$ of (12.4). For real $p' \in [-\pi,\pi]^4$ we have $\mathcal{D}(p') \le 4\pi^2(1+aC_*)+a$ by (12.8) and $|u_k(p')|^2 \ge c_u$, so that $\mathsf{N} \ge \gamma_\mathsf{N} := c_u(4\pi^2(1+aC_*)+a)^{-2}$. Lemma 14.1 gives $|\mathsf{N}^{-1}(y,y')| \le (2/\gamma_\mathsf{N})e^{-\theta_\mathsf{N}|y-y'|}$ with a $\theta_\mathsf{N} > 0$ which depends on a only.

The operator P is self-adjoint, and $P^2 = P$ because $Q_k'G'G'Q_k'^* = \mathsf{N}$. Its range is the range of $G'Q_k'^*$, whose orthogonal complement consists of the f with $Q_k'G'f = 0$, that is, of the $f = (-\Delta + aQ_k'^*Q_k')\lambda = -\Delta\lambda$ with $Q_k'\lambda = 0$. This is the second assertion, and P is the operator of [2, (1.38), (1.44)]. For a field A of finite support, $\partial^*A = \sum_{\nu'}\partial'^*_{\nu'}A_{\nu'}$, and a summation by parts gives $(Q_k'G'\partial^*A)(y') = \sum_{\nu'}\sum_{x'}\eta^4(\partial'_{\nu'}\mathcal{K})(x',y')A_{\nu'}(x')$; applying N^{-1}, then $G'Q_k'^*$ with its kernel $\mathcal{K}$, and ∂'_ν gives (15.1). The bounds (15.2) follow from (12.2) and the bound on N^{-1} by summation over y and y', with $\delta_\varpi := \frac{1}{2}\min(\delta_0,\theta_\mathsf{N})$. *q.e.d.*

The bounds (15.2) are [2, (1.126), (1.127)].

16 Proposition 1.2 for the Tilted Run

We can now prove the bounds of [2, Proposition 1.2] on the operator $G = \Delta_a^{-1}$ in the supremum and Hölder norms, with their exponential decay. Bałaban derives the decay from an expansion of G over a partition of unity [2, (1.119)–(1.131)], whose terms contain the commutators of the averages with functions. We obtain it more directly, from the stability of the lower bound (9.2) under the conjugation with an exponential weight, the method which he mentions first [2, p. 36]; the passage from L^2 to the supremum norm is then his, through the identities [2, (1.132), (1.133)].

In this section the lattice is a torus T_η/Γ' with $\Gamma' \subset \mathbb{Z}^4$, $d(x,x')$ is the distance on it, and $L \ge 64$. The operators of the whole lattice with the kernels of Sections 12–15 act on

the bounded functions, and they map periodic functions to periodic functions; since each of them is the inverse, or is built from inverses, of operators which are bounded below on the torus as on the whole lattice, their restrictions to the periodic functions are the corresponding operators of the torus. Propositions 14.3 and 14.4 and Lemma 15.1 therefore apply to the torus, with sums over the points of T_η and with the distance to the periodic support, which is the distance on the torus. For a subset X of the torus, $\|\cdot\|_X$ is the L^2-norm over the points, or the bonds starting, in X, and $\Delta(y) \coloneqq B_1(y)$ is the unit block of $y \in \mathbb{Z}^4$. We write $\|A\|_1^2 \coloneqq \langle A, (\Delta + I)A\rangle = \|\nabla A\|^2 + \|A\|^2$, and for a tuple $J = (J_\nu)$ of fields, $\nabla^* J \coloneqq \sum_\nu \partial'^*_\nu J_\nu$.

Lemma 16.1 (Decay in L^2)**.** *Let $a > 0$. There are constants $\theta_0 > 0$ and C, which depend on a and not on k, L, and the torus, such that for all $y, y' \in \mathbb{Z}^4$ and every field J, or tuple of fields J, supported in the bonds starting in $\{x \,|\, d(x, y') \le 2\}$,*

$$\begin{aligned} \|GJ\|_{\Delta(y)} + \|\nabla GJ\|_{\Delta(y)} &\le Ce^{-\theta_0 d(y,y')}\|J\| \,, \\ \|G\nabla^* J\|_{\Delta(y)} + \|\nabla G\nabla^* J\|_{\Delta(y)} &\le Ce^{-\theta_0 d(y,y')}\|J\| \,. \end{aligned} \tag{16.1}$$

The same holds for $G_0 \coloneqq (\Delta + aQ_k^* Q_k)^{-1}$ in place of G.

Proof. Let w be a real function on the torus with $|w(x) - w(x')| \le \theta d(x, x')$, $\theta \le \frac14$, acting on a field by multiplication with its value at the starting point of the positive bond. We compare the forms of $e^w \Delta_a e^{-w}$ and of Δ_a, term by term.

For the Laplace operator, $\langle A', e^w \Delta e^{-w} A\rangle = \sum_\nu \langle \partial'_\nu(e^w A'), \partial'_\nu(e^{-w}A)\rangle$. With $\varsigma \coloneqq w(x + \eta e'_\nu) - w(x)$, $|\varsigma| \le \theta\eta$, we have $\partial'_\nu(e^{\pm w} g)(x) = e^{\pm w(x)}(e^{\pm\varsigma}(\partial'_\nu g)(x) + \eta^{-1}(e^{\pm\varsigma} - 1)g(x))$, where $e^{|\varsigma|} \le 2$ and $\eta^{-1}|e^{\pm\varsigma} - 1| \le 2\theta$. The product of the two derivatives therefore differs from $(\partial'_\nu A')(\partial'_\nu A)$ by at most $4\theta(|\partial'_\nu A'||A| + |A'||\partial'_\nu A|) + 4\theta^2|A'||A|$ at every bond, and summing gives $|\langle A', (e^w \Delta e^{-w} - \Delta)A\rangle| \le 16\theta \|A'\|_1 \|A\|_1$.

For the average, $\langle A', e^w Q_k^* Q_k e^{-w} A\rangle = \sum_c (Q_k e^w A')(c)(Q_k e^{-w} A)(c)$, and for a bond c from y the two factors may be replaced by $Q_k(e^{w-w(y)}A')(c)$ and $Q_k(e^{-(w-w(y))}A)(c)$. The bonds which enter the average at c start at a distance of at most 4 from y, so that $|e^{\pm(w-w(y))} - 1| \le e^{4\theta} - 1 \le 8\theta$ on them, and by the estimate in the proof of Lemma 9.1(i), which bounds the square of each average by a sum over its own bonds, $|\langle A', (e^w Q_k^* Q_k e^{-w} - Q_k^* Q_k)A\rangle| \le (32\theta + 128\theta^2)\|A'\|\|A\| \le 64\theta\|A'\|\|A\|$.

For the projection, $e^w \partial P \partial^* e^{-w} - \partial P \partial^*$ has the kernel $\varpi(x, x')(e^{w(x)-w(x')} - 1)$ with respect to the sums over the points of T_η with the weights η^4, and $|e^{w(x)-w(x')} - 1| \le \theta|x - x'|e^{\theta|x-x'|}$. For $\theta \le \frac12 \delta_\varpi$, (15.2) and Schur's test bound its norm by $C_\varpi \theta$, with $C_\varpi \coloneqq 4c_\varpi \sup_x \sum_{x'} \eta^4 |x - x'| e^{-\delta_\varpi |x-x'|/2}$, which is finite and bounded uniformly in η.

Together, $|\langle A', (e^w \Delta_a e^{-w} - \Delta_a)A\rangle| \le C_w \theta \|A'\|_1 \|A\|_1$ with $C_w \coloneqq 16 + 64a + C_\varpi$. Let $\theta_0 \coloneqq \min(\frac14, \frac12\delta_\varpi, \gamma_0/(2C_w))$ with γ_0 of (9.2), and let $\theta \le \theta_0$. Then $\langle F, e^w \Delta_a e^{-w} F\rangle \ge \frac12 \gamma_0 \|F\|_1^2$ for every F, so that $e^w \Delta_a e^{-w}$ is invertible, and its inverse $G_w = e^w G e^{-w}$ satisfies $\|G_w J\|_1 \le (2/\gamma_0)\|J\|$ and, because $|\langle F, \nabla^* J\rangle| \le \|F\|_1 \|J\|$, also $\|G_w \nabla^* J\|_1 \le (2/\gamma_0)\|J\|$.

Now let $w(x) \coloneqq \theta_0 \max(0, d(x, y') - 3)$. It vanishes on the points at a distance of at most 3 from y', hence on the supports of J and of $\nabla^* J$, so that $GJ = e^{-w} G_w J$ and $G\nabla^* J = e^{-w} G_w \nabla^* J$; and $w \ge \theta_0(d(y, y') - 5)$ on $\Delta(y)$. Since $|\partial'_\nu(e^{-w}F)| \le 2e^{-w}(|\partial'_\nu F| + \theta_0|F|)$ by the formula above, this gives (16.1) with $C \coloneqq 8e^{5\theta_0}/\gamma_0$. For G_0 the term with the projection is absent, and the lower bound is $\Delta + aQ_k^* Q_k \ge \min(a/4, (2C_P + 65)^{-1})(\Delta + I)$, by Lemma 9.1(iv). *q.e.d.*

The lemma is [2, (1.114)] for the first derivatives.

Proposition 16.2 (Proposition 1.2 for the Tilted Run)**.** *Let $L \geq 64$ and $a > 0$. For $\alpha \in [0,1)$ and $\epsilon > 0$ with $\alpha + \epsilon < 1$ there are constants c and $\delta_G > 0$, which depend on a, L, α, and ϵ and not on k and the torus, such that for every bounded field J, or tuple of fields J, all points x, x' with $d(x,x') \leq 1$, and $D := d(\{x,x'\}, \operatorname{supp} J)$,*

(i) $|(GJ)(x)|$, $|(\nabla GJ)(x)|$, $|(G\nabla^* J)(x)|$, *and* $|(\Delta GJ)(x)|$ *are at most* $c\, e^{-\delta_G D} \sup|J|$;

(ii) the differences of the values of ∇GJ, *and of* $G\nabla^* J$, *at* x *and* x' *are at most* $c\, d(x,x')^\alpha e^{-\delta_G D} \sup|J|$ *in modulus;*

(iii) $|(\nabla G \nabla^* J)(x)| \leq c\, e^{-\delta_G D}(\|J\|_\epsilon + \sup|J|)$;

(iv) $|(\nabla G\nabla^* J)(x) - (\nabla G\nabla^* J)(x')| \leq c\, d(x,x')^\alpha e^{-\delta_G D}(\|J\|_{\alpha+\epsilon} + \sup|J|)$.

Here the values of a field at a point are those on the positive bonds from it, and the Hölder norms are those of its components.

Proof. Throughout, c and $\delta > 0$ denote constants with the dependence stated, which may change from line to line; sums of exponentials $\sum_{y'} e^{-\delta d(y,y')} e^{-\delta' \operatorname{dist}(y',X)}$ over the unit lattice are bounded by a constant times $e^{-\min(\delta,\delta')\operatorname{dist}(y,X)/2}$.

Two local bounds. For a field F and $y \in \mathbb{Z}^4$ let X_y be the set of the points at a distance of at most 9 from y, which contains every bond entering the following estimate: a bond of $\Delta(y)$ lies on staircases from points at a distance of less than 2 from it, these points lie in blocks whose corners are at a distance of less than 2 from them, and the staircases from a block stay within the distance 4 of its corner. First, with $P' := Q'^*_k Q'_k$ acting on each component,

$$\sup_{\Delta(y)} |(P' - Q_k^* Q_k)F| \leq 3 \|F\|_{X_y} \,. \tag{16.2}$$

Indeed $P'F$ is on $\Delta(y)$ the mean of each component of F over this block, at most $\|F\|_{\Delta(y)}$ in modulus. By the proof of Lemma 6.4(ii), the value of $Q_k^* B$ on a bond of T_η is η times a signed sum of at most $\frac{7}{5\eta}$ values of B, on bonds of the unit lattice from the corners of the blocks of the points whose staircases contain the bond; and by the estimate in the proof of Lemma 9.1(i) each value $(Q_k F)(c)$ is at most $\frac{7}{5}$ times the L^2-norm of F over the bonds of the staircases from the block of c. Second, by (15.1), (15.2), and the Cauchy–Schwarz inequality on each block,

$$\sup_{\Delta(y)} |\partial P \partial^* F| \leq c \sum_{y'} e^{-\delta_\varpi |y-y'|} \|F\|_{\Delta(y')} \,, \tag{16.3}$$

the sum over the unit lattice of the whole space, with F periodic.

The operator G_0. Let G' be the scalar operator $G_{(k)}$ of Section 13 for the constant a, acting on each component of a field. Then $G'^{-1} - G_0^{-1} = a(P' - Q_k^* Q_k)$, hence

$$G_0 = G' + aG'(P' - Q_k^* Q_k) G_0 \,, \tag{16.4}$$

which is [2, (1.133)] with the same constant a in both operators. Let J be bounded, and decompose it into its restrictions $J_{y'}$ to the bonds starting in the blocks $\Delta(y')$, so that $\|J_{y'}\| \leq 2 \sup|J|$. By Lemma 16.1 for G_0 and (16.2), the field $h := a(P' - Q_k^* Q_k) G_0 J$ satisfies $\sup_{\Delta(y)} |h| \leq c \sum_{y'} e^{-\theta_0 d(y,y')} \sup_{\Delta(y')} |J| \leq c\, e^{-\delta \operatorname{dist}(y, \operatorname{supp} J)} \sup|J|$. Applying Proposition 14.3 to J and to the restrictions of h to the blocks, and summing, we obtain the bounds (i) and (ii) for $G_0 J$ and $\nabla G_0 J$. For a tuple J we use (16.4) on $\nabla^* J$: the first term $G' \nabla^* J$ and its derivatives are bounded by Proposition 14.4, and in the second term $h := a(P' - Q_k^* Q_k) G_0 \nabla^* J$

is bounded as before, by the second half of (16.1), so that $G'h$ and $\nabla G'h$ are bounded by Proposition 14.3. This gives (i)–(iv) for G_0 in place of G, except the bound on $\Delta G_0 J$.

The operator G. Since $G_0^{-1} - G^{-1} = \partial P \partial^*$, we have $G = G_0 + G_0 \partial P \partial^* G$, which is [2, (1.132)]. For a bounded field J, Lemma 16.1 for G, applied to the restrictions $J_{y'}$, gives $\|GJ\|_{\Delta(y)} \le c\, e^{-\delta \operatorname{dist}(y, \operatorname{supp} J)} \sup|J|$, and then (16.3) gives the same bound for $\sup_{\Delta(y)}|g|$, $g := \partial P \partial^* G J$. The bounds for G_0, applied to J and to the restrictions of g to the blocks, give (i) and (ii) for GJ and ∇GJ. For a tuple J the same argument with $\nabla^* J$ in place of J, the second half of (16.1), and the bounds for $G_0 \nabla^* J$ and $\nabla G_0 \nabla^* J$ give the bounds for $G \nabla^* J$ and (iii), (iv); in $\nabla G_0 g$ with $g := \partial P \partial^* G \nabla^* J$ only the first derivative of G_0 on a bounded field enters. Finally $\Delta G = I + \partial P \partial^* G - a Q_k^* Q_k G$, and the last two terms applied to J are bounded in the supremum norm by (16.3), by the bound on $Q_k^* Q_k F$ in the proof of (16.2), and by the decay of $\|GJ\|_{\Delta(y)}$. *q.e.d.*

The proposition has the content of [2, (1.110)–(1.113)], with pointwise conditions on the supports in place of the cubes $\tilde{\Delta}(y)$ and the functions ζ there, and with constants which depend on L. With it, Proposition 9.3, and Lemma 16.1, the results of [2] on the operator G hold in the tilted run, with one exception, the bounds [2, (1.114)] on the second derivatives in L^2 with their decay. These are proved in the next section, together with the statements of [2] on the minimiser H_k and on the commutators of the averages with functions.

17 The Remaining Statements of the First Paper

Three statements of [2] were left aside in Section 10: the bounds on the minimiser H_k, the second derivatives in [2, (1.114)], and the commutators of the averages with functions, from which Bałaban builds his expansions. The second paper on the propagators uses all three, and we settle them here. For the first, Bałaban writes $Q_k G Q_k^*$ in the form [2, (1.99)] and reads its bounds off from the components [2, (1.101)], which are not available for the matrices of the tilted run. We use instead that the inverse of $Q_k G Q_k^*$ is the action Δ_k plus a constant, a fact which follows from the two variational problems defining these operators and does not involve the momentum representation.

The setting is that of Section 16: a torus T_η/Γ', $L \ge 64$, $a > 0$, and $G = \Delta_a^{-1}$. For a positive bond $c = \langle y, y + e_\mu \rangle$ of the unit lattice we write $c_- := y$ and $d(c, c') := d(c_-, c'_-)$, we let δ_c be the field which is 1 at c and 0 at the other positive bonds, and we put $J_c := Q_k^* \delta_c$. The kernel of an operator M on the fields of the unit lattice is $\mathsf{M}(c, c') := \langle \delta_c, \mathsf{M} \delta_{c'} \rangle$. By (6.3) and the weights of the adjoint, the value of J_c at a positive bond b of T_η is η times the number of the points $x \in B_1(c_-)$ whose staircase $\sigma_\mu^1(x)$ contains b, minus the number of those whose staircase contains b^{-1}. Hence

$$\sup|J_c| \le \tfrac{7}{5}\,, \qquad \|J_c\|^2 \le 2\,, \tag{17.1}$$

by the count in the proof of Lemma 9.1 and by part (i) of that lemma, and J_c vanishes at the bonds which start at a distance of 4 or more from c_-.

Lemma 17.1 (The Operator $Q_k G Q_k^*$)**.** *Let γ_0 and γ_ϕ be the constants of Proposition 9.3.*

(i) Q_k maps the fields on T_η/Γ' onto the fields on the unit lattice, $Q_k G Q_k^*$ is invertible, and $(Q_k G Q_k^*)^{-1} = \Delta_k + a$. The minimiser is $H_k = G Q_k^* (\Delta_k + a)$.

(ii) $(a + 24\gamma_\phi^{-1})^{-1} \le Q_k G Q_k^* \le a^{-1}$ and $Q_k Q_k^* \ge c_Q := \gamma_0 (a + 24\gamma_\phi^{-1})^{-1}$.

(iii) There are constants C and $\theta > 0$, which depend on a and not on k, L, and the torus, such that $|(Q_k G Q_k^*)(c,c')| \le C e^{-\theta d(c,c')}$ and $|(\Delta_k + a)(c,c')| \le C e^{-\theta d(c,c')}$ for all positive bonds c, c' of the unit lattice.

Proof. (i) For every field B on the unit lattice the last part of the proof of Proposition 9.3 exhibits a field A with $Q_k A = B$, so that Q_k is onto and Q_k^* is injective; since G is positive definite, so is $Q_k G Q_k^*$. Fix B and consider the minimum of $\langle A, \Delta_a A\rangle$ over the fields with $Q_k A = B$. It is attained at one field A_0, and $\Delta_a A_0$ is orthogonal to the null space of Q_k, hence of the form $Q_k^*\omega$. Then $A_0 = G Q_k^* \omega$, the constraint gives $\omega = (Q_k G Q_k^*)^{-1} B$, and the minimum is $\langle A_0, \Delta_a A_0\rangle = \langle \omega, Q_k G Q_k^* \omega\rangle = \langle B, (Q_k G Q_k^*)^{-1} B\rangle$. On the other hand, $\langle A, \Delta_a A\rangle = \|\partial A\|^2 + \|R\partial^* A\|^2 + a\|B\|^2$ on these fields, and by the gauge transformation in the proof of Proposition 9.3 every such field has a companion A_s with $Q_k A_s = B$, $\partial A_s = \partial A$, and $R\partial^* A_s = 0$. The minimum is therefore $a\|B\|^2$ plus the minimum of $\|\partial A\|^2$ over the fields with $Q_k A = B$, which is $\langle B, \Delta_k B\rangle$ by the same proof, and it is attained at a field with $R\partial^* A = 0$, that is, at $H_k B$. Two quadratic forms which agree define the same operator.

(ii) By (9.4), $0 \le \langle B, \Delta_k B\rangle \le \gamma_\phi^{-1} \|\partial_1 B\|^2$, and $\|\partial_1 B\|^2 \le 24 \|B\|^2$, because the value of $\partial_1 B$ at a plaquette is a signed sum of four values of B and a bond lies in six plaquettes. With (i) this gives the bounds on $Q_k G Q_k^*$. By (9.2), $G \le \gamma_0^{-1}$, so that $(a + 24\gamma_\phi^{-1})^{-1} \|\omega\|^2 \le \langle Q_k^*\omega, G Q_k^* \omega\rangle \le \gamma_0^{-1} \|Q_k^*\omega\|^2$.

(iii) We have $(Q_k G Q_k^*)(c,c') = \langle J_c, G J_{c'}\rangle$. A point at a distance of less than 4 from c_- lies in a block $\Delta(y'')$ with $d(y'', c_-) \le 6$, and there are at most 13^4 such blocks. Let $J_{c,y''}$ be the restriction of J_c to the bonds which start in $\Delta(y'')$, so that J_c is the sum of these restrictions and $\|J_{c,y''}\|^2 \le 2$ by (17.1). Lemma 16.1 gives $|\langle J_{c,y''}, G J_{c',y'''}\rangle| \le 2 C_0 e^{-\theta_0 d(y'', y''')}$ with the constant C_0 of that lemma, and $d(y'', y''') \ge d(c,c') - 12$; summing over the pairs of blocks we obtain the first bound, with $\theta = \theta_0$. For the second we apply Lemma 14.1 to $\mathsf{M} := Q_k G Q_k^*$ on the set of the positive bonds of the unit lattice with the function $d(c,c')$, which vanishes for different bonds with the same starting point; the proof of that lemma uses only the triangle inequality. By (ii), $\mathsf{M} \ge \gamma := (a + 24\gamma_\phi^{-1})^{-1}$, and by the first bound $\sum_{c'} |\mathsf{M}(c,c')|(e^{\theta d(c,c')} - 1)$ is at most a constant times $\sum_{y \in \mathbb{Z}^4} e^{-\theta_0 |y|}(e^{\theta |y|} - 1)$, which is at most $\frac{1}{2}\gamma$ for θ small. The lemma and (i) give the second bound. *q.e.d.*

The identity (i) replaces [2, (1.99)], the bounds (ii) are [2, (1.100)] with the conclusion drawn there from [2, (1.101)], and the representation of H_k is [2, (1.103)]. The bounds on H_k now follow from those on G.

Corollary 17.2 (The Minimiser)**.** *For $\alpha \in [0,1)$ there are constants c and $\delta > 0$, which depend on a, L, and α and not on k and the torus, such that for every field B on the unit lattice and all points x, x' of T_η/Γ' with $d(x,x') \le 1$,*

$$\begin{aligned} |(H_k B)(x)|\ ,\ |(\nabla H_k B)(x)| &\le c \sum_{c'} e^{-\delta d(x, c'_-)} |B(c')|\ , \\ |(\nabla H_k B)(x) - (\nabla H_k B)(x')| &\le c\, d(x,x')^\alpha \sum_{c'} e^{-\delta d(\{x,x'\}, c'_-)} |B(c')|\ . \end{aligned} \tag{17.2}$$

Proof. By Lemma 17.1(i), $H_k B = \sum_c ((\Delta_k + a)B)(c)\, G J_c$. By (17.1) and Proposition 16.2(i) and (ii), $G J_c$ and $\nabla G J_c$ are at most $\frac{7}{5} c\, e^{4\delta_G} e^{-\delta_G d(x, c_-)}$ in modulus at x, with the corresponding bound for the differences of $\nabla G J_c$, and by Lemma 17.1(iii), $|((\Delta_k + a)B)(c)| \le C \sum_{c'} e^{-\theta d(c,c')} |B(c')|$. The sum over c of the product of the two exponentials is at most a constant times $e^{-\delta d(x, c'_-)}$ with $\delta := \frac{1}{2}\min(\delta_G, \theta)$. *q.e.d.*

We turn to the second derivatives. The decay of Lemma 16.1 was obtained with a weight which is Lipschitz continuous, and such a weight does not commute with the Laplace operator up to a bounded error. The second derivatives are therefore localised afterwards, with a cutoff which has two bounded derivatives. We write $\nabla\nabla A \coloneqq (\partial'_\nu \partial'_{\nu'} A)_{\nu,\nu'}$, and $\nabla^*\nabla^* J \coloneqq \sum_{\nu,\nu'} \partial'^*_{\nu'} \partial'^*_\nu J_{\nu\nu'}$ for a family $J = (J_{\nu\nu'})$ of fields.

Lemma 17.3 (Second Derivatives in L^2)**.** *Let $a > 0$. There are constants $\theta_2 > 0$ and C, which depend on a and not on k, L, and the torus, such that for all $y, y' \in \mathbb{Z}^4$ and every field J, or family of fields J, supported in the bonds starting in $\{x \,|\, d(x,y') \le 2\}$,*

$$\|\nabla\nabla G J\|_{\Delta(y)} \le C e^{-\theta_2 d(y,y')} \|J\| \,, \qquad \|G\nabla^*\nabla^* J\|_{\Delta(y)} \le C e^{-\theta_2 d(y,y')} \|J\| \,. \tag{17.3}$$

Proof. Let $F \coloneqq GJ$, regarded as a periodic field on T_η, and let $\chi(t) \coloneqq \prod_\mu \psi(t_\mu - y_\mu - \frac{1}{2})$ for $t \in \mathbb{R}^4$, with a fixed twice continuously differentiable function ψ on $\mathbb{R}$ with values in $[0,1]$ which is 1 on $[-\frac{3}{2}, \frac{3}{2}]$ and 0 outside $[-\frac{5}{2}, \frac{5}{2}]$. Then χF has a finite support, and for a field A of finite support on T_η we have $\|\nabla\nabla A\| = \|\Delta A\|$, because the operators ∂'_ν commute with each other and with their adjoints. The points which enter $\partial'_\nu \partial'_{\nu'} F$ at a point of $\Delta(y)$ have coordinates within 2η of the cube $y + [0,1)^4$, where $\chi = 1$, so that $\|\nabla\nabla F\|_{\Delta(y)} \le \|\Delta(\chi F)\|$. For a point x and each component of F,

$$\begin{aligned}(\Delta(\chi F) - \chi \Delta F)(x) = &-\eta^{-2} \sum_\nu \sum_\pm (\chi(x \pm \eta e'_\nu) - \chi(x))(F(x \pm \eta e'_\nu) - F(x)) \\ &- \eta^{-2} \sum_\nu (\chi(x + \eta e'_\nu) + \chi(x - \eta e'_\nu) - 2\chi(x)) F(x) \,,\end{aligned} \tag{17.4}$$

as one sees by writing out the two second differences. The first differences of χ are at most $\varkappa\eta$ and the second ones at most $\varkappa_2\eta^2$ in modulus, with constants which depend on ψ only, so that the right side is at most $\varkappa \sum_\nu (|(\partial'_\nu F)(x)| + |(\partial'_\nu F)(x - \eta e'_\nu)|) + 4\varkappa_2 |F(x)|$. The support of χ is contained in the union of the blocks $\Delta(y'')$ with $y'' - y \in \{-2, \dots, 2\}^4$, each of which is mapped injectively to the torus. Hence $\|\Delta(\chi F)\|$ is at most a constant times the sum, over the y'' with $y'' - y \in \{-3, \dots, 3\}^4$, of $\|\Delta F\|_{\Delta(y'')} + \|\nabla F\|_{\Delta(y'')} + \|F\|_{\Delta(y'')}$, the norms being those of the torus.

By Lemma 16.1 the last two norms are at most $C_0 e^{-\theta_0 d(y'',y')} \|J\|$. For the first we use $\Delta F = J + \partial P \partial^* F - a Q_k^* Q_k F$. The norm of J over $\Delta(y'')$ is at most $\|J\|$, and it vanishes unless $d(y'', y') \le 4$. By (16.3) and Lemma 16.1, $\|\partial P \partial^* F\|_{\Delta(y'')} \le 2 \sup_{\Delta(y'')} |\partial P \partial^* F| \le 2c \sum_{y'''} e^{-\delta_\varpi |y'' - y'''|} C_0 e^{-\theta_0 d(y''', y')} \|J\|$, and by the estimates in the proof of (16.2), $\|Q_k^* Q_k F\|_{\Delta(y'')} \le 4 \|F\|_{X_{y''}}$, where $X_{y''}$ is contained in the blocks whose corners are at a distance of at most 11 from y''. Since $d(y'', y) \le 6$, all these terms are at most a constant times $e^{-\theta_2 d(y,y')} \|J\|$ with $\theta_2 \coloneqq \frac{1}{2} \min(\theta_0, \delta_\varpi)$, which is the first bound of (17.3).

For the second, let Φ be a field supported in the bonds starting in $\Delta(y)$. Since G is symmetric, $\langle \Phi, G\nabla^*\nabla^* J \rangle = \langle \nabla\nabla G \Phi, J \rangle$, and the support of J lies in the blocks $\Delta(y'')$ with $d(y'', y') \le 4$. The first bound, with y and y'' exchanged, bounds this by a constant times $e^{-\theta_2 d(y,y')} \|\Phi\| \|J\|$. *q.e.d.*

With Lemma 16.1 this is [2, (1.114)]: for a function ζ supported in $\Delta(y)$ the norms there are at most $\sup|\zeta|$ times the norms over $\Delta(y)$.

It remains to describe the commutators. For a function h on the torus and fields A on T_η/Γ' and B on the unit lattice we write $(hA)(b) \coloneqq h(b_-)A(b)$ and $(hB)(c) \coloneqq h(c_-)B(c)$ for positive bonds. Bałaban defines the commutator of Q_k with h through the contours and the segments of his average [2, (1.120)]. The sums of ∂h over paths which occur there are

differences of the values of h, so that the definition does not depend on the paths, and it can be stated for any average.

Lemma 17.4 (Commutators with a Function)**.** *Let h be a function on T_η/Γ' with $|h(x) - h(x')| \le \varkappa\, d(x,x')$ for all x, x' and $|h(x+\eta e'_\nu) + h(x-\eta e'_\nu) - 2h(x)| \le \varkappa_2 \eta^2$ for all x and ν, and put*

$$S(h) \coloneqq Q_k h - h Q_k \;, \qquad P_1(h) \coloneqq h\, \partial P \partial^* - \partial P \partial^* h \;. \tag{17.5}$$

(i) $Q_k^* Q_k h = h Q_k^* Q_k - S(h)^* Q_k + Q_k^* S(h)$*, and* $\Delta_a h = h \Delta_a - K(h)$ *with*

$$K(h) \coloneqq (h\Delta - \Delta h) + a S(h)^* Q_k - a Q_k^* S(h) - P_1(h) \;. \tag{17.6}$$

(ii) $|(S(h)A)(c)| \le \frac{28}{5} \varkappa \sup|A|$*, the supremum over the bonds which start at a distance of less than 4 from c_-, and $\|S(h)A\| \le 6\varkappa \|A\|$.*

(iii) $P_1(h)$ *has the kernel* $(h(x) - h(x'))\varpi_{\nu\nu'}(x,x')$*, which is at most* $\varkappa c_\varpi |x-x'| e^{-\delta_\varpi |x-x'|}$ *in modulus.*

(iv) $|((h\Delta - \Delta h)A)(x)| \le \varkappa \sum_\nu (|(\partial'_\nu A)(x)| + |(\partial'_\nu A)(x - \eta e'_\nu)|) + 4\varkappa_2 |A(x)|$ *for each component of A.*

Proof. (i) By the definition, $Q_k h = h Q_k + S(h)$, and taking adjoints, $h Q_k^* = Q_k^* h + S(h)^*$. Hence $Q_k^* Q_k h = Q_k^* h Q_k + Q_k^* S(h) = h Q_k^* Q_k - S(h)^* Q_k + Q_k^* S(h)$, and the formula for $K(h)$ follows from $\Delta_a = \Delta - \partial P \partial^* + a Q_k^* Q_k$. (ii) By (6.3), $(S(h)A)(c)$ is the average at c of the field $(h - h(c_-))A$. The bonds which enter it start at a distance of less than 4 from c_-, where $|h - h(c_-)| \le 4\varkappa$, and the two bounds follow from the proof of Lemma 9.1(i), in which the average of a field is at most $\frac{7}{5}$ times its supremum, and its square is bounded by a sum over its own bonds. (iii) is (15.1) and (15.2), and (iv) is (17.4) with h in place of χ. *q.e.d.*

Statement (i) is [2, (1.120), (1.121)]. For the functions h_z of [2, (1.118)], which are products of functions of the coordinates x_μ/M_0, the constants are $\varkappa = O(M_0^{-1})$ and $\varkappa_2 = O(M_0^{-2})$, and (ii)–(iv) give the bound [2, (1.128)] on $h_{z_1} K(h_{z_2})$ as in the text. The expansion [2, (1.119)–(1.123)] with its estimates [2, (1.124)–(1.131)] is therefore available in the tilted run, although we have proved Proposition 16.2 without it.

18 The Scalar Operator of Two Neighbouring Scales

In the second paper on the propagators the averages act at different scales in different regions of the lattice: there is a decreasing sequence of domains, the points of the scale j in the j-th region form a set Λ_j, and the scalar operator is $\Delta'_a = \Delta + Q'^* a Q'$ with $\langle \lambda, Q'^* a Q' \lambda \rangle = \sum_j \sum_{y \in \Lambda_j} a_j (L^j \eta)^{d-2} |(Q'_j \lambda)(y)|^2$ and $a_{j+1} = a a_j / (a L^{-2} + a_j)$, $a_1 = a$ [3, (2.14), p. 225]. These are the numbers $a_j = \alpha_j \ell_j^2$ of Section 13, because $1/a_{j+1} = 1/a + L^{-2}/a_j$. In Bałaban's formulae Δ stands for $\partial^* \partial$; in this section, as in Sections 13 and 14, Δ is the Laplace operator of the lattice, so that his Δ is our $-\Delta$. Bałaban's expansion of the inverse G' starts from local inverses $G'(\square)$ on cubes $\square$ of the size $2ML^j\eta$, each "an inverse of Δ'_a with some boundary conditions on the boundary of $\square$, e.g. with Neumann boundary conditions" [3, (2.37), p. 229]. A cube of the scale j meets the regions of the scales j and $j+1$ at most, and for a cube which meets both he expresses $G'(\square)$ through the operator of the scale j on the box and an operator $C_\Lambda^{(j)}(\square)$ on the points of the scale j in the region of the scale $j+1$ [3, (2.41), (2.42), p. 230]; the two bounds [3, (2.43), (2.44)] on which the expansion rests then follow from the properties of these operators on a box with Neumann conditions.

For a turned fine lattice the box with Neumann conditions is not available (Section 11), and we do not need it. The expansion uses of $G'(\square)$ only that it inverts the operator of the sequence on the support of the function $h_\square$ and that it satisfies [3, (2.43)]. Both hold for an operator on the whole lattice, without any boundary: the operator whose averages are those of the scale $j+1$ on an arbitrary set of cubes and those of the scale j everywhere else. This section constructs it and proves its bounds, by the argument of Sections 13 and 14 with a set of cubes in the place of the whole lattice.

We use the notation of Section 13: the unit is that of the last lattice, $\ell_j = L^{j-k}$, and α_j, $G_{(j)}$, $\Delta^{(j)}$ are as in (13.1) and (13.2). Fix $1 \le j \le k-1$, and write $\mathsf{q} \coloneqq Q'_{\ell_j}$, $Q'^{\mathrm{st}} \coloneqq Q'^{\mathrm{st}}_{\ell_j,\ell_{j+1}}$, and $\beta \coloneqq a\ell_{j+1}^{-2}$, as in the proof of Lemma 13.1. Let Λ' be any set of points of $\ell_{j+1}\mathbb{Z}^4$, let $\Lambda \subset \ell_j\mathbb{Z}^4$ be the union of the cubes of Q'^{st} at the points of Λ', and let

$$\Pi_{\Lambda'} \coloneqq Q'^{\mathrm{st}*}\mathbf{1}_{\Lambda'}Q'^{\mathrm{st}}\,, \qquad \mathcal{D}_{\Lambda'} \coloneqq -\Delta + \alpha_j\,\mathsf{q}^*\mathbf{1}_{\Lambda^c}\mathsf{q} + \alpha_{j+1}\,\mathsf{q}^*\Pi_{\Lambda'}\mathsf{q}\,. \tag{18.1}$$

Here $\Pi_{\Lambda'}$ is the orthogonal projection onto the functions on $\ell_j\mathbb{Z}^4$ which vanish outside Λ and are constant on each cube of Λ, and $\mathcal{D}_{\Lambda'}$ is the operator on the functions on T_η which has the average of the scale ℓ_{j+1} with the constant α_{j+1} on the blocks of Λ' and the average of the scale ℓ_j with the constant α_j on the other blocks; it is $G_{(j)}^{-1}$ for $\Lambda' = \emptyset$ and $G_{(j+1)}^{-1}$ for $\Lambda' = \ell_{j+1}\mathbb{Z}^4$. We write ι for the inclusion of the square-summable functions on Λ into those on $\ell_j\mathbb{Z}^4$.

Lemma 18.1 (Two Neighbouring Scales)**.** *Let $L \ge 64$ and $a > 0$. For every j and every Λ' as above the following hold, with constants which depend on a and L and not on k, j, or Λ'.*

(i) $\mathcal{D}_{\Lambda'} \ge G_{(j+1)}^{-1}$; in particular $\mathcal{D}_{\Lambda'}$ has a bounded inverse.

(ii) The operator $\iota^*\Delta^{(j)}\iota + \beta\Pi_{\Lambda'}$ on the functions on Λ has a bounded inverse C_Λ, and

$$\mathcal{D}_{\Lambda'}^{-1} = G_{(j)} + \alpha_j^2\, G_{(j)}\mathsf{q}^*\iota\, C_\Lambda\, \iota^*\mathsf{q}G_{(j)}\,. \tag{18.2}$$

(iii) The kernel C_Λ of C_Λ with respect to the sum over the points of Λ satisfies $|\mathsf{C}_\Lambda(y,y')| \le c_C\ell_j^2 e^{-\theta_C|y-y'|/\ell_j}$, with the constants of Lemma 14.2.

(iv) For every bounded function f on T_η, every $\alpha \in [0,1)$, all $x, x' \in T_\eta$ with $|x-x'| \le \ell_j$, and every ν, with $D \coloneqq \operatorname{dist}(\{x,x'\}, \operatorname{supp} f)$ and δ of Proposition 14.3,

$$\begin{gathered}|(\mathcal{D}_{\Lambda'}^{-1}f)(x)| \le c\,\ell_j^2 e^{-\delta D/\ell_j}\sup|f|\,, \qquad |(\partial'_\nu\mathcal{D}_{\Lambda'}^{-1}f)(x)| \le c\,\ell_j e^{-\delta D/\ell_j}\sup|f|\,,\\ |(\partial'_\nu\mathcal{D}_{\Lambda'}^{-1}f)(x) - (\partial'_\nu\mathcal{D}_{\Lambda'}^{-1}f)(x')| \le c'_\alpha\ell_j^{1-\alpha}|x-x'|^\alpha e^{-\delta D/\ell_j}\sup|f|\,.\end{gathered} \tag{18.3}$$

(v) With the same notation, $|(\mathcal{D}_{\Lambda'}^{-1}\partial'^*_\nu f)(x)| \le c\,\ell_j e^{-\delta D/\ell_j}\sup|f|$, the difference of the values of $\mathcal{D}_{\Lambda'}^{-1}\partial'^*_\nu f$ at x and x' is at most $c\,\ell_j^{1-\alpha}|x-x'|^\alpha e^{-\delta D/\ell_j}\sup|f|$, and $|(\Delta\mathcal{D}_{\Lambda'}^{-1}f)(x)| \le c\,e^{-\delta D_x/(2\ell_j)}\sup|f|$, where $D_x \coloneqq \operatorname{dist}(x, \operatorname{supp} f)$.

Proof. (i) Let $\Pi \coloneqq Q'^{\mathrm{st}*}Q'^{\mathrm{st}}$, the projection of the proof of Lemma 13.1. It is the sum of $\Pi_{\Lambda'}$ and of the corresponding projection for the complement of Λ', which acts on the functions supported in Λ^c and is therefore at most $\mathbf{1}_{\Lambda^c}$. Since $\alpha_j \ge \alpha_{j+1}$, this gives $\alpha_j\mathbf{1}_{\Lambda^c} + \alpha_{j+1}\Pi_{\Lambda'} \ge \alpha_{j+1}\Pi$, and $\mathsf{q}^*\Pi\mathsf{q} = P_{(j+1)}$ by Lemma 6.3, so that $\mathcal{D}_{\Lambda'} \ge -\Delta + \alpha_{j+1}P_{(j+1)} = G_{(j+1)}^{-1}$, which is bounded below by a positive number (Section 13).

(ii) On the functions on Λ the operator $\Pi_{\Lambda'}$ is an orthogonal projection, and, as in the proof of Lemma 13.1, $\alpha_j - \alpha_j^2(\alpha_j + \beta\Pi_{\Lambda'})^{-1}$ vanishes on its null space and equals α_{j+1} on its range. Hence $\alpha_j \mathbf{1}_{\Lambda^c} + \alpha_{j+1}\Pi_{\Lambda'} = \alpha_j - \alpha_j^2\, \iota(\alpha_j + \beta\Pi_{\Lambda'})^{-1}\iota^*$ on the functions on $\ell_j\mathbb{Z}^4$, and

$$\mathcal{D}_{\Lambda'} = G_{(j)}^{-1} - \alpha_j^2\, \mathsf{q}^*\iota(\alpha_j + \beta\Pi_{\Lambda'})^{-1}\iota^*\mathsf{q}\,. \tag{18.4}$$

We apply the inversion formula of the proof of Lemma 13.1 with $\mathsf{M} = G_{(j)}^{-1}$, $\mathsf{U} = \alpha_j\mathsf{q}^*\iota$, and $\mathsf{K} = (\alpha_j+\beta\Pi_{\Lambda'})^{-1}$. Then $\mathsf{M}-\mathsf{U}\mathsf{K}\mathsf{U}^* = \mathcal{D}_{\Lambda'}$ has a bounded inverse by (i), and $\mathsf{K}^{-1}-\mathsf{U}^*\mathsf{M}^{-1}\mathsf{U} = \alpha_j + \beta\Pi_{\Lambda'} - \alpha_j^2\iota^*\mathsf{q}G_{(j)}\mathsf{q}^*\iota = \iota^*\Delta^{(j)}\iota + \beta\Pi_{\Lambda'}$ by (13.2), which gives the existence of C_Λ and (18.2).

(iii) Since Λ is a union of cubes, Π commutes with $\mathbf{1}_\Lambda$ and $\iota^*\Pi\iota = \Pi_{\Lambda'}$, so that $C_\Lambda^{-1} = \iota^*(C^{(j)})^{-1}\iota$ is the compression of $(C^{(j)})^{-1}$ to the functions on Λ. In the unit ℓ_j, $(C^{(j)})^{-1} = \ell_j^{-2}\mathsf{M}$ with the operator M of the proof of Lemma 14.2, for which that proof gives $\mathsf{M} \geq L^{-2}\min(2c_\Delta, a)$ and $\sum_{y'}|\mathsf{M}(y,y')|(e^{\theta C|y-y'|}-1)$ at most half of this lower bound. Both properties pass to the compression: its quadratic form is the restriction of that of M, and its kernel is the restriction of the kernel of M to $\Lambda\times\Lambda$, whose sums over $y'\in\Lambda$ are not larger. Lemma 14.1 on the set Λ gives the bound, with the constants of Lemma 14.2.

(iv) Proposition 14.3, applied to the run of j steps and written in the unit of the last lattice of the run of k steps, gives the three bounds for $G_{(j)}$ in the place of $\mathcal{D}_{\Lambda'}^{-1}$: the operator $G_{(j)}$ is ℓ_j^2 times the operator of that proposition in the unit ℓ_j, a derivative costs ℓ_j^{-1} and a Hölder quotient of exponent α a further $\ell_j^{-\alpha}$. The second term of (18.2) is the operator T_j of (14.4) with C_Λ in the place of C_j and with the sums over y, y' restricted to Λ, and the estimate (14.5), with its counterparts for the derivative and for the Hölder quotient, uses of C_j only the bound (14.1), which C_Λ satisfies by (iii).

(v) The first two bounds are Proposition 14.4(i) for $G_{(j)}$, in the unit ℓ_j, and its proof for the second term of (18.2): the derivative of the source is moved to the last kernel $\mathcal{K}_j$ by a summation by parts, which replaces one factor ℓ_j^2 by ℓ_j. For the last bound, $-\Delta\mathcal{D}_{\Lambda'}^{-1}f = f - \alpha_j\mathsf{q}^*\mathbf{1}_{\Lambda^c}\mathsf{q}\mathcal{D}_{\Lambda'}^{-1}f - \alpha_{j+1}\mathsf{q}^*\Pi_{\Lambda'}\mathsf{q}\mathcal{D}_{\Lambda'}^{-1}f$ by (18.1). The operators $\mathsf{q}^*\mathbf{1}_{\Lambda^c}\mathsf{q}$ and $\mathsf{q}^*\Pi_{\Lambda'}\mathsf{q}$ replace a function by its means over blocks of diameter at most $4L\ell_j$, or by zero, and $\alpha_{j+1} \leq \alpha_j = a_j\ell_j^{-2} \leq a\ell_j^{-2}$; with the first bound of (18.3) the two averaging terms are at most $2ac\, e^{4L\delta}e^{-\delta D_x/\ell_j}\sup|f|$, and $f(x) = 0$ unless $D_x = 0$. *q.e.d.*

The identity (18.2) is [3, (2.42)] with the whole lattice in the place of the box, and (18.3), read in the unit ℓ_j, is [3, (2.43)]. To use the lemma in the expansion [3, (2.37), (2.38)], let $\square$ be a cube of the scale j of the cover, $1 \leq j \leq k-1$, and let $\Lambda' \coloneqq \Omega_{j+1}^{(j+1)}$ be the set of the points of $\ell_{j+1}\mathbb{Z}^4$ in Ω_{j+1}; we put $G'(\square) \coloneqq \mathcal{D}_{\Lambda'}^{-1}$, an operator which depends on the scale of the cube only. It has the average of the scale $j+1$ on Ω_{j+1} and that of the scale j elsewhere, so that the averaging terms of Δ'_a and of $\mathcal{D}_{\Lambda'}$, which at a point are determined by the block of that point, agree on the regions of the scales j and $j+1$; and the support of $h_\square$ lies in these two regions, for the cover which we fix in Section 20. For the cubes of the scale k, whose functions $h_\square$ are supported in the region of the scale k, we put $G'(\square) \coloneqq G_{(k)}$; the bounds (iv) and (v) of the lemma hold for it with $\ell_k = 1$, by Propositions 14.3 and 14.4(i) and by the proof of the last bound of (v). Hence $h_\square\Delta'_aG'(\square) = h_\square\mathcal{D}_{\Lambda'}\mathcal{D}_{\Lambda'}^{-1} = h_\square$, the computation which leads to [3, (2.38)] goes through with the commutator [3, (2.39)], and (18.3) gives [3, (2.44)], the factor M^{-1} coming from the derivatives of $h_\square$ as there. The commutator of h with the average of a block is the average of $h(x') - h(x)$ over the block, which is [3, (2.39)] with the contour replaced by any path from x to x', and it does not depend on the contours, as in Lemma 17.4. There remains the pair of the scales 0 and 1, in which the average of the

finer scale is the identity, $(Q_0'\lambda)(x) = \lambda(x)$ for $x \in \Lambda_0$ [3, p. 225], so that the region of the scale 0 carries a mass term, and the average of the scale 1 is the tilted one. Let $a_0 > 0$, let Λ' be a set of points of $\ell_1\mathbb{Z}^4$ and $\Lambda \subset T_\eta$ the union of the blocks $B_{\ell_1}(y)$, $y \in \Lambda'$, and let

$$\mathcal{D}^0_{\Lambda'} \coloneqq -\Delta + a_0\eta^{-2}\mathbf{1}_{\Lambda^c} + \alpha_1 Q'^*_{\ell_1}\mathbf{1}_{\Lambda'}Q'_{\ell_1} \tag{18.5}$$

on the functions on T_η.

Lemma 18.2 (The First Scale)**.** *For every Λ', $\eta^2\mathcal{D}^0_{\Lambda'} \geq \gamma_{00} \coloneqq \min(a_0, L^{-2}\min(C_P^{-1}, a))$, and there are constants c and $\theta_{00} > 0$, depending on a, a_0, and L only, such that the kernel g^0 of $(\mathcal{D}^0_{\Lambda'})^{-1}$ with respect to the sum over the points of T_η satisfies $|g^0(x,x')| \leq c\,\eta^2 e^{-\theta_{00}|x-x'|/\eta}$. Consequently $(\mathcal{D}^0_{\Lambda'})^{-1}$ satisfies* (18.3) *and the bounds of Lemma 18.1(v) with η in the place of ℓ_j and $\frac{1}{2}\theta_{00}$ in the place of δ. Moreover $\eta^2\mathcal{D}^0_{\Lambda'} \leq 16 + a_0 + a$.*

Proof. For a function ϕ on T_η, $\langle\phi, \eta^2\mathcal{D}^0_{\Lambda'}\phi\rangle$ is the sum of $\eta^2\|\partial\phi\|^2$, of $a_0\|\mathbf{1}_{\Lambda^c}\phi\|^2$, and of $(a/L^2)\|P\phi\|^2$, where P replaces ϕ by its mean on each block of Λ and by zero elsewhere, since $\eta^2\alpha_1 = a/L^2$. On a block of Λ, Lemma 8.3 for the scale ℓ_1 bounds the squared norm of ϕ minus its mean by $C_P L^2$ times η^2 times the sum of the squares of the differences along the bonds inside the block, so that the part of the form which belongs to the block is at least $L^{-2}\min(C_P^{-1}, a)$ times the squared norm of ϕ on the block, as in the proof of Lemma 14.2; on Λ^c the mass term gives a_0. The bonds inside different blocks are different, which gives the lower bound. The kernel of $\eta^2\mathcal{D}^0_{\Lambda'}$ with respect to the sum over the points is that of $-\eta^2\Delta$, which has the range η, plus a_0 on the diagonal in Λ^c and $(a/L^2)L^{-4}$ for two points of one block of Λ, whose distance is at most $2L\eta$. Hence, with $\mathsf{M} \coloneqq \eta^2\mathcal{D}^0_{\Lambda'}$ and the metric $|x-x'|/\eta$, $\sum_{x'}|\mathsf{M}(x,x')|(e^{\theta|x-x'|/\eta} - 1) \leq 8(e^\theta - 1) + (a/L^2)(e^{2L\theta} - 1) \leq \frac{1}{2}\gamma_{00}$ for $\theta = \theta_{00}$ small, and Lemma 14.1 gives the bound on the kernel with $c = 2/\gamma_{00}$. The bounds (18.3) and those of Lemma 18.1(v) follow as at the end of the proof of Proposition 14.3: the sum of $e^{-\theta_{00}|x-x'|/(2\eta)}$ over $x' \in T_\eta$ is bounded, a difference quotient of the result or of the source costs $2\eta^{-1}$, the Laplace operator costs $16\eta^{-2}$, and two different points of T_η are at least η apart. The upper bound holds because $-\eta^2\Delta \leq 16$, the mass term is at most a_0, and the last term is a/L^2 times a projection. *q.e.d.*

With $G'(\square) \coloneqq (\mathcal{D}^0_{\Lambda'})^{-1}$ for the cubes of the scale 0, and $\Lambda' \coloneqq \Omega_1^{(1)}$ the set of the points of $\ell_1\mathbb{Z}^4$ in Ω_1, so that $\mathcal{D}^0_{\Lambda'}$ has the mass term on the region of the scale 0 and the average of the scale 1 on Ω_1, the same reasoning gives [3, (2.38), (2.44)] for these cubes. The number a_0 is not fixed by the recursion of [3, (2.14)], which starts at $a_1 = a$; any $a_0 > 0$ serves.

19 The Distance of a Sequence of Domains

The decay of the operators of a sequence of domains is measured in [3] by a distance which counts every part of a path in the unit of the region in which it runs. Let $\mathfrak{B} = \bigcup_j \Lambda_j$ [3, (2.45)]. A contour is admissible if its part in the region $B^j(\Lambda_j)$ consists of bonds of the lattice Λ_j, and

$$d(y,y') \coloneqq \inf_{\Gamma_{y,y'}} \sum_{j=0}^{k} (L^j\eta)^{-1}|\Gamma_{y,y'} \cap B^j(\Lambda_j)| \tag{19.1}$$

for $y, y' \in \mathfrak{B}$, the infimum over the admissible contours from y to y', where $|\Gamma|$ is η times the number of fine bonds of Γ [3, (2.46), p. 231]. In the tilted run the definition is the same. The lattices Λ_j, $j \geq 1$, are those of the paper, and Λ_0 consists of points of the turned lattice T_η, so that the part of an admissible contour in the region of the scale 0 runs along the directions

e'_ν. Since we shall need upper bounds for this distance as well as lower ones, we state how we read the definition at the surfaces between the regions. Write $\ell_j \coloneqq L^j\eta$ for $0 \le j \le k$, as in Section 18, let $[\Omega_j] \subset \mathbb{R}^4$ be the union of the cubes $y + [0,\ell_j)^4$ over the points y of $\Omega_j^{(j)}$, so that $\Omega_j = T_\eta \cap [\Omega_j]$, put $[\Omega_0] \coloneqq \mathbb{R}^4$ and $[\Omega_{k+1}] \coloneqq \emptyset$, and let the *closed region* $\mathcal{R}_j$ of the scale j be the closure of $[\Omega_j] \setminus [\Omega_{j+1}]$. A *step* of the scale j is a pair of neighbours of the lattice $\ell_j\mathbb{Z}^4$, or of T_η for $j = 0$, both of which lie in $\mathcal{R}_j$. An admissible contour is a chain of steps, and $d(y,y')$ is the least number of steps of a chain from y to y'; a step of the scale j is a part of the length ℓ_j of the contour in the region of the scale j, which contributes 1 to (19.1). Two consecutive regions $\mathcal{R}_j$ and $\mathcal{R}_{j+1}$ meet in the boundary Σ_{j+1} of $[\Omega_{j+1}]$, the surface of [3, p. 231], and a chain passes from the one to the other at a point of $\ell_{j+1}\mathbb{Z}^4$ on this surface, as in the decomposition [3, (2.47)]. The triangle inequality [3, (2.54)] follows from the definition, since two chains with a common end form a chain. For points $x, x' \in T_\eta$ the distance $d(x,x')$ is that of the points of $\mathfrak{B}$ in whose blocks they lie.

Bałaban's Lemma 2.1 on this distance has two inputs in which the directions of the bonds enter. The first is that a contour is at least as long as the distance of its ends: in [3, (2.48), p. 232] the lengths of the parts of a contour are bounded below by the distances $|y - y_1|$, $|y_l - y'_l|$, $|y'_l - y_{l+1}|$ of their ends, and in [3, (2.58), p. 233] the sums over these ends are computed as products over the coordinates, so that the distance there is the sum of the moduli of the coordinates, which we write $|\cdot|_1$. The second is the lower bound [3, (2.57)] on a part which joins two of the surfaces Σ_j, which comes from the separation [3, (2.2), p. 224] of the domains. For a turned lattice both need a remark.

Lemma 19.1 (Lengths of Contours)**.** *Let Γ be a contour of n bonds of T_η from x to x'. Then $n\eta \ge |x - x'|_2 \ge |x - x'|_\infty$ and $n\eta \ge \frac{5}{7}|x - x'|_1$, where the norms are those of the coordinates of the paper. For a point $t \in \mathbb{R}^4$ and $c > 0$, the sum of $e^{-c|t-w|_1}$ over the points w of T_η is at most $(2/(1 - e^{-5c\eta/7}))^4$, and its sum over the points w of $\ell\mathbb{Z}^4$ at most $(2/(1 - e^{-c\ell}))^4$.*

Proof. The difference $x' - x$ is η times a sum of n vectors $\pm e'_\nu$, each of Euclidean length one, which gives the first inequality. By (2.2), $|e'_\nu|_1 \le \frac{7}{5}$ for every ν, so that $|x' - x|_1 \le \frac{7}{5}n\eta$. For the sums, $\sum_{z\in\mathbb{Z}} e^{-c|s-z|} \le 2\sum_{m\ge 0} e^{-cm} = 2/(1 - e^{-c})$ for every real s, which gives the statement for $\ell\mathbb{Z}^4$ coordinate by coordinate. The points of T_η are the points $\eta\sum_\nu w'_\nu e'_\nu$ with integers w'_ν, and since the matrix which expresses the coordinates of the paper through the coordinates w' is orthogonal with rows of $|\cdot|_1$-norm at most $\frac{7}{5}$, the sum of the moduli of the coordinates w' of a vector is at most $\frac{7}{5}$ of its norm $|\cdot|_1$; hence $e^{-c|t-w|_1} \le e^{-(5c/7)\eta\sum_\nu |t'_\nu - w'_\nu|}$, and the sum over the integers w'_ν is a product as before. *q.e.d.*

We read the separation [3, (2.2)] of the domains with the distance $|\cdot|_\infty$, the one in which the cubes are balls. The complement of $[\Omega_j]$ and the set $[\Omega_{j+1}]$ are unions of cubes of the side $M\ell_j$ of one grid, so that their distance in $|\cdot|_\infty$ is a multiple of $M\ell_j$. It is at least $RM\ell_j$: every cube of the side 5η contains a point of T_η at a distance of at most 3η from any given point of its closure, so that otherwise there would be points of Ω_j^c and of Ω_{j+1} at a distance of at most $(R-1)M\ell_j + 6\eta \le RM\ell_j$. A step of the scale j changes the position by at most ℓ_j in $|\cdot|_\infty$, by Lemma 19.1 for $j = 0$. A part of a chain which joins the surfaces Σ_j and Σ_{j+1} in $\mathcal{R}_j$ has therefore at least RM steps, whatever the directions of its bonds, which is [3, (2.57)] with $\ge$ in the place of $>$; the proof of Lemma 2.1 uses no more. In [3, (2.48)] the parts of a contour which run through the region of the scale 0 are bounded below by $\frac{5}{7}$ of the distance $|\cdot|_1$ of their ends, and the other parts by this distance itself. The sums of [3, (2.58)] run over the scaled images of the ends on unit lattices, which are lattices $\mathbb{Z}^4$ for the surfaces and for the points of the $\Lambda_{j'}$ with $j' \ge 1$, and the turned unit lattice for the points of Λ_0; by

the second part of the lemma each of them is at most the fourth power of $\bar{c}_0(\frac{25}{98}\alpha) \le \bar{c}_0(\frac{1}{4}\alpha)$, where

$$\bar{c}_0(\alpha) := \frac{2}{1 - e^{-\alpha\bar{\delta}_0}} < \frac{4}{\alpha\bar{\delta}_0} \tag{19.2}$$

is the bound which Bałaban uses for his $c_0(\alpha)$ [3, p. 233], and $\bar{\delta}_0 \in (0, 1]$ is the number which stands for the rate δ_0 in his formulae; we write it with a bar because δ_0 is the rate of Proposition 12.2. One factor $\frac{5}{7}$ comes from the length of the contour and one from the sum over a turned lattice, and $\frac{1}{2} \cdot \frac{5}{7} \cdot \frac{5}{7} = \frac{25}{98} \ge \frac{1}{4}$. The rest of the proof of Lemma 2.1 is a summation over the number m of the crossings and over the scales, in which the lattices do not enter. Hence:

Lemma 19.2 (Lemma 2.1 for the Tilted Run)**.** *Let $0 < \bar{\delta}_0 \le 1$ and $0 < \alpha < 1$, let the sequence of domains satisfy [3, (2.2)] with the distance $|\cdot|_\infty$, and let $\frac{1}{4}\alpha\bar{\delta}_0 RM > 8 \log \bar{c}_0(\frac{1}{4}\alpha) + 1$. Then [3, (2.60)–(2.63), p. 234] hold in the tilted run with $\bar{\delta}_0$ in the place of δ_0 and $c_1(\alpha) := 12\bar{c}_0(\frac{1}{4}\alpha)^4$.*

The rate enters the lemma through the product $\alpha\bar{\delta}_0$ only, and the condition on RM at one value of this product implies the condition at every larger value, because $\bar{c}_0$ decreases. We may therefore use the lemma for finitely many rates at once, provided RM is larger than a number which depends on the smallest of them.

The bound [3, (2.60)] uses only [3, (2.57)]. With this lemma the passage from the bound [3, (2.44)] on one term, which is Lemma 18.1(iv) and Lemma 18.2, to the convergence [3, (2.49), (2.50)] of the expansion of G' and to its decay [3, (2.51)–(2.56), (2.64)] is as in the text: these steps compose operators with kernels which decay in the distance (19.1) and sum over the points of $\mathfrak{B}$.

20 The Scalar Operators of a Sequence of Domains

With the local inverses of Section 18 and the distance of Section 19 we can follow Bałaban's construction of the scalar operator $G' = \Delta'^{-1}_a$ of a sequence of domains and of the inverse of $Q'G'^2Q'^*$, which defines the gauge-fixing projection [3, (2.37)–(2.88), pp. 229–238]. Throughout this section a sequence of domains as in [3, (2.1), (2.2), p. 224] is given, with the separation read in the distance $|\cdot|_\infty$, $R \ge 2$, and M is a multiple of L. We use the notation of Section 19: the scale j has the spacing $\ell_j = L^j\eta$, $[\Omega_j]$ is the subset of $\mathbb{R}^4$ which the domain Ω_j fills, $\mathcal{R}_j$ is the closed region of the scale j, and all distances between subsets of $\mathbb{R}^4$ are taken in $|\cdot|_\infty$.

The cover. Bałaban covers the region of each scale j by cubes of the size $2M\ell_j$ which are unions of 2^4 big blocks, takes the functions h of [2, (1.118)] "rescaled to proper scales", so that $\sum_\square h_\square^2 = 1$ [3, (2.36), p. 229], and uses that a cube of the scale j meets the regions of the scales j and $j+1$ only [3, p. 230]. The supports of these functions matter in what follows, and we fix a family with these properties. Let h be the function of one variable of [2, (1.118)], smooth, supported in $(-\frac{2}{3}, \frac{2}{3})$, and with $\sum_{n\in\mathbb{Z}} h(t-n)^2 = 1$. For $0 \le j \le k$ and $y \in M\ell_j\mathbb{Z}^4$ let $\square_{j,y} := y + [-M\ell_j, M\ell_j)^4$ and $h_y^j(x) := \prod_\mu h((x_\mu - y_\mu)/(M\ell_j))$, so that $\sum_y (h_y^j)^2 = 1$ and h_y^j vanishes outside the cube of the side $\frac{4}{3}M\ell_j$ around y. Let V_j be the set of the y with $\square_{j,y} \subset [\Omega_j]$, and

$$\zeta_j := \sum_{y\in V_j} (h_y^j)^2 \,, \qquad c_j := \sin(\tfrac{\pi}{2}\chi(\zeta_j)) \,, \qquad s_j := \cos(\tfrac{\pi}{2}\chi(\zeta_j)) \,, \tag{20.1}$$

with a fixed smooth function $\chi\colon [0,1] \to [0,1]$ which vanishes on $[0, \frac{1}{2}]$ and equals 1 at 1; for $j = k+1$ we put $\zeta_{k+1} \coloneqq 0$. Then $\zeta_0 = 1$, and for $j \geq 1$ the function ζ_j vanishes at the points whose distance from the complement of $[\Omega_j]$ is less than $\frac{1}{3}M\ell_j$, and it equals 1 at the points x whose distance from this complement is at least $\frac{5}{3}M\ell_j$, because every y with $h_y^j(x) \neq 0$ then belongs to V_j. For $y \in V_j$ we put

$$h_{j,y} \coloneqq h_y^j \, \zeta_j^{-1/2} \, c_j \, s_{j+1} \,, \tag{20.2}$$

which is a smooth function, because c_j vanishes where $\zeta_j \leq \frac{1}{2}$. The family $\mathscr{D}$ consists of the cubes $\square = \square_{j,y}$ with $0 \leq j \leq k$, $y \in V_j$, and $h_\square \coloneqq h_{j,y} \neq 0$; we call j the scale of $\square$. We record four properties.

(a) $\sum_{\square \in \mathscr{D}} h_\square^2 = 1$. Indeed the sum over the cubes of the scale j is $c_j^2 s_{j+1}^2 = c_j^2 - c_j^2 c_{j+1}^2$, and $c_j c_{j+1} = c_{j+1}$: the function c_{j+1} vanishes outside $[\Omega_{j+1}]$, whose distance from the complement of $[\Omega_j]$ is at least $RM\ell_j \geq 2M\ell_j$ (Section 19), and there $\zeta_j = 1$ and $c_j = 1$. The sum over j is $c_0^2 - c_{k+1}^2 = 1$.

(b) The support of $h_\square$ lies in the cube of the side $\frac{4}{3}M\ell_j$ with the centre of $\square$, at a distance of at least $\frac{1}{3}M\ell_j$ from the complement of $\square$ and of $[\Omega_j]$; and it lies at a distance of at least $\frac{1}{3}M\ell_{j+1}$ from $[\Omega_{j+2}]$, because $s_{j+1}(x) \neq 0$ requires $\zeta_{j+1}(x) < 1$, hence a distance of less than $\frac{5}{3}M\ell_{j+1}$ from the complement of $[\Omega_{j+1}]$, while $[\Omega_{j+2}]$ is at a distance of at least $RM\ell_{j+1} \geq 2M\ell_{j+1}$ from this complement.

(c) $\square \subset \mathcal{R}_j \cup \mathcal{R}_{j+1}$: by the definition of V_j the cube lies in $[\Omega_j]$, it contains a point at a distance of less than $\frac{5}{3}M\ell_{j+1}$ from the complement of $[\Omega_{j+1}]$, and its side is $2M\ell_j < \frac{1}{3}M\ell_{j+1}$. Since M is a multiple of L, the faces of $\square$ lie in hyperplanes of the grid of the blocks of the scale $j+1$, so that $\square$ is a union of blocks of the sequence.

(d) The function $h_\square$ is Lipschitz continuous with the constant $\varkappa(M\ell_j)^{-1}$, and its second differences along the bonds of T_η are at most $\varkappa_2(M\ell_j)^{-2}\eta^2$, with constants which depend on h and χ only: $h_\square$ is a smooth function of the coordinates of the paper, composed of h_y^j, of ζ_j, which is locally a sum of at most 2^4 terms, and of ζ_{j+1}, whose derivatives are smaller. Every point lies in the supports of at most $2 \cdot 2^4$ of the functions.

The local inverses. For a cube $\square$ of the scale j the operator $G'(\square)$ is the one defined in Section 18: $\mathcal{D}_{\Lambda'}^{-1}$ with $\Lambda' = \Omega_{j+1}^{(j+1)}$ for $1 \leq j \leq k-1$, $(\mathcal{D}_{\Lambda'}^0)^{-1}$ with $\Lambda' = \Omega_1^{(1)}$ for $j = 0$, and $G_{(k)}$ for $j = k$. By (b) the support of $h_\square$ lies in the regions of the scales j and $j+1$, on which the operator of the sequence and the inverse of $G'(\square)$ have the same averaging terms, so that $h_\square \Delta_a' G'(\square) = h_\square$.

The two distances. The bounds of Lemma 18.1 decay in the Euclidean distance, measured in the unit of the cube, and Bałaban's composition of the terms uses the distance (19.1); he passes from one to the other without comment [3, (2.44), (2.51)]. For the turned lattice, and for a cube which meets two regions, we check it.

Lemma 20.1 (The Two Distances in a Cube)**.** *Let $\square \in \mathscr{D}$ be a cube of the scale j, and let $y, y' \in \mathfrak{B} \cap \square$. Then $d(y,y') \leq 2|y-y'|_1/\ell_j + 34L$.*

Proof. The four hyperplanes through the centre of $\square$ divide it into 2^4 boxes of the side $M\ell_j$, which are big blocks of the scale j, and each of them lies in $[\Omega_{j+1}]$ or in its complement, because $[\Omega_{j+1}]$ is a union of big blocks of the scale $j+1$ and these are unions of big blocks

of the scale j. By (c) a box of the first kind lies in $\mathcal{R}_{j+1}$ and a box of the second kind in $\mathcal{R}_j$; we call them coarse and fine.

Let first $j \geq 1$. The points y and y' belong to $\ell_j\mathbb{Z}^4$, which contains $\ell_{j+1}\mathbb{Z}^4$. Let $p_0 = y, p_1, \dots, p_n = y'$ be the path of neighbours in $\ell_j\mathbb{Z}^4$ which changes the coordinates one after the other, each monotonically, so that $n = |y - y'|_1/\ell_j$ and the path crosses each of the four hyperplanes at most once. Let r be the map which replaces every coordinate of a point by the largest multiple of ℓ_{j+1} below or equal to it. Since the faces of the boxes lie in hyperplanes whose positions are multiples of $M\ell_j$, hence of ℓ_{j+1}, the map r maps every box to itself, and it maps the points of the path in the coarse boxes to points of $\mathfrak{B}$. We put $q_i \coloneqq p_i$ if p_i lies in a fine box and $q_i \coloneqq r(p_i)$ if it lies in a coarse one, so that $q_0 = y$ and $q_n = y'$, and we join each q_i to q_{i+1} by a chain. If both points lie in fine boxes, the pair is a step of the scale j. If both lie in coarse boxes, then $q_i = q_{i+1}$ or the pair is a step of the scale $j+1$, and along the part of the path which changes one coordinate the second case occurs at most once in L consecutive instances and once more. There remain the at most four instances in which the path passes between a fine and a coarse box, through a face in a hyperplane $\{x_\mu = t\}$. If the fine box is the one below the face, we move from the point of the fine box in its layer $\{x_\mu = t - \ell_j\}$, by at most $3(L-1)$ steps of the scale j, to the point whose other coordinates are those of the image under r of the point of the coarse box, and one more step of the scale j leads to this image, which lies on the face and hence in $\mathcal{R}_j \cap \mathcal{R}_{j+1}$. If the fine box is the one above, the point of the fine box lies on the face; we move within the face, by at most $3(L-1)$ steps of the scale j, to the point u whose other coordinates are multiples of ℓ_{j+1}, which belongs to $\ell_{j+1}\mathbb{Z}^4$ and to both regions, and the pair of u and $u - \ell_{j+1}e_\mu$, the image under r of the point of the coarse box, is a step of the scale $j+1$. Altogether the chain has at most $n + (n/L + 4) + 4 \cdot 3L$ steps.

For $j = 0$ the points of $\mathfrak{B}$ in the fine boxes are points of T_η, and we use the cells of Section 8. Since M is a multiple of 5, the boxes are unions of cells. A point of a fine box is joined to the corner of its cell by at most 17 bonds inside the cell (Definition 8.2), and two neighbouring corners z and $z + 5\eta e_\mu$ are joined by the connector from z, which has at most 7 bonds and stays in the closure of the cell of z (Lemma 8.1). We apply the construction above to the lattice $5\eta\mathbb{Z}^4$ of the corners in the place of $\ell_j\mathbb{Z}^4$, with $\ell_1 = (L/5) \cdot 5\eta$, starting from the corners of the cells of y and y', whose distance in $|\cdot|_1$ is at most $|y - y'|_1 + 20\eta$, and realising each step between corners in a fine box, or onto a face, by a connector. The path between the corners has $n_c \leq |y - y'|_1/(5\eta) + 4$ steps, and the chain has at most $34 + 7n_c + (5n_c/L + 4) + 4 \cdot 7 \cdot 3L/5$ steps. In both cases the number of steps is at most $2|y - y'|_1/\ell_j + 34L$, because $L \geq 64$. *q.e.d.*

Since $|v|_1 \leq 2|v|_2$ in four dimensions, the lemma gives $e^{-\delta|y-y'|_2/\ell_j} \leq e^{9L\delta}e^{-\frac{1}{4}\delta d(y,y')}$ for such points and every $\delta > 0$. We put

$$\bar{\delta}_0 \coloneqq \tfrac{1}{4}\min(\delta, \tfrac{1}{2}\theta_{00}, 1)\,, \tag{20.3}$$

with δ of Proposition 14.3 and θ_{00} of Lemma 18.2. This number takes the place of Bałaban's δ_0 in the rest of the section, as in Lemma 19.2; it depends on a, a_0, and L.

Proposition 20.2 (Proposition 2.2 for the Tilted Run)**.** *Let $L \geq 64$ be a multiple of 5, $a, a_0 > 0$, and $\bar{\delta}_0$ as in (20.3). There are numbers M_0 and ϱ_0, which depend on a, a_0, and L, such that for every sequence of domains with $R \geq 2$, $M \geq M_0$ a multiple of L, and $RM \geq \varrho_0$ the operator Δ'_a of the sequence has a bounded inverse G', given by the expansion [3, (2.50)] with the cover and the local inverses above, and G' satisfies the bounds [3, (2.67), p. 234] with $\bar{\delta}_0$ in the place of δ_0 and with constants which depend on a, a_0, L, and the Hölder exponent, and not on k, on the sequence, on M, or on R.*

Proof. We follow [3, (2.37)–(2.66)] and name the inputs. Since $h_\square \Delta'_a G'(\square) = h_\square$ and $\sum_\square h_\square^2 = 1$, we have $\Delta'_a G'_0 = I - R$ with G'_0 and R as in [3, (2.37), (2.38)] and the commutator $K(h)$ of [3, (2.39)]: its first two terms are those of (17.4), and its averaging term at a point x of a block of the scale j' is $a_{j'} \ell_{j'}^{-2}$ times the mean of $(h(x') - h(x))\lambda(x')$ over the points x' of this block. Let $\square$ be a cube of the scale j, and let λ be supported in a block of the sequence in $\square$. By (d), and because a block which meets the support of $h_\square$ has the scale j or $j+1$ and a diameter of at most $2L\ell_j$, the three terms of $K(h_\square)$ applied to a function ϕ are at most $\varkappa(M\ell_j)^{-1}$ times the sum of the moduli of the eight difference quotients of ϕ at the point, $4\varkappa_2(M\ell_j)^{-2}|\phi|$, and $2a\varkappa LM^{-1}\ell_j^{-2}$ times the supremum of $|\phi|$ over the block. With $\phi = G'(\square)h_\square\lambda$ and Lemma 18.1(iv), or Lemma 18.2, this gives $|(K(h_\square)G'(\square)h_\square\lambda)(x)| \le cLM^{-1}e^{-\delta D/\ell_j} \sup|\lambda|$, where D is the Euclidean distance of the block of x from the support of λ, which is [3, (2.44)]. Since D differs from the distance of the two points of $\mathfrak{B}$ by at most $4L\ell_j$, Lemma 20.1 and (20.3) turn it into [3, (2.51)] for R, the factor $e^{13L\delta}$ and the number of the cubes which contribute at a point entering the constant. The composition [3, (2.52)–(2.56)] uses the triangle inequality of the distance and Lemma 2.1, which is Lemma 19.2 at $\alpha = \frac{1}{2}$; it gives [3, (2.64), (2.65)] and the convergence of the series for M large. The series $G'_0 \sum_n R^n$ is then a bounded right inverse of Δ'_a on the bounded functions. The operator Δ'_a is bounded below on the square-summable functions, by [3, (2.11)] with Lemma 8.3 in the place of the bound for a cube, and the series applied to a function of finite support gives a square-summable function, by its decay, so that the series is the inverse G'. The first factor $h_\square G'(\square)h_\square$ of a term is bounded by Lemma 18.1(iv) and Lemma 20.1 in the same way; at a point of the region of the scale j the cubes which contribute have the scales j and $j-1$, and $\ell_{j-1} \le \ell_j$. This gives [3, (2.66)], the first bound of [3, (2.67)], and, with the second and third bounds of (18.3) for the first factor, the bounds on $\nabla G'\lambda$ and on its Hölder norm. For the Laplace operator on the left we use (17.4) and the last bound of Lemma 18.1(v).

For the two bounds with a derivative of the source the series $G'_0 \sum_n R^n$ cannot be used term by term, because its terms with $n \ge 1$ contain a derivative of $G'(\square)$ applied to a derivative of the source, which is not bounded in the supremum norm. We use the adjoint series instead. Since G', G'_0, and the operators $G'(\square)$ are symmetric, $G' = \sum_n (R^*)^n G'_0$ with $R^* = \sum_\square h_\square G'(\square)K(h_\square)^*$, and the adjoint of $K(h)$ consists of the operator $g \mapsto \sum_\nu \partial'^*_\nu((\partial'_\nu h)g)$ and its counterpart for the backward differences, of the multiplication by the second differences of h, and of an averaging term of the same form as that of $K(h)$. Every term of the series for $G'\partial'^*_\nu\lambda$ therefore ends with $G'(\square)h_\square\partial'^*_\nu\lambda$, in which $h_\square\partial'^*_\nu\lambda$ is $\partial'^*_\nu(h_\square\lambda)$ plus a difference quotient of $h_\square$ times a translate of λ, and it contains the operators $G'(\square)$ applied to bounded functions and to the adjoint derivatives ∂'^*_ν of bounded functions only. The first two bounds of Lemma 18.1(v) and Lemma 20.1 give the bound [3, (2.51)] for R^* and the bounds on $G'\nabla^*\lambda$ and on its Hölder norm as before. *q.e.d.*

Bałaban's bounds [3, (2.68), (2.83)] compare powers of $L^{j-j'}$ with the factor [3, (2.60)], and we do the same below. This requires $\bar\delta_0 RM$ to be larger than a multiple of $\log L$, a condition which he does not state and which appears among the hypotheses of Lemma 20.5 and of Proposition 20.6.

Conventions for the functions on $\mathfrak{B}$. The average Q' of the sequence maps a function λ on T_η to the function on $\mathfrak{B}$ whose value at $y \in \Lambda_j$ is the mean of λ over the block $B(y) := B_{\ell_j}(y)$ for $j \ge 1$, and $\lambda(y)$ for $j = 0$, in which case we put $B(y) := \{y\}$ [3, p. 225]. We write $j(y)$ for the scale of a point $y \in \mathfrak{B}$ and $\mathsf{w}(y) := \ell_{j(y)}^4$, which is η^4 times the number of the points of $B(y)$. The functions on $\mathfrak{B}$ carry the scalar product $\langle\omega, \omega'\rangle := \sum_{y\in\mathfrak{B}} \mathsf{w}(y)\omega(y)\omega'(y)$ of [3, (2.69)], so that $(Q'^*\omega)(x) = \omega(y)$ for $x \in B(y)$, $Q'Q'^* = I$, and $\|Q'^*\omega\| = \|\omega\|$. As in [3,

(2.68), (2.81), (2.87)], the *kernel* of an operator K on the functions on $\mathfrak{B}$ is the function $K(y,y')$ with $(K\omega)(y) = \sum_{y'} \mathsf{w}(y')K(y,y')\omega(y')$, and the kernel of a product KK' is then $\sum_{y''} \mathsf{w}(y'')K(y,y'')K'(y'',y')$. For two points, $|y-y'|$ without an index is the Euclidean distance, as in Section 18, and we put $\mathsf{z}(\vartheta) := \sum_{n\in\mathbb{Z}^4} e^{-\vartheta|n|}$ for $\vartheta > 0$.

The local operators. For a cube $\square \in \mathscr{D}$ of the scale j we write $A_\square := G'(\square)$, $\ell_\square := \ell_j$, and $\mathfrak{B}_\square := \mathfrak{B} \cap \square$, and we use the letter $\square$ also for the multiplication of a function on $\mathfrak{B}$ by the characteristic function of $\mathfrak{B}_\square$; the function $h_\square$ acts on the functions on $\mathfrak{B}$ through its values at the points of $\mathfrak{B}$. For the inverse of $Q'G'^2Q'^*$ Bałaban takes, for each cube, $C_\square := ((Q'G'(\tilde\square)^2Q'^*)\restriction_\square)^{-1}$ and $C := \sum_\square h_\square C_\square h_\square$, where $G'(\tilde\square)$ is the inverse on a box $\tilde\square$ of the size $4M$ around $\square$ [3, (2.70), p. 235]. We take $A_\square$ in the place of $G'(\tilde\square)$ and put

$$N_\square := \square\, Q' A_\square^2 Q'^*\, \square\,, \qquad C := \sum_{\square\in\mathscr{D}} h_\square C_\square h_\square\,, \tag{20.4}$$

where $C_\square$ is the inverse of $N_\square$ on the functions on $\mathfrak{B}_\square$, extended by zero to the functions on $\mathfrak{B}$; it exists by Lemma 20.4 below. We first collect what the cover gives for the distance (19.1).

Lemma 20.3 (The Support of $h_\square$ and the Distance)**.** *Let $\square \in \mathscr{D}$ be a cube of the scale j, and let $y'' \in \mathfrak{B}$ be a point of the support of $h_\square$.*

(i) Every point of $\mathfrak{B}_\square$ has the scale j or $j+1$, its block lies in $\square$, and $d(y,y') \le 16M+34L$ for all $y, y' \in \mathfrak{B}_\square$.

(ii) $d(y,y'') \ge M/(3L)$ for every $y \in \mathfrak{B}$ outside $\square$.

(iii) $d(u,y'') \ge M/3$ for every $u \in \mathfrak{B}$ whose scale is smaller than j or larger than $j+1$.

Proof. We use four facts. First, a point of $\mathfrak{B}$ of the scale m lies in $[\Omega_m]$ and not in $[\Omega_{m+1}]$, by the definition of Λ_m. Secondly, the closed region $\mathcal{R}_m$ lies in the closure of $[\Omega_m]$; here and below $[\Omega_m] := \emptyset$ for $m > k$ and $\ell_{k+1} := L\ell_k$. Thirdly, a step of the scale m changes the position by at most ℓ_m in $|\cdot|_\infty$ (Section 19), and the distance of a point from a fixed set, taken in $|\cdot|_\infty$, changes by at most as much. Fourthly, $\square$ does not meet the closure of $[\Omega_{j+2}]$: by the proof of (c) the cube contains a point at a distance of less than $\frac{5}{3}M\ell_{j+1}$ from the complement of $[\Omega_{j+1}]$ and its side is less than $\frac{1}{3}M\ell_{j+1}$, so that each of its points is at a distance of less than $2M\ell_{j+1}$ from this complement, while every point of the closure of $[\Omega_{j+2}]$ is at a distance of at least $RM\ell_{j+1} \ge 2M\ell_{j+1}$ from it (Section 19).

(i) Let $y \in \mathfrak{B}_\square$ have the scale m. Since $\square \subset [\Omega_j]$ by the definition of V_j, the point y lies in $[\Omega_j]$, which contains $[\Omega_{m+1}]$ if $m < j$; hence $m \ge j$ by the first fact. If $m \ge j+2$, then $y \in [\Omega_m] \subset [\Omega_{j+2}]$, against the fourth fact. By (c) the cube is a union of blocks of the sequence, and y lies in its block, so that this block lies in $\square$. For $y, y' \in \mathfrak{B}_\square$ we have $|y-y'|_1 \le 4|y-y'|_\infty < 8M\ell_j$, and Lemma 20.1 gives $d(y,y') \le 16M+34L$.

(ii) Let $q_0 = y'', q_1, \dots, q_n = y$ be a chain of steps, and let $\mathsf{d}(q)$ be the distance of q from the complement of $\square$. By (b), $\mathsf{d}(q_0) \ge \frac{1}{3}M\ell_j$, and $\mathsf{d}(q_n) = 0$. Let t be the first index with $\mathsf{d}(q_t) = 0$. For $i < t$ the point q_i lies in $\square$, hence outside the closure of $[\Omega_{j+2}]$ by the fourth fact and in no region $\mathcal{R}_m$ with $m \ge j+2$ by the second, so that the step from q_i to q_{i+1} has a scale of at most $j+1$ and, by the third fact, lowers d by at most ℓ_{j+1}. Hence $t\ell_{j+1} \ge \frac{1}{3}M\ell_j$ and $n \ge t \ge M/(3L)$.

(iii) Let $q_0 = y'', \dots, q_n = u$ be a chain of steps. Suppose first that the scale of u is smaller than j, so that u does not lie in $[\Omega_j]$ by the first fact, and let $\mathsf{d}(q)$ be the distance of

q from the complement of $[\Omega_j]$. Then $\mathsf{d}(q_n) = 0$ and $\mathsf{d}(q_0) \geq \frac{1}{3}M\ell_j$ by (b). Let s be the last index for which q_s lies in the closure of $[\Omega_{j+1}]$, and $s := 0$ if there is none. In the first case $\mathsf{d}(q_s) \geq RM\ell_j$, because the distance of $[\Omega_{j+1}]$ from the complement of $[\Omega_j]$ is at least $RM\ell_j$ (Section 19); in both cases $\mathsf{d}(q_s) \geq \frac{1}{3}M\ell_j$. For $i > s$ the point q_i lies outside the closure of $[\Omega_{j+1}]$, hence in no region $\mathcal{R}_m$ with $m \geq j+1$, so that the step from q_{i-1} to q_i has a scale of at most j and lowers d by at most ℓ_j. Hence $(n-s)\ell_j \geq \frac{1}{3}M\ell_j$ and $n \geq M/3$. Suppose now that the scale of u is larger than $j+1$, so that $u \in [\Omega_{j+2}]$, and let $\mathsf{d}(q)$ be the distance of q from $[\Omega_{j+2}]$. Then $\mathsf{d}(q_n) = 0$ and $\mathsf{d}(q_0) \geq \frac{1}{3}M\ell_{j+1}$ by (b). Let t be the first index with $\mathsf{d}(q_t) = 0$. For $i < t$ the point q_i lies outside the closure of $[\Omega_{j+2}]$, the step from q_i to q_{i+1} has a scale of at most $j+1$ as in (ii), and it lowers d by at most ℓ_{j+1}. Hence $t\ell_{j+1} \geq \frac{1}{3}M\ell_{j+1}$ and $n \geq M/3$. *q.e.d.*

The next lemma contains the three properties of the local operators which the construction uses. Bałaban proves the lower bound [3, (2.78)] with Bessel's inequality and a comparison of the box with the whole lattice [3, (2.73), (2.74)]; for the operators of Section 18 there is no box, and the comparison disappears.

Lemma 20.4 (The Local Inverses)**.** *Let $L \geq 64$ be a multiple of 5 and $a, a_0 > 0$. There are constants c_A, c_N, $c_\square$ and a rate $\theta_\square \in (0, \frac{1}{384}\bar{\delta}_0]$, which depend on a, a_0, and L only, such that for every sequence of domains with $R \geq 2$ and M a multiple of L, and every cube $\square \in \mathscr{D}$ of the scale j, the following hold.*

(i) For every bounded, square-summable function f on T_η and every $x \in T_\eta$, with $D := \operatorname{dist}(x, \operatorname{supp} f)$,

$$|(A_\square f)(x)| \leq c_A \ell_j^2 e^{-4\bar{\delta}_0 D/\ell_j} \sup|f| \,, \qquad |(A_\square^2 f)(x)| \leq c_N \ell_j^4 e^{-2\bar{\delta}_0 D/\ell_j} \sup|f| \,. \tag{20.5}$$

(ii) For every function ω on $\mathfrak{B}_\square$,

$$\langle \omega, N_\square \omega \rangle \geq \gamma_j^2 \, \ell_j^4 \|\omega\|^2 \,, \qquad \gamma_{\mathsf{T}} := \frac{t_0}{1 + a t_0} \,, \qquad \gamma_0 := \frac{1}{16 + a_0 + a} \,, \tag{20.6}$$

where $\gamma_j := \gamma_{\mathsf{T}}$ for $j \geq 1$ and $t_0 := c_u/(4\pi^2)$ with c_u of Section 12.

(iii) The operator $N_\square$ on the functions on $\mathfrak{B}_\square$ has an inverse $C_\square$, whose kernel satisfies

$$|C_\square(y, y')| \leq c_\square \ell_j^{-8} e^{-\theta_\square |y - y'|/\ell_j} \tag{20.7}$$

for all $y, y' \in \mathfrak{B}_\square$.

Proof. (i) The first bound is Lemma 18.1(iv) for $1 \leq j \leq k-1$, Proposition 14.3 for $j = k$, where $\ell_k = 1$, and Lemma 18.2 for $j = 0$, where $\ell_0 = \eta$; the rates of these statements are δ, δ, and $\frac{1}{2}\theta_{00}$, each of which is at least $4\bar{\delta}_0$ by (20.3), and c_A is the largest of their constants. For the second bound let S be the support of f, put $g := A_\square f$ and $r := 4\bar{\delta}_0$, and let g_n, $n \geq 0$, be the product of g and of the characteristic function of the set of the points z with $n\ell_j \leq \operatorname{dist}(z, S) < (n+1)\ell_j$. By the first bound $\sup|g_n| \leq c_A \ell_j^2 e^{-rn} \sup|f|$, and a point of the support of g_n is at a distance of less than $(n+1)\ell_j$ from S, so that $\operatorname{dist}(x, \operatorname{supp} g_n) \geq D - (n+1)\ell_j$. The functions g_n have disjoint supports and their sum g is square-summable, so that $g = \sum_n g_n$ converges in the space of the square-summable functions, on which $A_\square$ is bounded; hence $A_\square g = \sum_n A_\square g_n$ at every point, and the first bound, applied to each g_n, gives with $t := D/\ell_j$

$$|(A_\square^2 f)(x)| \leq c_A^2 \ell_j^4 \sup|f| \sum_{n \geq 0} e^{-rn} e^{-r \max(t-n-1, 0)} \,. \tag{20.8}$$

Each of the at most t terms with $n+1 \le t$ equals $e^r e^{-rt}$, and the terms with $n > t-1$ add up to at most $e^{-r(t-1)}/(1-e^{-r})$. Since $te^{-rt/2} \le 2/(er)$, the sum is at most $e^r(2/(er)+1/(1-e^{-r}))e^{-rt/2}$, which is the second bound with $c_N := c_A^2 e^r(2/(er)+1/(1-e^{-r}))$.

(ii) Let first $j \ge 1$, and let $\mathsf{q} = Q'_{\ell_j}$ as in Section 18. By Lemma 20.3(i) a point of $\mathfrak{B}_\square$ has the scale j or $j+1$. Let ω_1 be the function on $\ell_j\mathbb{Z}^4$ which equals $\omega(y)$ at the points y of $\mathfrak{B}_\square$ of the scale j, equals $\omega(y)$ at the L^4 points z with $y_\mu \le z_\mu < y_\mu + \ell_{j+1}$ for each point y of $\mathfrak{B}_\square$ of the scale $j+1$, and vanishes elsewhere; this is the function $Q''^*\omega$ of [3, p. 236]. Since $\ell_{j+1}^4 = L^4\ell_j^4$, we have $\|\omega_1\| = \|\omega\|$, and since the block $B_{\ell_{j+1}}(y)$ is the union of the blocks $B_{\ell_j}(z)$ over these z (Lemma 6.2(i)), $Q'^*\omega = \mathsf{q}^*\omega_1$. Hence

$$\langle \omega, N_\square \omega\rangle = \|A_\square \mathsf{q}^*\omega_1\|^2 \ge \|\mathsf{q}A_\square\mathsf{q}^*\omega_1\|^2 \ge \frac{\langle \omega_1, \mathsf{q}A_\square\mathsf{q}^*\omega_1\rangle^2}{\|\omega_1\|^2}\ . \tag{20.9}$$

The equality is [3, (2.71), (2.72)]; the first inequality holds because $\mathsf{q}^*\mathsf{q} = P_{(j)}$ is a projection, so that $\|\mathsf{q}\phi\|^2 = \langle\phi, P_{(j)}\phi\rangle \le \|\phi\|^2$ for every ϕ, and the second is Schwarz's inequality, which replaces Bessel's inequality of [3, (2.73)]. For $j \le k-1$ the second term of (18.2), multiplied by q on the left and by q^* on the right, is of the form $\mathsf{X}^*C_\Lambda\mathsf{X}$, and C_Λ is the inverse of a nonnegative operator, hence nonnegative; therefore $\mathsf{q}A_\square\mathsf{q}^* \ge \mathsf{q}G_{(j)}\mathsf{q}^*$, and for $j = k$ the two sides are equal. In the unit ℓ_j the operator $\mathsf{q}G_{(j)}\mathsf{q}^*$ is the multiplication by $\mathsf{T}(p')/(1+a_j\mathsf{T}(p'))$ in the momentum representation, with the symbols of the run of j steps, as in the proof of Lemma 14.2. For $p' \in [-\pi,\pi]^4$ the term $l=0$ of the sum which defines T gives $\mathsf{T}(p') \ge |u_k(p')|^2/\Delta(p') \ge c_u/(4\pi^2) = t_0$, by the bounds $|u_k|^2 \ge c_u$ and $\Delta \le 4\pi^2$ on this cube which the proof of Proposition 12.2 uses; at $p'=0$, where Δ vanishes, the multiplier is $1/a_j$. Since $t \mapsto t/(1+a_jt)$ increases and $a_j \le a$, the multiplier is at least γ_T, so that $\langle\omega_1, \mathsf{q}G_{(j)}\mathsf{q}^*\omega_1\rangle \ge \gamma_\mathsf{T}\ell_j^2\|\omega_1\|^2$ in the unit of the last lattice, which is [3, (2.76)]; with (20.9) this gives (20.6). For $j=0$ we have $A_\square = (\mathcal{D}^0_{\Lambda'})^{-1} \ge \gamma_0\eta^2$ by the upper bound of Lemma 18.2, hence $A_\square^2 \ge \gamma_0^2\eta^4$, and $\langle\omega, N_\square\omega\rangle = \|A_\square Q'^*\omega\|^2 \ge \gamma_0^2\eta^4\|Q'^*\omega\|^2 = \gamma_0^2\eta^4\|\omega\|^2$.

(iii) By (ii) the operator $N_\square$, which acts on a space of finite dimension, has an inverse. For $y, y' \in \mathfrak{B}_\square$ its kernel is $N_\square(y,y') = \mathsf{w}(y')^{-1}\mathsf{n}(y,y')$, where $\mathsf{n}(y,y')$ is the mean of $A_\square^2\mathbf{1}_{B(y')}$ over $B(y)$, because $Q'^*\omega = \sum_{y'}\omega(y')\mathbf{1}_{B(y')}$. By Lemma 20.3(i) the two blocks have scales of at most $j+1$, so that each of them lies in a cube of the side $L\ell_j$, whose Euclidean diameter is $2L\ell_j$, and contains its point of $\mathfrak{B}$. A point of $B(y)$ is therefore at a distance of at least $|y-y'| - 4L\ell_j$ from $B(y')$, and (20.5) gives

$$|\mathsf{n}(y,y')| \le c_N e^{8L\bar\delta_0}\ell_j^4 e^{-2\bar\delta_0|y-y'|/\ell_j}\ . \tag{20.10}$$

The functions $\mathsf{w}(y)^{-1/2}\mathbf{1}_{\{y\}}$, $y \in \mathfrak{B}_\square$, form an orthonormal basis, in which $\ell_j^{-4}N_\square$ has the matrix $\mathsf{M}(y,y') := \ell_j^{-4}\mathsf{w}(y)^{1/2}\mathsf{n}(y,y')\mathsf{w}(y')^{-1/2}$. This matrix is symmetric, because $N_\square$ is self-adjoint, it is bounded below by $\gamma_*^2 := \min(\gamma_\mathsf{T},\gamma_0)^2$ by (ii), and since the weights of two points of $\mathfrak{B}_\square$ differ by the factor L^4 at most, $|\mathsf{M}(y,y')| \le L^2c_Ne^{8L\bar\delta_0}e^{-2\bar\delta_0|y-y'|/\ell_j}$. The differences of the points of $\mathfrak{B}_\square$ lie in $\ell_j\mathbb{Z}^4$ for $j \ge 1$ and in $\eta\rho\mathbb{Z}^4$ for $j=0$, and ρ preserves the Euclidean length, so that for every y and every $\theta \le \bar\delta_0$

$$\sum_{y'\in\mathfrak{B}_\square}|\mathsf{M}(y,y')|(e^{\theta|y-y'|/\ell_j}-1) \le L^2c_Ne^{8L\bar\delta_0}\sum_{n\in\mathbb{Z}^4}e^{-2\bar\delta_0|n|}(e^{\theta|n|}-1)\ . \tag{20.11}$$

The terms of the last sum are at most $e^{-\bar\delta_0|n|}$ and tend to zero with θ, so that the sum tends to zero by dominated convergence, and we let $\theta_\square \in (0, \frac{1}{384}\bar\delta_0]$ be so small that the right side of (20.11) is at most $\frac12\gamma_*^2$ for $\theta = \theta_\square$. Lemma 14.1 on the set $\mathfrak{B}_\square$ with the metric $|y-y'|/\ell_j$ then gives $|\mathsf{M}^{-1}(y,y')| \le (2/\gamma_*^2)e^{-\theta_\square|y-y'|/\ell_j}$; this is the second of the two methods

which Bałaban names for [3, (2.79)]. In the same basis $C_\square$ has the matrix $\ell_j^{-4}\mathsf{M}^{-1}(y,y') = \mathsf{w}(y)^{1/2}C_\square(y,y')\mathsf{w}(y')^{1/2}$, and $\mathsf{w} \ge \ell_j^4$ on $\mathfrak{B}_\square$, which gives (20.7) with $c_\square := 2/\gamma_*^2$. It is [3, (2.81)]. *q.e.d.*

There remains the difference between the squares of G' and of a local operator, for which Bałaban refers to "the usual estimate of the type (1.12) [3] connected with a change of a domain" [3, p. 238]. Here the two operators are the inverses of operators which differ only far from the support of $h_\square$, in the distance of the sequence, and the estimate follows from the resolvent identity. The difference of the two operators themselves cannot be made small in this way at every point, since the decay of $A_\square$ is Euclidean and does not pay the powers of L which the finer regions produce; the identity is therefore arranged so that G', whose decay is measured in the distance of the sequence, stands next to the source.

Lemma 20.5 (The Change of the Domain)**.** *Suppose that the hypotheses of Proposition 20.2 hold, that the condition of Lemma 19.2 holds for $\alpha = \frac{1}{16}$, and that $\bar\delta_0 RM \ge 32 \log L$. There is a constant c_E, which depends on a, a_0, and L only, such that for every cube $\square \in \mathscr{D}$ of the scale j, every bounded function f on T_η which is supported in the union of the blocks $B(y'')$ of the points $y'' \in \mathfrak{B}$ of the support of $h_\square$, and every $x \in T_\eta$,*

$$|((G'^2 - A_\square^2)f)(x)| \le c_E\, \ell_j^4 e^{-\bar\delta_0 M/48} \sup|f| \,. \tag{20.12}$$

Proof. Let S be the set of the points of $\mathfrak{B}$ in the support of $h_\square$. It is finite, so that f has a finite support, and its points have the scales j and $j+1$ by Lemma 20.3(i). For $u \in \mathfrak{B}$ we write $d(u,S)$ for the least of the distances $d(u,y'')$, $y'' \in S$. We write c_G for the constant of the bounds [3, (2.67)] in Proposition 20.2, $\bar a := \max(a, a_0)$, and $c_1(\alpha)$ for the constant of Lemma 19.2; by the remark which follows that lemma, [3, (2.61)] holds for every $\alpha \in [\frac{1}{16}, 1)$.

The resolvent identity. Let $E := \Delta'_a - A_\square^{-1}$. Then $A_\square - G' = A_\square E G'$, and therefore

$$A_\square^2 - G'^2 = A_\square(A_\square - G') + (A_\square - G')G' = A_\square^2 E G' + A_\square E G'^2 \,. \tag{20.13}$$

Both Δ'_a and $A_\square^{-1}$ are the sum of $-\Delta$ and of an averaging term, which at a point z is a constant times the mean over a block which contains z. For Δ'_a it is the block of the sequence, of some scale m, with the constant $a_m \ell_m^{-2} \le \bar a \ell_m^{-2}$. For $A_\square^{-1}$ it is the block of the scale j with the constant $\alpha_j \le a\ell_j^{-2}$ if $z \notin \Omega_{j+1}$, and the block of the scale $j+1$ with the constant $\alpha_{j+1} \le a\ell_{j+1}^{-2}$ if $z \in \Omega_{j+1}$ (Section 18); for $j = 0$ the first of these is the mass $a_0\eta^{-2}$ at the point, and for $j = k$ there is only the first. On $\Omega_j \setminus \Omega_{j+2}$ the two averaging terms agree. Hence $(Eg)(z) = 0$ for $z \in \Omega_j \setminus \Omega_{j+2}$, and at every other point

$$|(Eg)(z)| \le \bar a\, \ell_m^{-2} \sup_B |g| + a\, \ell_{j'}^{-2} \sup_{B'} |g| \,, \tag{20.14}$$

where B is the block of the sequence which contains z and m is its scale, and where B' is the block of the scale $j' := j$ which contains z if $z \notin \Omega_j$, and the block of the scale $j' := j+1$ which contains z if $z \in \Omega_{j+2}$. Since Ω_j and Ω_{j+2} are unions of blocks of the scales j and $j+2$, the block B' is in the first case a union of blocks of the sequence of scales smaller than j, and in the second case it lies in one block of the sequence, of a scale of at least $j+2$. In both cases every block of the sequence which meets B or B' is the block of a point $u \in \mathfrak{B}$ whose scale is smaller than j or larger than $j+1$, so that

$$d(u,S) \ge \tfrac{1}{3}M \tag{20.15}$$

by Lemma 20.3(iii).

The function $G'f$. Let z' lie in the block of a point $u \in \mathfrak{B}$ of the scale m. We write f as the sum of its restrictions to the blocks $B(y'')$, $y'' \in S$, and apply [3, (2.67)] to each of them:

$$|(G'f)(z')| \le c_G \ell_m^2 \sum_{y'' \in S} e^{-\frac{1}{2}\bar\delta_0 d(u,y'')} \sup|f| \le c_G c_1(\tfrac14)\, \ell_m^2 e^{-\frac14 \bar\delta_0 d(u,S)} \sup|f| \,. \tag{20.16}$$

The second inequality takes the factor $e^{-\frac14\bar\delta_0 d(u,y'')} \le e^{-\frac14\bar\delta_0 d(u,S)}$ out of the sum and uses [3, (2.61)] with $\alpha = \frac14$ for the rest.

The powers of L. For $u \in \mathfrak{B}$ of the scale m and $y'' \in S$ we have $m - j \le 2 + \max(|m - j(y'')| - 1, 0)$, because $j(y'') \le j + 1$. By [3, (2.60)] with $\alpha = \frac{1}{16}$ and by $\frac{1}{16}\bar\delta_0 RM \ge 2 \log L$, taking for y'' a point of S nearest to u,

$$L^{2(m-j)} \le L^4 e^{\frac{1}{16}\bar\delta_0 d(u,S)} \,. \tag{20.17}$$

The first term of (20.13). We insert (20.16) into (20.14) with $g = G'f$. For the block B the factor ℓ_m^{-2} cancels ℓ_m^2. For B' in the first case the blocks of the sequence in B' have scales $m' < j$, and $\ell_j^{-2}\ell_{m'}^2 \le 1$. In the second case B' lies in a block of the sequence of the scale $m' \ge j + 2$, and $\ell_{j+1}^{-2}\ell_{m'}^2 = L^{-2}L^{2(m'-j)} \le L^2 e^{\frac{1}{16}\bar\delta_0 d(u,S)}$ by (20.17). With (20.15) and $(\frac14 - \frac{1}{16})\frac13 = \frac{1}{16}$ this gives

$$\sup|EG'f| \le 2\bar a c_G c_1(\tfrac14) L^2 e^{-\bar\delta_0 M/16} \sup|f| \,. \tag{20.18}$$

The function $EG'f$ is bounded and square-summable, and the first bound of (20.5), applied twice, gives $|(A_\square^2 EG'f)(x)| \le c_A^2 \ell_j^4 \sup|EG'f|$.

The second term. Let again z' lie in the block of $u \in \mathfrak{B}$ of the scale m. We write $G'f$ as the sum of its restrictions to the blocks $B(v)$, $v \in \mathfrak{B}$, which converges in the space of the square-summable functions, and apply [3, (2.67)] to each of them, with (20.16) for its supremum:

$$|(G'^2 f)(z')| \le c_G^2 c_1(\tfrac14)\, \ell_m^2 \sum_{v \in \mathfrak{B}} e^{-\frac12\bar\delta_0 d(u,v)}\, \ell_{j(v)}^2 e^{-\frac14\bar\delta_0 d(v,S)} \sup|f| \,. \tag{20.19}$$

By (20.17), $\ell_{j(v)}^2 = \ell_j^2 L^{2(j(v)-j)} \le L^4 \ell_j^2 e^{\frac{1}{16}\bar\delta_0 d(v,S)}$. The triangle inequality gives $e^{-\frac18\bar\delta_0 d(u,v)} e^{-\frac18\bar\delta_0 d(v,S)} \le e^{-\frac18\bar\delta_0 d(u,S)}$, and the remaining factors are at most $e^{-\frac38\bar\delta_0 d(u,v)}$, whose sum over v is at most $c_1(\frac38)$ by [3, (2.61)]. Hence

$$|(G'^2 f)(z')| \le c_G^2 c_1(\tfrac14) c_1(\tfrac38) L^4\, \ell_j^2 \ell_m^2 e^{-\frac18\bar\delta_0 d(u,S)} \sup|f| \,. \tag{20.20}$$

We insert this into (20.14) with $g = G'^2 f$. The three kinds of blocks give the same factors as before, namely 1, $\ell_j^{-2}\ell_{m'}^2 \le 1$, and $L^{-2}L^{2(m'-j)} \le L^2 e^{\frac{1}{16}\bar\delta_0 d(u,S)}$, and with (20.15) and $(\frac18 - \frac{1}{16})\frac13 = \frac{1}{48}$,

$$\sup|EG'^2 f| \le 2\bar a c_G^2 c_1(\tfrac14) c_1(\tfrac38) L^6\, \ell_j^2 e^{-\bar\delta_0 M/48} \sup|f| \,. \tag{20.21}$$

By the first bound of (20.5), $|(A_\square EG'^2 f)(x)| \le c_A \ell_j^2 \sup|EG'^2 f|$. Adding the two terms of (20.13) gives (20.12) with $c_E := 2\bar a c_A c_G c_1(\frac14) L^6 (c_A + c_G c_1(\frac38))$. *q.e.d.*

Proposition 20.6 (Proposition 2.3 for the Tilted Run)**.** *Let $L \ge 64$ be a multiple of 5, $a, a_0 > 0$, $\theta_\square$ as in Lemma 20.4, and $\delta_1 := \frac14\theta_\square$. There are numbers $M_1 \ge M_0$, $\varrho_1 \ge \varrho_0$, and c, which depend on a, a_0, and L only, such that for every sequence of domains with $R \ge 2$, $M \ge M_1$ a multiple of L, and $RM \ge \varrho_1$ the following hold for all $y, y' \in \mathfrak{B}$, of the scales j and j'.*

(i) The kernel of $N \coloneqq Q'G'^2Q'^*$ *satisfies* $|N(y,y')| \le c\,\ell_j^4\ell_{j'}^{-4}e^{-\frac14\bar\delta_0 d(y,y')}$, *which is [3, (2.68), p. 235].*

(ii) The operator N *has a bounded inverse on the square-summable functions on* $\mathfrak{B}$. *On the functions of finite support it is given by the series [3, (2.86)] with the operators* (20.4), *and* $|N^{-1}(y,y')| \le c\,\ell_j^{-4}\ell_{j'}^{-4}e^{-\frac12\delta_1 d(y,y')}$, *which is [3, (2.87), p. 238].*

(iii) Let $P \coloneqq G'Q'^*N^{-1}Q'G'$. *For* $x \in B(y)$, $x' \in B(y')$, *and all* μ *and* ν, *the kernel of* $\partial'_\mu P \partial'^*_\nu$ *with respect to the sum over the points of* T_η *with the weights* η^4 *is at most* $c\,\ell_j^{-2}\ell_{j'}^{-4}e^{-\frac18\delta_1 d(y,y')}$ *in modulus, which is [3, (2.88)] with* $\delta_2 = \frac18\delta_1$.

Proof. We write c_G, $\bar a$, and $c_1(\alpha)$ as in the proof of Lemma 20.5, and we put $\alpha_1 \coloneqq \delta_1/(2\bar\delta_0)$ and $\alpha_2 \coloneqq \delta_1/(8\bar\delta_0)$. Let $\varrho_1 \ge \varrho_0$ be such that $RM \ge \varrho_1$ implies the condition of Lemma 19.2 for $\alpha = \alpha_2$ and the inequality $\delta_1 RM \ge 32\log L$. Since $\delta_1 \le \bar\delta_0/1536$, we have $\alpha_2 < \frac{1}{16}$, so that [3, (2.61)] holds for every $\alpha \in [\alpha_2, 1)$ by the remark which follows Lemma 19.2, and the hypotheses of Lemma 20.5 are satisfied. We use [3, (2.60)] in the following form: if $p > 0$ and $\alpha\bar\delta_0 RM \ge p\log L$, then for all $u, v \in \mathfrak{B}$

$$L^{p(j(u)-j(v))} \le L^p e^{\alpha\bar\delta_0 d(u,v)}\,, \tag{20.22}$$

because $j(u) - j(v) \le 1 + \max(|j(u) - j(v)| - 1, 0)$. By the choice of ϱ_1 it applies with every $p \le 4$ and every $\alpha \ge \alpha_2$.

(i) Since $Q'^*\omega = \sum_{y'} \omega(y')\mathbf{1}_{B(y')}$, the kernel is $N(y,y') = \mathsf{w}(y')^{-1}\mathsf{n}(y,y')$, where $\mathsf{n}(y,y')$ is the mean of $G'^2\mathbf{1}_{B(y')}$ over $B(y)$. By [3, (2.67)], $|G'\mathbf{1}_{B(y')}| \le c_G\ell^2_{j(v)}e^{-\frac12\bar\delta_0 d(v,y')}$ on the block of $v \in \mathfrak{B}$. We write $G'\mathbf{1}_{B(y')}$ as the sum of its restrictions to the blocks, which converges in the space of the square-summable functions, and apply [3, (2.67)] to each of them:

$$|\mathsf{n}(y,y')| \le c_G^2\ell_j^2 \sum_{v\in\mathfrak{B}} e^{-\frac12\bar\delta_0 d(y,v)}\,\ell^2_{j(v)}e^{-\frac12\bar\delta_0 d(v,y')}\,. \tag{20.23}$$

By (20.22) with $p = 2$ and $\alpha = \frac16$, $\ell^2_{j(v)} \le L^2\ell_j^2 e^{\frac16\bar\delta_0 d(y,v)}$. Of the remaining exponent $\frac13\bar\delta_0 d(y,v) + \frac12\bar\delta_0 d(v,y')$, the part $\frac14\bar\delta_0(d(y,v) + d(v,y'))$ is at least $\frac14\bar\delta_0 d(y,y')$ by the triangle inequality, and the part $\frac{1}{12}\bar\delta_0 d(y,v)$ is summed by [3, (2.61)]. This gives (i) with the constant $c_{N'} \coloneqq c_G^2L^2c_1(\frac{1}{12})$.

The operator N is bounded and self-adjoint on the square-summable functions on $\mathfrak{B}$, because Q' and Q'^* have the norm one and G' is bounded. It is bounded below: the operator Δ'_a is the sum of $-\Delta \le 16\eta^{-2}$ and of an averaging term which is at most $\bar a\eta^{-2}$, so that $\|G'\psi\| \ge (16+\bar a)^{-1}\eta^2\|\psi\|$ for every ψ, and $\langle\omega, N\omega\rangle = \|G'Q'^*\omega\|^2 \ge (16+\bar a)^{-2}\eta^4\|\omega\|^2$. Hence N has a bounded inverse, and it remains to identify its kernel.

(ii) *The identity.* The operator C of (20.4) is bounded: $\|C_\square\| \le \gamma_*^{-2}\eta^{-4}$ by (20.6), every point lies in the supports of at most $2\cdot 2^4$ functions $h_\square$ by (d), and $\sum_\square\|h_\square\phi\|^2 = \|\phi\|^2$ by (a), so that $\|C\phi\|^2 \le 32\sum_\square\|h_\square C_\square h_\square\phi\|^2 \le 32\gamma_*^{-4}\eta^{-8}\|\phi\|^2$. Let $\square \in \mathscr{D}$. The support of $h_\square$ lies in $\square$ by (b), so that $h_\square = \square h_\square$ and $Nh_\square = \square N\square h_\square + (1-\square)Nh_\square$. By the definition of $N_\square$, $\square N\square = N_\square + \square Q'(G'^2 - A_\square^2)Q'^*\square$, and $N_\square h_\square C_\square h_\square = h_\square N_\square C_\square h_\square - [h_\square, N_\square]C_\square h_\square = h_\square^2 - [h_\square, N_\square]C_\square h_\square$, because $N_\square C_\square = \square$. Summing over the cubes, with $\sum_\square h_\square^2 = 1$, we obtain

$$\begin{gathered} NC = I - \mathsf{R}\,, \qquad \mathsf{R} \coloneqq \mathsf{R}^{(1)} + \mathsf{R}^{(2)} + \mathsf{R}^{(3)}\,, \qquad \mathsf{R}^{(1)} \coloneqq \sum_\square [h_\square, N_\square]C_\square h_\square\,, \\ \mathsf{R}^{(2)} \coloneqq \sum_\square \square\, Q'(A_\square^2 - G'^2)Q'^* h_\square C_\square h_\square\,, \qquad \mathsf{R}^{(3)} \coloneqq -\sum_\square (1-\square)Nh_\square C_\square h_\square\,, \end{gathered} \tag{20.24}$$

which is [3, (2.82)]; there the last term carries a further partition of unity, which we do not need. We bound the kernels of the three operators. In each of them the term of a cube $\square$ vanishes unless $h_\square(y') \neq 0$, which for a given y' happens for at most 32 cubes, and for such a cube of the scale $j_\square$ we have $\ell_\square^{-4} \le L^4 \mathsf{w}(y')^{-1}$ by Lemma 20.3(i). For two points of $\mathfrak{B}_\square$, Lemma 20.1 gives, as after that lemma,

$$e^{-\theta_\square |y-y'|/\ell_\square} \le e^{9L\theta_\square} e^{-\delta_1 d(y,y')} \,. \tag{20.25}$$

The commutators. The kernel of $[h_\square, N_\square]$ is $(h_\square(y) - h_\square(y''))\mathsf{w}(y'')^{-1}\mathsf{n}_\square(y,y'')$ with the function $\mathsf{n}_\square$ of (20.10), so that the term of $\square$ in $\mathsf{R}^{(1)}(y,y')$ is $\sum_{y'' \in \mathfrak{B}_\square} (h_\square(y) - h_\square(y''))\mathsf{n}_\square(y,y'')C_\square(y'',y')h_\square(y')$, for $y \in \mathfrak{B}_\square$. By (d), $|h_\square(y) - h_\square(y'')| \le \varkappa |y-y''|/(M\ell_\square)$. With (20.10), the inequality $te^{-2\bar\delta_0 t} \le (e\bar\delta_0)^{-1} e^{-\bar\delta_0 t}$, and (20.7), the modulus of this term is at most

$$\frac{\varkappa c_N e^{8L\bar\delta_0} c_\square}{e\bar\delta_0 M} \ell_\square^{-4} \sum_{y'' \in \mathfrak{B}_\square} e^{-\bar\delta_0 |y-y''|/\ell_\square} e^{-\theta_\square |y''-y'|/\ell_\square} \le \frac{\varkappa c_N e^{8L\bar\delta_0} c_\square \mathsf{z}(\frac12\bar\delta_0)}{e\bar\delta_0 M} \ell_\square^{-4} e^{-\theta_\square |y-y'|/\ell_\square} \,, \tag{20.26}$$

where we have used $\theta_\square \le \frac12 \bar\delta_0$, the triangle inequality, and the fact that the differences $y - y''$ lie in $\ell_\square \mathbb{Z}^4$, or in $\eta\rho\mathbb{Z}^4$ for the scale 0. By (20.25) and the remarks before it, $|\mathsf{R}^{(1)}(y,y')| \le c^{(1)} M^{-1} e^{-\delta_1 d(y,y')} \mathsf{w}(y')^{-1}$ with $c^{(1)} := 32L^4 e^{9L\theta_\square} \varkappa c_N e^{8L\bar\delta_0} c_\square \mathsf{z}(\frac12\bar\delta_0)/(e\bar\delta_0)$.

The change of the domain. The kernel of $h_\square C_\square h_\square$ is $h_\square(y'')C_\square(y'',y')h_\square(y')$, and Q'^* maps a function ψ on $\mathfrak{B}$ to $\sum_{y''} \psi(y'')\mathbf{1}_{B(y'')}$. For $y \in \mathfrak{B}_\square$ the term of $\square$ in $\mathsf{R}^{(2)}(y,y')$ is therefore the mean over $B(y)$ of $(A_\square^2 - G'^2)f$ with

$$f := \sum_{y''} h_\square(y'')C_\square(y'',y')h_\square(y')\mathbf{1}_{B(y'')} \,. \tag{20.27}$$

This function is supported in the blocks of the points of the support of $h_\square$, and $\sup|f| \le c_\square \ell_\square^{-8}$ by (20.7). Lemma 20.5 bounds the term by $c_E c_\square \ell_\square^{-4} e^{-\bar\delta_0 M/48}$. This bound does not decay in the distance of y and y', but both points lie in $\square$, so that $\delta_1 d(y,y') \le \delta_1(16M + 34L) \le \bar\delta_0 M/96 + 34L\delta_1$ by Lemma 20.3(i) and $16\delta_1 = 4\theta_\square \le \bar\delta_0/96$. Hence $|\mathsf{R}^{(2)}(y,y')| \le c^{(2)} e^{-\bar\delta_0 M/96} e^{-\delta_1 d(y,y')} \mathsf{w}(y')^{-1}$ with $c^{(2)} := 32L^4 c_E c_\square e^{34L\delta_1}$.

The terms outside the cube. For $y \notin \square$ the term of $\square$ in $\mathsf{R}^{(3)}(y,y')$ is

$$-\sum_{y''} \mathsf{w}(y'')N(y,y'')h_\square(y'')C_\square(y'',y')h_\square(y') \,, \tag{20.28}$$

where y'' runs over the points of $\mathfrak{B}$ in the support of $h_\square$. By (i), (20.7), and (20.25), its modulus is at most

$$c_{N'} c_\square e^{9L\theta_\square} \sum_{y''} \ell_j^4 \ell_\square^{-8} e^{-\frac14 \bar\delta_0 d(y,y'')} e^{-\delta_1 d(y'',y')} \,. \tag{20.29}$$

Since $j(y'') \le j_\square + 1$, (20.22) with $p = 4$ and $\alpha = \frac{1}{16}$ gives $\ell_j^4 \le L^4 \ell_{j(y'')}^4 e^{\frac{1}{16}\bar\delta_0 d(y,y'')} \le L^8 \ell_\square^4 e^{\frac{1}{16}\bar\delta_0 d(y,y'')}$. Of the remaining exponent $\frac{3}{16}\bar\delta_0 d(y,y'')$, one third is at least $\bar\delta_0 M/(48L)$ by Lemma 20.3(ii), one third is at least $\delta_1 d(y,y'')$, because $\delta_1 \le \frac{1}{16}\bar\delta_0$, and gives $e^{-\delta_1 d(y,y')}$ with the last factor of (20.29) by the triangle inequality, and one third is summed over y'' by [3, (2.61)] with $\alpha = \frac{1}{16}$. Hence $|\mathsf{R}^{(3)}(y,y')| \le c^{(3)} e^{-\bar\delta_0 M/(48L)} e^{-\delta_1 d(y,y')} \mathsf{w}(y')^{-1}$ with $c^{(3)} := 32L^{12} c_{N'} c_\square e^{9L\theta_\square} c_1(\frac{1}{16})$.

The series. Altogether, since $e^{-\bar\delta_0 M/96} \le e^{-\bar\delta_0 M/(48L)}$,

$$|\mathsf{R}(y,y')| \le \varepsilon_M \, e^{-\delta_1 d(y,y')} \mathsf{w}(y')^{-1} \,, \qquad \varepsilon_M := c^{(1)} M^{-1} + (c^{(2)} + c^{(3)}) e^{-\bar\delta_0 M/(48L)} \,, \tag{20.30}$$

which is [3, (2.85)]. Two remarks on kernels are needed before we sum the series. The operator $\mathsf{R} = I - NC$ is bounded, and the computations above give its values on the functions of finite support, for which all the sums over the cubes are finite. If a bounded operator K is given on these functions by a kernel with $\sup_y \sum_{y'} \mathsf{w}(y')|K(y,y')| < \infty$, then it is given by this kernel on every bounded, square-summable function ϕ: the truncations of ϕ to finite sets converge to ϕ in the space of the square-summable functions, hence their images converge at every point, and the sums $\sum_{y'} \mathsf{w}(y')K(y,y')\phi(y')$ converge absolutely. By (20.30) and [3, (2.61)] this applies to R. We now show by induction that $|\mathsf{R}^n(y,y')| \le \varepsilon_M^n c_1(\alpha_1)^{n-1} e^{-\frac12\delta_1 d(y,y')}\mathsf{w}(y')^{-1}$ for $n \ge 1$. In the kernel $\sum_v \mathsf{w}(v)\mathsf{R}^n(y,v)\mathsf{R}(v,y')$ of R^{n+1} the weight $\mathsf{w}(v)$ cancels against the factor $\mathsf{w}(v)^{-1}$ of the bound for $\mathsf{R}^n(y,v)$; the triangle inequality gives $e^{-\frac12\delta_1 d(y,v)}e^{-\frac12\delta_1 d(v,y')} \le e^{-\frac12\delta_1 d(y,y')}$, and the remaining factor $e^{-\frac12\delta_1 d(v,y')}$ is summed over v by [3, (2.61)] with $\alpha = \alpha_1$. Let $M_1 \ge M_0$ be such that $\varepsilon_M c_1(\alpha_1) \le \frac12$ for $M \ge M_1$. Then the kernels of the operators R^n, $n \ge 1$, add up to a kernel $\mathsf{S}'(y,y')$ with $|\mathsf{S}'(y,y')| \le 2\varepsilon_M e^{-\frac12\delta_1 d(y,y')}\mathsf{w}(y')^{-1}$, because $\sum_{n\ge1}\varepsilon_M^n c_1(\alpha_1)^{n-1} \le 2\varepsilon_M$. Let ω have a finite support, and let $\phi \coloneqq \omega + \mathsf{S}'\omega$, where $\mathsf{S}'\omega$ is defined through the kernel. Then $|\phi|$ is at most a constant times $\sum_{y'} e^{-\frac12\delta_1 d(\cdot,y')}$, the sum over the support of ω, so that ϕ is bounded, and it is summable, hence square-summable, by [3, (2.61)] and $\mathsf{w} \le 1$. Since the series converges absolutely, $\mathsf{R}\phi = \sum_{n\ge1}\mathsf{R}^n\omega = \phi - \omega$, and (20.24) gives $NC\phi = \omega$. As N is injective, $N^{-1}\omega = C\phi = C(I + \mathsf{S}')\omega$, which is the series [3, (2.86)].

The kernel of the inverse. By (20.7), (20.25), and Lemma 20.3(i), by which $\ell_\square^{-8} \le L^8\mathsf{w}(y)^{-1}\mathsf{w}(v)^{-1}$ for two points of $\mathfrak{B}_\square$, the kernel of C satisfies $|C(y,v)| \le c_{\text{inv}}\mathsf{w}(y)^{-1}\mathsf{w}(v)^{-1}e^{-\delta_1 d(y,v)}$ with $c_{\text{inv}} \coloneqq 32L^8 c_\square e^{9L\theta_\square}$. In the kernel $\sum_v \mathsf{w}(v)C(y,v)\mathsf{S}'(v,y')$ of $C\mathsf{S}'$ the weight $\mathsf{w}(v)$ cancels again, and the triangle inequality and [3, (2.61)] with $\alpha = \alpha_1$ give

$$|N^{-1}(y,y')| \le c_{\text{inv}}(1 + 2\varepsilon_M c_1(\alpha_1))\mathsf{w}(y)^{-1}\mathsf{w}(y')^{-1}e^{-\frac12\delta_1 d(y,y')} \le 2c_{\text{inv}}\,\ell_j^{-4}\ell_{j'}^{-4}e^{-\frac12\delta_1 d(y,y')}\,. \tag{20.31}$$

Since $\mathsf{w} \ge \eta^4$, the sums $\sum_{y'}\mathsf{w}(y')|N^{-1}(y,y')|$ are bounded uniformly in y by [3, (2.61)], and by the remark on kernels above this kernel represents the bounded operator N^{-1} on every bounded, square-summable function.

(iii) Let λ be a function of finite support on T_η. The mean of a function over $B(y_2)$ is $\mathsf{w}(y_2)^{-1}$ times its sum over $B(y_2)$ with the weights η^4, and G' is symmetric, so that

$$(Q'G'\partial_\nu'^*\lambda)(y_2) = \mathsf{w}(y_2)^{-1}\sum_{x'\in T_\eta}\eta^4(\partial_\nu' G'\mathbf{1}_{B(y_2)})(x')\lambda(x')\,. \tag{20.32}$$

This function of y_2 is bounded and square-summable. Applying N^{-1} through its kernel, in which the weight $\mathsf{w}(y_2)$ cancels, then Q'^*, which maps a function ψ to $\sum_{y_1}\psi(y_1)\mathbf{1}_{B(y_1)}$ with convergence in the space of the square-summable functions, and then the bounded operator $\partial_\mu' G'$, we find that $\partial_\mu' P\partial_\nu'^*$ has the kernel

$$K(x,x') = \sum_{y_1,y_2\in\mathfrak{B}}(\partial_\mu' G'\mathbf{1}_{B(y_1)})(x)\,N^{-1}(y_1,y_2)\,(\partial_\nu' G'\mathbf{1}_{B(y_2)})(x') \tag{20.33}$$

with respect to the sum over x' with the weights η^4. By [3, (2.67)], $|(\partial_\mu' G'\mathbf{1}_{B(y_1)})(x)| \le c_G\ell_j e^{-\frac12\bar\delta_0 d(y,y_1)}$ for $x \in B(y)$, and likewise for the last factor with $\ell_{j'}$ and $d(y',y_2)$. With (20.31),

$$|K(x,x')| \le 2c_G^2 c_{\text{inv}}\,\ell_j\ell_{j'}\sum_{y_1,y_2}\mathsf{w}(y_1)^{-1}\mathsf{w}(y_2)^{-1}e^{-\frac12\bar\delta_0 d(y,y_1)}e^{-\frac12\delta_1 d(y_1,y_2)}e^{-\frac12\bar\delta_0 d(y_2,y')}\,. \tag{20.34}$$

By (20.22) with $p = 4$ and $\alpha = \alpha_2$, $\mathsf{w}(y_1)^{-1} \le L^4 \ell_j^{-4} e^{\frac{1}{8}\delta_1 d(y,y_1)}$ and $\mathsf{w}(y_2)^{-1} \le L^4 \ell_{j'}^{-4} e^{\frac{1}{8}\delta_1 d(y_2,y')}$. Since $\bar\delta_0 \ge \delta_1$, each of the three exponents which remain is at least $\frac{3}{8}\delta_1$ times its distance. The parts $\frac{1}{4}\delta_1$ give $e^{-\frac{1}{4}\delta_1 d(y,y')}$ by the triangle inequality, and the parts $\frac{1}{8}\delta_1 d(y,y_1)$ and $\frac{1}{8}\delta_1 d(y_2,y')$ are summed by [3, (2.61)] with $\alpha = \alpha_2$. Hence $|K(x,x')| \le 2c_G^2 c_{\text{inv}} L^8 c_1(\alpha_2)^2\, \ell_j^{-3} \ell_{j'}^{-3} e^{-\frac{1}{4}\delta_1 d(y,y')}$. Finally $\ell_j^{-3}\ell_{j'}^{-3} = \ell_j^{-2}\ell_{j'}^{-4} L^{j'-j} \le L\, \ell_j^{-2}\ell_{j'}^{-4} e^{\frac{1}{8}\delta_1 d(y,y')}$ by (20.22) with $p = 1$ and $\alpha = \alpha_2$, which gives (iii). The constant c of the proposition is the largest of $c_{N'}$, $2c_{\text{inv}}$, and $2c_G^2 c_{\text{inv}} L^9 c_1(\alpha_2)^2$. *q.e.d.*

The two propositions and [3, (2.88)] are all that Section B of [3] contains. In the tilted run they hold with constants which depend on L, as those of Section 14 do, and the conditions that M and RM be large are conditions in terms of a, a_0, and L. Three of the numbers differ from Bałaban's in kind and not only in value. The factor $e^{-\frac{1}{8}\delta_0 M}$ of [3, (2.83)] becomes $e^{-\bar\delta_0 M/(48L)}$, because a chain which leaves a cube of the scale j may run through the part of the cube in the region of the scale $j+1$, where a step is L times longer; the rate δ_1 is chosen small compared with $\bar\delta_0$, so that the term of the change of the domain, which is bounded without decay inside a cube, decays at the rate δ_1 at the cost of a part of its small factor; and RM has to be large compared with $\delta_1^{-1} \log L$.

21 The Operators of the Gauge Functions

In Section C of [3] the propagator of a cube which meets the regions of two scales is expressed through operators of one scale and of the unit lattice, by a change of the gauge [3, (2.95)–(2.129), pp. 240–246]. The gauge transformations which leave the gauge-fixing term of the finer scale unchanged are the functions $\lambda = H'_j \mu$ of [3, (2.101), p. 241], the minimisers of $\|\Delta\lambda\|^2$ at given averages μ, and the integral over their parameters is Gaussian with the quadratic form $\Delta'_j = H'^*_j \Delta^2 H'_j$ [3, (2.107), p. 242]. Bałaban reads the properties of these two operators off their momentum representations [3, (2.132), (2.108)], "together with the analyticity method described in [3]" [3, p. 246]. For the tilted run the representations are the same with the symbols of Section 7, and the method is that of Proposition 12.2 with the square of the Laplace operator in the place of the operator itself. This section carries it out. In Bałaban's formulae Δ stands for $\partial^*\partial$; here, as in Section 12, $\Delta(p)$ is the nonnegative symbol (7.1), and only squares of the Laplace operator occur, so that the sign plays no role.

The setting is that of Section 12: the fine lattice is the infinite lattice T_η, the unit lattice is $\mathbb{Z}^4$, $Q' \coloneqq Q'_k$, the adjoints are taken with the weights η^4 and 1, a momentum of the fine lattice is written $p' + l$ with $p' \in [-\pi,\pi]^4$ and $l \in \mathfrak{L}$, and U is the complex neighbourhood of $[-\pi,\pi]^4$ of the proof of Proposition 12.2. For $p' \in U$ let

$$\mathsf{S}_*(p') \coloneqq \sum_{l \ne 0} \frac{u_k(p'+l) u_k(-p'-l)}{\Delta(p'+l)^2}\,, \qquad \mathcal{D}'(p') \coloneqq u_k(p') u_k(-p') + \Delta(p')^2\, \mathsf{S}_*(p')\,. \tag{21.1}$$

For real $p' \ne 0$ we have $\mathcal{D}'(p') = \Delta(p')^2 \mathsf{S}(p')$ with the function S of Lemma 9.2, and $\mathcal{D}'(p') \ge |u_k(p')|^2 \ge c_u$ for every real $p' \in [-\pi,\pi]^4$, by (7.8) and with $c_u = \frac{1}{4}(2/\pi)^8$ as in Section 12. We define an operator H' from the square-summable functions on $\mathbb{Z}^4$ to those on T_η, and an operator Δ' on the former, by their momentum representations

$$\widetilde{(H'\mu)}(p'+l) \coloneqq \frac{u_k(-p'-l)\,\Delta(p')^2}{\Delta(p'+l)^2\,\mathcal{D}'(p')}\,\tilde\mu(p')\,, \qquad \widetilde{(\Delta'\mu)}(p') \coloneqq \frac{\Delta(p')^2}{\mathcal{D}'(p')}\,\tilde\mu(p')\,, \tag{21.2}$$

where for $l = 0$ the first quotient is read as $u_k(-p')/\mathcal{D}'(p')$. For $p' \ne 0$ these are [3, (2.132), (2.108)], written with $\mathcal{D}'$ so that the point $p' = 0$ needs no separate treatment.

Lemma 21.1 (The Minimiser of the Gauge Functions)**.** *Let* $10\pi\eta \le \frac{1}{2}$.

(i) The operators H' *and* Δ' *are bounded,* $Q'H' = I$, $\Delta^2 H' = Q'^*\Delta'$, *and* $\Delta' = H'^*\Delta^2 H'$.

(ii) For every square-summable μ *on* $\mathbb{Z}^4$ *and every square-summable* λ *on* T_η *with* $Q'\lambda = \mu$, $\|\Delta\lambda\|^2 = \langle\mu, \Delta'\mu\rangle + \|\Delta(\lambda - H'\mu)\|^2$.

(iii) With c_{S} *of Lemma 9.2 and* $C_{\mathsf{S}} := \frac{\pi^4}{16} + 400(1 + \frac{\pi^2}{4})^4$, *and with the Laplace operator* Δ_0 *of the unit lattice,* $C_{\mathsf{S}}^{-1}\|\Delta_0\omega\|^2 \le \langle\omega, \Delta'\omega\rangle \le c_{\mathsf{S}}^{-1}\|\Delta_0\omega\|^2 \le 256\, c_{\mathsf{S}}^{-1}\|\omega\|^2$ *for every square-summable* ω *on* $\mathbb{Z}^4$.

Proof. (i) For real p' the multiplier of Δ' is at most $(4\pi^2)^2/c_u$, since $\Delta(p') \le |p'|^2 \le 4\pi^2$. The squared norm of $H'\mu$ is the integral of $|\tilde\mu(p')|^2\mathcal{D}'(p')^{-2}$ times $|u_k(p')|^2 + \Delta(p')^4\sum_{l\neq 0}|u_k(p'+l)|^2\Delta(p'+l)^{-4}$, and this sum is bounded because $|u_k| \le 1$, $\Delta(p'+l) \ge 4$ for $l \neq 0$, and $\sum_{l\neq 0}|u_k(p'+l)|^2 \le 25(1+\pi^2/4)^4$ (proof of Lemma 7.3). By Lemma 7.1(i), as in the proof of Proposition 12.2, $\widetilde{(Q'f)}(p') = \sum_l u_k(p'+l)\tilde f(p'+l)$ and $\widetilde{(Q'^*g)}(p'+l) = u_k(-p'-l)\tilde g(p')$. Hence $\widetilde{(Q'H'\mu)}(p') = \mathcal{D}'(p')^{-1}(u_k(p')u_k(-p') + \Delta(p')^2\mathsf{S}_*(p'))\tilde\mu(p') = \tilde\mu(p')$, and $\widetilde{(\Delta^2 H'\mu)}(p'+l) = u_k(-p'-l)\Delta(p')^2\mathcal{D}'(p')^{-1}\tilde\mu(p') = \widetilde{(Q'^*\Delta'\mu)}(p'+l)$. The third identity follows from the first two: $H'^*\Delta^2 H' = H'^*Q'^*\Delta' = (Q'H')^*\Delta' = \Delta'$.

(ii) Let $\nu := \lambda - H'\mu$, so that $Q'\nu = 0$ by (i). The operator Δ is bounded and symmetric, and by (i) $\langle\Delta H'\mu, \Delta\nu\rangle = \langle\Delta^2 H'\mu, \nu\rangle = \langle Q'^*\Delta'\mu, \nu\rangle = \langle\Delta'\mu, Q'\nu\rangle = 0$ and $\|\Delta H'\mu\|^2 = \langle\mu, H'^*\Delta^2 H'\mu\rangle = \langle\mu, \Delta'\mu\rangle$. Expanding $\|\Delta(H'\mu + \nu)\|^2$ gives the assertion.

(iii) For real $p' \neq 0$ the multiplier of Δ' is $\mathsf{S}(p')^{-1}$, and $\mathsf{S} \ge c_{\mathsf{S}}\Delta_0^{-2}$ by Lemma 9.2. For the upper bound of S we use the inequalities of the proof of that lemma: the term $l = 0$ is at most $\Delta(p')^{-2} \le (\pi^2/4)^2|p'|^{-4} \le (\pi^4/16)\Delta_0(p')^{-2}$, and the terms with $l \neq 0$ add up to at most $\frac{1}{16}\cdot 25(1+\pi^2/4)^4 \le 400(1+\pi^2/4)^4\Delta_0(p')^{-2}$, because $\Delta(p'+l) \ge 4$ and $\Delta_0 \le 16$. Hence $C_{\mathsf{S}}^{-1}\Delta_0^2 \le \mathsf{S}^{-1} \le c_{\mathsf{S}}^{-1}\Delta_0^2$, and $\Delta_0 \le 16$ gives the last inequality. *q.e.d.*

Statement (ii) says that $H'\mu$ is the solution of the variational problem behind [3, (2.102)], and (iii) is [3, (2.109)]. The identity $\Delta^2 H' = Q'^*\Delta'$ contains [3, (2.99)]: the function $\Delta H'\mu$ is the image of $Q'^*\Delta'\mu$ under the inverse of Δ.

Proposition 21.2 (The Kernels of H' and Δ')**.** *Let* $32\eta \le \frac{1}{2}$. *There are constants* $\delta_0' \in (0, \frac{1}{8}]$ *and* c', *and for every* $\alpha \in [0,1)$ *a constant* c'_α, *which do not depend on* k *and* L, *with the following properties. Let* $\mathcal{H}(x,y)$, $x \in T_\eta$, $y \in \mathbb{Z}^4$, *be the kernel of* H', *so that* $(H'\mu)(x) = \sum_y \mathcal{H}(x,y)\mu(y)$, *and let* $\partial^{(n)}$ *be a product of* n *of the operators* ∂'_ν, *acting on the first argument.*

(i) For $0 \le n \le 3$, $|(\partial^{(n)}\mathcal{H})(x,y)| \le c' e^{-\delta_0'|x-y|}$, *and for* $n = 3$ *and* $|x - x'| \le 1$,

$$|(\partial^{(3)}\mathcal{H})(x,y) - (\partial^{(3)}\mathcal{H})(x',y)| \le c'_\alpha|x-x'|^\alpha e^{-\delta_0'\min(|x-y|,|x'-y|)}\,. \tag{21.3}$$

(ii) The operator Δ' *commutes with the translations of* $\mathbb{Z}^4$, *and its kernel satisfies* $|\Delta'(y,y')| \le c' e^{-\delta_0'|y-y'|}$.

Proof. The representation. As in (12.3), by (21.2),

$$\begin{aligned} \mathcal{H}(x,y) &= (2\pi)^{-4}\int_{[-\pi,\pi]^4}\Psi(p')\,\mathrm{d}p'\,, \\ \Psi(p') &:= \frac{1}{\mathcal{D}'(p')}\Big(e^{ip'\cdot(x-y)}u_k(-p') + \sum_{l\neq 0} e^{i(p'+l)\cdot(x-y)}u_k(-p'-l)\frac{\Delta(p')^2}{\Delta(p'+l)^2}\Big)\,, \end{aligned} \tag{21.4}$$

and Ψ is periodic with respect to $2\pi\mathbb{Z}^4$, a translation of p' permuting the classes.

The terms $l \neq 0$ on U. Let $p' \in U$ and $l \neq 0$, and let $\hat{p}$ and q be defined from the real part of $p'+l$ as in Section 12. By (12.6), $|\Delta(p'+l)| \geq (2/\pi^2)|\hat{p}|^2$ and $|\hat{p}| \geq 2.8$, and by Lemma 12.1(i), applied to $p'+l$ and to $-p'-l$, $|u_k(p'+l)u_k(-p'-l)| \leq e^{9/2}\prod_\mu \min(1, \pi/|q_\mu|)$. Hence S_* and $\mathcal{D}'$ are analytic on U, and with the sum Σ_0 of (12.7) and $|\hat{p}|^{-4} \leq |\hat{p}|^{-2}/7.84$,

$$|\mathsf{S}_*(p')| \leq \frac{\pi^4}{4} e^{9/2} \sum_{l\neq 0} |\hat{p}|^{-4} \prod_\mu \min(1, \pi/|q_\mu|) \leq C'_* := \frac{\pi^4}{31} e^{9/2} C_0 \,. \tag{21.5}$$

The denominator. Let $s \in [-\pi, \pi]^4$ and $\tau \in \mathbb{R}^4$ with $|\tau_\mu| \leq \delta \leq \frac{1}{8}$, and put $p' := s + i\tau$. As in the proof of Proposition 12.2, Cauchy's estimate on the disc of the radius $\frac{1}{4}$ gives $|\mathsf{S}_*(p') - \mathsf{S}_*(s)| \leq 8C'_*\delta$, Lemma 12.1(ii) and (i) give $|u_k(p')u_k(-p') - |u_k(s)|^2| \leq 8\delta(e^{9/4}+1)$, and Lemma 12.1(iii) gives $|\Delta(p') - \Delta(s)| \leq 52\delta$ with $0 \leq \Delta(s) \leq 4\pi^2$, so that $|\Delta(p')| \leq 4\pi^2 + 7$ and $|\Delta(p')^2 - \Delta(s)^2| \leq 52\delta(8\pi^2+7)$. Therefore

$$|\mathcal{D}'(p') - \mathcal{D}'(s)| \leq \delta(8(e^{9/4}+1) + 52(8\pi^2+7)C'_* + 128\pi^4 C'_*) \,, \tag{21.6}$$

where the second term bounds $|\Delta(p')^2 - \Delta(s)^2||\mathsf{S}_*(p')|$ and the third $\Delta(s)^2|\mathsf{S}_*(p') - \mathsf{S}_*(s)|$. Let $\delta'_0 \leq \frac{1}{8}$ be so small that the right side with $\delta = \delta'_0$ is at most $\frac{1}{2}c_u$. Since $\mathcal{D}'(s) \geq c_u$, we have $|\mathcal{D}'(s+i\tau)| \geq \frac{1}{2}c_u$ whenever $|\tau_\mu| \leq \delta'_0$; the function Ψ is analytic in a neighbourhood of the set of these $s + i\tau$, and by its periodicity and Cauchy's theorem in each variable $\int \Psi(s)\,\mathrm{d}s = \int \Psi(s+i\tau)\,\mathrm{d}s$ over $[-\pi,\pi]^4$.

The bounds. Let $\tau_\mu := \delta'_0$ if $x_\mu \geq y_\mu$ and $\tau_\mu := -\delta'_0$ otherwise, so that the exponentials in (21.4) at $p' = s + i\tau$ are at most $e^{-\delta'_0|x-y|}$ in modulus. A difference operator ∂'_ν multiplies the term l by a factor of modulus at most $2(|\hat{p}|+1)$, as in the proof of Proposition 12.2, where $\hat{p}$ belongs to $p'+l$; for $l=0$ this is at most $2(2\pi+1)$, and for $l \neq 0$ at most $3|\hat{p}|$, because $|\hat{p}| \geq 2.8$. With n such operators the term $l=0$ is at most $e^{9/4}(4\pi+2)^n$, and by (12.6) and Lemma 12.1(i) the terms $l \neq 0$ add up to at most

$$e^{9/4}(4\pi^2+7)^2 \frac{\pi^4}{4} 3^n \sum_{l\neq 0} |\hat{p}|^{n-4} \prod_\mu \min(1, \pi/|q_\mu|) \,, \tag{21.7}$$

where the sums are those of (12.7): for $n \leq 2$ the sum is at most $\Sigma_0 \leq C_0$, since $|\hat{p}| \geq 1$, and for $n = 3$ it is $\Sigma_1 \leq C_1$. Dividing by $|\mathcal{D}'| \geq \frac{1}{2}c_u$ gives the first part of (i). For (21.3) we may assume $|x'-y| \leq |x-y|$ and choose τ for the pair (x', y); the difference has the terms of $(\partial^{(3)}\mathcal{H})(x', y)$ multiplied by $e^{i(p'+l)\cdot(x-x')} - 1$, of modulus at most $4(|\hat{p}|+1)^\alpha |x-x'|^\alpha$, and the terms $l \neq 0$ are bounded through $\Sigma_{1+\alpha} \leq C_{1+\alpha}$.

(ii) The operator Δ' is a multiplication in the momentum representation of the unit lattice, so that it commutes with the translations and $\Delta'(y, y') = (2\pi)^{-4} \int e^{ip'\cdot(y-y')} \Delta(p')^2 \mathcal{D}'(p')^{-1}\,\mathrm{d}p'$. For real $p' \neq 0$ the integrand is $e^{ip'\cdot(y-y')}\mathsf{S}(p')^{-1}$, which is periodic with respect to $2\pi\mathbb{Z}^4$, and it is analytic where $|\tau_\mu| \leq \delta'_0$. Moving the integration as before, we obtain $|\Delta'(y, y')| \leq 2(4\pi^2+7)^2 c_u^{-1} e^{-\delta'_0|y-y'|}$. *q.e.d.*

The proposition gives the properties which Bałaban states after [3, (2.132)]: the derivatives of H'_j up to the third order and the Hölder norms of the third are bounded and decay exponentially, uniformly. Three is the largest number of derivatives for which the sum (21.7) converges. For the run of j steps, in the unit ℓ_j, H'_j and Δ'_j are the operators H' and Δ' with the fine spacing L^{-j}. On a torus the operators are defined by the same multipliers on the dual of the torus, their kernels are the sums of the kernels above over the periods,

and Lemma 21.1 holds as it stands; the constants are mapped to themselves, because the multiplier of H' at $p' = 0$ is 1 for $l = 0$ and 0 for $l \neq 0$.

The integral over the parameters of the gauge transformations runs over the functions ω on a set of cubes with vanishing means, $\int \mathrm{d}\omega \restriction_\Lambda \delta(Q'_1\omega)$ in [3, (2.106)], and its covariance $C'^{(j)}_\Lambda$ is the inverse of Δ'_j on this space. Bałaban obtains its decay from [3, (2.110)] and from the theorem on the operators of the unit lattice in [1]; we use Lemma 14.1. Let Λ' be a set of points of $L\mathbb{Z}^4$, let $\Lambda \subset \mathbb{Z}^4$ be the union of the cubes $\{z \,|\, y_\mu \leq z_\mu < y_\mu + L\}$, $y \in \Lambda'$, let $\mathcal{S}_\Lambda$ be the space of the square-summable functions on $\mathbb{Z}^4$ which vanish outside Λ and have the mean zero on each of these cubes, and let E be the orthogonal projection onto $\mathcal{S}_\Lambda$. It is $\mathbf{1}_\Lambda - \Pi_{\Lambda'}$, where $\Pi_{\Lambda'}$ replaces a function by its means over the cubes of Λ and by zero elsewhere, as in (18.1).

Lemma 21.3 (The Covariance of the Gauge Functions)**.** *Let $32\eta \leq \frac{1}{2}$ and $L \geq 2$. For every Λ' and every $\omega \in \mathcal{S}_\Lambda$,*

$$\gamma' \|\omega\|^2 \leq \langle \omega, \Delta'\omega \rangle \leq 256\, c_{\mathsf{S}}^{-1} \|\omega\|^2 \,, \qquad \gamma' \coloneqq \frac{4}{C_{\mathsf{S}} L^4} \,. \tag{21.8}$$

The operator $E\Delta'E$ on $\mathcal{S}_\Lambda$ has an inverse C'_Λ, and there are constants c'' and $\theta' > 0$, which depend on L and not on k, η, or Λ', such that the kernel of $C'_\Lambda E$ on the functions on $\mathbb{Z}^4$ satisfies $|(C'_\Lambda E)(y, y')| \leq c'' e^{-\theta'|y-y'|}$.

Proof. Let Π be the orthogonal projection onto the functions which are constant on each cube of the side L of the grid $L\mathbb{Z}^4$. As in the proof of Lemma 14.2, Δ_0 is at least the sum of the Laplace operators of these cubes with Neumann conditions, each of which is at least $2/L^2$ on the functions of mean zero, so that $\Delta_0 + (2/L^2)\Pi \geq 2/L^2$ and $\|(\Delta_0 + (2/L^2)\Pi)\omega\| \geq (2/L^2)\|\omega\|$ for every ω. A function $\omega \in \mathcal{S}_\Lambda$ has the mean zero on every cube of the grid, so that $\Pi\omega = 0$ and $\|\Delta_0\omega\| \geq (2/L^2)\|\omega\|$. With Lemma 21.1(iii) this gives (21.8); this is [3, (2.110)].

Let $N \coloneqq E\Delta'E + (1-E)$ on the square-summable functions on $\mathbb{Z}^4$. Then $N \geq \min(\gamma', 1) = \gamma'$, and N commutes with E, so that $C'_\Lambda \coloneqq N^{-1} \restriction_{\mathcal{S}_\Lambda}$ maps $\mathcal{S}_\Lambda$ to itself and inverts $E\Delta'E$ there. The kernel of E is $\mathbf{1}_\Lambda(y)$ on the diagonal minus L^{-4} for two points of one cube of Λ, whose distance is at most $2L$; hence $\sum_v |E(y,v)| \leq 2$ for every y, and by Proposition 21.2(ii) the kernel of $E\Delta'E$ is at most $4c' e^{4L\delta'_0} e^{-\delta'_0|y-y'|}$ in modulus, while that of $1 - E$ vanishes for $|y - y'| > 2L$ and is at most 1. Therefore $\sum_{y'} |N(y,y')|(e^{\theta|y-y'|} - 1)$ is at most $4c' e^{4L\delta'_0} \sum_{n \in \mathbb{Z}^4} e^{-\delta'_0|n|}(e^{\theta|n|} - 1) + 2(e^{2L\theta} - 1)$ for every y, which tends to zero with θ, and we let $\theta' \in (0, \frac{1}{2}\delta'_0]$ be such that it is at most $\frac{1}{2}\gamma'$ for $\theta = \theta'$. Lemma 14.1 gives $|N^{-1}(y,y')| \leq (2/\gamma') e^{-\theta'|y-y'|}$, and $C'_\Lambda E = N^{-1}E$ has the kernel $\sum_v N^{-1}(y,v)E(v,y')$, which is at most $(4/\gamma') e^{2L\theta'} e^{-\theta'|y-y'|}$ in modulus. *q.e.d.*

For the pair of the scales 0 and 1 the average of the finer scale is the identity, so that $H'_0 = I$ and $\Delta'_0 = \Delta^2$ on the functions on T_η, and the cubes of Λ are the blocks $B_{\ell_1}(y)$ of the tilted run. The lemma then holds in the following form, with the same proof. For a function ω on T_η which vanishes outside the union Λ of the blocks $B_{\ell_1}(y)$, $y \in \Lambda'$, and has the mean zero on each of them, the operator $\mathcal{D}^0_{\Lambda'}$ of (18.5) with $a = a_0 = 1$ acts as $-\Delta$, so that Lemma 18.2 gives $\eta^2 \|\Delta\omega\| \geq L^{-2} \min(C_P^{-1}, 1)\|\omega\|$; and $\eta^4\Delta^2 \leq 256$. The operator $\eta^4 E\Delta^2 E + (1-E)$ has a kernel which vanishes for points at a distance of more than $(4L+2)\eta$, and Lemma 14.1 with the metric $|y - y'|/\eta$ gives the decay of the inverse, at a rate and with a constant which depend on L only.

22 The Axial Gauge on the Turned Lattice

In Section C of [3], and in the fluctuation integral of every renormalisation step [3, (2.152)], the fields of the unit lattice are restricted by the axial gauge condition in the blocks, and the lower bounds of the quadratic forms rest on the inequality [3, (2.123), p. 244], by which a field in this gauge is bounded inside a block by its curl. For the cubes of a scale $j \geq 1$ the unit lattice is a lattice of the paper, with its blocks of side L, and these arguments are as printed. For the pair of the scales 0 and 1, and for the first step, the unit lattice is the turned lattice T_η and the blocks are those of Definition 6.1. Bałaban's proof of [3, (2.123)] expresses the value of the field at a bond through the plaquettes of a strip which joins the bond to the contours of the gauge condition; on the turned lattice the same strip exists in the coordinates of the lattice, and it may leave the block. We prove the inequality in this form.

For $x \in T_\eta$ we write $x'_\nu := \eta^{-1} x \cdot e'_\nu$ for its coordinates in the basis $\eta e'_\nu$, which are integers. For $o, x \in T_\eta$ let $\pi_o(x)$ be the path of bonds of T_η from o to x which changes these coordinates one after the other, in the order $\nu = 3, 2, 1, 0$, each monotonically. For $o = y \in a\mathbb{Z}^4$ and $x \in B_a(y)$ these are the paths of the axial gauge condition of Section 10, and we say that a field A is *axial on the block* $B_a(y)$ if $A(\pi_y(x)) = 0$ for every $x \in B_a(y)$; this is [2, (1.10)], and [3, (2.121)] for the turned lattice. As in Section 9, $(\partial A)(p)$ is η^{-1} times the sum of A over the boundary of the plaquette p, with an orientation which we do not need to fix.

Lemma 22.1 (The Strip of a Bond)**.** *Let A be a field, $o \in T_\eta$, and $b = \langle x, x + \eta e'_j \rangle$ a bond of T_η. There are a set $F_o(b)$ of $\sum_{i<j} |x'_i - o'_i|$ plaquettes and signs $\epsilon_p = \pm 1$, which do not depend on A, such that*

$$A(\pi_o(x)) + A(b) - A(\pi_o(x + \eta e'_j)) = \eta \sum_{p \in F_o(b)} \epsilon_p (\partial A)(p) \; . \tag{22.1}$$

Each plaquette of $F_o(b)$ lies in a plane spanned by e'_j and one of the vectors e'_i, $i < j$, and each coordinate of each of its corners lies between the corresponding coordinates of o and of an end of b.

Proof. The paths $\pi_o(x)$ and $\pi_o(x + \eta e'_j)$ change the coordinates $\nu > j$ in the same way, and they coincide up to the point w at which the coordinate j has been brought to the value which is nearer to o'_j among x'_j and $x'_j + 1$. From w on, one of the two paths makes one more step in the direction $\pm e'_j$, and then both change the coordinates $i < j$ by the same sequence ς of $\sum_{i<j} |x'_i - o'_i|$ steps, the one at the level x'_j of the coordinate j and the other at the level $x'_j + 1$. The path $\pi_o(x)$ followed by b and by the reverse of $\pi_o(x + \eta e'_j)$ is therefore, after the cancellation of the common part from o to w, the boundary of the strip which the sequence ς sweeps between the two levels: its two long sides are the two copies of ς, and its two short sides are b and the additional step at w. The strip is the union of the plaquettes spanned by a step of ς and the vector $\eta e'_j$, one for each step, and two consecutive plaquettes share the bond in the direction e'_j between them. The sum of A over the boundary of the strip is the sum over these plaquettes of the sums of A over their boundaries, all oriented alike, since the shared bonds are run through once in each sense. This gives (22.1), with $F_o(b)$ the set of these plaquettes. Their corners are points of the two copies of ς, whose coordinates lie between those of o and of x or $x + \eta e'_j$. *q.e.d.*

Lemma 22.2 (The Axial Gauge in a Block)**.** *Let a be a scale, $N := a/\eta$, and $y \in a\mathbb{Z}^4$.*

(i) For a field A let $\lambda_A(x) \coloneqq \eta A(\pi_{y(x)}(x))$, where $y(x) \in a\mathbb{Z}^4$ is the point with $x \in B_a(y(x))$, and $\mathfrak{a}_a A \coloneqq A - \partial((1 - Q_a'^* Q_a')\lambda_A)$. Then $\mathfrak{a}_a A$ is axial on every block $B_a(y)$, $Q_a \mathfrak{a}_a A = Q_a A$, $\partial \mathfrak{a}_a A = \partial A$ on the plaquettes, and $\mathfrak{a}_a A = A$ if A is axial on every block.

(ii) If A is axial on $B_a(y)$, then

$$\sum_b |A(b)|^2 \le \frac{17}{5} N^5 \eta^2 \sum_p |(\partial A)(p)|^2 \,, \tag{22.2}$$

where b runs over the bonds $\langle x, x+\eta e_\nu' \rangle$ with both ends in $B_a(y)$ and p over the plaquettes of T_η whose corners are at a Euclidean distance of less than $3a$ from y.

Proof. (i) Let $\mu \coloneqq (1 - Q_a'^* Q_a')\lambda_A$, which on each block is λ_A minus its mean over the block. For every function μ on T_η and every path γ from y to x the sum of $\partial\mu$ over γ is $\eta^{-1}(\mu(x) - \mu(y))$, wherever the path runs. For $x \in B_a(y)$ this is $\eta^{-1}(\lambda_A(x) - \lambda_A(y)) = A(\pi_y(x))$, because $\lambda_A(y) = 0$, so that $(\mathfrak{a}_a A)(\pi_y(x)) = 0$. By Lemma 6.4(i), $Q_a \partial\mu = \partial^a Q_a' \mu = 0$, since $Q_a'(1 - Q_a'^* Q_a') = 0$; and the sum of $\partial\mu$ over the boundary of a plaquette vanishes. If A is axial on every block, then $\lambda_A = 0$.

(ii) Let $b = \langle x, x + \eta e_j' \rangle$ have both ends in $B_a(y)$. Since $A(\pi_y(x)) = A(\pi_y(x + \eta e_j')) = 0$, Lemma 22.1 with $o = y$ gives $A(b) = \eta \sum_{p \in F_y(b)} \epsilon_p (\partial A)(p)$. The components of $x - y$ in the basis e_μ lie in $[0, a)$, so that by (2.2) $0 \le x_\nu' - y_\nu' < N$ for $\nu \in \{0, 1\}$, $0 \le x_2' - y_2' < \frac{7}{5}N$, and $-\frac{3}{5}N < x_3' - y_3' < \frac{4}{5}N$. Hence $F_y(b)$ has at most $N + N + \frac{7}{5}N = \frac{17}{5}N$ plaquettes, and by Schwarz's inequality $|A(b)|^2 \le \frac{17}{5} N \eta^2 \sum_{p \in F_y(b)} |(\partial A)(p)|^2$. A plaquette in a plane spanned by e_i' and e_j', $i < j$, belongs to $F_y(b)$ only for bonds b of the direction e_j', and there are N^4 bonds of each direction which start in $B_a(y)$; summing over b gives (22.2). By Lemma 22.1 the coordinates of a corner of a plaquette of $F_y(b)$ differ from those of y by at most N, N, $\frac{7}{5}N + 1 \le \frac{8}{5}N$, and $\frac{4}{5}N + 1 \le N$, because $N \ge 5$, so that its Euclidean distance from y is at most $(1 + 1 + \frac{64}{25} + 1)^{1/2} N\eta < 3a$. *q.e.d.*

Statement (ii) is [3, (2.123)] for the turned lattice, with the plaquettes of a neighbourhood of the block in the place of those of the block, which suffices for [3, Lemma 2.4], where the field vanishes outside a union of blocks and all the plaquettes are summed. The operator $\mathfrak{a}_a$ of (i) is a projection onto the fields which are axial on every block; it changes a field by a gauge transformation $\partial\mu$ with $Q_a'\mu = 0$, and $(\mathfrak{a}_a A)(b)$ depends on A only on the paths $\pi_y(x)$ of the blocks of the two ends of b.

For the bonds which join two blocks Bałaban compares, in a slab at the common face of two blocks, the average over a bond of the coarse lattice with the mean of the field over the bonds which cross the face, and he bounds the deviation from this mean by an inequality for a lattice of one dimension less [3, (2.124)–(2.127), p. 245]. On the turned lattice the bonds which leave a block do not form a slab, and a bond in the directions e_2' and e_3' may join two blocks which are neighbours across an edge. We compare paths instead: the sum of the field over a staircase differs from its value at any bond which crosses the same face by the sum over a closed path, which is a sum of values of the curl.

Lemma 22.3 (Closed Paths)**.** *Let γ be a closed path of n bonds of T_η, and let $o \in T_\eta$ and r be such that $|s_\nu' - o_\nu'| \le r$ for every site s of γ and every ν. There are integers $n_\gamma(p)$, which do not depend on the field, with $A(\gamma) = \eta \sum_p n_\gamma(p)(\partial A)(p)$ for every field A, $|n_\gamma(p)| \le n$, and $\sum_p |n_\gamma(p)| \le 3rn$; and $n_\gamma(p) = 0$ unless every corner s of p satisfies $|s_\nu' - o_\nu'| \le r$ for every ν.*

Proof. Let $s_0, s_1, \dots, s_n = s_0$ be the sites of γ. For the step from s_{m-1} to s_m, Lemma 22.1, applied to this step or to its reverse, gives $A(\pi_o(s_{m-1})) + A(\langle s_{m-1}, s_m\rangle) - A(\pi_o(s_m)) = \pm\eta \sum_{p\in F} \epsilon_p (\partial A)(p)$ with a set F of at most $3r$ plaquettes whose corners have the stated property. Since the path is closed, the terms $A(\pi_o(s_m))$ cancel in the sum over m, which is therefore $A(\gamma)$. Each step contributes each plaquette at most once. *q.e.d.*

Lemma 22.4 (Lemma 2.4 for the Turned Lattice)**.** *Let a be a scale with $N := a/\eta \geq 10$, let Λ' be a set of points of $a\mathbb{Z}^4$ and Λ the union of the blocks $B_a(y)$, $y \in \Lambda'$, and let A be a field which is axial on $B_a(y)$ for every $y \in \Lambda'$. Then*

$$\sum_b \eta^4 |A(b)|^2 \leq 216 N^2 \sum_c a^4 |(Q_a A)(c)|^2 + C_1 N^5 a^2 \sum_p \eta^4 |(\partial A)(p)|^2 + C_2 N^5 \sum_{b'} \eta^4 |A(b')|^2 \,, \tag{22.3}$$

with $C_1 := 2.2 \cdot 10^{12}$ and $C_2 := 3.1 \cdot 10^9$, where b runs over the bonds of T_η with at least one end in Λ, c over the bonds of $a\mathbb{Z}^4$ with at least one end in Λ', p over the plaquettes of T_η, and b' over the bonds of T_η with no end in Λ.

Proof. Paths in the blocks. For $y \in a\mathbb{Z}^4$ and $z, w \in B_a(y)$ let $\kappa_y(z,w)$ be the path from z to w which consists of the reverse of $\pi_y(z)$ and of $\pi_y(w)$ if $y \in \Lambda'$, and of the reverse of the contour $\Gamma^a_{y,z}$ of Definition 8.2 and of $\Gamma^a_{y,w}$ if $y \notin \Lambda'$. In the first case $A(\kappa_y(z,w)) = 0$, because A is axial on the block. In the second case all the sites of the path lie in $B_a(y)$, so that its bonds have no end in Λ. A path $\pi_y(z)$ has at most $N + N + \frac{7}{5}N + \frac{4}{5}N$ bonds, by the bounds on the coordinates in the proof of Lemma 22.2, and a contour $\Gamma^a_{y,z}$ at most $\frac{28}{5}N + 17$; hence $\kappa_y(z,w)$ has at most $18N$ bonds. Its sites lie at a Euclidean distance of less than $3a$ from y.

Three numbers. Fix $y \in a\mathbb{Z}^4$. All the closed paths below consist of paths κ of $B_a(y)$ and of at most two blocks $B_a(y_1)$ with $|y_1 - y|_\infty \leq a$ in the plane of e_2 and e_3 or with $y_1 = y + ae_\mu$, of bonds between these blocks, and of staircases which start in $B_a(y)$. Their sites lie at a Euclidean distance of less than $5a$ from y, so that Lemma 22.3 applies with $o = y$ and $r = 5N$, and the plaquettes which occur have their corners at a Euclidean distance of at most $10a$ from y. We write $\mathcal{P}(y)$ for the sum of $|(\partial A)(p)|^2$ over the plaquettes whose corners are at a distance of at most $12a$ from y, and $\mathcal{O}(y)$ for the sum of $|A(b')|^2$ over the bonds b' with no end in Λ whose ends are at a distance of at most $12a$ from y. For a closed path γ of n bonds of this kind, Schwarz's inequality and the two bounds of Lemma 22.3 give

$$\Big|\eta \sum_p n_\gamma(p)(\partial A)(p)\Big|^2 \leq \eta^2 \sum_p |n_\gamma(p)| \sum_p |n_\gamma(p)||(\partial A)(p)|^2 \leq 15 N n^2 \eta^2 \, \mathcal{P}(y) \,, \tag{22.4}$$

since $\sum_p |n_\gamma(p)| \leq 3rn = 15Nn$ and $|n_\gamma(p)| \leq n$.

Bonds across a face. Let $c = \langle y, y + ae_\mu\rangle$, and let $b = \langle z, z_+\rangle$ be a bond with $z \in B_a(y)$ and $z_+ \in B_a(y + ae_\mu)$, or the reverse of such a bond. For $x \in B_a(y)$ the staircase $\sigma^a_\mu(x)$, which has at most $\frac{7}{5}N$ bonds, followed by the reverse of the path which consists of $\kappa_y(x,z)$, of b, and of $\kappa_{y+ae_\mu}(z_+, x + ae_\mu)$, is a closed path γ_x of at most $39N$ bonds. The sum of A over the two paths κ is a sum S_x over at most $36N$ bonds with no end in Λ, so that $|S_x|^2 \leq 36N\mathcal{O}(y)$. Hence $A(\sigma^a_\mu(x)) = A(b) + S_x + A(\gamma_x)$, and the mean over the N^4 points x of the block is, by (6.3),

$$N(Q_a A)(c) = A(b) + N^{-4} \sum_{x \in B_a(y)} (S_x + A(\gamma_x)) \,. \tag{22.5}$$

By (22.4) with $n = 39N$ and $15 \cdot 39^2 \leq 24000$, $|A(\gamma_x)|^2 \leq 24000 N^3 \eta^2 \mathcal{P}(y)$, and the square of a sum of three terms is at most three times the sum of their squares, so that

$$|A(b)|^2 \leq 3N^2 |(Q_a A)(c)|^2 + 72000 N^3 \eta^2 \, \mathcal{P}(y) + 108 N \, \mathcal{O}(y) \,. \tag{22.6}$$

In this bound only plaquettes at a distance of at most $10a$ from y and bonds b' at a distance of less than $5a$ from y occur. If c has no end in Λ', then b has no end in Λ and $|A(b)|^2$ is one of the terms of $\mathcal{O}(y)$, so that (22.6) holds in this case with the first term omitted.

Bonds across an edge. A bond in the direction e_0 or e_1 changes one coordinate, and a bond in the direction $\pm e_2'$ or $\pm e_3'$ changes the coordinates 2 and 3 by less than η each, by (2.2). A bond $b = \langle z, z_+ \rangle$ with $z \in B_a(y)$ whose ends lie in two different blocks which are not neighbours across a face therefore ends in $B_a(y')$ with $y' = y + a(\epsilon_2 e_2 + \epsilon_3 e_3)$, $\epsilon_2, \epsilon_3 = \pm 1$. Let $y'' := y + a\epsilon_2 e_2$. There is a bond b_1 from $B_a(y)$ to $B_a(y'')$ and a bond b_2 from $B_a(y'')$ to $B_a(y')$: for the point $x_0 := y + 5\eta\lfloor N/10 \rfloor (e_0 + e_1 + e_2 + e_3)$ every site of the staircase $\sigma_2^a(x_0)$ is at a distance of at most $\frac{3}{5}\eta$ from the segment $[x_0, x_0 + ae_2]$ (Lemma 6.2(ii)), whose points have their coordinates other than the second at a distance of at least 5η from the faces of the blocks, so that the staircase runs from $B_a(y)$ to $B_a(y + ae_2)$ through sites of these two blocks and contains a bond which joins them; the same holds for the other pairs. The bond b followed by the reverse of the path which consists of $\kappa_y(z, \cdot)$, b_1, $\kappa_{y''}$, b_2, and $\kappa_{y'}(\cdot, z_+)$ is a closed path of at most $55N$ bonds, the three paths κ contribute a sum over at most $54N$ bonds with no end in Λ, and by (22.4) with $15 \cdot 55^2 \leq 46000$,

$$|A(b)|^2 \leq 4|A(b_1)|^2 + 4|A(b_2)|^2 + 184000N^3\eta^2\,\mathcal{P}(y) + 216N\,\mathcal{O}(y)\,. \tag{22.7}$$

The bonds b_1 and b_2 cross the faces of the bonds c_1 and c_2 of $a\mathbb{Z}^4$ which join y to y'' and y'' to y', and (22.6) applies to them with the initial point $y_1 \in \{y, y'', y'\}$ of c_1 or c_2 in the place of y. Since $|y_1 - y| \leq 2a$, the plaquettes and the bonds which occur in it are among those of $\mathcal{P}(y)$ and $\mathcal{O}(y)$, by the remark after (22.6). Hence, with $8 \cdot 72000 + 184000 = 760000$ and $8 \cdot 108 + 216 = 1080$,

$$|A(b)|^2 \leq 12N^2(|(Q_a A)(c_1)|^2 + |(Q_a A)(c_2)|^2) + 760000N^3\eta^2\,\mathcal{P}(y) + 1080N\,\mathcal{O}(y)\,, \tag{22.8}$$

where a term with c_i is omitted if c_i has no end in Λ'.

The sum. A bond with an end in Λ has both ends in one block of Λ, or it crosses a face, or an edge. For the first kind we use (22.2). A bond c of $a\mathbb{Z}^4$ is crossed by at most $8N^4$ bonds of T_η, counted without orientation, and every y is the initial point of four bonds c. At most $4N^4$ bonds which cross an edge start in a given block, and a bond c occurs as c_1 or c_2 for at most four pairs of a block and one of its four neighbours across an edge. Finally a plaquette, or a bond b', is counted in $\mathcal{P}(y)$, or in $\mathcal{O}(y)$, for at most 25^4 points y. Adding (22.2), (22.6), and (22.8) we obtain

$$\begin{aligned}\sum_b |A(b)|^2 \leq (24 + 192)N^6 \sum_c |(Q_a A)(c)|^2 + 25^4 \cdot 5.4 \cdot 10^6 N^7 \eta^2 \sum_p |(\partial A)(p)|^2 \\ + 25^4 \cdot 7776 N^5 \sum_{b'} |A(b')|^2\,.\end{aligned} \tag{22.9}$$

Here $24N^6 = 8N^4 \cdot 3N^2$ and $192N^6 = 4 \cdot 4N^4 \cdot 12N^2$; for each y the coefficient of $\eta^2\mathcal{P}(y)$ is at most $\frac{17}{5}N^5 + 4 \cdot 8N^4 \cdot 72000N^3 + 4N^4 \cdot 760000N^3 \leq 5.4 \cdot 10^6 N^7$, and that of $\mathcal{O}(y)$ is at most $4 \cdot 8N^4 \cdot 108N + 4N^4 \cdot 1080N = 7776N^5$. Since $25^4 \cdot 5.4 \cdot 10^6 \leq C_1$ and $25^4 \cdot 7776 \leq C_2$, multiplying by η^4 and using $N\eta = a$ gives (22.3). *q.e.d.*

Bałaban uses his Lemma 2.4 twice: for the lower bound of the quadratic form [3, (2.120)] of the fields of the unit lattice between two scales, whose inverse is the covariance $\tilde{C}_\Lambda^{(j)}$ of [3, (2.129)], and for the lower bound [3, (2.153)] of the quadratic form of the fluctuation integral of a renormalisation step. In both cases the Gaussian integral runs over the fields in the axial gauge, in the second with the further condition that the averages vanish, and he obtains the

covariance by solving the conditions for some of the variables [3, (2.155), (2.156)]. On the turned lattice the conditions are not solved by leaving out variables, since a bond lies on staircases of several bonds of the coarse lattice, and we describe the covariance through a projection onto the space of the conditions. The next lemma is the general statement; in a space of finite dimension it says, by completing the square, that $\tilde{C}$ is the covariance of the Gaussian integral over $\mathcal{S}$ with the quadratic form M.

Lemma 22.5 (Constrained Covariances)**.** *Let $\mathcal{X}$ be a real Hilbert space, $\mathcal{S}$ a closed subspace, Π a bounded operator with $\Pi^2 = \Pi$ and the range $\mathcal{S}$, and M a bounded self-adjoint operator with $\langle v, \mathsf{M}v\rangle \geq \gamma\|v\|^2$ for every $v \in \mathcal{S}$, where $\gamma > 0$. Then the operator $\mathsf{N} \coloneqq \Pi^*\mathsf{M}\Pi + (1-\Pi)^*(1-\Pi)$ satisfies $\mathsf{N} \geq \frac{1}{2}\min(\gamma, 1)$. For every $J \in \mathcal{X}$ the vector $\tilde{C}J \coloneqq \mathsf{N}^{-1}\Pi^*J$ lies in $\mathcal{S}$, it is the only $B_0 \in \mathcal{S}$ with $\langle B, \mathsf{M}B_0\rangle = \langle B, J\rangle$ for every $B \in \mathcal{S}$, and for every $B \in \mathcal{S}$*

$$2\langle J, B\rangle - \langle B, \mathsf{M}B\rangle = \langle J, \tilde{C}J\rangle - \langle B - \tilde{C}J, \mathsf{M}(B - \tilde{C}J)\rangle \,. \tag{22.10}$$

In particular $\tilde{C} = \Pi\mathsf{N}^{-1}\Pi^*$.

Proof. For $v \in \mathcal{X}$, $\langle v, \mathsf{N}v\rangle = \langle \Pi v, \mathsf{M}\Pi v\rangle + \|(1-\Pi)v\|^2 \geq \min(\gamma, 1)(\|\Pi v\|^2 + \|(1-\Pi)v\|^2)$, because $\Pi v \in \mathcal{S}$, and $\|v\|^2 \leq 2\|\Pi v\|^2 + 2\|(1-\Pi)v\|^2$. Let $v \coloneqq \mathsf{N}^{-1}\Pi^*J$. Applying $(1-\Pi)^*$ to the equation $\mathsf{N}v = \Pi^*J$ and using $(1-\Pi)^*\Pi^* = 0$ and $((1-\Pi)^*)^2 = (1-\Pi)^*$, we find $(1-\Pi)^*(1-\Pi)v = 0$, hence $\|(1-\Pi)v\|^2 = 0$ and $v = \Pi v \in \mathcal{S}$. The equation then reads $\Pi^*\mathsf{M}v = \Pi^*J$, and for $B \in \mathcal{S}$ we have $B = \Pi B$ and $\langle B, \mathsf{M}v\rangle = \langle \Pi B, \mathsf{M}v\rangle = \langle B, \Pi^*J\rangle = \langle B, J\rangle$. If B_0 and B_1 in $\mathcal{S}$ both have this property, then $\langle B, \mathsf{M}(B_0 - B_1)\rangle = 0$ for $B = B_0 - B_1$, so that $B_0 = B_1$. For (22.10) we insert $\langle J, B\rangle = \langle B, \mathsf{M}\tilde{C}J\rangle$ and $\langle J, \tilde{C}J\rangle = \langle \tilde{C}J, \mathsf{M}\tilde{C}J\rangle$. Finally $\tilde{C}J = \Pi\tilde{C}J$. *q.e.d.*

We apply the lemma on the turned lattice for the scale $\ell_1 = L\eta$, whose blocks have $N = L$ points along each side. The fields are the square-summable fields on the bonds of T_η, with the scalar product of the weights η^4; the bonds of $\ell_1\mathbb{Z}^4$ carry the weights ℓ_1^4, and $Q \coloneqq Q_{\ell_1}$. A *local operator of the range ϱ* is an operator on these fields, or between them and the functions on the bonds of $\ell_1\mathbb{Z}^4$, whose kernel vanishes for two bonds at a distance of more than $\varrho\ell_1$ and has sums over either argument which are bounded by a constant depending on L only. Products and adjoints of local operators are local.

Proposition 22.6 (The Fields of the First Scale)**.** *Let $L \geq 64$ be a multiple of 5, let $a, a_0 > 0$, and let C_1, C_2 be the constants of Lemma 22.4.*

(i) Let Λ' be a set of points of $\ell_1\mathbb{Z}^4$, Λ the union of their blocks, and for a field A let

$$\mathfrak{q}_\Lambda(A) \coloneqq a_0\eta^{-2}\sum_{b'}\eta^4|A(b')|^2 + a\ell_1^{-2}\sum_c \ell_1^4|(QA)(c)|^2 + \|\partial A\|^2 \,, \tag{22.11}$$

with b' and c as in Lemma 22.4. If A is axial on every block of Λ, then $\mathfrak{q}_\Lambda(A) \geq \gamma''\eta^{-2}\|A\|^2$ with $\gamma'' \coloneqq 1/\max(216L^4/a, C_1L^7, (C_2L^5+1)/a_0)$.

(ii) If A is axial on every block $B_{\ell_1}(y)$, $y \in \ell_1\mathbb{Z}^4$, and $QA = 0$, then $\|\partial A\|^2 \geq (C_1L^7)^{-1}\eta^{-2}\|A\|^2$.

(iii) In both cases let $\mathcal{S}$ be the space of the fields with the stated conditions and M the operator with $\langle A, \mathsf{M}A\rangle = \eta^2\mathfrak{q}_\Lambda(A)$, respectively $\eta^2\|\partial A\|^2$. There is a bounded projection Π onto $\mathcal{S}$ such that the operator $\tilde{C} = \Pi\mathsf{N}^{-1}\Pi^*$ of Lemma 22.5 has a kernel, with respect to the sum over the positive bonds, with $|\tilde{C}(b, b')| \leq c\,e^{-\theta|b_- - b'_-|/\eta}$, where c and $\theta > 0$ depend on L, a, and a_0 and not on k, η, or Λ'.

Proof. (i) By (22.3) with a replaced by the scale ℓ_1 and $N = L$, and with $\ell_1 = L\eta$,

$$\|A\|^2 \le \frac{216L^4\eta^2}{a} \cdot a\ell_1^{-2} \sum_c \ell_1^4 |(QA)(c)|^2 + C_1 L^7 \eta^2 \|\partial A\|^2 + \frac{(C_2 L^5 + 1)\eta^2}{a_0} \cdot a_0 \eta^{-2} \sum_{b'} \eta^4 |A(b')|^2 \,, \tag{22.12}$$

where the term 1 accounts for the bonds b' themselves. The right side is at most $\eta^2 \mathfrak{q}_\Lambda(A)/\gamma''$. (ii) This is (22.3) for $\Lambda' = \ell_1 \mathbb{Z}^4$, where no bond b' occurs and the first term vanishes.

(iii) *The projections.* For a set Λ' let $\mathfrak{a}_\Lambda A \coloneqq A - \partial((1 - Q'^*_{\ell_1} Q'_{\ell_1})\lambda)$ with $\lambda(x) \coloneqq \eta A(\pi_{y(x)}(x))$ for $x \in \Lambda$ and $\lambda(x) \coloneqq 0$ otherwise. By the proof of Lemma 22.2(i), which treats each block by itself, $\mathfrak{a}_\Lambda A$ is axial on the blocks of Λ, $\mathfrak{a}_\Lambda A = A$ if A is, and $Q\mathfrak{a}_\Lambda = Q$. The value $\lambda(x)$ is η times a signed sum of at most $\frac{21}{5}L$ values of A at bonds at a distance of less than $3\ell_1$ from $y(x)$, and a bond lies on the paths of at most L^4 points of each of at most 7^4 blocks; hence $\mathfrak{a}_\Lambda$ is a local operator of the range 6. In the first case we take $\Pi \coloneqq \mathfrak{a}_\Lambda$. In the second case let $\Pi \coloneqq \mathfrak{a}_\Lambda(1 - Q^*(QQ^*)^{-1}Q)$ with $\Lambda' = \ell_1 \mathbb{Z}^4$. The operator Q is local of the range 3: $(QA)(c)$ is L^{-5} times a sum of the values of A at the at most $\frac{7}{5}L^5$ bonds of the staircases from the points of $B_{\ell_1}(c_-)$, and a bond lies on at most $\frac{7}{5}L$ staircases of each direction (proof of Lemma 6.4(ii)). By Lemma 17.1(ii), applied to the run of one step with $a = 1$ there, in the unit ℓ_1, the operator QQ^* is bounded below by a positive constant which does not depend on k and L; the lemma is stated for a torus, and it gives the bound on the infinite lattice, because for a function ω of finite support on the bonds of $\ell_1 \mathbb{Z}^4$ the norm of $Q^*\omega$ is the same on the infinite lattice and on a torus with large periods. Since QQ^* is local, the kernel of its inverse decays exponentially by Lemma 14.1. The operator $1 - Q^*(QQ^*)^{-1}Q$ is the orthogonal projection onto the null space of Q, and $\mathfrak{a}_\Lambda$ maps this space to itself and to the axial fields, on which it is the identity; hence Π is a projection onto $\mathcal{S}$.

The decay. The operator M is local: the first term of $\eta^2 \mathfrak{q}_\Lambda$ is diagonal, the second is (a/L^2) times the form of $Q^*\mathbf{1}Q$ with the characteristic function of a set of bonds of $\ell_1 \mathbb{Z}^4$, and $\eta^2 \|\partial A\|^2 \le 24 \|A\|^2$ has a kernel of the range 2η. By (i), (ii), and Lemma 22.5, $\mathsf{N} \ge \frac{1}{2} \min(\gamma, 1)$ with $\gamma = \gamma''$, respectively $(C_1 L^7)^{-1}$. In the first case N is local, and in the second its kernel is bounded by a constant times $e^{-\theta_Q |b_- - b'_-|/\ell_1}$, with constants which depend on L. In both cases $\sum_{b'} |\mathsf{N}(b, b')| (e^{\theta |b_- - b'_-|/\eta} - 1)$ tends to zero with θ, uniformly in b, η, and Λ', and Lemma 14.1 gives the exponential decay of the kernel of N^{-1}. The same decay follows for $\Pi \mathsf{N}^{-1} \Pi^*$, because the kernel of Π has bounded sums and vanishes beyond a bounded distance, or decays exponentially. *q.e.d.*

Statement (i) is the lower bound which Bałaban asserts for the form [3, (2.122)], here for the pair of the scales 0 and 1, where the form of the finer scale is $\|\partial A\|^2$, and $\eta^2 \tilde{C}$ is the covariance $\tilde{C}^{(0)}_\Lambda$ of [3, (2.129)]. Statement (ii) is [3, (2.153)] for the first step of the tilted run, and (iii) replaces [3, (2.154)–(2.157)] for this step. For the later steps and scales the unit lattice is a lattice of the paper and these statements are as printed.

23 The Propagator of Two Neighbouring Scales

We come to Section C of [3]. Its first result is a representation of the propagator whose averages are those of the scale $j + 1$ on a set of blocks and those of the scale j elsewhere, through the operators of the scale j alone and operators of the unit lattice [3, (2.129), p. 246]. Bałaban derives it from a sequence of transformations of Gaussian integrals [3, (2.95)–(2.119)]: he replaces the gauge-fixing term of the two scales by that of the scale j together with an axial gauge condition for the averages, by the method of Faddeev and Popov, and then integrates

in stages. Each of these transformations is a statement about the minimum of a quadratic form over a subspace or over the orbits of a group, and we prove the representation in this form. The proof uses the averages only through the identities of Section 6, so that it holds for the tilted run; the lattice enters the result through the operators H' and Δ' of Section 21 and, for the first scale, through Section 22.

The setting. We fix $1 \le j \le k-1$ and take the spacing of the lattice of the scale j as the unit of length; the pair of the scales 0 and 1 is treated at the end of the section. The fine lattice is a torus T_η/Γ' as in Section 9, with $\eta = L^{-j}$ in this unit and with periods which are multiples of L, the unit lattice is $\mathbb{Z}^4/\Gamma'$, and $Q \coloneqq Q_k$, $Q' \coloneqq Q'_k$ are the averages of the run of j steps. We write $Q_1 \coloneqq Q^{\mathrm{st}}_{1,L}$ and $Q'_1 \coloneqq Q'^{\mathrm{st}}_{1,L}$ for Bałaban's averages (6.4) from the unit lattice to $L\mathbb{Z}^4/\Gamma'$; by Lemma 6.3, in the original unit $Q_1 Q$ and $Q'_1 Q'$ are the tilted averages of the scale $j+1$. For a function ω on the unit lattice, (6.4) gives

$$(Q_1\partial_1\omega)(\langle y, y+Le_\mu\rangle) = L^{-5}\sum_z(\omega(z+Le_\mu)-\omega(z)) = L^{-1}((Q'_1\omega)(y+Le_\mu)-(Q'_1\omega)(y)) \;, \quad (23.1)$$

the sum over the points z of the cube of y. All the spaces of this section have finite dimension, and $\Delta \coloneqq \partial^*\partial$ on the functions, as in [3]. Let Λ' be a set of points of $L\mathbb{Z}^4/\Gamma'$, Λ the union of their cubes in the unit lattice, $\mathcal{S}_\Lambda$, E, and C'_Λ as in Lemma 21.3, which holds on the torus with the same proof, the kernels being summed over the periods. As in [3, p. 224], the bonds of a set of points are the bonds with at least one end in the set: $\mathfrak{b}(\Lambda^c)$ is the set of the bonds of the unit lattice with no end in Λ, and $\mathfrak{b}(\Lambda')$ the set of the bonds of $L\mathbb{Z}^4/\Gamma'$ with at least one end in Λ'. For $a > 0$ and a field B on the unit lattice let

$$\mathfrak{q}''(B) \coloneqq a\sum_{b\in\mathfrak{b}(\Lambda^c)}|B(b)|^2 + aL^2\sum_{c\in\mathfrak{b}(\Lambda')}|(Q_1B)(c)|^2 \;, \quad (23.2)$$

so that $\mathfrak{q}''(QA)$ is the form $\langle QA, aQA\rangle$ of [3, (2.97)]. The gauge functions of the two scales and of the scale j are the spaces

$$\mathcal{N} \coloneqq \{\lambda \,|\, Q'\lambda \in \mathcal{S}_\Lambda\} \;, \qquad \mathcal{N}_j \coloneqq \{\lambda \,|\, Q'\lambda = 0\} \;, \quad (23.3)$$

of functions on the fine lattice; the first is the space $N(Q')$ of [3, (2.10)] for the two scales, since $Q'\lambda \in \mathcal{S}_\Lambda$ means that $Q'\lambda$ vanishes outside Λ and $Q'_1Q'\lambda$ vanishes on Λ'. Neither space contains a constant other than zero. Let R and R_j be the orthogonal projections onto $\Delta\mathcal{N}$ and $\Delta\mathcal{N}_j$ [3, p. 225], so that $1-R_j$ is the projection P of Lemma 15.1, and let

$$\mathfrak{g}(A) \coloneqq \|\partial A\|^2 + \|R\partial^*A\|^2 + \mathfrak{q}''(QA) \;, \qquad \mathfrak{h}_j(A) \coloneqq \|\partial A\|^2 + \|R_j\partial^*A\|^2 \;. \quad (23.4)$$

The first is the quadratic form of the operator Δ_a of [3, (2.19)] for the two scales, whose inverse is the operator $G_\square$ of [3, (2.90)], and $\mathfrak{h}_j(A) + a\|QA\|^2$ is the form of the operator Δ_a of Proposition 9.3, whose inverse we denote by G_j. With Lemma 17.1 we put

$$H_j \coloneqq G_jQ^*(QG_jQ^*)^{-1} \;, \qquad \Delta_j \coloneqq (QG_jQ^*)^{-1} - a \;, \qquad \tilde{G}_j \coloneqq G_j - G_jQ^*(QG_jQ^*)^{-1}QG_j \;, \quad (23.5)$$

which are [3, (2.130), (2.131)] and the operators H_k and Δ_k of Section 17. A field B on the unit lattice is *axial on the cubes of* Λ if its sum over the contour $\Gamma_{y,x}$ of [2, (1.7)] vanishes for every $y \in \Lambda'$ and every x in the cube of y; these contours lie in the cube and form a tree, so that this is the condition [3, (2.121)].

Lemma 23.1 (Five Facts)**.** *In the setting above the following hold.*

(i) For every field A and every $\lambda \in \mathcal{N}$, $\partial(\partial\lambda) = 0$ and $\mathfrak{q}''(Q(A-\partial\lambda)) = \mathfrak{q}''(QA)$.

(ii) $\mathcal{N}$ *is the direct sum of* $\mathcal{N}_j$ *and* $H'\mathcal{S}_\Lambda$*, and* $\|\Delta(\nu + H'\omega)\|^2 = \|\Delta\nu\|^2 + \langle\omega, \Delta'\omega\rangle$ *for* $\nu \in \mathcal{N}_j$ *and* $\omega \in \mathcal{S}_\Lambda$*. Consequently* $\|Rf\|^2 = \|R_j f\|^2 + \langle EH'^*\Delta f, C'_\Lambda EH'^*\Delta f\rangle$ *for every function* f*.*

(iii) For every field A *there is exactly one* $\omega \in \mathcal{S}_\Lambda$ *such that* $Q(A - \partial H'\omega)$ *is axial on the cubes of* Λ*.*

(iv) For every field B *on the unit lattice and every field* A' *with* $QA' = 0$*,* $\mathfrak{h}_j(H_jB + A') = \langle B, \Delta_j B\rangle + \mathfrak{h}_j(A')$*, and* $\sup\{2\langle J, A'\rangle - \mathfrak{h}_j(A') \,|\, QA' = 0\} = \langle J, \tilde{G}_j J\rangle$ *for every field* J*.*

(v) There is a constant $\gamma''_0 > 0$*, which depends on* a *and* L *only, such that* $\mathfrak{q}''(B) + \langle B, \Delta_j B\rangle \geq \gamma''_0\|B\|^2$ *for every* B *which is axial on the cubes of* Λ*.*

Proof. (i) The curl of a gradient vanishes. By Lemma 6.4(i), $Q\partial\lambda = \partial_1\omega$ with $\omega := Q'\lambda \in \mathcal{S}_\Lambda$. On a bond of $\mathfrak{b}(\Lambda^c)$ the field $\partial_1\omega$ vanishes, because ω vanishes at both ends. The function $Q'_1\omega$ vanishes at every point of $L\mathbb{Z}^4/\Gamma'$: at the points of Λ' because ω has the mean zero on their cubes, and at the other points because ω vanishes on their cubes. Hence $Q_1\partial_1\omega = 0$ by (23.1).

(ii) By Lemma 21.1(i), $Q'H'\omega = \omega$, so that $H'\mathcal{S}_\Lambda \subset \mathcal{N}$, and for $\lambda \in \mathcal{N}$ the function $\nu := \lambda - H'Q'\lambda$ lies in $\mathcal{N}_j$; the sum is direct because $Q'(\nu + H'\omega) = \omega$. The identity of the norms is Lemma 21.1(ii) for $\lambda = \nu + H'\omega$ and $\mu = \omega$. It shows that $\Delta\mathcal{N}$ is the orthogonal sum of $\Delta\mathcal{N}_j$ and of the range of the operator $\mathsf{X} := \Delta H'$ on $\mathcal{S}_\Lambda$, for which $\mathsf{X}^*\mathsf{X} = E\Delta'E$ is the inverse of C'_Λ on $\mathcal{S}_\Lambda$. The orthogonal projection onto this range is $\mathsf{X}C'_\Lambda\mathsf{X}^* = \Delta H'C'_\Lambda EH'^*\Delta$, and R is its sum with R_j.

(iii) Let $B := QA$. By Lemma 6.4(i) and $Q'H' = I$, $Q(A - \partial H'\omega) = B - \partial_1\omega$. Let $\lambda_B(x)$ be the sum of B over $\Gamma_{y,x}$ for x in the cube of $y \in \Lambda'$, and let ω be equal to λ_B minus its mean over the cube on each cube of Λ, and to zero outside Λ. Then $\omega \in \mathcal{S}_\Lambda$, and the sum of $B - \partial_1\omega$ over $\Gamma_{y,x}$ is $\lambda_B(x) - (\omega(x) - \omega(y)) = 0$, because $\lambda_B(y) = 0$. If $\partial_1\omega$ is axial for some $\omega \in \mathcal{S}_\Lambda$, then $\omega(x) = \omega(y)$ on each cube, and $\omega = 0$ because its means vanish; this gives the uniqueness.

(iv) Let H be the operator of the form $\mathfrak{h}_j$, so that $G_j^{-1} = \mathsf{H} + aQ^*Q$, and let $A_0 := H_jB$. Then $QA_0 = B$, and for $QA' = 0$ we have $\langle A', \mathsf{H}A_0\rangle = \langle A', G_j^{-1}A_0\rangle - a\langle QA', QA_0\rangle = \langle QA', (QG_jQ^*)^{-1}B\rangle = 0$, which gives the first identity with $\mathfrak{h}_j(A_0) = \langle A_0, G_j^{-1}A_0\rangle - a\|B\|^2 = \langle B, \Delta_j B\rangle$. On the null space of Q the form $\mathfrak{h}_j$ is that of G_j^{-1}, which is positive by (9.2). The field $A'_0 := \tilde{G}_jJ$ satisfies $QA'_0 = 0$ and $G_j^{-1}A'_0 = J - Q^*(QG_jQ^*)^{-1}QG_jJ$, so that $\langle A', G_j^{-1}A'_0\rangle = \langle A', J\rangle$ whenever $QA' = 0$; hence $2\langle J, A'\rangle - \mathfrak{h}_j(A') = \langle J, \tilde{G}_jJ\rangle - \mathfrak{h}_j(A' - A'_0)$, as in (22.10).

(v) By (9.4), $\langle B, \Delta_j B\rangle \geq \frac{1}{24}\|\partial_1 B\|^2$, and $\mathfrak{q}''$ is at least $\min(a, 1)$ times its value for $a = 1$. For $a = 1$ the form $\mathfrak{q}''(B) + \gamma_0\|\partial_1 B\|^2$ is [3, (2.122)], a form of the unit lattice of the paper with its cubes and Bałaban's averages, and it is bounded below by a positive multiple of $\|B\|^2$ on the fields which satisfy [3, (2.121)], with a constant which depends on γ_0 and L [3, p. 246]. *q.e.d.*

Proposition 23.2 (The Representation (2.129))**.** *In the setting above the form* $\mathfrak{g}$ *is positive definite. Let* G *be the inverse of its operator, let* $\tilde{C}_\Lambda$ *be the operator of Lemma 22.5 for the space of the fields on the unit lattice which are axial on the cubes of* Λ *and for the form* $\mathfrak{q}''(B) + \langle B, \Delta_j B\rangle$*, and let* $\mathsf{W} := \partial\Delta H'C'_\Lambda EH'^*\partial^*$*. Then*

$$G = \partial H'C'_\Lambda EH'^*\partial^* + (1 - \mathsf{W})^*(\tilde{G}_j + H_j\tilde{C}_\Lambda H_j^*)(1 - \mathsf{W}) \,, \tag{23.6}$$

which is [3, (2.129)]. Moreover, let Q'' map a field B on the unit lattice to the function on $\mathfrak{b}(\Lambda^c) \cup \mathfrak{b}(\Lambda')$ which is B on the bonds of the first set and $Q_1 B$ on those of the second, so that $Q''Q$ is the average of the two scales. Then $Q''QGQ^*Q''^* = Q''\tilde{C}_\Lambda Q''^*$, which is [3, (2.146)].

Proof. The gauge-fixing term. By Lemma 23.1(ii) with $f = \partial^* A$,

$$\mathfrak{g}(A) = \mathfrak{g}_j(A) + \langle \mathsf{V}A, C'_\Lambda \mathsf{V}A \rangle \,, \qquad \mathfrak{g}_j(A) \coloneqq \mathfrak{h}_j(A) + \mathfrak{q}''(QA) \,, \qquad \mathsf{V} \coloneqq EH'^* \Delta \partial^* \,. \quad (23.7)$$

The slice. By Lemma 23.1(iii) every field is $A = A_2 + \partial H'\omega$ with a unique $\omega \in \mathcal{S}_\Lambda$ and a unique field A_2 for which QA_2 is axial on the cubes of Λ. The form $\mathfrak{g}_j$ does not change if $\partial H'\omega$ is added to its argument: the first and the last term by Lemma 23.1(i), since $H'\omega \in \mathcal{N}$, and the second because $\partial^* \partial H'\omega = \Delta H'\omega$ is orthogonal to $\Delta \mathcal{N}_j$ by Lemma 23.1(ii). By Lemma 21.1(i), $\mathsf{V}\partial H'\omega = EH'^* \Delta^2 H'\omega = E\Delta'\omega$, which is the image of ω under the inverse of C'_Λ. With $\tilde{\omega} \coloneqq \omega + C'_\Lambda \mathsf{V} A_2 \in \mathcal{S}_\Lambda$ we obtain

$$\mathfrak{g}(A) = \mathfrak{g}_j(A_2) + \langle \tilde{\omega}, \Delta' \tilde{\omega} \rangle \,, \qquad A = (1 - \partial H' C'_\Lambda \mathsf{V}) A_2 + \partial H' \tilde{\omega} \,, \quad (23.8)$$

and the pairs of a field A_2 with axial average and a function $\tilde{\omega} \in \mathcal{S}_\Lambda$ parametrise all fields.

Positivity. Let $B \coloneqq QA_2$ and $A_2 = H_j B + A'$ with $QA' = 0$. By Lemma 23.1(iv), $\mathfrak{g}_j(A_2) = \mathfrak{q}''(B) + \langle B, \Delta_j B \rangle + \mathfrak{h}_j(A')$. The three terms of $\mathfrak{g}(A)$ which result are nonnegative, and they vanish only for $\tilde{\omega} = 0$ by (21.8), $B = 0$ by Lemma 23.1(v), and $A' = 0$ by (9.2), that is, only for $A = 0$.

The supremum. For a field J, $\langle J, GJ \rangle$ is the supremum of $2\langle J, A \rangle - \mathfrak{g}(A)$ over all fields. By (23.8), with $J'' \coloneqq (1 - \partial H' C'_\Lambda \mathsf{V})^* J = (1 - \mathsf{W})J$, it is the sum of the suprema of

$$2\langle EH'^* \partial^* J, \tilde{\omega} \rangle - \langle \tilde{\omega}, \Delta' \tilde{\omega} \rangle \qquad \text{and} \qquad 2\langle J'', H_j B + A' \rangle - \mathfrak{q}''(B) - \langle B, \Delta_j B \rangle - \mathfrak{h}_j(A') \quad (23.9)$$

over $\tilde{\omega} \in \mathcal{S}_\Lambda$, and over the axial fields B and the fields A' with $QA' = 0$. The first supremum is $\langle EH'^* \partial^* J, C'_\Lambda EH'^* \partial^* J \rangle$. In the second, the supremum over A' is $\langle J'', \tilde{G}_j J'' \rangle$ by Lemma 23.1(iv), and the supremum over B is $\langle H_j^* J'', \tilde{C}_\Lambda H_j^* J'' \rangle$ by (22.10). The two sides of (23.6) are symmetric operators with the same quadratic form.

The averages. Let $J = Q^* Q''^* B_0$. For $\omega \in \mathcal{S}_\Lambda$, $\langle H'^* \partial^* J, \omega \rangle = \langle B_0, Q'' Q \partial H' \omega \rangle = \langle B_0, Q'' \partial_1 \omega \rangle = 0$ by the proof of Lemma 23.1(i), so that $EH'^* \partial^* J = 0$; hence the first term of (23.6) and W vanish on J. Further $\tilde{G}_j Q^* = 0$ by (23.5), and $H_j^* Q^* = (QH_j)^* = I$. Therefore $\langle J, GJ \rangle = \langle Q''^* B_0, \tilde{C}_\Lambda Q''^* B_0 \rangle$. *q.e.d.*

The fields $J = Q^* Q''^* B_0$ are those against which Bałaban evaluates the form for the inverse of $QG_\square Q^*$ [3, (2.144)–(2.146)]; that the terms with W drop out is his remark that only gauge transformations with $Q'\lambda = 0$ were made.

The scales 0 *and* 1. For the pair of the scales 0 and 1 the unit lattice is the fine lattice T_η / Γ' itself, and the statements above hold with the following substitutions, each of which makes the corresponding step of the proofs a tautology or replaces its reference: Q and Q' are the identity, Q_1 and Q'_1 are the tilted averages Q_{ℓ_1} and Q'_{ℓ_1}, for which (23.1) is Lemma 6.4(i); $H' = I$ and $\Delta' = \Delta^2$; $\mathcal{N}_0 = \{0\}$, $R_0 = 0$, and $\mathfrak{h}_0(A) = \|\partial A\|^2$; $H_0 = I$, $\langle B, \Delta_0 B \rangle = \|\partial B\|^2$, and $\tilde{G}_0 = 0$; the cubes of Λ are the blocks $B_{\ell_1}(y)$, the space $\mathcal{S}_\Lambda$ and the covariance C'_Λ are those of the remark after Lemma 21.3, a field is axial in the sense of Section 22, and in (iii) the function λ_B is that of Lemma 22.2(i); the form $\mathfrak{q}''$ has the constant $a\eta^{-2}$ on the bonds of $\mathfrak{b}(\Lambda^c)$ and $a\ell_1^{-2}$, with the weights ℓ_1^4, on those of $\mathfrak{b}(\Lambda')$, and (v) is Proposition 22.6(i)

with $a_0 = a$. The representation (23.6) then reads $G = \partial C'_\Lambda E \partial^* + (1 - \mathsf{W})^* \tilde{C}_\Lambda (1 - \mathsf{W})$ with $\mathsf{W} = \partial \Delta C'_\Lambda E \partial^*$.

It remains to draw the bounds from the representation. All the operators in it have been bounded, and what has to be checked is that their composition has the regularity of the propagator of one scale, although W contains three derivatives of H'.

Proposition 23.3 (Proposition 2.5 for the Tilted Run)**.** *Let $L \geq 64$ be a multiple of 5 and $a > 0$. The operator G of Proposition 23.2 satisfies the bounds (i)–(iv) of Proposition 16.2 and the bounds* (9.3)*, with constants c, $\delta_G > 0$, γ_0, and C which depend on a, L, and the Hölder exponents, and not on j, k, the torus, or the set Λ'. For the scales 0 and 1 the same holds with the unit η.*

Proof. We write c and δ for constants with the stated dependence. An operator between bounded functions or fields on the two lattices is *of the class* $\mathcal{K}$ if it has a kernel, with respect to the sum over the points of the lattice of its argument with the weight η^4 for the fine lattice and 1 for the unit lattice, which is at most $c\, e^{-\delta d}$ in modulus, d being the distance of the two points; and it is *of the class* $\mathcal{B}$ if the modulus of its value at a point x is at most $c \sum_{y'} e^{-\delta d(x,y')} \sup_{\Delta(y')} |f|$ for every argument f, the sum over the points y' of the unit lattice and the supremum over the block of y', or the value at y' for an argument on the unit lattice. An operator of the class $\mathcal{K}$ is of the class $\mathcal{B}$, and it is bounded on the square-summable functions, by Schur's test. Sums and products of operators of either class are of the same class, and an operator of the class $\mathcal{B}$ maps a function supported at the distance D from a point to a function which is at most $c\, e^{-\delta D/2} \sup |f|$ there.

The factors. (a) By Proposition 21.2(i), $\partial^{(n)} H'$ for $n \leq 3$ and the adjoints $H'^* (\partial^{(n)})^*$ are of the class $\mathcal{K}$; in particular $\partial H'$, $\nabla \partial H'$, $\Delta \partial H'$, $H'^* \partial^*$, $H'^* \partial^* \nabla^*$, and $H'^* \Delta \partial^*$ are. The differences of the values of $\nabla \partial H' \omega$ at two points x, x' with $d(x, x') \leq 1$ are at most $c\, d(x, x')$ times a sum of the same form, because the two points are joined by a path of at most $2d(x, x')/\eta$ bonds and a third difference quotient of the kernel is bounded. (b) $C'_\Lambda E$ is of the class $\mathcal{K}$ by Lemma 21.3. (c) G_j, ∇G_j, $G_j \nabla^*$, and ΔG_j are of the class $\mathcal{B}$ by Proposition 16.2(i) applied to the restrictions of a field to the blocks, and Q, Q^* are of the class $\mathcal{K}$, being averages over blocks and staircases. By Lemma 17.1, $(Q G_j Q^*)^{-1} = \Delta_j + a$ is of the class $\mathcal{K}$. Hence $H_j = G_j Q^* (\Delta_j + a)$, ∇H_j, ΔH_j, $H_j^* = (\Delta_j + a) Q G_j$, and $H_j^* \nabla^*$ are of the class $\mathcal{B}$, and the differences of $\nabla H_j B$ satisfy the Hölder bound of Proposition 16.2(ii). (d) $\tilde{C}_\Lambda$ is of the class $\mathcal{K}$. Indeed the form $\mathfrak{q}''(B) + \langle B, \Delta_j B \rangle$ is bounded below on the axial fields by Lemma 23.1(v); its operator has a kernel which decays exponentially, by Lemma 17.1(iii) and because $\mathfrak{q}''$ is the form of an operator whose kernel vanishes beyond the distance $3L$; and the map $B \mapsto B - \partial_1 \omega$ of Lemma 23.1(iii) is a projection onto the axial fields whose kernel vanishes beyond the distance $2L$ and has bounded sums. Lemma 22.5 and Lemma 14.1 give the decay of $\tilde{C}_\Lambda$, as in the proof of Proposition 22.6(iii).

The composition. Let $\mathsf{G}_1 := \tilde{G}_j + H_j \tilde{C}_\Lambda H_j^*$, where $\tilde{G}_j = G_j - H_j Q G_j$. By (c) and (d), G_1, $\nabla \mathsf{G}_1$, $\mathsf{G}_1 \nabla^*$, and $\Delta \mathsf{G}_1$ are of the class $\mathcal{B}$, and $\nabla \mathsf{G}_1 \nabla^* J$ is the sum of $\nabla G_j \nabla^* J$, which satisfies Proposition 16.2(iii) and (iv), and of an operator of the class $\mathcal{B}$ applied to J, whose values satisfy the Hölder bound of (ii). By (a) and (b), $\mathsf{W} = \partial \Delta H' C'_\Lambda E H'^* \partial^*$, $\mathsf{W} \nabla^*$, $\mathsf{W}^* = \partial H' C'_\Lambda E H'^* \Delta \partial^*$, $\nabla \mathsf{W}^*$, and $\Delta \mathsf{W}^*$ are of the class $\mathcal{K}$, and the values of $\nabla \mathsf{W}^* F$ are Lipschitz continuous in the sense of (a); the same holds for the first term $\mathsf{G}_0 := \partial H' C'_\Lambda E H'^* \partial^*$ of (23.6) with $\nabla \mathsf{G}_0$, $\mathsf{G}_0 \nabla^*$, $\nabla \mathsf{G}_0 \nabla^*$, and $\Delta \mathsf{G}_0$. In no case is a derivative applied to W from the left or to W^* from the right. Now $G = \mathsf{G}_0 + (1 - \mathsf{W}^*) \mathsf{G}_1 (1 - \mathsf{W})$. For a bounded field J the fields $(1 - \mathsf{W}) J$ and $(1 - \mathsf{W}) \nabla^* J = \nabla^* J - \mathsf{W} \nabla^* J$ are J, respectively $\nabla^* J$, plus the image of J under an operator of the class $\mathcal{B}$. Applying G_1 and then $1 - \mathsf{W}^*$, with or without

a derivative on the left, we obtain in each of the cases (i)–(iv) the corresponding expression for G_1, which is that for G_j up to operators of the class $\mathcal{B}$, plus images of J under operators of the class $\mathcal{B}$ whose values are Hölder continuous with every exponent $\alpha < 1$. This gives (i)–(iv).

The bounds in L^2. The operators of the class $\mathcal{K}$ are bounded on the square-summable functions. The operators G_0, W, and W^* are of this class, and they remain so with up to two derivatives on the left of G_0 and W^* and on the right of G_0 and W, because this gives at most three derivatives of H' on either side. The operator G_1 satisfies (9.3): for G_j this is Proposition 9.3, and $H_j = G_j Q^*(\Delta_j + a)$ and its adjoint are products of G_j, with the derivatives on it, and of operators of the class $\mathcal{K}$. Inserting these facts into $G = \mathsf{G}_0 + (1 - \mathsf{W}^*)\mathsf{G}_1(1 - \mathsf{W})$ gives (9.3) for G, with other constants.

For the scales 0 and 1 all the operators of the representation act on the lattice T_η, in whose unit a difference quotient is a bounded operator of the range one, and the statement follows from the decay of $C'_\Lambda E$ and of $\tilde{C}_\Lambda$, which is the remark after Lemma 21.3 and Proposition 22.6(iii). *q.e.d.*

24 The Propagator of a Sequence of Domains

The rest of Section C of [3] passes from the propagators of two scales to the propagator G of a sequence of domains, by the expansion [3, (2.91)–(2.93), (2.141)], and then to the inverse of QGQ^* and to the minimiser H [3, (2.142)–(2.151)]. The scheme is that of Section 20, and the lattice enters it through the statements of Sections 20 to 23 only. There is one step which Bałaban does not carry out: the remainder of the expansion contains the difference of the gauge-fixing projection of the sequence and that of the local operator [3, (2.92)], and its smallness is asserted with the others [3, (2.134)]. We prove it, as a statement on two sequences of domains which agree near a cube (Lemma 24.2). A second step needs an argument on the turned lattice: the lower bound [3, (2.147)] uses that every function on the bonds of the two scales is the average of a field in the axial gauge of comparable norm, which in [3, (2.146)] is read off from the bonds between the blocks (Lemma 24.4).

The setting. The fine lattice is a torus $\mathbb{T} := T_\eta/\Gamma'$, in the unit of the last lattice as in Section 20, with a sequence of domains which satisfies the hypotheses of Proposition 20.6; the statements of Section 20 are used on $\mathbb{T}$, with the kernels summed over the periods. As in [3, (2.3)], $\mathfrak{b}(\Lambda_j)$ is the set of the bonds of the lattice of the scale j with at least one end in Ω_j and no end in Ω_{j+1}, $\mathfrak{b}(\mathfrak{B})$ is the union of these sets, with the weights $\mathsf{w}(b) := \ell_j^4$, and $(QA)(b) := (Q_{\ell_j}A)(b)$ for $b \in \mathfrak{b}(\Lambda_j)$, $j \geq 1$, and $(QA)(b) := A(b)$ for $j = 0$ [3, (2.20)]. The operator $P = G'Q'^*N^{-1}Q'G'$ of Proposition 20.6 is the orthogonal projection onto the orthogonal complement of $\Delta N(Q')$: it is the orthogonal projection onto the range of $\mathsf{X} := G'Q'^*$, since $\mathsf{X}^*\mathsf{X} = N$; for $\lambda \in N(Q')$ we have $\Delta\lambda = \Delta'_a\lambda$ and $\langle \Delta\lambda, G'Q'^*\omega\rangle = \langle Q'\lambda, \omega\rangle = 0$; and if f is orthogonal to the range of X, then $\lambda := G'f$ satisfies $Q'\lambda = 0$ and $f = \Delta'_a\lambda = \Delta\lambda$. This is [3, (2.17)]. With $R := 1 - P$ let

$$\mathfrak{g}(A) := \|\partial A\|^2 + \|R\partial^* A\|^2 + \sum_{b \in \mathfrak{b}(\mathfrak{B})} a\, \ell_{j(b)}^{-2} \mathsf{w}(b) |(QA)(b)|^2 \,, \tag{24.1}$$

the quadratic form of the operator $\Delta_a = \partial^*\partial + \partial R\partial^* + Q^* a Q$ of [3, (2.19)], whose inverse, when it exists, is the propagator G.

The local operators. For $0 \leq j \leq k$ let $\mathcal{S}^{(j)}$ be the sequence in which the domains of the indices up to j are replaced by $\mathbb{T}$, the domain Ω_{j+1} is kept, and the later ones are omitted.

For $j < k$ it is, in the unit ℓ_{j+1}, a sequence of domains for the run of $j+1$ steps, of the kind which [3, p. 224] admits when he allows the first domains to be the whole lattice, and for $j = k$ it is the sequence all of whose domains are $\mathbb{T}$. It satisfies the hypotheses of Proposition 20.6 with the same M and R, the separation [3, (2.2)] being void for it, so that the statements of Section 20 hold for it, with its own distance. We mark its objects by the index $\square$, for a cube $\square \in \mathscr{D}$ of the scale j: its distance $d_\square$, its points $\mathfrak{B}_{(\square)}$, its average $Q_\square$, its scalar operators $Q'_\square$, $A_\square = (\Delta'_{a,\square})^{-1}$, which is the operator of Section 18, $N_{(\square)} \coloneqq Q'_\square A_\square^2 Q'^*_\square$, its projection $P_\square$, and its form $\mathfrak{g}_\square$; they depend on j only. The form $\mathfrak{g}_\square$ is, in the unit ℓ_j, the form (23.4) with $\Lambda' = \Omega_{j+1}^{(j+1)}$, respectively its version for the scales 0 and 1, and for $j = k$ the form of Proposition 9.3. By Proposition 23.2 it is positive definite, and we let $G_\square$ be the inverse of its operator. By Proposition 23.3, written in the unit of the last lattice, for every bounded field J and every x, with $D \coloneqq \operatorname{dist}(x, \operatorname{supp} J)$,

$$|(G_\square J)(x)| \le c\,\ell_j^2 e^{-\delta_G D/\ell_j} \sup|J| \;, \qquad |(\nabla G_\square J)(x)| \;,\; |(G_\square \nabla^* J)(x)| \le c\,\ell_j e^{-\delta_G D/\ell_j} \sup|J| \;, \tag{24.2}$$

together with the Hölder bounds and the bounds for $\Delta G_\square$ and $\nabla G_\square \nabla^*$ of Proposition 16.2 with the corresponding powers of ℓ_j, which is [3, (2.133)]. Throughout this section c denotes constants which depend on a, L, and the Hölder exponents, and which may change from line to line, and we use Lemma 19.2 and the rule (20.22) as in the proof of Proposition 20.6, for both sequences.

The zones of a cube. Let $\square \in \mathscr{D}$ be a cube of the scale j. For $0 < \vartheta \le \frac{1}{4}$ let Z_ϑ be the union of the blocks of the points of $\mathfrak{B}$ whose distance from the support of $h_\square$, in $|\cdot|_\infty$, is less than $\vartheta M \ell_j$.

Lemma 24.1 (The Zones)**.** *Let $M \ge 240L$ and $0 < \vartheta < \vartheta' \le \frac{1}{4}$.*

(i) $Z_{1/4}$ lies in $\square$; its blocks are blocks of both sequences, of the scales j and $j+1$; and on $Z_{1/4}$ the averaging terms of Δ'_a and of $\Delta'_{a,\square}$, and those of Δ_a and of the operator of $\mathfrak{g}_\square$, coincide.

(ii) If $u \in \mathfrak{B}$ has a scale smaller than j or larger than $j+1$ and $v \in \mathfrak{B}$ lies in Z_ϑ, then $d(u,v) \ge (\frac{1}{3} - \vartheta)M - L$.

(iii) If v lies in Z_ϑ and v' does not lie in $Z_{\vartheta'}$, then $d(v,v') \ge (\vartheta' - \vartheta)M/L - 2$ for $v, v' \in \mathfrak{B}$, and $d_\square(v,v') \ge (\vartheta' - \vartheta)M/L - 2$ for $v, v' \in \mathfrak{B}_{(\square)}$.

Proof. A block of a scale of at most $j+1$ has the diameter $L\ell_j$ at most in $|\cdot|_\infty$. By property (b) of the cover the support of $h_\square$ is at a distance of at least $\frac{1}{3}M\ell_j$ from the complement of $\square$ and of $[\Omega_j]$ and of at least $\frac{1}{3}M\ell_{j+1}$ from $[\Omega_{j+2}]$. Hence the points of $Z_{1/4}$ are at a distance of at least $(\frac{1}{12}M - L)\ell_j \ge 19\ell_{j+1}$ from these three sets, which gives (i): the blocks of the two sequences coincide in $[\Omega_j] \setminus [\Omega_{j+2}]$, and the averaging terms at a point involve the blocks and the staircases within the distance $9\ell_{j+1}$ of it, as in the proof of Proposition 16.2. (ii) and (iii) are proved as Lemma 20.3(iii) and (ii): a chain from v to u, or to v', starts at a distance of at least $(\frac{1}{3} - \vartheta)M\ell_j - L\ell_j$ from the complement of $[\Omega_j]$, of at least $\frac{1}{3}M\ell_{j+1} - \vartheta M\ell_j - L\ell_j$ from $[\Omega_{j+2}]$, and of less than $(\vartheta M + L)\ell_j$ from the support of $h_\square$, while v' is at a distance of at least $(\vartheta' M - L)\ell_j$ from it; the steps which lower the first distance have the length ℓ_j at most, and all the steps inside $\square$, and all the steps of $\mathcal{S}^{(j)}$, the length ℓ_{j+1} at most. *q.e.d.*

Lemma 24.2 (The Projections of Two Sequences)**.** *Under the hypotheses of Proposition 20.6, and for $M \ge 240L$, there is a constant c such that for every cube $\square \in \mathscr{D}$ of the scale j, every bounded field g supported in $Z_{1/6}$, and every point x of $Z_{1/6}$,*

$$|(\partial(P - P_\square)\partial^* g)(x)| \le c\,\ell_j^{-2} e^{-\delta_1 M/(50L)} \sup|g| \;. \tag{24.3}$$

Proof. We normalise $\sup|g| = 1$ and put $f := \partial^* g$, $A := A_\square$, $\varepsilon := e^{-\delta_1 M/(50L)}$. All the exponential factors in M below have exponents of at least $\delta_1 M/(48L) - c$, by Lemma 24.1 and $\delta_1 \le \bar\delta_0/1536$, and are bounded by $c\,\varepsilon$. Let $\omega_2 := N_{(\square)}^{-1} Q'_\square A f$, so that $P_\square f = A Q'^*_\square \omega_2$, and $\omega := N^{-1} Q' G' f$, so that $P f = G' Q'^* \omega$. Let $\mathbf{1}$ be the multiplication by the characteristic function of the set of the points of $\mathfrak{B}$ in $Z_{1/4}$, which by Lemma 24.1(i) are points of both sequences with the same blocks, so that $Q'^* \mathbf{1} = Q'^*_\square \mathbf{1}$. With $\rho := N \mathbf{1} \omega_2 - Q' G' f$ we have $\omega - \mathbf{1}\omega_2 = -N^{-1}\rho$, and

$$P f - P_\square f = -G' Q'^* N^{-1} \rho + (G' - A) Q'^* \mathbf{1} \omega_2 - A Q'^*_\square (1 - \mathbf{1}) \omega_2 \,. \tag{24.4}$$

We bound the derivative of each term at x.

The function ω_2. By [3, (2.67)] for $\mathcal{S}^{(j)}$, whose scales are j and $j+1$, $|Q'_\square A \partial^* g| \le c\,\ell_j e^{-\frac14 \bar\delta_0 d_\square(\cdot, Z_{1/6})}$, and by [3, (2.87)] for $\mathcal{S}^{(j)}$ and [3, (2.61)], $|\omega_2(v)| \le c\,\ell_j^{-3} e^{-\frac14 \delta_1 d_\square(v, Z_{1/6})}$. For v outside $Z_{1/4}$ this is at most $c\,\ell_j^{-3}\varepsilon$ by Lemma 24.1(iii). The third term of (24.4) is the sum over these v of $\omega_2(v) A \mathbf{1}_{B(v)}$, and $|\partial A \mathbf{1}_{B(v)}| \le c\,\ell_j e^{-\frac12 \bar\delta_0 d_\square(\cdot, v)}$; its derivative is therefore at most $c\,\ell_j^{-2}\varepsilon$ at every point.

The change of the domain. The function $\psi := Q'^* \mathbf{1} \omega_2$ is supported in $Z_{1/4}$, and $\sup|\psi| \le c\,\ell_j^{-3}$. Let $E := \Delta'_a - A^{-1}$, so that $G' - A = -AEG'$. The proof of Lemma 20.5 uses of the support of the source only the inequality (20.15); with Lemma 24.1(ii) in its place, that is, with $d(u,v) \ge \frac{1}{12} M - L$ for the blocks u on which E does not vanish and the points v of $Z_{1/4}$, it gives

$$\sup|EG'\psi| \le c\, e^{-\bar\delta_0 M/64} \sup|\psi| \,, \qquad \sup|(G'^2 - A^2)\psi| \le c\,\ell_j^4 e^{-\bar\delta_0 M/192} \sup|\psi| \,, \tag{24.5}$$

and in the same way, with the bound $c\,\ell_m$ of [3, (2.67)] for $G'\nabla^*$ on a block of the scale m in the place of $c\,\ell_m^2$ and with (20.22) for $p = 1$, $\sup|EG'\partial^* g| \le c\,\ell_j^{-1} e^{-\bar\delta_0 M/32}$. Since $|\partial A \phi| \le c\,\ell_j \sup|\phi|$ and $|A\phi| \le c\,\ell_j^2 \sup|\phi|$ by Lemma 20.4(i) and Lemma 18.1(iv), the derivative of the second term of (24.4) is at most $c\,\ell_j^{-2}\varepsilon$ at every point, and $\sup|(G' - A) f| \le c\,\ell_j \varepsilon$.

The function ρ. Let $v \in \mathfrak{B}$. If v lies in $Z_{7/24}$, defined as the sets Z_ϑ, then Q' and $Q'_\square$ have the same block at v, and $N_{(\square)}\omega_2 = Q'_\square A f$ gives

$$\rho(v) = (Q'(G'^2 - A^2)\psi)(v) - (Q'(G' - A) f)(v) - (N_{(\square)}(1 - \mathbf{1})\omega_2)(v) \,. \tag{24.6}$$

The first two terms are at most $c\,\ell_j \varepsilon$ by the bounds just proved, and so is the third, by [3, (2.68)] for $\mathcal{S}^{(j)}$ and the bound on ω_2 outside $Z_{1/4}$. If v does not lie in $Z_{7/24}$, then $d(v, v') \ge M/(24L) - 2$ for the points v' of $Z_{1/4}$ by Lemma 24.1(iii), and [3, (2.68), (2.67)] with (20.22) give $|(N\mathbf{1}\omega_2)(v)| + |(Q'G'\partial^* g)(v)| \le c\,\ell_j \varepsilon\, e^{-\frac{1}{32}\bar\delta_0 d(v, Z_{1/4})}$. Hence $|\rho(v)| \le c\,\ell_j \varepsilon$ for every v.

The first term. For x in the block of $y \in \mathfrak{B}$, [3, (2.67)] and [3, (2.87)] give

$$|(\partial G' Q'^* N^{-1} \rho)(x)| \le c\,\ell_{j(y)} \sum_{w, v \in \mathfrak{B}} e^{-\frac12 \bar\delta_0 d(y,w)} \mathsf{w}(w)^{-1} e^{-\frac12 \delta_1 d(w,v)} |\rho(v)| \,. \tag{24.7}$$

For $x \in Z_{1/6}$ the scale of y is j or $j+1$, $\mathsf{w}(w)^{-1} \le L^4 \ell_j^{-4} e^{\frac{1}{16}\bar\delta_0 d(y,w)}$ by (20.22), and the two sums are bounded by [3, (2.61)]; the right side is at most $c\,\ell_j^{-2}\varepsilon$. *q.e.d.*

The expansion. Let $G_0 := \sum_{\square \in \mathscr{D}} h_\square G_\square h_\square$. By Lemma 24.1(i), $h_\square \Delta_a = h_\square \Delta_{a,\square} - h_\square \partial (P - P_\square) \partial^*$, where $\Delta_{a,\square}$ is the operator of $\mathfrak{g}_\square$, and therefore, with $\sum_\square h_\square^2 = 1$,

$$\Delta_a G_0 = I - \mathsf{R}_G \,, \qquad \mathsf{R}_G := \sum_{\square \in \mathscr{D}} (h_\square \partial (P - P_\square) \partial^* - [\Delta_a, h_\square]) G_\square h_\square \,, \tag{24.8}$$

which is [3, (2.91)–(2.93)] without the auxiliary functions $\zeta_\square$. The commutator is $[\Delta, h_\square] + [Q^* a Q, h_\square] - [\partial P \partial^*, h_\square]$. Its first two parts are the operators of the first two lines of [3, (2.92)]: first-order difference operators with coefficients of the order $(M\ell_j)^{-1}$ and averaging terms with the differences of $h_\square$ over a block, as in (17.4) and Lemma 17.4, which holds for the average of a block of any scale. The third has the kernel of $\partial P \partial^*$ multiplied by $h_\square(x') - h_\square(x)$.

Proposition 24.3 (Proposition 2.6 for the Tilted Run)**.** *Let $L \ge 64$ be a multiple of 5 and $a > 0$. There are numbers $M_2 \ge \max(M_1, 240L)$, $\varrho_2 \ge \varrho_1$, $\delta_3 > 0$, and c, which depend on a, L, and the Hölder exponents, such that for every sequence of domains with $R \ge 2$, $M \ge M_2$ a multiple of L, and $RM \ge \varrho_2$ the form $\mathfrak{g}$ is positive definite, its propagator is $G = G_0 \sum_{n\ge 0} \mathsf{R}_G^n$, which is [3, (2.141)], and G satisfies the bounds [3, (2.136)–(2.140), p. 247] with the rate δ_3.*

Proof. The rates. Let δ_G be the rate of (24.2), $\delta_2 = \frac18 \delta_1$ that of Proposition 20.6(iii), and $\delta_3' \coloneqq \min(\frac{1}{224}\delta_G, \frac14 \delta_2, \delta_1/(1600L))$. Let $\varrho_2 \ge \varrho_1$ be such that $RM \ge \varrho_2$ implies the condition of Lemma 19.2 for $\alpha = \delta_3'/(8\bar\delta_0)$ and $\delta_3' RM \ge 32 \log L$, so that [3, (2.61)] and (20.22) with $p \le 4$ are available for every $\alpha \ge \delta_3'/(8\bar\delta_0)$.

The remainder. Let J be supported in the block of $y' \in \mathfrak{B}$, let $\square$ be a cube of the scale j with $h_\square \ne 0$ on this block, and $F \coloneqq G_\square h_\square J$, which satisfies (24.2). We bound the three parts of the term of $\square$ in $\mathsf{R}_G J$ at a point x of the block of y. The local part of the commutator vanishes outside $\square$ and is at most $c(L/M) e^{-\delta_G \operatorname{dist}(x, B(y'))/\ell_j} \sup|J|$, as in the proof of Proposition 20.2. For the part with $\partial P \partial^*$ we use the kernel bound of Proposition 20.6(iii) and the inequality $|h_\square(x') - h_\square(x)| \le c(L/M)(d(y, y_1) + 1)$ for x' in the block of y_1: if $d(y, y_1) < M/(3L)$ and one of the two points lies in the support of $h_\square$, then both lie in $\square$ by Lemma 20.3(ii), their distance in $|\cdot|_\infty$ is at most $L\ell_j(d(y, y_1) + 2)$, and the inequality is property (d) of the cover; otherwise it holds because the left side is at most 1. With (24.2), (20.22), and [3, (2.61)] this part is at most $c(L/M) \sup|J|$ at every point, and it carries the factor $e^{-\frac12 \delta_2 d(y,\square)}$ if y lies outside $\square$. For the part with $P - P_\square$, which vanishes unless x lies in the support of $h_\square$, we split F into its restriction F_1 to $Z_{1/6}$ and the rest F_2. By Lemma 24.2, $|(\partial(P - P_\square)\partial^* F_1)(x)| \le c\,\ell_j^{-2} e^{-\delta_1 M/(50L)} \sup|F_1| \le c\, e^{-\delta_1 M/(50L)} \sup|J|$. The field F_2 vanishes within the distance $(\frac16 M - L)\ell_j$ of the support of $h_\square$, so that $\sup|F_2| \le c\,\ell_j^2 e^{-\delta_G M/7} \sup|J|$ by (24.2), and the kernel bounds of Proposition 20.6(iii) for the two sequences, summed by [3, (2.61)], give $|\partial P \partial^* F_2| + |\partial P_\square \partial^* F_2| \le c\,\ell_j^{-2} \sup|F_2|$ on the support of $h_\square$. Altogether, with $\varepsilon_M' \coloneqq c(L/M + e^{-\delta_1 M/(100L)} + e^{-\delta_G M/14})$,

$$|(\mathsf{R}_G J)(x)| \le \varepsilon_M'\, e^{-\delta_3' d(y,y')} \sup|J| \,. \tag{24.9}$$

Indeed, for y in $\square$ the decay in the Euclidean distance becomes a decay in $d(y, y')$ by Lemma 20.1, or is bought from one half of the exponent of the factor in M as in the proof of Proposition 20.6, since $d(y, y') \le 16M + 34L$ there and $16\delta_3' \le \min(\delta_1/(100L), \delta_G/14)$; and for y outside $\square$ we use $d(y, y') \le d(y, \square) + 16M + 34L$, where $d(y, \square)$ is the least distance from y to a point of $\mathfrak{B}_\square$. This is [3, (2.135)].

The series. By (24.9) and [3, (2.61)], the series $\sum_n \mathsf{R}_G^n$ converges for $M \ge M_2$, in the norms of the proof of Proposition 20.2, and $\Delta_a G_0 \sum_n \mathsf{R}_G^n = I$ by (24.8). On the finite torus the operator Δ_a, which is symmetric and nonnegative, is therefore invertible, so that $\mathfrak{g}$ is positive definite and G is the series. The bounds [3, (2.136)–(2.139)] follow as in the proof of Proposition 20.2, with (24.2) and its Hölder counterparts for the first factor $h_\square G_\square h_\square$ in the place of Lemma 18.1(iv), and with the adjoint series $G = \sum_n (\mathsf{R}_G^*)^n G_0$ for the bounds with a

derivative of the source: $\mathsf{R}^*_G = \sum_\square h_\square G_\square(\partial(P - P_\square)\partial^* h_\square + [\Delta_a, h_\square])$ satisfies (24.9) by the same estimates, a derivative which the commutator applies to its argument being moved to $G_\square$ by a summation by parts. The bounds [3, (2.140)] in L^2 follow in the same way from the bounds (9.3) of Proposition 23.3 for $G_\square$ and from Schur's test for the remainder. *q.e.d.*

For the inverse of QGQ^* the functions on $\mathfrak{b}(\mathfrak{B})$ carry the scalar product with the weights w, and kernels are taken with respect to it, as in Section 20. For a cube $\square$ of the scale j let $\mathfrak{b}_\square$ be the set of the bonds of $\mathfrak{b}(\mathfrak{B})$ with both ends in $\square$; by Lemma 24.1(i) and its proof they are bonds of both sequences, with the same averages. Following [3, (2.143)] we put $N^Q_\square := \square\, Q_\square G_\square Q^*_\square\, \square$ on the functions on $\mathfrak{b}_\square$, where $\square$ now denotes the restriction to $\mathfrak{b}_\square$. The lower bound for this operator rests on the following fact.

Lemma 24.4 (A Right Inverse of the Averages)**.** *In the setting of Section 23, for the scales j and $j+1$ with $j \geq 1$ as for the scales 0 and 1, there is a constant c_r, which depends on L only, such that every function B on $\mathfrak{b}(\Lambda^c) \cup \mathfrak{b}(\Lambda')$ is $Q''B'$ for a field B' on the unit lattice which is axial on the cubes of Λ and satisfies $\|B'\| \leq c_r\|B\|$. Consequently $\langle B, Q''\tilde{C}_\Lambda Q''^* B\rangle \geq \gamma_Q \|B\|^2$ with a constant $\gamma_Q > 0$ which depends on a and L.*

Proof. We write B^0 and B^1 for the restrictions of B to $\mathfrak{b}(\Lambda^c)$ and $\mathfrak{b}(\Lambda')$, and we extend B^0 by zero to a field of the unit lattice.

The scales $j \geq 1$. For a bond $c = \langle y, y + Le_\mu\rangle$ of $\mathfrak{b}(\Lambda')$ let f_c be the field which is 1 on the L^3 bonds $\langle z, z + e_\mu\rangle$ with z in the cube of y and $z_\mu = y_\mu + L - 1$, and zero elsewhere. Each of these bonds lies on the segments $[z', z' + Le_\mu]$ of L points z' of the cube of y and on no other segment of (6.4), so that $(Q_1\mathsf{f}_c)(c) = L^{-5}\cdot L^4 = L^{-1}$ and $(Q_1\mathsf{f}_c)(c') = 0$ for $c' \neq c$. These bonds join two cubes of which one lies in Λ; they are therefore not in $\mathfrak{b}(\Lambda^c)$ and on no contour of the axial gauge. Hence $B' := B^0 + \sum_c L(B^1(c) - (Q_1B^0)(c))\mathsf{f}_c$ is axial on the cubes of Λ, on whose inner bonds it vanishes, $Q''B' = B$, and $\|B'\|^2 \leq \|B^0\|^2 + 2L\sum_c L^4(|B^1(c)|^2 + |(Q_1B^0)(c)|^2) \leq (1 + 4L)\|B\|^2$, because $\sum_c L^4|(Q_1B^0)(c)|^2 \leq \|B^0\|^2$ by Schwarz's inequality.

The scales 0 and 1. Here the unit lattice is T_η/Γ', $Q_1 = Q_{\ell_1}$, and a staircase through a bond need not start in one block. Let $c = \langle y, y + \ell_1 e_\mu\rangle \in \mathfrak{b}(\Lambda')$ and $t := y_\mu + \ell_1$. If $y \in \Lambda'$, let X_c be the set of the bonds $\langle z, z'\rangle$ of T_η with $z_\mu < t \leq z'_\mu$ and $z \in B_{\ell_1}(y)$; otherwise let it be the set of these bonds with $z' \in B_{\ell_1}(y + \ell_1 e_\mu)$. Let f_c be the field which is 1 on the bonds of X_c, hence -1 on their reverses, and zero elsewhere; all its bonds have an end in Λ. Along a staircase $\sigma^{\ell_1}_\mu(x)$ the coordinate μ increases at every step, by (2.2), so that the staircase contains at most one bond which crosses the level t, and exactly one if $x \in B_{\ell_1}(y')$ with $y'_\mu = y_\mu$. Its sites are within $\frac{3}{5}\eta$ of the segment from x (Lemma 6.2(ii)), and they have the coordinates 0 and 1 of x unless μ is one of these. Hence the sum of f_c over $\sigma^{\ell_1}_\mu(x)$ is 0 or 1 for all x, and it is 1 for the points x of $B_{\ell_1}(y)$ whose coordinate ν is at a distance of at least $\frac{3}{5}\eta$ from the faces of the block, where $\{\mu, \nu\} = \{2, 3\}$, and for all the points of the block if $\mu \in \{0, 1\}$: the two ends of the crossing bond then lie in $B_{\ell_1}(y)$ and $B_{\ell_1}(y + \ell_1 e_\mu)$. For x in another block the sum vanishes unless $y' = y \pm \ell_1 e_\nu$ and x_ν is within $\frac{3}{5}\eta$ of the common face. A strip of the width $\frac{6}{5}\eta$ and the length ℓ_1 in the plane is covered by $2(L + 1)$ squares of the side η, each of which contains at most four points of the turned lattice, whose points are at least η apart; hence at most $9L^3$ points of a block have a planar coordinate in a given interval of the length $\frac{6}{5}\eta$, and at most $5L^3$ in one of the length $\frac{3}{5}\eta$. A staircase $\sigma^{\ell_1}_\nu(x)$ of the other turned direction crosses the level t only if x_μ is within $\frac{3}{5}\eta$ of t; its crossings alternate in sense, and those which belong to X_c are consecutive, because the coordinate ν increases along it and decides the block of the relevant end. The sum of f_c over it is therefore 0 or ± 1, and it vanishes unless x lies in one of four blocks, in each of which at most $9L^3$ points

qualify. The staircases of the directions e_0 and e_1 do not cross a level of a turned direction, and conversely. Since $(Q_{\ell_1}\mathsf{f}_c)(c')$ is L^{-5} times the sum over the points of the block of c'_-, we have

$$(Q_{\ell_1}\mathsf{f}_c)(c) \geq L^{-5}(L^4 - 10L^3) \geq \tfrac{5}{6}L^{-1} , \qquad \sum_{c' \neq c} |(Q_{\ell_1}\mathsf{f}_c)(c')| \leq (2 \cdot 5 + 4 \cdot 9)L^{-2} = 46L^{-2} , \tag{24.10}$$

where we have used $L \geq 65$, and the same bound holds for the sum over c at fixed c', by the same count. The matrix $\mathsf{T}(c', c) := (Q_{\ell_1}\mathsf{f}_c)(c')/(Q_{\ell_1}\mathsf{f}_c)(c)$ on $\mathfrak{b}(\Lambda')$ is therefore the identity plus a matrix of norm at most $56/L \leq \frac{7}{8}$, by Schur's test, and it has an inverse of norm at most 8. With $\beta := \mathsf{T}^{-1}(B^1 - Q_{\ell_1}B^0)$ the field $A'' := B^0 + \sum_c \beta(c)\mathsf{f}_c/(Q_{\ell_1}\mathsf{f}_c)(c)$ equals B^0 on $\mathfrak{b}(\Lambda^c)$, has the average B^1 on $\mathfrak{b}(\Lambda')$, and satisfies $\|A''\| \leq c\,L\|B\|$, since each f_c has at most $8L^4$ bonds, a bond belongs to at most eight of the sets X_c, and Q_{ℓ_1} is bounded. The field $B' := \mathfrak{a}_\Lambda A''$ of the proof of Proposition 22.6 is axial on the blocks of Λ, has the same averages, equals A'' on the bonds with no end in Λ, and $\mathfrak{a}_\Lambda$ is bounded by a constant which depends on L.

The lower bound. Let $\mathfrak{f}$ be the form of which $\tilde{C}_\Lambda$ is the covariance on the space $\mathcal{Y}$ of the axial fields. By Lemma 22.5, for every $B_2 \in \mathcal{Y}$, $\langle Q''^* B, \tilde{C} Q''^* B\rangle \geq 2\langle B, Q'' B_2\rangle - \mathfrak{f}(B_2)$. The form $\mathfrak{f}$ is bounded above by a constant $c_\mathfrak{f}$ times the squared norm, by (9.4), $\|\partial_1 B_2\|^2 \leq 24\|B_2\|^2$, and the boundedness of Q_1, respectively by (22.11) in the unit η. Taking $B_2 := tB'$ with $Q''B' = B$ and optimising in t gives the lower bound with $\gamma_Q := 1/(c_\mathfrak{f} c_r^2)$. *q.e.d.*

Proposition 24.5 (Proposition 2.7 and Corollary 2.8 for the Tilted Run)**.** *Under the hypotheses of Proposition 24.3, and for M and RM larger than numbers which depend on a and L, the following hold with constants c and rates $\delta_4, \delta_5 > 0$ which depend on a and L; the condition on RM is that of Lemma 19.2 and of* (20.22) *for the products of the rates δ_3, δ_4 with fixed fractions.*

(i) $\langle B, N_\square^Q B\rangle \geq \gamma_Q \ell_j^2 \|B\|^2$ *for every cube $\square$ of the scale j and every function B on $\mathfrak{b}_\square$, and the kernel of $C_\square^Q := (N_\square^Q)^{-1}$ is at most $c\,\ell_j^{-6} e^{-\delta_4|b_- - b'_-|/\ell_j}$ in modulus, which is [3, (2.147), (2.148)].*

(ii) The operator QGQ^* on the functions on $\mathfrak{b}(\mathfrak{B})$ has an inverse, given by the expansion [3, (2.86)] with $C^Q := \sum_\square h_\square C_\square^Q h_\square$, and $|(QGQ^*)^{-1}(b, b')| \leq c\,\ell_{j(b)}^{-2} \mathsf{w}(b')^{-1} e^{-\frac{1}{2}\delta_4 d(b,b')}$, which is [3, (2.149)].

(iii) The operator $H := GQ^*(QGQ^*)^{-1}$ maps a function B on $\mathfrak{b}(\mathfrak{B})$ to the field which minimises $\|\partial A\|^2$ among the fields with $QA = B$ and $R\partial^* A = 0$, and its kernel satisfies [3, (2.151)] with the rate δ_5.

Proof. (i) A function B on $\mathfrak{b}_\square$, extended by zero, is a function on the bonds of $\mathcal{S}^{(j)}$, and $\langle B, N_\square^Q B\rangle = \langle Q_\square^* B, G_\square Q_\square^* B\rangle$. By Proposition 23.2 this is $\langle Q''^* B, \tilde{C}_\Lambda Q''^* B\rangle$ in the unit ℓ_j, and Lemma 24.4 gives the lower bound; the factor ℓ_j^2 is that of the change of the unit, as in [3, (2.144)]. For $j = k$ there is one scale, and the bound is Lemma 17.1(ii). The kernel of $N_\square^Q$ is at most $c\,\ell_j^2 \mathsf{w}(b')^{-1} e^{-\delta_G|b_- - b'_-|/(2\ell_j)}$ by (24.2), the average of a field over the staircases of a bond being at most its supremum over them. The decay of the inverse follows from Lemma 14.1 after the symmetrisation with the weights, as in the proof of Lemma 20.4(iii).

(ii) By [3, (2.136)], which is Proposition 24.3, the kernel of QGQ^* satisfies [3, (2.142)], as in the proof of Proposition 20.6(i). On the finite torus it is a symmetric matrix, which is invertible as soon as it has a right inverse. As in (20.24),

$$QGQ^* C^Q = I - \sum_\square \Big([h_\square, N_\square^Q] + \square\, Q(G_\square - G)Q^* h_\square - (1 - \square) QGQ^* h_\square \Big) C_\square^Q h_\square , \tag{24.11}$$

where we have used that Q and $Q_\square$ have the same averages on the bonds of $\mathfrak{b}_\square$. The first and the third term of the remainder are bounded as the corresponding terms in the proof of Proposition 20.6(ii), with the powers ℓ^2 and ℓ^{-2} in the place of ℓ^4 and ℓ^{-4} and with (i) and [3, (2.142)]. For the second term we show that for every field J supported in $Z_{1/24}$, and every x,

$$|((G_\square - G)J)(x)| \le c\,\ell_j^2 e^{-\delta M/L} \sup|J| \tag{24.12}$$

with a rate $\delta > 0$. Indeed $G_\square - G = G_\square(\Delta_a - \Delta_{a,\square})G$, and $\Delta_a - \Delta_{a,\square}$ is the sum of the difference E_Q of the two averaging terms, which by Lemma 24.1(i) and (ii) vanishes within the distance $\frac{1}{12}M - L$, in the distance d, of the zone $Z_{1/4}$, and of $-\partial(P - P_\square)\partial^*$. Let $F := GJ$, which by Proposition 24.3 is at most $c\,\ell_m^2 e^{-\delta_3 d(u,y')} \sup|J|$ on the block of a point u of the scale m, for J supported in the block of y'. Then $\sup|\mathsf{E}_Q F| \le c\,e^{-\delta_3 M/13} \sup|J|$, by (20.22) as in the proof of Lemma 20.5. We split F into its restriction F_1 to $Z_{1/12}$ and the rest F_2; by Lemma 24.1(iii) and (20.22), $\sup|F_2| \le c\,\ell_j^2 e^{-\delta_3 M/(49L)} \sup|J|$, also as a function on the blocks of $\mathcal{S}^{(j)}$. By Lemma 24.2, $\partial(P - P_\square)\partial^* F_1$ is at most $c\,e^{-\delta_1 M/(50L)} \sup|J|$ on $Z_{1/6}$; outside $Z_{1/6}$ the two terms $\partial P \partial^* F_1$ and $\partial P_\square \partial^* F_1$ are each at most $c\,e^{-\delta_2 M/(25L)} \sup|J|$, by the kernel bounds of Proposition 20.6(iii) for the two sequences, Lemma 24.1(iii), and (20.22); and by the same kernel bounds $\partial P \partial^* F_2$ and $\partial P_\square \partial^* F_2$ are at most $c\,\ell_j^{-2} \sup|F_2|$ times a power of L which (20.22) pays. Hence $(\Delta_a - \Delta_{a,\square})GJ$ is bounded by $c\,e^{-\delta M/L} \sup|J|$ at every point, and the first bound of (24.2) for $G_\square$ gives (24.12). With it the second term of the remainder is bounded as the term of the change of the domain in the proof of Proposition 20.6(ii), the fields $Q^* h_\square C_\square^Q h_\square B$ being supported in the blocks and on the staircases of the bonds of the support of $h_\square$, which lie in $Z_{1/24}$ because $\frac{1}{24}M \ge 10L$. The remainder then has a kernel which is at most a small multiple of $e^{-\delta_4 d(b,b')} \mathsf{w}(b')^{-1}$, the series converges as in that proof, it gives a right inverse of QGQ^*, hence the inverse, and the bound on its kernel follows as there.

(iii) By the argument of Lemma 23.1(iv), with $\mathfrak{g}$, Q, and the form $\|\partial A\|^2 + \|R\partial^* A\|^2$ in the place of the forms of the scale j, the field HB minimises this form among the fields with $QA = B$. Its minimum over the fields with a given average and a given curl is attained where $R\partial^* A = 0$, because for $\lambda \in N(Q')$ the field $A - \partial\lambda$ has the same curl and the same average, by the argument of Lemma 23.1(i) for each pair of neighbouring scales, and $R\partial^*(A - \partial\lambda) = R\partial^* A - \Delta\lambda$ vanishes for one λ. This is the characterisation [3, (2.5), (2.6), (2.12)] of H. The bounds [3, (2.151)] are the composition of the bounds of Proposition 24.3 for GQ^*, ∇GQ^*, and the Hölder norm, with (ii), by [3, (2.61)] and (20.22), as in the proof of Proposition 20.6(iii). *q.e.d.*

The remaining formulae of Section A of [3] are identities between the operators of a sequence, which Bałaban obtains from Gaussian integrals by the method of Faddeev and Popov [3, (2.24)–(2.34)]. They follow from the definitions.

Lemma 24.6 (The Identities of Section A)**.** *Under the hypotheses of Proposition 24.3 the following hold.*

(i) For every $\lambda \in N(Q')$, $Q\partial\lambda = 0$, $\Delta_a \partial\lambda = \partial\Delta\lambda$, and $G\partial\Delta\lambda = \partial\lambda$.

(ii) $R\partial^* G \partial R = R$, $QG\partial R = 0$, and $R\partial^* G Q^* = 0$, which are [3, (2.31), (2.34)].

(iii) The operator $\mathscr{G} := G'^2 - G'^2 Q'^* N^{-1} Q' G'^2$ of [3, (2.27)] maps into $N(Q')$, and $R = \Delta\mathscr{G}\Delta$, which is [3, (2.26)].

Proof. (i) Let $b \in \mathfrak{b}(\Lambda_j)$ with $j \geq 1$. By Lemma 6.4(i), $(Q\partial\lambda)(b) = \ell_j^{-1}((Q'_{\ell_j}\lambda)(b_+) - (Q'_{\ell_j}\lambda)(b_-))$. An end of b in Ω_j is a point of Λ_j, because b has no end in Ω_{j+1}, and there $Q'_{\ell_j}\lambda$ vanishes by the definition of $N(Q')$. The block of the scale j of an end outside Ω_j is a union of blocks of the sequence, of smaller scales, on each of which the mean of λ vanishes, or of points of Λ_0, at which λ vanishes; hence $Q'_{\ell_j}\lambda$ vanishes there as well. For $j = 0$ both ends of b lie in Λ_0, or one lies in a block of Λ_1; in the first case $(\partial\lambda)(b) = 0$, and the second does not occur for a bond of $\mathfrak{b}(\Lambda_0)$, which has no end in Ω_1. Hence $Q\partial\lambda = 0$. The curl of $\partial\lambda$ vanishes, $\partial^*\partial\lambda = \Delta\lambda$ lies in $\Delta N(Q')$, on which R is the identity, and therefore $\Delta_a\partial\lambda = \partial R\Delta\lambda = \partial\Delta\lambda$. Applying G gives the third identity.

(ii) For a function f let $\lambda \in N(Q')$ be the function with $\Delta\lambda = Rf$. By (i), $G\partial Rf = \partial\lambda$, so that $R\partial^*G\partial Rf = R\Delta\lambda = Rf$ and $QG\partial Rf = Q\partial\lambda = 0$. The third identity is the adjoint of the second.

(iii) Since $Q'G'^2Q'^* = N$, we have $Q'\mathscr{G} = 0$. For $\mu \in N(Q')$ and a function J, $\langle\Delta'_a\mu, \Delta'_a\mathscr{G}J\rangle = \langle\mu, J\rangle - \langle Q'\mu, N^{-1}Q'G'^2J\rangle = \langle\mu, J\rangle$, and $\Delta'_a = \Delta$ on $N(Q')$; hence $\langle\Delta\mu, \Delta\mathscr{G}J\rangle = \langle\mu, J\rangle$. For $J = \Delta f$ this says that $f - \Delta\mathscr{G}\Delta f$ is orthogonal to $\Delta N(Q')$, while $\Delta\mathscr{G}\Delta f$ lies in this space, so that $\Delta\mathscr{G}\Delta f = Rf$. *q.e.d.*

With (ii), the equations [3, (2.23)] for the minimiser reduce to $QGQ^*\omega = B$, as in [3, p. 228], and Proposition 24.5(iii) is [3, (2.35)]. The variant of the operators with Dirichlet conditions outside a neighbourhood of Ω_1, for which Bałaban states that all the results hold "with minor and obvious changes" [3, p. 228], has not been examined for the tilted run.

With Propositions 23.2 to 24.5 the results of Section C of [3] hold for the tilted run. Three points of this section go beyond a transcription of Bałaban's argument: the representation [3, (2.129)] is proved by minimising quadratic forms; the comparison of the gauge-fixing projections of the sequence and of a cube, which [3, (2.134)] uses, is Lemma 24.2; and the local operators are those of a sequence of two scales on the whole torus, not of a torus $T_\square$ cut out of the lattice, so that the tori of [3, p. 238] and their compatibility with the turned lattice do not enter.

25 The Operators with Dirichlet Conditions

The later papers do not use the operators of a sequence of domains on the whole lattice. They use the operators "with Dirichlet boundary conditions" outside a neighbourhood Ω_0 of Ω_1, the inverses of $\Omega_0\Delta'_a\Omega_0$ and of $\Omega_0\Delta_a\Omega_0$, because these depend on a background field through its restriction to Ω_0 only [3, p. 228], [5, pp. 393–395]; of their properties Bałaban says that they hold "with minor and obvious changes" [3, p. 228]. For the turned lattice we do not go through the proofs again. The region outside Ω_1 is the region of the scale 0, in which the averages are the identity, and for this reason the operators with Dirichlet conditions are given by the operators of Sections 20 and 24 through exact formulae, from which their bounds follow.

The setting. We are in the setting of Section 24: a torus $\mathbb{T}$, a sequence of domains which satisfies the hypotheses of Proposition 24.5, and the constant $a_0 = a$ in the region of the scale 0, as in [5, (3.24)]. Let Z be a set of points of Λ_0 no point of which has a neighbour in Ω_1, and $Y := \mathbb{T} \setminus Z$; let $\mathfrak{z}$ be a set of bonds of $\mathfrak{b}(\Lambda_0)$, that is, of bonds of $\mathbb{T}$ with no end in Ω_1. In Bałaban's setting $Y = \Omega_0$, and $\mathfrak{z}$ is the set of the bonds outside Ω_0. The points of Z are points of $\mathfrak{B}$ of the scale 0, and we write $\mathbf{1}_Z$ and $\mathbf{1}_Y$ for the multiplications by the characteristic functions, on the functions on $\mathbb{T}$ and on those on $\mathfrak{B}$, where $\mathbf{1}_Y$ is the characteristic function

of $\mathfrak{B} \setminus Z$; likewise $\mathbf{1}_\mathfrak{z}$ and $\mathbf{1}_\mathfrak{y} := 1 - \mathbf{1}_\mathfrak{z}$ on the fields and on the functions on $\mathfrak{b}(\mathfrak{B})$. Since the block of a point of Z is the point itself, Q' and Q'^* commute with these multiplications, and so do Q and Q^* with $\mathbf{1}_\mathfrak{z}$, because $(QA)(b) = A(b)$ on $\mathfrak{b}(\Lambda_0)$. For an operator A and one of these characteristic functions χ such that $\chi\mathsf{A}\chi$ is invertible on the range of χ, we write $(\chi\mathsf{A}\chi)^{-1}$ for the operator which is this inverse on the range of χ and zero on the range of $1 - \chi$. The operators with Dirichlet conditions are

$$G'_Y := (\mathbf{1}_Y \Delta'_a \mathbf{1}_Y)^{-1} , \qquad N_Y := \mathbf{1}_Y Q' G'^2_Y Q'^* \mathbf{1}_Y , \qquad G_\mathfrak{y} := (\mathbf{1}_\mathfrak{y} \Delta_a \mathbf{1}_\mathfrak{y})^{-1} , \tag{25.1}$$

where Δ_a is the operator of the form (24.1). Two remarks justify the notation. The averaging term of $\mathbf{1}_Y \Delta'_a \mathbf{1}_Y$ is that of the sequence which starts with $\Omega_0 = Y$ [5, (3.24)], since the two differ by the terms of the points of Z, at which the functions vanish. And the projection R is the same for both: the space $N(Q')$ consists of functions which vanish on Λ_0, the functions $\Delta\lambda$ with $\lambda \in N(Q')$ vanish at the points of Λ_0 none of whose neighbours lies in Ω_1, hence on Z, so that $R\mathbf{1}_Z = \mathbf{1}_Z R = 0$, and the orthogonal projection onto $\Delta N(Q')$ in the space of the functions on Y is the restriction of R. Hence $\mathbf{1}_\mathfrak{y} \Delta_a \mathbf{1}_\mathfrak{y}$ is the operator $\Omega_0 \Delta_a \Omega_0$ of [3, p. 228] and of [5, (3.26), (3.27)] at the background field $U = 1$.

For $y, y' \in \mathfrak{B}$ let $d_Z(y, y')$ be the least value of $d(y, z) + d(z, y')$ over the points z of Z, and $d_\mathfrak{z}(y, y')$ the least value of $d(y, b_-) + d(b_-, y')$ over the bonds b of $\mathfrak{z}$; both are at least $d(y, y')$.

Proposition 25.1 (The Operators with Dirichlet Conditions)**.** *In the setting above, and for RM larger than a number which depends on a and L, the operators* (25.1) *exist, N_Y is invertible on the functions on $\mathfrak{B} \setminus Z$, and the following hold, with the constant $c_\Delta := 40 + 16a$ and with constants c and rates which depend on a, L, and the Hölder exponents, and not on k, the torus, the sequence, Z, or $\mathfrak{z}$. The condition on RM is that of Lemma 19.2 and of* (20.22) *for the rates which occur in the proof.*

(i) The operator $\mathsf{D}'_Z := \mathbf{1}_Z G' \mathbf{1}_Z$ satisfies $\mathsf{D}'_Z \geq (16 + a)^{-1} \eta^2$ on the functions on Z, and

$$G'_Y = \mathbf{1}_Y (G' - G'(\mathsf{D}'_Z)^{-1} G') \mathbf{1}_Y . \tag{25.2}$$

The values of $(\mathsf{D}'_Z)^{-1}$ on the function which is 1 at a point z' of Z and zero elsewhere are at most $2(16 + a)\eta^{-2} e^{-\theta_Z d(z, z')}$ in modulus, with a rate $\theta_Z > 0$.

(ii) $N_Y^{-1} = \mathbf{1}_Y (N^{-1} - Q' \Delta'_a \mathbf{1}_Z \Delta'_a Q'^*) \mathbf{1}_Y$, where $N = Q' G'^2 Q'^*$, and $G'_Y Q'^* N_Y^{-1} Q' G'_Y = \mathbf{1}_Y P \mathbf{1}_Y = P - \mathbf{1}_Z$.

(iii) The operator $\mathsf{D}_\mathfrak{z} := \mathbf{1}_\mathfrak{z} G \mathbf{1}_\mathfrak{z}$ satisfies $\mathsf{D}_\mathfrak{z} \geq c_\Delta^{-1} \eta^2$ on the fields on $\mathfrak{z}$, and

$$G_\mathfrak{y} = \mathbf{1}_\mathfrak{y} (G - G(\mathsf{D}_\mathfrak{z})^{-1} G) \mathbf{1}_\mathfrak{y} . \tag{25.3}$$

The values of $(\mathsf{D}_\mathfrak{z})^{-1}$ on the field which is 1 at a bond b' of $\mathfrak{z}$ and zero elsewhere are at most $2c_\Delta \eta^{-2} e^{-\theta_\mathfrak{z} d(b_-, b'_-)}$ in modulus, with a rate $\theta_\mathfrak{z} > 0$.

(iv) The operator $\mathbf{1}_\mathfrak{y} Q G_\mathfrak{y} Q^* \mathbf{1}_\mathfrak{y}$ is invertible on the functions on $\mathfrak{b}(\mathfrak{B}) \setminus \mathfrak{z}$, its inverse is $\mathbf{1}_\mathfrak{y} (QGQ^*)^{-1} \mathbf{1}_\mathfrak{y}$, and $G_\mathfrak{y} Q^* (\mathbf{1}_\mathfrak{y} Q G_\mathfrak{y} Q^* \mathbf{1}_\mathfrak{y})^{-1} = H \mathbf{1}_\mathfrak{y}$. For a function B on $\mathfrak{b}(\mathfrak{B}) \setminus \mathfrak{z}$ the field HB vanishes on $\mathfrak{z}$, and it minimises $\|\partial A\|^2$ among the fields A which vanish on $\mathfrak{z}$ and satisfy $QA = B$ and $R\partial^* A = 0$.

(v) The operator G'_Y satisfies the bounds [3, (2.67)], the kernel of N_Y^{-1} the bound [3, (2.87)], the operator $G_\mathfrak{y}$ the bounds [3, (2.136)–(2.140)], and the operators of (iv) the bounds [3, (2.149), (2.151)], as in Propositions 20.2, 20.6, 24.3, and 24.5, with other constants and rates. The differences $G'_Y - \mathbf{1}_Y G' \mathbf{1}_Y$ and $G_\mathfrak{y} - \mathbf{1}_\mathfrak{y} G \mathbf{1}_\mathfrak{y}$ satisfy the same bounds with $d_Z(y, y')$, respectively $d_\mathfrak{z}(y, y')$, in the place of $d(y, y')$.

Proof. All the spaces have finite dimension. We use one identity of linear algebra. Let A be a symmetric positive definite operator with the inverse G, and let χ and $\chi' = 1 - \chi$ be complementary orthogonal projections. Then $\chi\mathsf{A}\chi$ and $\chi'\mathsf{G}\chi'$ are positive definite on the ranges of χ and χ', and

$$(\chi\mathsf{A}\chi)^{-1} = \chi(\mathsf{G} - \mathsf{G}(\chi'\mathsf{G}\chi')^{-1}\mathsf{G})\chi \,. \tag{25.4}$$

Indeed, multiplying $\mathsf{A}(\chi+\chi')\mathsf{G} = 1$ by χ from the left and by χ, respectively χ', from the right gives $\chi\mathsf{A}\chi\,\chi\mathsf{G}\chi + \chi\mathsf{A}\chi'\,\chi'\mathsf{G}\chi = \chi$ and $\chi\mathsf{A}\chi\,\chi\mathsf{G}\chi' + \chi\mathsf{A}\chi'\,\chi'\mathsf{G}\chi' = 0$; the second equation gives $\chi\mathsf{A}\chi' = -\chi\mathsf{A}\chi\,\mathsf{G}(\chi'\mathsf{G}\chi')^{-1}$, and inserting this into the first gives $\chi\mathsf{A}\chi\,\chi(\mathsf{G} - \mathsf{G}(\chi'\mathsf{G}\chi')^{-1}\mathsf{G})\chi = \chi$.

(i) The operator Δ'_a is positive definite (Proposition 20.2), and (25.4) with $\chi = \mathbf{1}_Y$ is (25.2). For a function λ, $\langle\lambda, \Delta'_a\lambda\rangle \le 16\eta^{-2}\|\lambda\|^2 + a\eta^{-2}\|Q'\lambda\|^2 \le (16+a)\eta^{-2}\|\lambda\|^2$, because $a_j \le a$, $\ell_j \ge \eta$, and Q' is a contraction, its adjoint being an isometry; hence $G' \ge (16+a)^{-1}\eta^2$, and the same holds for its compression D'_Z. For $z, z' \in Z$ the value at z of G' applied to the function which is 1 at z' and zero elsewhere is at most $c\,\eta^2 e^{-\frac{1}{2}\bar\delta_0 d(z,z')}$ in modulus, by the first bound of [3, (2.67)] for two blocks of the scale 0. The set Z with the distance d and the matrix $\mathsf{M}(z,z')$ of these values satisfy the hypotheses of Lemma 14.1 with $\gamma = (16+a)^{-1}\eta^2$ and a rate $\theta_Z \le \frac{1}{8}\bar\delta_0$: since $e^t - 1 \le te^t$ and $te^{-t} \le 1$,

$$\sum_{z'}|\mathsf{M}(z,z')|(e^{\theta d(z,z')} - 1) \le c\,\eta^2\,\frac{8\theta}{\bar\delta_0}\sum_{z'\in\mathfrak{B}} e^{-\frac{1}{4}\bar\delta_0 d(z,z')} \le c\,\eta^2\,\frac{8\theta}{\bar\delta_0}\,c_1(\tfrac{1}{4}) \tag{25.5}$$

for $\theta \le \frac{1}{8}\bar\delta_0$, by [3, (2.61)] (Lemma 19.2), and this is at most $\frac{1}{2}\gamma$ for $\theta = \theta_Z$ small. The lemma gives the bound on $(\mathsf{D}'_Z)^{-1}$.

(ii) Let $\mathsf{X} \coloneqq G'Q'^*$, so that $N = \mathsf{X}^*\mathsf{X}$ and $P = \mathsf{X}N^{-1}\mathsf{X}^*$ (Section 24), and $\mathsf{X}_Y \coloneqq G'_Y Q'^*\mathbf{1}_Y$, so that $N_Y = \mathsf{X}_Y^*\mathsf{X}_Y$. We have seen that $R\mathbf{1}_Z = 0$, that is, $P\mathbf{1}_Z = \mathbf{1}_Z$. Applying $Q'\Delta'_a$ to $\mathsf{X}N^{-1}\mathsf{X}^*\mathbf{1}_Z = \mathbf{1}_Z$ and using $Q'Q'^* = 1$ we obtain

$$N^{-1}\mathsf{X}^*\mathbf{1}_Z = Q'\Delta'_a\mathbf{1}_Z \,, \qquad \mathbf{1}_Z\mathsf{X}N^{-1}\mathsf{X}^*\mathbf{1}_Z = \mathbf{1}_Z \,, \tag{25.6}$$

so that $\mathsf{V} \coloneqq Q'\Delta'_a\mathbf{1}_Z\Delta'_aQ'^* = N^{-1}\mathsf{X}^*\mathbf{1}_Z\mathsf{X}N^{-1}$. Let $\mathsf{J} \coloneqq \mathbf{1}_Y + \mathbf{1}_Z\Delta'_a\mathsf{X}_Y$ on the functions on $\mathfrak{B}$, where the second term maps a function on $\mathfrak{B}$ to a function on $Z \subset \mathfrak{B}$. For a function ω on $\mathfrak{B} \setminus Z$ the function $f \coloneqq \mathsf{X}_Y\omega$ vanishes on Z and satisfies $\mathbf{1}_Y\Delta'_af = Q'^*\omega$, hence $\Delta'_af = Q'^*\omega + \mathbf{1}_Z\Delta'_af = Q'^*\mathsf{J}\omega$ and $f = \mathsf{X}\mathsf{J}\omega$. Therefore $\mathsf{X}_Y = \mathsf{X}\mathsf{J}\mathbf{1}_Y$, $N_Y = \mathbf{1}_Y\mathsf{J}^*N\mathsf{J}\mathbf{1}_Y$, $\mathbf{1}_Z\mathsf{X}\mathsf{J}\mathbf{1}_Y = 0$, and $\mathbf{1}_Y\mathsf{J}\mathbf{1}_Y = \mathbf{1}_Y$. The last property shows that J is injective on the functions on $\mathfrak{B} \setminus Z$, and the last but one that it maps them into the null space of $\mathbf{1}_Z\mathsf{X}$; by the second identity of (25.6) the operator $\mathbf{1}_Z\mathsf{X}$ maps onto the functions on Z, so that its null space has the dimension of the space of the functions on $\mathfrak{B} \setminus Z$ and is the range of $\mathsf{J}\mathbf{1}_Y$. The orthogonal complement of this range is therefore the range of $\mathsf{X}^*\mathbf{1}_Z$, on which $N^{-1} - \mathsf{V}$ vanishes by (25.6). Since $1 - \mathbf{1}_Y\mathsf{J}^*$ maps into the null space of $\mathbf{1}_Y\mathsf{J}^*$, which is this complement, $(N^{-1} - \mathsf{V})\mathbf{1}_Y\mathsf{J}^* = N^{-1} - \mathsf{V}$, and

$$\mathbf{1}_Y(N^{-1} - \mathsf{V})\mathbf{1}_Y\,N_Y = \mathbf{1}_Y(N^{-1} - \mathsf{V})N\mathsf{J}\mathbf{1}_Y = \mathbf{1}_Y\mathsf{J}\mathbf{1}_Y - \mathbf{1}_YN^{-1}\mathsf{X}^*\,\mathbf{1}_Z\mathsf{X}\mathsf{J}\mathbf{1}_Y = \mathbf{1}_Y \,. \tag{25.7}$$

A square matrix with a left inverse is invertible, which gives the formula for N_Y^{-1}. By the argument of Section 24 in the space of the functions on Y, $\mathsf{X}_YN_Y^{-1}\mathsf{X}_Y^*$ is the orthogonal projection onto the orthogonal complement of $\mathbf{1}_Y\Delta'_aN(Q') = \Delta N(Q')$ in this space, which is $\mathbf{1}_YP\mathbf{1}_Y$.

(iii) By Proposition 24.3 the operator Δ_a is positive definite, and (25.4) with $\chi = \mathbf{1}_\mathfrak{y}$ is (25.3). In the form (24.1), $\|\partial A\|^2 \le 24\eta^{-2}\|A\|^2$, since the curl at a plaquette is η^{-1} times a

signed sum of four values and a bond lies on six plaquettes; $\|R\partial^* A\|^2 \le \|\partial^* A\|^2 \le 16\eta^{-2}\|A\|^2$, since the divergence at a point is η^{-1} times a signed sum of eight values and a bond has two ends; and the last term is at most $a\eta^{-2}\|QA\|^2 \le 16a\eta^{-2}\|A\|^2$. For the last inequality, the average of the scale $\ell = N\eta$ at a bond c of the direction μ is N^{-5} times the sum of the values of A at the at most $\frac{7}{5}N^5$ bonds of the staircases from the points of the block of c_-, so that its square is at most $\frac{7}{5}N^{-5}$ times the sum of the squares of these values, and a bond of T_η lies on at most $\frac{7}{5}N$ staircases of each of the four directions (proof of Lemma 6.4(ii)); hence the sum of $\ell^4|(QA)(c)|^2$ over the bonds of one scale is at most $\frac{196}{25}\|A\|^2$, and a bond of T_η lies on staircases of bonds of at most two scales, the regions of two scales which are not neighbours being separated by more than the length of a staircase. Hence $\Delta_a \le c_\Delta \eta^{-2}$ and $\mathsf{D}_\mathfrak{z} \ge c_\Delta^{-1}\eta^2$. The matrix of $\mathsf{D}_\mathfrak{z}$ is bounded by $c\,\eta^2 e^{-\delta_3 d(b_-, b'_-)}$, by the first bound of [3, (2.136)] for two blocks of the scale 0, and Lemma 14.1 applies as in (i), with the distance $d(b_-, b'_-) + 1$ for two different bonds.

(iv) Let B be a function on $\mathfrak{b}(\mathfrak{B})$ which vanishes on $\mathfrak{z}$. A field A with $QA = B$ vanishes on $\mathfrak{z}$, because $(QA)(b) = A(b)$ there; conversely the fields which vanish on $\mathfrak{z}$ and whose averages on $\mathfrak{b}(\mathfrak{B}) \setminus \mathfrak{z}$ are those of B are the fields with $QA = B$. On these fields $\mathfrak{g}(A)$ differs from $\|\partial A\|^2 + \|R\partial^* A\|^2$ by a constant. Since QGQ^* is invertible (Proposition 24.5(ii)), Q^* is injective; it commutes with $\mathbf{1}_\mathfrak{y}$, and $G_\mathfrak{y}$ is positive definite on the fields which vanish on $\mathfrak{z}$, so that $\mathsf{Q} \coloneqq \mathbf{1}_\mathfrak{y} Q G_\mathfrak{y} Q^* \mathbf{1}_\mathfrak{y}$ is positive definite on the functions which vanish on $\mathfrak{z}$. The field $A_0 \coloneqq G_\mathfrak{y} Q^* \mathsf{Q}^{-1} B$ vanishes on $\mathfrak{z}$, has the averages B, and satisfies $\mathfrak{g}(A_0, A') = \langle Q^* \mathsf{Q}^{-1} B, A'\rangle = 0$ for every field A' which vanishes on $\mathfrak{z}$ with $QA' = 0$, where $\mathfrak{g}(\cdot,\cdot)$ is the bilinear form of $\mathfrak{g}$; it is therefore the minimiser of $\mathfrak{g}$ among the fields with $QA = B$, and $\mathfrak{g}(A_0) = \langle B, \mathsf{Q}^{-1} B\rangle$. The same computation on the whole torus shows that this minimiser is $GQ^*(QGQ^*)^{-1}B = HB$ and that the minimum is $\langle B, (QGQ^*)^{-1}B\rangle$. Hence $A_0 = HB$, the two quadratic forms agree on the functions which vanish on $\mathfrak{z}$, and so do the operators. The last statement is Proposition 24.5(iii).

(v) For N_Y^{-1}: the kernel of V vanishes unless both points are neighbours of one point of Z, hence points of the scale 0 at a distance d of at most 2, and it is then at most $8\eta^{-8}$, because the kernel of Δ'_a between two neighbours is $-\eta^{-6}$ with respect to the weights η^4 and a point has eight neighbours; this is of the form [3, (2.87)] for two points of the scale 0, and N^{-1} satisfies this bound by Proposition 20.6. For the operators of (iv) the bounds are those of Proposition 24.5, restricted. For G'_Y let λ be a bounded function supported in the block of $y' \in \mathfrak{B}$. By [3, (2.67)], $|(G'\lambda)(z')| \le c\,\eta^2 e^{-\frac{1}{2}\bar\delta_0 d(z',y')} \sup|\lambda|$ for $z' \in Z$; by (i) and [3, (2.61)] the function $\phi \coloneqq (\mathsf{D}'_Z)^{-1} G'\lambda$, which is supported in Z, satisfies $|\phi(z)| \le c\, e^{-\theta d(z,y')} \sup|\lambda|$ with $\theta \coloneqq \frac{1}{2}\theta_Z$; and $G'\phi$ is the sum over $z \in Z$ of $\phi(z)$ times G' applied to the function which is 1 at z and zero elsewhere, to which the bounds [3, (2.67)] apply, with or without a derivative or a Hölder quotient on the left. Summing over z with [3, (2.61)] we find at a point of the block of y a bound of the form [3, (2.67)] with the factor $e^{-\theta' d_Z(y,y')}$, $\theta' \coloneqq \frac{1}{2}\theta$, in the place of $e^{-\frac{1}{2}\bar\delta_0 d(y,y')}$. For the bounds with a derivative of the source the function $G'\nabla^*\lambda$ is at most $c\,\eta\, e^{-\frac{1}{2}\bar\delta_0 d(z',y')} \sup|\lambda|$ on Z, so that ϕ carries a factor η^{-1} and the result a factor $\ell_j^2 \eta^{-1} = L^j \ell_j$ at a point of the scale j, of which L^j is paid by a part of the decay between the scales 0 and j, by (20.22). This gives the statements on G'_Y. Those on $G_\mathfrak{y}$ follow in the same way from (iii) and [3, (2.136)–(2.139)]; in [3, (2.138)] and in the Hölder bound for $\nabla G \nabla^* J$ the term with $(\mathsf{D}_\mathfrak{z})^{-1}$ needs the supremum of J only. For the bounds [3, (2.140)] in L^2 we use in addition that $(\mathsf{D}_\mathfrak{z})^{-1}$ is bounded by $c_\Delta \eta^{-2}$ on the square-summable fields and Schur's test for the sums over the blocks. *q.e.d.*

The last statement of (v) is the form which the comparison of the operators of two domains

takes here [1, (1.11), (1.12)], [5, Theorem 3.14]: the change of the operator caused by the condition on Z decays in the distance from the two arguments to Z. The formula of [3, p. 228] for the minimiser in terms of $G(\Omega)$ reduces to (iv) when B vanishes outside Ω_1, the case which he uses.

26 The Averaging Operations for Fields with Values in the Group

The paper [4] defines the average of a configuration with values in G over the blocks and proves its regularity and analyticity. Its results are of three kinds. Section A concerns functions on the Lie algebra, in which no lattice occurs. The Propositions 1, 3, and 9, and the identities of Section C, concern one step: a configuration on a lattice, the blocks of this lattice of the side L, a contour $\Gamma_{y,x}$ from the corner y of each block to each of its points x, and for each bond c of the lattice of the blocks a path $[x, x(c)]$ from each point x of the block of c_- to its translate $x(c)$ in the block of c_+ [4, (14), (15), p. 19]. The remaining propositions are inductions over the steps, in which the one-step statements are applied to the lattices $\Omega^{(j)}$ one after the other [4, (43), (53), (127)]. In the tilted run all the lattices but the finest are those of the paper, so that every step but the first is a step of [4] as printed. The first step averages a configuration on the turned lattice T_η over the blocks $B_{\ell_1}(y)$, and for it the one-step statements have to be proved. We do this for the first two propositions in detail and then go through the rest of the paper.

The first step. As in Section 22 we use the paths $\pi_y(x)$ as the contours $\Gamma_{y,x}$ of a block, and the staircases as the paths between two blocks. For a bond $c = \langle y, y + \ell_1 e_\mu \rangle$ of $\ell_1\mathbb{Z}^4$ and $x \in B_{\ell_1}(y)$ let $\Gamma_{c,x}$ be the path from c_- to c_+ which consists of $\pi_{c_-}(x)$, of the staircase $\sigma_\mu^{\ell_1}(x)$, and of the reverse of $\pi_{c_+}(x + \ell_1 e_\mu)$, and let $U(c) \coloneqq U(\sigma_\mu^{\ell_1}(c_-))$, which is $U(\Gamma_{c,c_-})$. For a configuration U on the bonds of T_η with values in G we define, as in [4, (15)],

$$\bar{U}_c \coloneqq \exp\Big[i \sum_{x \in B_{\ell_1}(c_-)} L^{-4} \frac{1}{i} \log(U(\Gamma_{c,x}) U(c)^{-1}) \Big] U(c) \; . \tag{26.1}$$

Under a gauge transformation u every factor $U(\Gamma_{c,x})U(c)^{-1}$ is conjugated by $u(c_-)$, so that $\overline{U^u} = (\bar{U})^u$, by the argument of [4, p. 24]; this is the condition [4, (11)]. For $U = e^{iA}$ with A small the linear part of $\frac{1}{i} \log \bar{U}_c$ is $L^{-4} \sum_x A(\Gamma_{c,x})$, which for a field in the axial gauge of Section 22 is L times the average $(Q_{\ell_1} A)(c)$ of Definition 6.1; the factor L is the ratio of the spacings of the two lattices, as between [4, (14)] and [4, (15)]. The average $\bar{U}_c$ depends on U on the bonds of the paths $\Gamma_{c,x}$, which lie at a Euclidean distance of less than $4\ell_1$ from c_-; it is local in this sense, and not in the sense of [4, p. 24], where the bonds lie in the two blocks of c. The averages of the later steps are those of [4, (43)], applied to $\bar{U}$.

Bałaban's estimates start from a local axial gauge, in which the deviation of a bond from the identity is bounded by the deviations of the plaquettes of a strip [4, pp. 24–25]. On the turned lattice the strip is that of Lemma 22.1.

Lemma 26.1 (Strips and Closed Paths for Unitary Configurations)**.** *Let V be a configuration on the bonds of T_η with values in the unitary group, let $o \in T_\eta$, and let $\alpha \geq 0$ and $r \geq 0$ be such that $|V(\partial p) - 1| \leq \alpha$ for every plaquette p all of whose corners s satisfy $|s'_\nu - o'_\nu| \leq r$ for every ν.*

(i) For every bond $b = \langle x, x + \eta e'_j \rangle$ both of whose ends satisfy this condition, $|V(\pi_o(x))V(b)V(\pi_o(x + \eta e'_j))^{-1} - 1| \leq \alpha \sum_{i<j} |x'_i - o'_i|$.

(ii) *For every closed path γ of n bonds all of whose sites satisfy this condition, $|V(\gamma)-1| \le 3rn\alpha$; more precisely, it is at most α times the sum over the steps of γ of the numbers of the plaquettes of their strips.*

(iii) *If $V_0 := V^{v_0}$ with $v_0(x) := V(\pi_o(x))$, then $V_0(\pi_o(x)) = 1$ for every x, and $|V_0(b)-1| \le 3r\alpha$ for the bonds of (i).*

Proof. We use that $|W_1 \cdots W_m - 1| \le \sum_i |W_i - 1|$ for unitary matrices and that the conjugation by a unitary matrix preserves the norm. (i) By the proof of Lemma 22.1, the closed path which consists of $\pi_o(x)$, of b, and of the reverse of $\pi_o(x + \eta e'_j)$ is, after the cancellation of the common part of the two paths, the boundary of a strip of $m = \sum_{i<j} |x'_i - o'_i|$ plaquettes $p_1, \dots, p_m$, of which consecutive ones share a bond. The product of V over the boundary of the strip, from the point where the two paths separate, is the ordered product of the matrices $g_i V(\partial p_i)^{\pm 1} g_i^{-1}$, where g_i is the product of V over the part of the first side of the strip up to p_i; this follows by induction on m, the boundary of the strip of m plaquettes being the boundary of the strip of $m-1$ plaquettes followed by the boundary of p_m with the path to it and back. The common part conjugates the product once more. (ii) With the sites $s_0, \dots, s_n = s_0$ of γ, the matrix $V(\pi_o(s_0))V(\gamma)V(\pi_o(s_0))^{-1}$ is the ordered product over m of the matrices $V(\pi_o(s_{m-1}))V(\langle s_{m-1}, s_m\rangle)V(\pi_o(s_m))^{-1}$, each of which is bounded by (i), applied to the step or to its reverse, with at most $3r$ plaquettes. (iii) By the definition of the gauge transformation, $V_0(\gamma) = v_0(\gamma_-)V(\gamma)v_0(\gamma_+)^{-1}$ for a path from γ_- to γ_+; for $\gamma = \pi_o(x)$ this is 1, since $v_0(o) = 1$, and for $\gamma = b$ it is the matrix of (i), in which the sum is at most $3r$. *q.e.d.*

Proposition 26.2 (Propositions 1 and 2 for the Tilted Run)**.** *Let $L \ge 64$ be a multiple of 5. There are constants C'_0 and $c''_2 > 0$, which depend on L, with the following properties.*

(i) *Let V be a configuration on T_η with values in G, let p' be a plaquette of $\ell_1\mathbb{Z}^4$ with the corner y, and let $|V(\partial p) - 1| < \alpha_0 \le c''_2$ for every plaquette p of T_η whose corners are at a Euclidean distance of at most $8\ell_1$ from y. Then $|\bar V(\partial p') - 1| < 4L^2\alpha_0 + C'_0(L^2\alpha_0)^2$.*

(ii) *Let $k \ge 1$, $\eta = L^{-k}$, and let U be a configuration on T_η with $|U(\partial p) - 1| < \alpha_0\eta^2$ for every plaquette, where $\alpha_0 \le \min(c''_2, 1/C'_0, \frac{1}{5}c_2)$ with the constant c_2 of [4, Proposition 2]. Then the average $\bar U^k$ of the tilted run satisfies $|\bar U^k(\partial p) - 1| < 10\alpha_0$ for every plaquette p of the unit lattice. The statement is local: for a plaquette p it suffices to assume the bound on the plaquettes of T_η at a distance of at most $8\ell_1$ from the blocks of the scale 1 of the four corners of p.*

Proof. (i) We follow [4, (44)–(50), pp. 24–25], in the unit η. Let y_0, y_1, y, y_2 be the corners of p'. The paths $\Gamma_{c,x}$ of the four bonds c of $\partial p'$ consist of paths π of the four blocks of these corners and of staircases between them; their sites have coordinates in the basis $\eta e'_\nu$ which differ from those of y by at most $r := 4L$, they are at a Euclidean distance of less than $8\ell_1$ from y, and each path has at most $10L$ bonds, by the bounds on the coordinates in the proof of Lemma 22.2. Let V_0 be the configuration of Lemma 26.1(iii) for $o = y$; since the average is covariant, $|\bar V(\partial p') - 1| = |\bar V_0(\partial p') - 1|$. By that lemma $|V_0(b) - 1| \le 12L\alpha_0$ on the bonds concerned, in the place of the bound $dL\alpha_0$ of [4, p. 25], so that $V_0(b) = e^{iA_b}$ with $|A_b| \le 24L\alpha_0$, and the three estimates which lead to [4, (47)] hold with other constants:

$$\Big|\bar V_0(\partial p') - 1 - i \sum_{c \subset \partial p'} \sum_{x \in B_{\ell_1}(c_-)} L^{-4} A(\Gamma_{c,x})\Big| \le C(L^2\alpha_0)^2 \,, \tag{26.2}$$

where a bond c which $\partial p'$ runs through against its orientation contributes with the sign $-$, and C depends on nothing once $L^2\alpha_0$ is small. In the double sum the terms $A(\pi_{c_\pm}(\cdot))$ cancel:

the sum of $A(\pi_{y_1}(x+\ell_1 e_\mu))$ over $x \in B_{\ell_1}(y_0)$ from the first bond is the sum of $A(\pi_{y_1}(x))$ over $x \in B_{\ell_1}(y_1)$ from the second, and likewise at the other three corners. There remains, as in [4, (48)],

$$\sum_{c \subset \partial p'} \sum_x L^{-4} A(\Gamma_{c,x}) = \sum_{x \in B_{\ell_1}(y_0)} L^{-4} A(\partial(p')_x) \ , \tag{26.3}$$

where $\partial(p')_x$ is the closed path of the staircases $\sigma_\mu(x)$, $\sigma_\nu(x+\ell_1 e_\mu)$, and the reverses of $\sigma_\mu(x+\ell_1 e_\nu)$ and $\sigma_\nu(x)$, for the directions $\mu < \nu$ of p'. By Lemma 22.3, $A(\partial(p')_x) = \sum_p n_x(p) A(\partial p)$ with integers $n_x(p)$, and we claim that $\sum_p |n_x(p)| \le 4L^2$. If $\mu, \nu \in \{0,1\}$, the path is the boundary of a square of L^2 plaquettes. If $\mu \in \{0,1\}$ and $\nu \in \{2,3\}$, it is the boundary of the surface which the staircase $\sigma_\nu(x)$ sweeps under the translations by $t\eta e_\mu$, $0 \le t \le L$, which consists of $\frac{7}{5}L \cdot L$ plaquettes. If $\mu = 2$ and $\nu = 3$, the path lies in a plane, and in Lemma 22.1 with $o = x$ only its bonds in the direction e_3' have plaquettes, at most $|s_2' - x_2'| \le \frac{7}{5}L + 1$ each; there are $2(\frac{4}{5}L + \frac{3}{5}L)$ such bonds, and $\frac{14}{5}L(\frac{7}{5}L+1) \le 4L^2$. Since $|A(\partial p)| \le |V_0(\partial p) - 1| + \frac{1}{2}(4 \cdot 24 L\alpha_0)^2$, as in [4, (49)], the modulus of (26.3) is at most $4L^2\alpha_0 + C(L^2\alpha_0)^2$, and with (26.2) this gives (i).

(ii) By (i), applied to U with $\alpha_0\eta^2$ in the place of α_0, $|\bar U(\partial p') - 1| < 4\alpha_0(L\eta)^2 + C_0'\alpha_0^2(L\eta)^4 \le 5\alpha_0(L\eta)^2$, because $C_0'\alpha_0 \le 1$ and $L\eta \le 1$. For $k = 1$ this is the claim. For $k \ge 2$ the configuration $\bar U$ lives on the lattice $\ell_1\mathbb{Z}^4$ of the paper, whose spacing is $L\eta = L^{-(k-1)}$, the remaining $k-1$ averages are those of [4], and $\bar U$ satisfies the hypothesis [4, (52)] of his Proposition 2 for $k-1$ steps with $5\alpha_0 \le c_2$ in the place of α_0. That proposition gives $|\bar U^k(\partial p) - 1| < 2 \cdot 5\alpha_0$, and its locality together with that of the first step gives the last statement. *q.e.d.*

The factor 4 in (i) is an upper bound for the number of the plaquettes of T_η, counted with multiplicities, which fill a plaquette of the coarse lattice bounded by staircases, divided by L^2; for the lattice of the paper it is 1, and this is the only place in which the constant of the leading term of [4, (51)] is used. Since the first step occurs once, the loss is a factor in the constants and not in the scaling.

The rest of the paper. We go through [4] in its order.

Section A, (16)–(41). These are properties of the exponential map and of the function $\log(e^X e^Y)$ on matrices; no lattice occurs.

Section C, (55)–(108). The averages $\tilde U'$ and $\bar{\bar U}_1$, the gauge transformation [4, (62)] which passes from the axial gauge conditions [4, (58)] to the condition [4, (61)], the average [4, (78)] of a gauge transformation, and the identities [4, (63), (88), (92), (97), (99), (104)–(108)] are consequences of the definitions and of the covariance of the average. They use of the geometry that $\Gamma_{y,x}$ is a path from y to x for x in the block of y, that $\Gamma_{c,x}$ is the path from c_- to c_+ through $\Gamma_{c_-,x}$, a path from x to $x(c)$, and the reverse of $\Gamma_{c_+,x(c)}$, and that $x \mapsto x(c)$ maps the block of c_- onto that of c_+. These hold for the first step of the tilted run with $x(c) = x + \ell_1 e_\mu$, and the identities hold for it with the same proofs. The axial gauge condition [4, (58)] for the first step is that of Section 22, with the transports of V_0 along the paths π_y.

Section D, Propositions 3 to 5. Proposition 3 is the one-step statement for $\bar{\bar V}_1$. Its proof [4, (109)–(126)] uses the matrices $Y_x = \frac{1}{i}\log V_0(\Gamma_{c,x} \cup (-c))$, which for the first step of the tilted run are the logarithms of the products of V_0 over closed paths of at most $12L$ bonds and are at most $2 \cdot 3 \cdot 4L \cdot 12L\alpha_0$ in modulus by Lemma 26.1(ii), in the place of his $O(L^2\alpha_0)$; the lengths of the contours, at most $10L$ in the place of $(2d+1)L$; and the functions of Section A. The linear form [4, (124)] has the main term [4, (125)], the mean over x of the sums of the transported field over the paths from x to $x(c)$, divided by L, which is at most $\frac{7}{5}|A|$ in

modulus for a staircase of $\frac{7}{5}L$ bonds, in the place of $|A|$; at $V_0 = 1$ it is the average Q_{ℓ_1} of Definition 6.1. Hence Proposition 3 holds for the first step with the bound $(\frac{7}{5} + O(L^2\alpha_0))\alpha_1$ in [4, (126)] and with other constants C_1, c_3, which depend on L. In the induction [4, (127)–(133)] the first function of the sequence [4, (127)] is that of the first step and the others are Bałaban's, taken at the averages $\bar{U}_0^j$, which satisfy his hypotheses by Proposition 26.2(ii) with $10\alpha_0$ in the place of $2\alpha_0$. The bound [4, (128)] acquires the factor $\frac{7}{5}$, the hypothesis [4, (131)] becomes a bound by $3\alpha_1$ in the place of $2\alpha_1$, and Proposition 4 follows with other constants C_2, c_4, which do not depend on k. For Proposition 5 the operators Q and Q'' of [4, (139), (140)] are, at the first step, the mean over the staircases and the mean over the bonds of all the paths $\Gamma_{c,x}$ of a bond c; the inequalities [4, (139), (142)] and $QQ''_j \le 2Q''_{j+1}$, $Q''Q_j \le Q''_{j+1}$ hold with constants which depend on L if Q''_k in [4, (141)] is the mean over the bonds on which $\bar{U}_c^k$ depends, and the induction [4, (143)–(155)] gives [4, (156), (157)] with other constants.

Section E, Propositions 6 and 7. These follow from Proposition 4 and from the representation [4, (159), (160)] as printed, with the bound $3\alpha_1 L^j \eta$ in [4, (161)].

Section F, Propositions 8 to 10. The classes $\Lambda_k(U_0, \alpha_3)$ of gauge transformations [4, (166), (167)] are defined through the averages of Section C. Proposition 8 rests on the one-step identity [4, (170)], whose proof [4, (168), (169)] uses the definition [4, (78)] and the functions of Section A only. Proposition 9 is the one-step statement for $\tilde{v}'$; in its proof [4, (181)–(198)] the contours enter through their lengths and through the matrices $V_0(\Gamma_{c,x} \cup (-c))$, as in Proposition 3, and the main term of [4, (196)] is at most $\frac{7}{5}L\alpha'_4$ in the place of $L\alpha'_4$. The induction of Proposition 10 then holds with $\frac{7}{5}\alpha_4$ in the place of α_4 in [4, (203)] from the first step on, and with other constants. The expansion [4, (212)–(214)] has the linear term $\sum_{x \in B^j(y)} L^{-4j} R(U_0(\Gamma^{(j)}_{y,x}))\lambda(x)$, which at $U_0 = 1$ is the average Q' of Definition 6.1.

In sum, the averaging operations of the tilted run have the properties of [4], with constants which depend on L through the first step, with the factors 4 and $\frac{7}{5}$ named above, and with the locality of the first step relaxed from the two blocks of a bond to their neighbourhood of the radius $4\ell_1$. The statements for which only the changes are named, Propositions 3 to 10, have not been written out line by line.

27 The Propagators in a Background Field

The paper [5] extends the results of [3] to the operators which are obtained when the action is expanded around a regular configuration U with values in the group. Its objects are of four kinds, and the turned lattice enters each in a different way. The covariant derivatives and the operator $\Delta(U)$ are defined on the bonds and plaquettes of the fine lattice alone. The averages $Q_j(U)$ and $Q'_j(U)$ are the linear parts of the averaging operations of [4], which we have treated in Section 26. The propagators at $U = 1$, with Dirichlet conditions, are those of Proposition 25.1. Everything else is derived from these by expansions: in the field, around $U = 1$ in a cube in which U is a pure gauge up to a small field, and over walks of cubes. We first complete the statement at $U = 1$, which in [5] contains more bounds on the scalar operator than [3, Proposition 2.2], and then go through the paper.

Lemma 27.1 (Further Bounds on the Scalar Operator of a Sequence)**.** *Under the hypotheses of Proposition 25.1 there are constants c and a rate $\delta'' > 0$, which depend on a, L, and the Hölder exponents, such that the operators G' and G'_Y satisfy, in the place of $G'(U)$ at $U = 1$, the bounds [5, (3.44)–(3.47)] with δ'' in the place of δ_0. In particular, for $y, y' \in \mathfrak{B}$ of the scales j, j' and every square-summable λ supported in the block of y', the norms over the*

block of y satisfy

$$\|G'\lambda\|_{\Delta(y)} \le c\,\ell_j^2 e^{-\delta'' d(y,y')}\|\lambda\|\,, \qquad \|\nabla G'\lambda\|_{\Delta(y)}\,,\ \|G'\nabla^*\lambda\|_{\Delta(y)} \le c\,\ell_j e^{-\delta'' d(y,y')}\|\lambda\|\,, \quad (27.1)$$

and $\|\nabla\nabla G'\lambda\|_{\Delta(y)}$, $\|\nabla G'\nabla^*\lambda\|_{\Delta(y)}$, and $\|G'\nabla^*\nabla^*\lambda\|_{\Delta(y)}$ are at most $c\,e^{-\delta'' d(y,y')}\|\lambda\|$.

Proof. It suffices to treat G': by (25.2) the operator G'_Y differs from it by $G'(\mathsf{D}'_Z)^{-1}G'$, in which the two factors G' act from and to functions on Z, whose points are blocks of the scale 0, and $(\mathsf{D}'_Z)^{-1}$ has the matrix bound of Proposition 25.1(i), so that every bound below passes to G'_Y by the argument of Proposition 25.1(v), with Schur's test for the bounds in L^2. We write c and δ for constants with the stated dependence.

The first-order bounds in L^2. Let $K(x,x')$ be the kernel of G' with respect to the weights η^4. The first bound of [3, (2.67)], applied to the functions of modulus one on the block of y', gives $\sum_{x'\in\Delta(y')}\eta^4|K(x,x')| \le c\,\ell_j^2 e^{-\frac12\bar\delta_0 d(y,y')}$ for $x\in\Delta(y)$, and since K is symmetric, $\sum_{x\in\Delta(y)}\eta^4|K(x,x')| \le c\,\ell_{j'}^2 e^{-\frac12\bar\delta_0 d(y,y')}$ for $x'\in\Delta(y')$. By Schur's test the norm of $\mathbf{1}_{\Delta(y)}G'\mathbf{1}_{\Delta(y')}$ is at most $c\,\ell_j\ell_{j'}e^{-\frac12\bar\delta_0 d(y,y')}$, and (20.22) replaces $\ell_{j'}$ by ℓ_j at the cost of a part of the rate. For $\nabla G'$ the sums of the difference quotients of K in the first argument over $x'\in\Delta(y')$ are bounded by the second bound of [3, (2.67)], and their sums over $x\in\Delta(y)$ by the third, applied with the roles of the two blocks exchanged, because $(G'\nabla^*\lambda)(x') = \sum_x \eta^4(\nabla_x K)(x,x')\lambda(x)$ for a tuple of functions λ; Schur's test gives the bound for $\nabla G'$, and that for $G'\nabla^*$ is the bound for the adjoint. This is (27.1). If an operator T satisfies $\|\mathbf{1}_{\Delta(y)}\mathsf{T}\mathbf{1}_{\Delta(y')}\| \le m(y,y')$ for all $y,y'\in\mathfrak{B}$, then $\|\mathsf{T}\|^2 \le \sup_y\sum_{y'} m(y,y')\cdot\sup_{y'}\sum_y m(y,y')$, by Schur's test for the matrix m, the blocks being disjoint; we refer to this as Schur's test for the blocks.

The second-order bounds in L^2, without decay. For a function ϕ on the torus, $\sum_{\nu,\nu'}\|\partial'_\nu\partial'_{\nu'}\phi\|^2 = \|\Delta\phi\|^2$, because in the Fourier representation of the group $\mathbb{T}$ the operator ∂'_ν is the multiplication by a function s_ν and $-\Delta$ that by $\sum_\nu|s_\nu|^2$. With $\phi = G'\lambda$ we have $-\Delta\phi = \lambda - Q'^* a Q'\phi$. The operator $Q'^* a Q'$ is $a_j\ell_j^{-2}$ times the orthogonal projection onto the functions which are constant on the blocks in the region of the scale j, summed over j, so that $\|Q'^* a Q'\phi\|_{\Delta(y)} \le a\ell_j^{-2}\|\phi\|_{\Delta(y)}$, and by (27.1), [3, (2.61)], and Schur's test for the blocks, $\|Q'^* a Q' G'\lambda\| \le c\|\lambda\|$. Hence $\|\nabla\nabla G'\lambda\| \le c\|\lambda\|$ for every λ, and by taking the adjoint $\|G'\nabla^*\nabla^*\lambda\| \le c\|\lambda\|$. Since $\Delta'_a \ge -\Delta = \nabla^*\nabla$, the supremum over ϕ of $2\langle\nabla^*\lambda,\phi\rangle - \langle\phi,\Delta'_a\phi\rangle$, which is $\langle\nabla^*\lambda, G'\nabla^*\lambda\rangle$, is at most that of $2\langle\lambda,\nabla\phi\rangle - \|\nabla\phi\|^2 \le \|\lambda\|^2$ for a tuple λ, and $\|\nabla G'\nabla^*\lambda\|^2 = \langle G'\nabla^*\lambda, -\Delta G'\nabla^*\lambda\rangle \le \langle G'\nabla^*\lambda, \nabla^*\lambda\rangle \le \|\lambda\|^2$.

The decay of the second-order bounds. Let y have the scale j, and let χ be a function of the coordinates of the paper which equals 1 at the points at a distance of at most $2\ell_j$ from $\Delta(y)$ in $|\cdot|_\infty$, vanishes at those at a distance of more than $4\ell_j$, and has differences and second differences along the bonds of T_η bounded by $c\,\eta/\ell_j$ and $c\,\eta^2/\ell_j^2$. The points of its support and their neighbours lie in the union E of the blocks of at most $c\,L^4$ points y_1 of $\mathfrak{B}$ with $d(y,y_1)\le c\,L$, of the scales between $j-1$ and $j+1$. If the block of y' and its neighbours do not meet E, then for λ supported in $\Delta(y')$ and $\phi = G'\lambda$ we have $\chi\Delta\phi = \chi Q'^* a Q'\phi$, and $\nabla\nabla(\chi\phi)$ equals $\nabla\nabla\phi$ on $\Delta(y)$, so that, by the identity above and (17.4),

$$\|\nabla\nabla\phi\|_{\Delta(y)} \le \|\Delta(\chi\phi)\| \le c\,\ell_j^{-2}\|\phi\|_E + c\,\ell_j^{-1}\|\nabla\phi\|_E \le c\,e^{-\delta d(y,y')}\|\lambda\| \qquad (27.2)$$

by (27.1) for the blocks of E and (20.22). For $\phi = G'\nabla^*\lambda$, which satisfies $\Delta'_a\phi = 0$ on the support of χ, we take the scalar product of this equation with $\chi^2\phi$: $\langle\nabla(\chi^2\phi),\nabla\phi\rangle = -\langle\chi^2\phi, Q'^* a Q'\phi\rangle$, hence $\|\chi\nabla\phi\|^2 \le c\,\ell_j^{-1}\|\phi\|_E\|\chi\nabla\phi\| + c\,\ell_j^{-2}\|\phi\|_E^2$, where we have used that the difference of χ^2 along a bond is at most $c(\eta/\ell_j)(\chi + c\,\eta/\ell_j)$ at either end and that a

difference quotient of ϕ is at most $2\eta^{-1}$ times its values, and $\|\nabla\phi\|_{\Delta(y)} \le c\,\ell_j^{-1}\|\phi\|_E \le c\,e^{-\delta d(y,y')}\|\lambda\|$ by (27.1). If the block of y' or one of its neighbours meets E, then $d(y,y') \le c\,L$, and the bounds without decay suffice. The bound for $G'\nabla^*\nabla^*$ between two blocks is that for the adjoint of $\nabla\nabla G'$. This gives the inequalities [5, (3.46)], and [5, (3.47)] follows from [3, (2.67)] and [3, (2.61)], as Bałaban remarks [5, p. 398].

The bounds [5, (3.44), (3.45)]. The local operators satisfy them: for a cube $\square$ of a scale $j \ge 1$, by (18.2), $\nabla A_\square \nabla^* f$ is the sum of $\nabla G_{(j)}\nabla^* f$, for which they are Proposition 14.4(ii) and (iii) in the unit ℓ_j, and of $\alpha_j^2 \nabla G_{(j)}\mathsf{q}^*\iota\, C_\Lambda\, \iota^*\mathsf{q}G_{(j)}\nabla^* f$, in which $G_{(j)}\nabla^* f$ is at most $c\,\ell_j \sup|f|$ by Proposition 14.4(i), the operator C_Λ contributes $c\,\ell_j^2$ by Lemma 18.1(iii), $\alpha_j^2 \le a^2\ell_j^{-4}$, and $\nabla G_{(j)}$ contributes $c\,\ell_j$ and is Hölder continuous by Proposition 14.3, each with its decay; for the scale 0 a difference quotient is a bounded operator in the unit η. For the sequence we use $G' = G'_0 + G'\mathsf{R}$ and $G' = \sum_n (\mathsf{R}^*)^n G'_0$ with the operators of the proof of Proposition 20.2, so that

$$\nabla G'\nabla^*\lambda = \nabla G'_0\nabla^*\lambda + \sum_{n\ge 0} \nabla(\mathsf{R}^*)^n G'_0\, \mathsf{R}\nabla^*\lambda\,. \tag{27.3}$$

The first term is a sum of terms $\nabla h_\square A_\square h_\square \nabla^*\lambda$, to which the local bounds apply, with the Hölder norm of λ. In the second, $\mathsf{R}\nabla^*\lambda = \sum_\square K(h_\square)A_\square h_\square\nabla^*\lambda$ is bounded by the supremum of λ and is Hölder continuous with every exponent smaller than one, by Lemma 18.1(iv) and (v), and each further factor $h_\square A_\square K(h_\square)^*$ of R^* maps bounded Hölder continuous functions to such functions, with the factor $c\,L/M$: the term of $K(h_\square)^*$ which is an adjoint derivative of a function is treated by Lemma 18.1(v), and the derivative on the left of the first factor meets $A_\square\nabla^*$ applied to a Hölder continuous function, for which we use the local bounds again. The composition is as in the proof of Proposition 20.2. *q.e.d.*

We now go through [5] in its order.

Section A, the definitions (3.1)–(3.34). The expansion [5, (3.1)–(3.12)] of the action around U_0, the covariant derivatives [5, (3.3), (3.4), (3.8), (3.9)], and the operators $\Delta(U)$, $\Delta'(U)$, and J [5, (3.10), (3.11)] involve the bonds and plaquettes of one lattice with four orthogonal directions, and they are unchanged on T_η with the directions e'_ν. The averages [5, (3.13)–(3.15)] are $Q_j(U) = Q(\bar U^{j-1})\cdots Q(\bar U)Q(U)$ with the one-step operator $Q(V)$ of [4, (124)]; in the tilted run the last factor $Q(U)$ is that of the first step (Section 26), which at $U = 1$ and in the normalisation of [5, (3.14)] is the average Q_{ℓ_1}, and the others are Bałaban's. The same holds for $Q'_j(U)$ [5, (3.19)], in which the contour $\Gamma_{x_1,x}$ of the first step is the path $\pi_{x_1}(x)$. The form [5, (3.16)], the gauge-fixing density [5, (3.17), (3.20)], the projection $R(U)$ with its formula [5, (3.21)–(3.25)], and the operators $\Delta_a(U)$ and $G(U)$ [5, (3.26), (3.27)] are defined from these as printed; that the Dirichlet conditions do not change R is the remark after (25.1), which holds for every U. The transformation laws [5, (3.28)–(3.34)] follow from the invariance of the action and from the covariance of the averages, which holds for the first step of the tilted run (Section 26).

Section A, the conditions and the theorems (3.35)–(3.49). The cubes of the regularity conditions [5, (3.35), (3.36)] are unions of big blocks, hence cubes of the paper, and the conditions bound a field A with $U^u = e^{i\eta A}$ and its differences along the bonds of T_η; they are conditions on a configuration of the turned lattice which involve the coarse lattices through the cubes only, and so are [5, (3.37), (3.38)]. In the Hölder norms [5, (3.40)] the contour $\Gamma_{x,x'}$ is a path of T_η with the least number of bonds. Theorems 3.1, 3.2, and 3.3 are statements about the operators so defined. At $U = 1$ they are, for the tilted run, Proposition 25.1 with Propositions 20.2, 20.6, and 24.3 and Lemma 27.1; the bounds [5, (3.49)] on P at $U = 1$ are Proposition 20.6(iii).

Section B, (3.50)–(3.86). The expansions [5, (3.50)–(3.54), (3.69)–(3.75)] of $\Delta_{U'U}$, of $\Delta'(U'U)$, and of the two second-order operators in the field A are computations at a point and its neighbours on one lattice and are unchanged. The expansion [5, (3.55)–(3.59)] of $Q'_j(U'U)$ uses the identities [4, (97), (102), (103)] and the bounds [4, (161), (162)] and Proposition 4 of that paper, which hold for the tilted run with the changes named in Section 26; the bound [5, (3.58)] follows with another constant. The expansion [5, (3.78)–(3.81)] of $Q_j(U'U)$ uses [4, (124), (140), (141), (143)] and Proposition 6 there, with the operators Q'' of Section 26, and gives [5, (3.81)] with a constant which depends on L. The operator $P_2(A)$ of [5, (3.82), (3.83)] is semi-local in the sense that its value at a bond of the region of the scale j depends on the fields in the blocks of the scale j around the bond and, through the first step, in their neighbourhood of the radius $4\ell_1$. The remaining steps, [5, (3.60)–(3.68), (3.76), (3.77), (3.84)–(3.86)], are Neumann series whose terms are composed with Lemma 2.1 of [3], which is Lemma 19.2, and with the comparison of the powers of L which we have made explicit in (20.22). Hence Theorem 3.4 and Corollary 3.5 hold for the tilted run with the same proofs, given Theorems 3.1 to 3.3 for U, respectively at $U = 1$. In Corollary 3.6 the sequence $\{\Omega'_j\}$ lies in a cube $\square$ of the condition [5, (3.35)], and its operators are those of Proposition 25.1 for the sequence $\Omega'_1 \supset \cdots \supset \Omega'_k$ on a torus which contains $\square$, with $Y = \Omega'_0$; they do not depend on the torus, being inverses of operators on the functions and fields on Ω'_0.

Section C, (3.87)–(3.108). The cover, the sequences $\{\Omega_n(\square)\}$ of cubes around a cube $\square$ [5, p. 408], and the partition of unity are defined by the big blocks and are those of the paper; a partition of unity with the properties used is that of Section 20. The local operators $G'_\square(U)$, $C_\square(U)$, and $G_\square(U)$ are those of Corollary 3.6. The commutator [5, (3.88)] of Δ'_a with a function has an averaging term with the differences of the function over a block, weighted by the transports along the contours, and the bound [5, (3.89)] uses the size of the block only. The commutators [5, (3.100)–(3.104)] for the fields are those of Lemma 17.4 with the transports; in [5, (3.102)] the kernel of Q_j is supported, for the tilted run, in the neighbourhood of the radius $4\ell_1$ of the two blocks of the bond. The convergence of the expansions [5, (3.90), (3.96), (3.98), (3.106), (3.107)] and the bounds [5, (3.94), (3.99), (3.108)] on their terms follow from these and from Lemma 19.2 as in the text. Two inputs are quoted there from the earlier papers. The first is the smallness of the differences of the operators of two sequences which agree near a cube [5, (3.97)]; for the tilted run at $U = 1$ these are Lemma 20.5, Lemma 24.2, (24.12), and Proposition 25.1(v), all of which are proved from the resolvent identity and the bounds on the kernels of the two operators, and the passage to a regular U is that of Corollary 3.5. The second is the cancellation [5, p. 415] of the term with $P_\square$, which is algebra. Theorems 3.7, 3.9, and 3.10, and with them Theorems 3.1 to 3.3 for a configuration U which satisfies [5, (3.35)], therefore hold for the tilted run, with constants which depend on L also through Sections 20–26. The proof of Theorem 3.11 is an argument on a symmetric invertible operator and its expansion and is unchanged. We have not written out the expansions of this section for the tilted run line by line.

Section D, (3.109)–(3.154). The minimiser H of [5, (3.109), (3.110)] and the operators H_1, G_1, $\mathfrak{P}$, and $\mathfrak{G}$ are defined by variational problems and Gaussian integrals in which the lattice does not occur beyond the operators $\Delta(U)$, D, Q, Q', and R. The derivations [5, (3.112)–(3.130), (3.139)–(3.153)] use four properties of these operators: the identity [5, (3.114), (3.115)], $Q_j D\lambda = \bar{D}^j Q'_j \lambda$; the invariance [5, (3.117)] of the action; the definition of R; and $Q'\mathscr{G} = 0$. The first is the linear part of the transformation law of $\bar{\bar{U}}'$ under a gauge transformation, [5, p. 418], whose derivation uses the covariance of the average and the definition [5, (3.113)], in which the product of $R_y U'$ along the contour $\Gamma_{y,x}$ telescopes for every path from y to x; it holds for the first step of the tilted run with the paths $\pi_y(x)$ and the staircases, and at $U = 1$ it is Lemma 6.4(i). The others are as printed. The bounds [5,

(3.131)–(3.133), (3.137)] use Theorems 3.1 to 3.3 and [4, (149)], which holds for the tilted run with another constant (Section 26), and Theorems 3.12 and 3.13 follow as in the text. For the inverse of QGQ^* Bałaban refers to the analysis of $(Q'G'^2Q'^*)^{-1}$ [5, p. 422]; at $U = 1$ it is Proposition 24.5(ii) with Proposition 25.1(iv), and its lower bound on a cube rests on Lemma 24.4. Theorem 3.14 is proved from the expansions, as in the text; at $U = 1$ and for the change of the set Z it is Proposition 25.1(v).

Section E, (3.155)–(3.187). The propagator $C^{(k)}(\Lambda)$ is the covariance of the fluctuation field of the unit lattice after k steps, in the axial gauge of the blocks of the next step and with the averages of the next step equal to zero. For $k \geq 1$ the unit lattice, its blocks, the contours of the axial gauge, the bonds $B(c) \cap c$, and the average Q_1 are those of the paper, and the derivation [5, (3.155)–(3.186)] uses, besides them, the operators of the sequence with the additional domain $\Omega_{k+1} = B^k(\Lambda)$ and the identities of Section D; it is unchanged, and Theorem 3.15 holds with the operators of the tilted run. The lower bound of $C^*\Delta_k C$ at $U = 1$ is Bałaban's Lemma 2.4 on a lattice of the paper. For $k = 0$ the fluctuation field lives on the turned lattice itself. Then $H_1 = 1$, and the quadratic form [5, (3.156)] is that of $\Delta(U)$ and of a term with J. We take it on the space of the fields B which vanish on the bonds with no end in Λ, are axial on the blocks of Λ in the sense of Section 22, and whose averages $(Q_{\ell_1}B)(c)$ vanish for the bonds c of $\mathfrak{b}(\Lambda')$; how the conditions are to be read at the boundary of Λ is a matter of the first step of the paper, which Section 34 lists. At $U = 1$ the lower bound of the form on this space is the inequality (22.12), in which the first and the third term on the right vanish for these fields, so that $\|\partial B\|^2 \geq (C_1 L^7)^{-1}\eta^{-2}\|B\|^2$. The decay of the covariance follows from Lemma 22.5 as in Proposition 22.6(iii), with the following projection onto this space: the restriction to the bonds with an end in Λ, followed by the orthogonal projection, within the fields supported on these bonds, onto those whose averages vanish on $\mathfrak{b}(\Lambda')$, followed by $\mathfrak{a}_\Lambda$, which preserves the averages, makes a field axial on the blocks of Λ, and changes it on bonds with an end in Λ only. The orthogonal projection has a kernel which decays exponentially, because the average, restricted to the fields supported on these bonds, has a bounded right inverse on the functions on $\mathfrak{b}(\Lambda')$: the fields $\mathfrak{f}_c$ of the proof of Lemma 24.4 are supported on bonds with an end in Λ, and the matrix (24.10) is inverted as there. For a regular U the form differs from that at $U = 1$, after the gauge transformation of [5, (3.35)] on the cubes, by an operator which is small in the unit η, and the perturbation argument of Section B applies. The appendix on the chiral models is not used.

28 The Spaces of Regular Configurations

The paper [6] shows that a configuration which satisfies the gauge-invariant regularity conditions [6, (1.7)–(1.9)] and whose averages are close to those of a regular configuration U_0 can be brought, by a unique gauge transformation, into the Landau gauge relative to U_0, in which it is $e^{i\eta A}U_0$ with a small and regular field A (Theorems 2 and 4), and it removes with this the additional regularity condition [5, (3.35)] from the hypotheses of the theorems on the propagators (Proposition 6). The proof consists of computations at a plaquette of the fine lattice, of contraction arguments in spaces of functions with weighted norms, of applications of [4] and of Theorems 3.1 to 3.3 of [5], and of one geometric statement, Lemma 1, on a configuration in the axial gauge of one step. Only the last has to be proved anew for the first step of the tilted run.

In the axial gauge relative to V_0 of [6, (1.19), (1.24)] the product of the matrices $R(V_0(\Gamma_{y,b_-}))V'_b$ along the contour $\Gamma_{y,x}$ is 1 for every point x of the block of y. Since $(V'V_0)(\Gamma) = \prod_{b\subset\Gamma} R(V_0(\Gamma_{y,b_-}))V'_b \cdot V_0(\Gamma)$ for a path Γ from y, the condition says that $V'V_0$

and V_0 have the same product along $\Gamma_{y,x}$. For the first step of the tilted run the contours are the paths $\pi_y(x)$, and in this form the condition does not require that they stay in the block or that the configuration V' be trivial on their bonds, which Bałaban uses [6, p. 79]. The norm of a matrix is the Hilbert–Schmidt norm with the normalised trace, as in [5]; it is invariant under the multiplication by unitary matrices from either side.

Lemma 28.1 (Lemma 1 for the First Step of the Tilted Run)**.** *Let $L \geq 64$ be a multiple of 5, let V_0 and $V = V'V_0$ be configurations on the bonds of T_η with values in the unitary group, let $y_1 \in \ell_1\mathbb{Z}^4$, and let $\alpha, \alpha_1 \geq 0$. Suppose that*

(a) $|V_0(\partial p) - 1| \leq \alpha$ and $|V(\partial p) - 1| \leq \alpha$ for every plaquette p of T_η all of whose corners s satisfy $|s'_\nu - y'_{1,\nu}| \leq 5L$ for every ν;

(b) $V(\pi_y(x)) = V_0(\pi_y(x))$ for every $y \in \ell_1\mathbb{Z}^4$ with $|y - y_1|_\infty \leq \ell_1$ and every $x \in B_{\ell_1}(y)$;

(c) $|\bar V_c \bar V_{0,c}^{-1} - 1| \leq \alpha_1$ for every bond c of $\ell_1\mathbb{Z}^4$ between two of these points y, with the average (26.1).

Then for every bond b of T_η which starts in $B_{\ell_1}(y_1)$,

$$|V'_b - 1| \leq \begin{cases} \frac{42}{5} L\alpha & \text{if } b \text{ ends in } B_{\ell_1}(y_1), \\ \alpha_1 + 1710 L^2 \alpha & \text{if } b \text{ ends in } B_{\ell_1}(y_1 \pm \ell_1 e_\mu) \text{ for some } \mu, \\ 2\alpha_1 + 2850 L^2 \alpha & \text{otherwise.} \end{cases} \tag{28.1}$$

Proof. For $x \in B_{\ell_1}(y)$ the coordinates of $x - y$ in the basis $\eta e'_\nu$ are smaller than L, L, $\frac{7}{5}L$, and $\frac{4}{5}L$ in modulus, so that $\pi_y(x)$ has at most $\frac{21}{5}L$ bonds; a staircase has at most $\frac{7}{5}L$ bonds; and the sites of all the paths below have coordinates which differ from those of y_1 by at most $5L$, by the bounds in the proof of Lemma 22.2. We use Lemma 26.1 for V and for V_0, with $o = y$ in (i) and with $o = y_1$, $r = 5L$ in (ii), and we put $\kappa := 285 L^2 \alpha$.

A bond inside the block. Let $b = \langle x, x^+ \rangle$ with $x, x^+ \in B_{\ell_1}(y_1)$, and $g := V(\pi_{y_1}(x))$, $g^+ := V(\pi_{y_1}(x^+))$, which by (b) are also the products of V_0. By Lemma 26.1(i), $gV_b(g^+)^{-1}$ and $gV_{0,b}(g^+)^{-1}$ differ from 1 by at most $\frac{21}{5}L\alpha$ each, the strip of b having at most $\frac{21}{5}L$ plaquettes. Hence $|V_b - V_{0,b}| \leq \frac{42}{5}L\alpha$, and $|V'_b - 1| = |V_b V_{0,b}^{-1} - 1|$ is the same number.

A bond across a face. Let $x \in B_{\ell_1}(y_1)$ and $x^+ \in B_{\ell_1}(y_2)$ with $y_2 = y_1 + \ell_1 e_\mu$; the case $y_2 = y_1 - \ell_1 e_\mu$ follows by exchanging the roles of the two blocks, since $|V'_b - 1|$ does not change under the reversal of b up to a conjugation. Let $c := \langle y_1, y_2 \rangle$, and for $x_0 \in B_{\ell_1}(y_1)$ let Γ_{c,x_0} be the path of (26.1) and $W(x_0) := V(\Gamma_{c,x_0}) V_0(\Gamma_{c,x_0})^{-1}$. The path γ which consists of $\pi_{y_1}(x)$, of b, of the reverse of $\pi_{y_2}(x^+)$, and of the reverse of Γ_{c,x_0} is closed and has at most $4 \cdot \frac{21}{5}L + \frac{7}{5}L + 1 \leq 19L$ bonds, so that $|V(\gamma) - 1| \leq \kappa$ and $|V_0(\gamma) - 1| \leq \kappa$ by Lemma 26.1(ii). With $g_1 := V(\pi_{y_1}(x))$ and $g_2 := V(\pi_{y_2}(x^+))$, which by (b) are the products of V_0 as well, $V(\gamma) = g_1 V_b g_2^{-1} V(\Gamma_{c,x_0})^{-1}$ and $V_0(\gamma) = g_1 V_{0,b} g_2^{-1} V_0(\Gamma_{c,x_0})^{-1}$, and therefore

$$g_1 V'_b g_1^{-1} = V(\gamma)\, W(x_0)\, V_0(\gamma)^{-1}\,, \qquad |g_1 V'_b g_1^{-1} - W(x_0)| \leq 2\kappa\,. \tag{28.2}$$

We compare $W(y_1)$ with the averages. The closed path which consists of Γ_{c,x_0} and of the reverse of Γ_{c,y_1}, the staircase from y_1, has at most $12L$ bonds, so that $u := V(\Gamma_{c,x_0}) V(c)^{-1}$ satisfies $|u - 1| \leq \kappa$. The principal logarithm of a unitary matrix u satisfies $|\frac{1}{i} \log u| \leq \frac{\pi}{2}|u - 1|$, because $|e^{i\theta} - 1| \geq \frac{2}{\pi}|\theta|$ for $|\theta| \leq \pi$, and $|e^{iX} - 1| \leq |X|$ for a Hermitian matrix X. Hence the mean over x_0 of the logarithms in (26.1) is at most 2κ in norm, $|\bar V_c V(c)^{-1} - 1| \leq 2\kappa$, and in the

same way $|\bar V_{0,c} V_0(c)^{-1} - 1| \le 2\kappa$. Since $V(c)V_0(c)^{-1} = W(y_1)$, we find $|\bar V_c \bar V_{0,c}^{-1} - W(y_1)| \le 4\kappa$, and with (c) and (28.2) for $x_0 = y_1$,

$$|W(y_1) - 1| \le \alpha_1 + 4\kappa\,, \qquad |V_b' - 1| \le \alpha_1 + 6\kappa\,. \tag{28.3}$$

A bond across an edge. A bond of the directions e_2' or e_3' may lead from $B_{\ell_1}(y_1)$ to the block of $y_2 = y_1 + \ell_1(\epsilon_2 e_2 + \epsilon_3 e_3)$ with $\epsilon_2, \epsilon_3 \in \{1, -1\}$; the bonds of the directions e_0' and e_1' change one coordinate, and no other case occurs. Let $y_3 := y_1 + \ell_1 \epsilon_2 e_2$, and let Γ be the path from y_1 over y_3 to y_2 along the staircases of the two bonds of $\ell_1 \mathbb{Z}^4$, run through backwards where $\epsilon = -1$. The closed path of $\pi_{y_1}(x)$, b, the reverse of $\pi_{y_2}(x^+)$, and the reverse of Γ has at most $12L$ bonds, and as in (28.2), $|g_1 V_b' g_1^{-1} - V(\Gamma)V_0(\Gamma)^{-1}| \le 2\kappa$. The matrix $V(\Gamma)V_0(\Gamma)^{-1}$ is the product of the matrix W of the first bond and of a conjugate of that of the second, or of their inverses, each of which differs from 1 by at most $\alpha_1 + 4\kappa$ by (28.3). Hence $|V_b' - 1| \le 2\alpha_1 + 10\kappa$. *q.e.d.*

With $\alpha = \alpha_0 L^{-2}$, the regularity of [6, (1.7)] for one step, the lemma gives $|V' - 1| < 2850\alpha_0 + 2\alpha_1$ in the place of Bałaban's $4d^2\alpha_0 + \alpha_1$ [6, (1.25)], and it uses the hypotheses in the neighbourhood of the radius $5\ell_1$, in the coordinates of T_η, of the corner of the block, where he uses them on the two blocks of a bond [6, p. 80]. We go through the paper with this change.

Sections A and B, (1.1)–(1.32). The covariant derivatives [6, (1.1), (1.2)], the domains and the spaces $\mathfrak{U}_k(\{\Omega_j\}, \alpha_0)$ [6, (1.3)–(1.11)], in which the conditions concern the plaquettes and bonds of T_η in the domains, and the spaces $\mathfrak{V}_k$ [6, (1.12)–(1.14)] are defined as printed, with the averages of the tilted run. The axial gauge [6, (1.15), (1.19)] is that of [4, (64)–(72)], whose conditions at the first level are those along the paths $\pi_y(x)$; that they fix one element in each orbit of the transformations [6, (1.14)] uses only that there is one condition for each point $x \ne y$ of a block, on a path from y to x, as in [6, (1.17)]. The identities [6, (1.21), (1.22)] concern one plaquette. The Landau gauge [6, (1.27)] and the formulation [6, (1.28)–(1.32)] of the problem use the operators of [5] and the identities [4, (87), (92), (97), (99), (105)], which hold for the tilted run (Section 26).

Section C, (1.33)–(1.74). The basic estimate [6, (1.43)–(1.54)] is an expansion at a bond and the plaquettes which contain it, and it holds on T_η with the same constants. Proposition 3 follows from it, from [4, Proposition 4], which gives [6, (1.56)] with the constant C_2 of the tilted run, and from Theorem 3.3 of [5] with the bound [5, (3.133)] on $H(U_0)$, which give [6, (1.59)] with the constants of Section 27. In [6, (1.37)] the bound $2dL\alpha_1$ on the configuration B of [6, (1.31)] counts the bonds of a contour of a block; for the bonds between Λ_0 and Λ_1 the contour is a path π, and the bound is $9L\alpha_1$. The reduction of [6, (1.35)] to [6, (1.66)] applies Lemma 1 to the pairs $\bar U_0^n$, $(\overline{U'U_0})^n$ for $n = k-1, \dots, 0$, with the bound on the plaquettes of the averages from [4, Proposition 2]. For $n \ge 1$ these are configurations on a lattice of the paper and the lemma is Bałaban's, with the bound $10\alpha_0$ of Proposition 26.2(ii) in the place of $2\alpha_0$; for $n = 0$ it is Lemma 28.1. Hence [6, (1.65)] holds with a bound $C\alpha_0 + 2\alpha_1$, C a number, and [6, (1.66)] is assumed with this bound renamed. In the induction of Theorem 4, the base $k = 1$ uses Lemma 1 and requires that B_1 be not too small [6, p. 89]; here $B_1 \ge 4 \cdot 2850$ suffices. The bounds [6, (1.70)–(1.74)] on the gauge transformation u_1 use [4, (105), (106), (108), (130)] and sum over the bonds of the contours of the blocks, at most $\frac{21}{5}L$ at the first level in the place of dL, which changes the constant $16d$ of [6, (1.72)].

Sections D and E, (1.75)–(1.125). The equations for the gauge transformation u' are derived from the functions of [4, Section A], in which no lattice occurs, and from the expansion [6, (1.86)–(1.89)] at a point. Their solution uses the linearisation [4, (213), (214)] of the

average of a gauge transformation and Propositions 8 and 10 of that paper, which hold for the tilted run with other constants; the bounds [5, (3.47)] on G' and the boundedness of R in the weighted norms; and the bound [6, (1.92)] on $H' = G'^2 Q'^* (Q' G'^2 Q'^*)^{-1}$, which is a composition of the bounds of Theorems 3.1 and 3.2 of [5] by Lemma 19.2. The rest is the contraction mapping theorem in the spaces [6, (1.102), (1.119)] and Cauchy's formula for analytic functions, in which the lattice does not occur. Proposition 5, and with it Theorems 2 and 4, hold for the tilted run.

Sections F to H, (1.126)–(1.147). The cubes of Section F are cubes of the paper, the global axial gauge on $\tilde{\square}^{(k)}$ [6, (1.129)] is taken on the unit lattice, and [6, (1.128)] is Proposition 26.2(ii) with 10 in the place of 2. The repeated application of Lemma 1 in [6, (1.130)] is as above, so that the numerical factors in [6, (1.130), (1.133), (1.134), (1.136)] and in the condition $7dL^2 M\alpha_0 \le c_1$ of Proposition 6 change, and the statements hold with other numbers in the place of $6d$ and $7d$. Proposition 7 follows from [6, (1.21), (1.47), (1.54)] and [4, Proposition 4], and Theorem 8 from the proofs of Theorems 2 and 4 with a function f on the right of the gauge condition, as in the text.

In sum, the results of [6] hold for the tilted run with other constants. One statement, Lemma 1 at the first level, has a new proof; the others have Bałaban's, with the inputs from Sections 26 and 27, which are themselves classified there and not written out line by line.

29 The Variational Problem

The paper [7] constructs the background field: the minimum $U_k(V)$ of the action among the regular configurations with the averages V, unique up to gauge transformations, with its regularity and its analytic dependence on V (Theorem 1, Propositions 2 to 9). The proof is an induction on the number k of the levels. At each stage it expands the action around a configuration U_0 close to the minimum, linearises the averages, and solves the equation of the critical point by a contraction in weighted norms; its inputs are the expansion of the action at a plaquette, the results of [4], [5], and [6], and, at the start of each stage, a configuration U_0 with averages close to V. In the tilted run every stage has the turned lattice T_η as its finest lattice, and the lattices of the levels 1 to k are those of the paper. The stage $k \ge 2$ starts from the minimum for $k-1$ levels and a configuration V_0 on the lattice of the level $k-1$ with the averages V on the level k [7, (11), p. 279], which Bałaban obtains by copying $V_{b'}$ to the bonds which join the blocks of b'; both levels are lattices of the paper. The stage $k = 1$ starts from $U_0 = V_0$ [7, p. 280], a configuration on the fine lattice, and for it we need a configuration on the turned lattice.

Lemma 29.1 (A Regular Configuration with Given Averages)**.** *Let $L \ge 64$ be a multiple of 5, let V be a configuration on the bonds of $\ell_1\mathbb{Z}^4$ with values in G, and let $\varepsilon_1 \ge 0$. For two points y, y^+ of $\ell_1\mathbb{Z}^4$ let $\mathsf{v}(y, y^+)$ be 1 if $y^+ = y$; $V(\langle y, y^+\rangle)$ if the two points are neighbours; and, if $y^+ - y = \ell_1(\epsilon_2 e_2 + \epsilon_3 e_3)$ with $\epsilon_2, \epsilon_3 \in \{1, -1\}$, the product of V along the path of two bonds from y to y^+ through the point m which has the smaller of the coordinates 3 of the two points and the coordinate 2 of that one of them whose coordinate 3 is the larger. Let $U_0(b) \coloneqq \mathsf{v}(y, y^+)$ for a bond b of T_η from $B_{\ell_1}(y)$ to $B_{\ell_1}(y^+)$. Let $y_1 \in \ell_1\mathbb{Z}^4$, and suppose that $|V(\partial p') - 1| \le \varepsilon_1$ for every plaquette p' of $\ell_1\mathbb{Z}^4$ whose corners are at a distance of at most $2\ell_1$ from y_1 in $|\cdot|_\infty$. Then*

(i) U_0 is a configuration, that is, $U_0(b^{-1}) = U_0(b)^{-1}$, and its product along a path of T_η is the product of V along the path of $\ell_1\mathbb{Z}^4$ through the corners of the blocks which the path visits, with the paths of two bonds above for the diagonal steps;

(ii) $|U_0(\partial p) - 1| \le 24\varepsilon_1$ *for every plaquette* p *of* T_η *with a corner in* $B_{\ell_1}(y_1)$*, and* $|(D^{\eta*}_{U_0}\partial U_0)(b)| \le 144\eta^{-1}\varepsilon_1$ *for every bond* b *which starts in* $B_{\ell_1}(y_1)$*, if the hypothesis holds also at the neighbours of* y_1*;*

(iii) $|\bar U_{0,c} - V_c| \le 137\varepsilon_1$ *for every bond* c *of* $\ell_1\mathbb{Z}^4$ *which starts at* y_1*, with the average* (26.1).

Proof. (i) A bond of T_η joins two points of one block, of two blocks with a common face, or, if its direction is e'_2 or e'_3, of two blocks which differ in the coordinates 2 and 3, as in the proof of Lemma 28.1; so U_0 is defined. In the diagonal case the point m is the same for the pair (y, y^+) and for the pair (y^+, y), so that the two paths are the reverses of each other. Hence $\mathsf{v}(y^+, y) = \mathsf{v}(y, y^+)^{-1}$, and the statement on the paths follows from the definition.

(ii) We use Lemma 26.1(ii) on the lattice $\ell_1\mathbb{Z}^4$, for which it holds with the same proof: the product of V along a closed path of n bonds whose sites differ from a point o by at most $r\ell_1$ in every coordinate differs from 1 by at most $3rn$ times the bound on the plaquettes of these sites. The four corners of a plaquette p of T_η are within the distance $\sqrt{2}\eta$ of each other, so that their blocks have corners which differ by at most ℓ_1 in every coordinate, and by (i) $U_0(\partial p)$ is the product of V along a closed path of at most 8 bonds through these corners and the corners of the diagonal steps, which differ from one of them by at most ℓ_1 in every coordinate. This gives $|U_0(\partial p) - 1| \le 3 \cdot 8\,\varepsilon_1$. The value of $D^{\eta*}_{U_0}\partial U_0$ at a bond is η^{-1} times a signed sum of the values of $\partial U_0 - 1$, transported, at six plaquettes [6, (1.1), (1.2)], which gives the second bound.

(iii) Let $c = \langle y_1, y_2\rangle$ have the direction μ and $x \in B_{\ell_1}(y_1)$. The path $\pi_{y_1}(x)$ consists of four straight parts in the directions e'_3, e'_2, e'_1, e'_0, of lengths smaller than $\frac45\ell_1$, $\frac75\ell_1$, ℓ_1, ℓ_1. Along the first the coordinates 2 and 3 change by less than $\frac{16}{25}\ell_1$ and $\frac{12}{25}\ell_1$, along the second by less than $\frac{21}{25}\ell_1$ and $\frac{28}{25}\ell_1$, so that the path passes from one block to another at most $1+1+1+2+1+1 = 7$ times; it starts and ends in $B_{\ell_1}(y_1)$ and stays in the blocks around it. By (i) and (ii), $U_0(\pi_{y_1}(x))$ is the product of V along a closed path of at most 14 bonds and differs from 1 by at most $42\varepsilon_1$; the same holds for $\pi_{y_2}(x+\ell_1 e_\mu)$. Along the staircase $\sigma^{\ell_1}_\mu(x)$ the coordinate μ increases at every step, so that it passes once from the blocks below the face between y_1 and y_2 to those above; if $\mu \in \{2,3\}$ it may in addition pass several times between $B_{\ell_1}(y_1)$ and one neighbouring block in the other of these two directions, and between their translates by $\ell_1 e_\mu$, since its sites stay within $\frac35\eta$ of a segment (Lemma 6.2(ii)). The corners of the blocks which it visits are therefore corners of one plaquette of $\ell_1\mathbb{Z}^4$ which contains c, steps back and forth cancel in the product, and the product of V along the path of (i) is V_c, or V_c times a conjugate of the product around this plaquette or of its inverse, because the path passes once from the two corners below the face to the two above. Hence $|U_0(\sigma^{\ell_1}_\mu(x)) - V_c| \le \varepsilon_1$, and $|U_0(\Gamma_{c,x}) - V_c| \le 85\varepsilon_1$ for every x, in particular $|U_0(c) - V_c| \le \varepsilon_1$. Then $|U_0(\Gamma_{c,x})U_0(c)^{-1} - 1| \le 86\varepsilon_1$, the mean of the logarithms in (26.1) is at most $\frac{\pi}{2} \cdot 86\varepsilon_1 \le 136\varepsilon_1$ in norm, as in the proof of Lemma 28.1, and $|\bar U_{0,c} - V_c| \le (136+1)\varepsilon_1$. *q.e.d.*

With $\eta = L^{-1}$, as at the stage $k = 1$, the lemma gives [7, (14)] for this U_0 on the blocks of Λ_1 away from the boundary of Ω_1: the plaquettes of U_0 are bounded by $24L^2\varepsilon_1 \cdot L^{-2}$ and $D^*\partial U_0$ by $144L^3\varepsilon_1 \cdot L^{-2}$, where [7, (2)] requires $\varepsilon_0 L^{-2}$ for both, and the averages differ from V by at most $137\varepsilon_1$. Bałaban's configuration has the averages V exactly and satisfies the conditions with $C_1 = L^3$; the generalised hypothesis [7, (14)], under which the rest of the paper is written, requires closeness only, with a constant C_1 which depends on L, and it holds here with $C_1 B_3 = 144L^3$. At the bonds between Λ_0 and the blocks of Λ_1 the configuration V_0 has to be joined to the given configuration on Λ_0; the text does not describe this for its own lattice [7, p. 279], and we add nothing to it.

We go through the paper.

The statement and Section A, (1)–(21). The spaces [7, (2), (3), (6)], the action [7, (5)], and the condition [7, (7)] on V are defined as printed, with the plaquettes of T_η on Λ_0 and with the tilted average of the first step in the plaquettes which meet the boundary between Λ_0 and Λ_1. The reduction to the space [7, (19)–(21)] is Theorem 2 of [6] with Proposition 6 there, which hold for the tilted run (Section 28); the bound $2dLC_1\varepsilon_1$ on B becomes $9LC_1\varepsilon_1$, as in [6, (1.37)].

Sections B to D, (22)–(112). The expansion [7, (22)–(40)] of the action up to the fourth order is a computation at one plaquette, with the formula of Baker, Campbell, and Hausdorff, and it is unchanged. The linearising transformation [7, (44)–(62)] is a fixed point of a contraction built from C_j of [4, Proposition 4] and from the operator H with the bounds [7, (46)] of [5, Theorem 3.12]; its pseudo-locality [7, (63)–(73)] uses [4, Proposition 5, (157)], the kernel bound [5, (3.133)], and Lemma 2.1 of [3]. In [7, (69)] the function C_j at a bond c depends, in the tilted run, on the field in the neighbourhood of the radius $4\ell_1$ of the two blocks of c, which changes the constant. The functional derivative of the interaction [7, (74)–(98)] is estimated with [5, (3.132)], the bounds just named, and the identities [6, (1.50), (1.52)] at a plaquette; the equations [7, (99)–(112)] are those of the critical point in terms of H_1, $\mathfrak{P}$, and $\mathfrak{G}$ of [5, Section D]. Propositions 3, 4, and 5 hold for the tilted run with the same proofs and other constants.

Section E, (113)–(143). The solution of [7, (111)] is a contraction with [5, Theorem 3.13]. The passage to the gauge conditions [7, (123)] is Theorem 8 of [6]. The second-order bounds [7, (130)–(140)] use the bound of [5, (3.42)] with the covariant Laplace operator, the identity [7, (135)] between $D^*D + DD^*$ and the covariant Laplace operator on the fields, which is a computation at a bond and its plaquettes on one cubic lattice, and the bounds [5, (3.49), (3.69), (3.137)]. The return to the axial gauge is Proposition 7 of [6]. Propositions 6 and 7 follow.

Section F, (144)–(169). The cubes and the sequences of cubes are those of [6, Section F]; the operators are those at the background field 1, that is, of Sections 10–25. In [7, (145), (146)] the factor $2L^2\varepsilon_0$ is $10L^2\varepsilon_0$ by Proposition 26.2(ii), and $11d^2\varepsilon_0$ is the constant of [6, (1.65)] for the tilted run. Lemma 1 of [6] is applied in the derivation of [7, (160)] on two neighbouring blocks of the level $j-1$; for $j \geq 2$ these are blocks of a lattice of the paper, and for $j = 1$ the lemma is Lemma 28.1, which gives [7, (160)] with other numbers. The sum [7, (161), (162)] which defines B_3 is bounded by Lemma 19.2 and (20.22). Proposition 8 and the regularity [7, (9), (10)] follow as in the text, and with them Theorem 1.

Section G, (170)–(190). The analyticity of $U_k(V)$ and the expansion [7, (174)–(180)] follow from the analyticity of the fixed points; the law [7, (181)] uses the covariance of the averages. For the decay [7, (190)] of the functional derivative Bałaban states the bound [7, (189)] on the second derivative of the interaction without carrying out the computation [7, p. 308]; we accept it as his, and it involves the lattice through the same inputs as [7, (85)–(96)]. Proposition 9 holds for the tilted run on this basis.

30 The Vacuum Polarisation under the Symmetries of the Tilted Run

The first of the papers on the renormalisation transformations, [9], defines its averages and its gauge fixing so that the effective actions are invariant under the symmetries of the lattice [9, pp. 252–254, 269], and it uses this invariance for one purpose, in its Sections 4 and 5: the term of the second order of the effective action of a step, the vacuum polarisation Π, is written

as a multiple β of the quadratic part of the action plus terms with three derivatives [9, (5.37), (5.38), (5.43)], and the coefficient β, which the renormalisation of the coupling removes, is identified [9, (5.42)]. The argument uses the covariance of Π under the permutations of the coordinates and under the reflection of each coordinate [9, (5.6), (5.7)]. In the tilted run the covariance holds at most for the group $\mathfrak{G}_0$ of Lemma 5.1, which contains neither the reflection of the coordinate 2 alone nor the permutations which mix the coordinates $0, 1$ with $2, 3$. We determine what remains of the representation. This makes the remark at the end of Section 5 precise; it does not remove the difficulty stated there.

The setting. Let $\Pi_{\mu\nu}(x)$, $x \in \mathbb{Z}^4$, $\mu, \nu \in \{0,1,2,3\}$, be real numbers with $|\Pi_{\mu\nu}(x)| \leq E\, e^{-\delta_1 |x|_1}$, where $|x|_1 = \sum_\mu |x_\mu|$; this is the function of [9, (5.5), (5.8), (5.10)]: $\Pi_{\mu\nu}(x-y)$ is the value $\Pi(b,b')$ at the bonds $b = \langle x, x+e_\mu\rangle$ and $b' = \langle y, y+e_\nu\rangle$, the function Π is extended to the reversed bonds by $\Pi(-b,b') = -\Pi(b,b') = \Pi(b,-b')$, and $\Pi_{\mu\nu}(x) = \Pi_{\nu\mu}(-x)$. We write $\Pi_{\mu\nu}(p) \coloneqq \sum_x e^{-ip\cdot x}\Pi_{\mu\nu}(x)$ and $\partial_\mu(p) \coloneqq e^{ip_\mu} - 1$, as in [9, (5.11), (5.15)], and we assume the two identities of the gauge invariance [9, (5.15)],

$$\sum_\mu \overline{\partial_\mu(p)}\, \Pi_{\mu\nu}(p) = 0\,, \qquad \sum_\nu \partial_\nu(p)\, \Pi_{\mu\nu}(p) = 0\,. \tag{30.1}$$

A linear isometry r of $\mathbb{Z}^4$ maps e_μ to $s_\mu e_{\pi(\mu)}$ with a permutation π and signs s_μ, and it maps the bond $\langle x, x+e_\mu\rangle$ to the bond from rx in the direction $s_\mu e_{\pi(\mu)}$. We say that Π is *covariant under* r if $\Pi(rb, rb') = \Pi(b,b')$ for all bonds, which is [9, (5.4)]. A *plane* is a set $\{\kappa,\lambda\}$ of two indices, and the planes form three classes: the plane $\{0,1\}$, the plane $\{2,3\}$, and the four mixed planes $\{a,j\}$ with $a \in \{0,1\}$ and $j \in \{2,3\}$. For a class a let $T^{\mathsf{a}}_{\mu\nu}(x)$ be the function of finite support with

$$\sum_{x,y}\sum_{\mu,\nu} T^{\mathsf{a}}_{\mu\nu}(x-y)\, B'_\mu(x) B_\nu(y) = \sum_x \sum_{\{\kappa,\lambda\}\in\mathsf{a}} (\partial B')_{\kappa\lambda}(x)\, (\partial B)_{\kappa\lambda}(x) \tag{30.2}$$

for all real fields B, B' of finite support on the bonds of $\mathbb{Z}^4$, where $(\partial B)_{\kappa\lambda}(x) = B_\lambda(x+e_\kappa) - B_\lambda(x) - B_\kappa(x+e_\lambda) + B_\kappa(x)$. Its transform is $T^{\mathsf{a}}_{\mu\nu}(p) = \sum_{\{\kappa,\lambda\}\in\mathsf{a}} (\delta_{\mu\lambda}\overline{\partial_\kappa(p)} - \delta_{\mu\kappa}\overline{\partial_\lambda(p)})(\delta_{\nu\lambda}\partial_\kappa(p) - \delta_{\nu\kappa}\partial_\lambda(p))$, and the sum of the three is $\delta_{\mu\nu}\Delta(p) - \partial_\mu(p)\overline{\partial_\nu(p)}$ with $\Delta(p) = \sum_\kappa |\partial_\kappa(p)|^2$.

Proposition 30.1 (The Three Coefficients)**.** *Let Π be as above and covariant under the elements of $\mathfrak{G}_0$. Then*

$$\Pi_{\mu\nu}(p) = \sum_{\mathsf{a}} \beta^{\mathsf{a}}\, T^{\mathsf{a}}_{\mu\nu}(p) + \Pi'_{\mu\nu}(p)\,, \qquad \beta^{\mathsf{a}} \coloneqq \sum_x \Pi_{\kappa\lambda}(x)\, x_\kappa x_\lambda\,, \tag{30.3}$$

where the sum runs over the three classes, $\{\kappa,\lambda\}$ is any plane of the class a, on whose choice β^{a} does not depend, and Π' has the following property: it is a finite sum of products of three of the functions $\partial_\kappa(p)$, $\overline{\partial_\kappa(p)}$ with coefficients which are periodic and analytic in the domain $|\operatorname{Im} p_\mu| < \delta_1$ and bounded by a constant times E in every domain $|\operatorname{Im} p_\mu| \leq \delta_1' < \delta_1$, the constant depending on δ_1 and δ_1'. If Π is covariant under all the isometries of $\mathbb{Z}^4$, the three coefficients are equal, and (30.3) *is [9, (5.37), (5.38)].*

Proof. The moments. Let $m^0_{\mu\nu}$, $m^1_{\mu\nu,\kappa}$, and $m_{\mu\nu,\kappa\lambda}$ be the sums over x of $\Pi_{\mu\nu}(x)$, of $x_\kappa\Pi_{\mu\nu}(x)$, and of $x_\kappa x_\lambda \Pi_{\mu\nu}(x)$, so that $\Pi_{\mu\nu}(p) = m^0_{\mu\nu} - i\sum_\kappa m^1_{\mu\nu,\kappa} p_\kappa - \frac{1}{2}\sum_{\kappa,\lambda} m_{\mu\nu,\kappa\lambda} p_\kappa p_\lambda + O(|p|^3)$. By $\Pi_{\mu\nu}(x) = \Pi_{\nu\mu}(-x)$, m^1 is antisymmetric and m symmetric in μ,ν. We expand the second identity of (30.1) in powers of p, with $\partial_\nu(p) = ip_\nu - \frac{1}{2}p_\nu^2 + \dots$. The terms of the first order give $\sum_\nu p_\nu m^0_{\mu\nu} = 0$ for all p, hence $m^0 = 0$. The terms of the second order then give $\sum_{\nu,\kappa} m^1_{\mu\nu,\kappa} p_\nu p_\kappa = 0$, so that m^1 is antisymmetric in ν,κ, hence in all three indices.

The covariance. Let $r \in \mathfrak{G}_0$, with π and s as above. The bond $r\langle x, x+e_\mu\rangle$ is $\langle rx - t_\mu, rx - t_\mu + e_{\pi(\mu)}\rangle$ with $t_\mu = 0$ if $s_\mu = 1$, and the reverse of this bond with $t_\mu = e_{\pi(\mu)}$ if $s_\mu = -1$. The covariance therefore reads $\Pi_{\pi(\mu)\pi(\nu)}(rx - t_\mu + t_\nu) = s_\mu s_\nu \Pi_{\mu\nu}(x)$. Multiplying by $(rx - t_\mu + t_\nu)_{\pi(\kappa)} = s_\kappa x_\kappa + \text{const}$ and summing over x, we find $m^1_{\pi(\mu)\pi(\nu),\pi(\kappa)} = s_\mu s_\nu s_\kappa m^1_{\mu\nu,\kappa}$, the constant being multiplied by $m^0 = 0$. For $r\colon z \mapsto -z$, where $s_2 = s_3 = -1$ and π is the identity, this shows that $m^1_{\mu\nu,\kappa}$ vanishes if exactly one or all three of its indices lie in $\{2,3\}$, and for the element which changes the signs of x_0 and x_1 it shows the same for $\{0,1\}$. Three different indices contain one or three elements of one of the two sets, so that $m^1 = 0$. With $m^0 = m^1 = 0$ the same computation for the second moments gives

$$m_{\pi(\mu)\pi(\nu),\pi(\kappa)\pi(\lambda)} = s_\mu s_\nu s_\kappa s_\lambda \, m_{\mu\nu,\kappa\lambda} \;, \tag{30.4}$$

and the terms of the third order of the second identity of (30.1) give $\sum_{\nu,\kappa,\lambda} m_{\mu\nu,\kappa\lambda} p_\nu p_\kappa p_\lambda = 0$, that is,

$$m_{\mu\nu,\kappa\lambda} + m_{\mu\kappa,\nu\lambda} + m_{\mu\lambda,\nu\kappa} = 0 \tag{30.5}$$

for all indices.

The tensors with these properties. Let m be a tensor which is symmetric in its first two and in its last two indices and satisfies (30.4) for the elements of $\mathfrak{G}_0$ and (30.5). We show that it is determined by the three numbers $m_{01,01}$, $m_{23,23}$, and $m_{02,02}$. By (30.4) for the reflections of x_0 and of x_1 and for $z \mapsto -z$, a component vanishes unless the index 0 and the index 1 occur an even number of times and the indices of $\{2,3\}$ together an even number of times. From (30.5) with $\nu = \kappa = \lambda$ we have $m_{\mu\nu,\nu\nu} = 0$, and with $\kappa = \lambda$,

$$m_{\mu\nu,\kappa\kappa} = -2\, m_{\mu\kappa,\nu\kappa} \;. \tag{30.6}$$

All indices in $\{0,1\}$. The components $m_{00,00}$ and $m_{11,11}$ vanish, and $m_{00,11} = m_{11,00} = -2m_{01,01}$ by (30.6). *Two indices equal to* $a \in \{0,1\}$ *and two indices* $j, j' \in \{2,3\}$. By (30.6), $m_{aa,jj} = m_{jj,aa} = -2m_{aj,aj}$. For $j \neq j'$, (30.6) gives $m_{jj',aa} = -2m_{ja,j'a}$, and since the left side is symmetric in j, j', $m_{a2,a3} = m_{a3,a2}$; the element $z \mapsto iz$, for which π exchanges 2 and 3 and $s_2 s_3 = -1$, gives $m_{a3,a2} = -m_{a2,a3}$ by (30.4). Hence $m_{a2,a3} = 0$ and $m_{23,aa} = 0$, and (30.5) with $(\mu,\nu,\kappa,\lambda) = (a,a,2,3)$ gives $m_{aa,23} = 0$. The same element gives $m_{a3,a3} = m_{a2,a2}$, and the exchange of x_0 and x_1 gives $m_{1j,1j} = m_{0j,0j}$. *All indices in* $\{2,3\}$. The components with three or four equal indices in the positions ν, κ, λ vanish: $m_{22,22} = m_{33,33} = m_{23,33} = m_{23,22} = 0$; (30.5) with $(\mu,\nu,\kappa,\lambda) = (2,2,2,3)$ gives $2m_{22,23} + m_{23,22} = 0$, hence $m_{22,23} = 0$, and likewise $m_{33,23} = 0$; and $m_{22,33} = m_{33,22} = -2m_{23,23}$ by (30.6). This proves the claim.

The second order. The functions T^{a} satisfy the hypotheses of the proposition: (30.1) holds for them because the right side of (30.2) vanishes if B or B' is the gradient of a function, and they are covariant under $\mathfrak{G}_0$ because an element of $\mathfrak{G}_0$ permutes the planes of each class and maps the curl at a plaquette to the curl at its image, up to a sign which occurs twice. The component $T^{\mathsf{a}}_{\kappa\lambda}(p)$ is $-\overline{\partial_\lambda(p)}\partial_\kappa(p) = -p_\kappa p_\lambda + O(|p|^3)$ if the plane $\{\kappa,\lambda\}$ belongs to a and 0 otherwise, so that the second moments of T^{a} are 1 at the component $(\kappa\lambda,\kappa\lambda)$ of the planes of a and 0 at those of the other planes. The second moments of $\Pi' := \Pi - \sum_{\mathsf{a}} \beta^{\mathsf{a}} T^{\mathsf{a}}$ form a tensor with the properties above whose three determining numbers vanish; hence they vanish, and so do the moments of the orders 0 and 1. That β^{a} does not depend on the plane of the mixed class was shown above.

The remainder. Let $z_\mu := e^{ip_\mu}$ and $f(z) := \sum_x \Pi'_{\mu\nu}(x) z^{-x}$ for fixed μ, ν, a function which is analytic for $e^{-\delta_1} < |z_\mu| < e^{\delta_1}$, because Π' decays as Π does, and which vanishes with its derivatives of the first and the second order at the point $z = (1,\dots,1)$. We split the sum over x into the 2^4 sums over the points with prescribed signs of the coordinates, $x_\mu \leq 0$ or

$x_\mu > 0$, as in [9, (5.22)]: $f(z) = \sum_\epsilon g^\epsilon(z^\epsilon)$ with $z^\epsilon = (z_\mu^{\epsilon_\mu})$ and functions g^ϵ which are power series converging for $|w_\mu| < e^{\delta_1}$. This domain is convex and contains the point $(1, \dots, 1)$, and Taylor's formula with the integral form of the remainder, as in [9, (5.30)], writes each $g^\epsilon(w)$ as a polynomial of the degree two in the numbers $w_\kappa - 1$ plus a sum of products of three of these numbers with analytic coefficients. Inserting $w = z^\epsilon$ we obtain $f(z) = P(z) + f_3(z)$, where P is a polynomial of the degree two in the eight quantities $u_\kappa := z_\kappa - 1$ and $v_\kappa := z_\kappa^{-1} - 1$, and f_3 is a sum of products of three of them with coefficients analytic in the domain of f. Since $u_\kappa + v_\kappa + u_\kappa v_\kappa = 0$, we may replace v_κ by $-u_\kappa - u_\kappa v_\kappa$ in P: a term of the first degree produces a term of the first degree in u and the term $-u_\kappa v_\kappa$, to which we apply the replacement once more, and every term of the second degree produces a term of the second degree in u and products of three or four of the quantities. Hence $P = P_0 + P_3$ with a polynomial P_0 of the degree at most two in the u_κ alone and a sum P_3 of products of three of the quantities with polynomial coefficients. The functions f, f_3, and P_3 vanish with their derivatives up to the second order at $u = 0$, hence so does P_0, and a polynomial of the degree two with this property is zero. Thus $f = f_3 + P_3$, which is the stated form of Π' with $u_\kappa = \partial_\kappa(p)$ and $v_\kappa = \overline{\partial_\kappa(p)}$ for real p. The bounds on the coefficients follow from Cauchy's estimates for the derivatives of the functions g^ϵ, which are bounded by a constant times E on every smaller polydisc.

The symmetric case. If Π is covariant under all the isometries of $\mathbb{Z}^4$, then (30.4) for a permutation which maps the plane $\{0, 1\}$ to another plane shows that the three numbers are equal. *q.e.d.*

The proof shows more than the statement: the moments of Π of the orders 0 and 1 vanish, and its second moments are those of $\sum_{\mathsf{a}} \beta^{\mathsf{a}} T^{\mathsf{a}}$. It also shows that the representation of the remainder needs no symmetry beyond the decay.

The vacuum polarisation is compared in [9] with the corresponding term of the action, whose kernel is that of the form $\langle B, \Delta_j B\rangle$, the minimum of $\|\partial A\|^2$ over the fields with the averages B, and of which Bałaban uses the expansion [9, (4.41)] at the momentum zero, read off from the formula [2, (1.66)]. In the tilted run this formula contains the matrix (7.9), and one may ask whether the expansion acquires three coefficients as well. It does not.

Lemma 30.2 (The Second Order of the Action)**.** *Let $32\eta \le \frac{1}{2}$, and for $p' \in [-\pi, \pi]^4$ and $b \in \mathbb{C}^4$ let $\langle b, \Delta_k(p')b\rangle$ be the form of the action Δ_k of the tilted run on the whole lattice in the momentum representation, that is, the minimum of $\sum_l \langle a_l, \mathcal{D}(p' + l)a_l\rangle$ over the families of vectors $a_l \in \mathbb{C}^4$ with $\sum_l u_k(p' + l)V(p' + l)a_l = b$, where l runs through the classes of (7.9), $\partial'_\nu(p) := \eta^{-1}(e^{i\eta p \cdot e'_\nu} - 1)$, and $\langle a, \mathcal{D}(p)a\rangle := \sum_{\kappa<\lambda} |\partial'_\kappa(p)a_\lambda - \partial'_\lambda(p)a_\kappa|^2$ is the symbol of $\|\partial A\|^2$. There is a number C, which does not depend on k and L, such that*

$$|\langle b, \Delta_k(p')b\rangle - (|p'|^2|b|^2 - |p' \cdot b|^2)| \le C|p'|^3|b|^2 \,. \tag{30.7}$$

Proof. We write $O(t)$ for a quantity bounded by a number times t, the number not depending on k, L, p', and the vectors. Three facts are used. First, $u_k(p') = 1 + O(|p'|)$ and $V(p') = R_0 + O(|p'|)$ with the orthogonal matrix $R_0 = (e_\mu \cdot e'_\nu)$, by (7.4), (7.5), and the proof of Lemma 7.2(iii), the functions D being means of exponentials over points at a distance of at most 1 from the origin. Secondly, $\partial'_\nu(p') = i(p' \cdot e'_\nu)(1 + O(\eta|p'|))$, so that $\Delta(p') = |p'|^2(1 + O(\eta|p'|))$ and $\langle a, \mathcal{D}(p')a\rangle = |p'|^2|a|^2 - |p' \cdot \mathbf{a}|^2 + O(|p'|^3)|a|^2$ with the vector $\mathbf{a} := \sum_\nu a_\nu e'_\nu$, the sum of the squares of the components of an antisymmetric tensor being the same in every orthonormal basis; moreover, by Lagrange's identity, $\mathcal{D}(p) = \Delta(p)(1 - \mathsf{P}(p))$ with the orthogonal projection $\mathsf{P}(p)$ onto the vector $\partial'(p) = (\partial'_\nu(p))_\nu$.

Thirdly, $u_k(p)V(p)\partial'(p) = u_k(p)\partial^1(p)$ with $\partial^1_\mu(p) = e^{ip_\mu} - 1$, which is Lemma 6.4(i) applied to the function e_p, by Lemma 7.1(i) and (ii).

The upper bound. Let $a_0 := (u_k(p')V(p'))^{-1}b$ and $a_l := 0$ for $l \neq 0$; the inverse exists by Lemma 7.2(iii). By the first fact $\mathbf{a}_0 = b + O(|p'|)|b|$ as vectors of $\mathbb{C}^4$, and by the second $\langle a_0, \mathcal{D}(p')a_0\rangle = |p'|^2|b|^2 - |p'\cdot b|^2 + O(|p'|^3)|b|^2$.

The lower bound. Let $j \in \mathbb{C}^4$ be orthogonal to $\partial^1(p')$, and $c_l := \overline{u_k(p'+l)}\, V(p'+l)^* j$. By the third fact, and because ∂^1 has the periods 2π, c_l is orthogonal to $\partial'(p'+l)$. For an admissible family, $\langle j, b\rangle = \sum_l \langle c_l, a_l\rangle$, and since c_l lies in the range of $1 - \mathsf{P}(p'+l)$, Schwarz's inequality for the form $\mathcal{D}(p'+l)$ gives $|\langle c_l, a_l\rangle|^2 \leq \Delta(p'+l)^{-1}|c_l|^2\langle a_l, \mathcal{D}(p'+l)a_l\rangle$. Hence

$$|\langle j, b\rangle|^2 \leq \langle j, \phi(p')j\rangle \sum_l \langle a_l, \mathcal{D}(p'+l)a_l\rangle \tag{30.8}$$

with the matrix ϕ of (7.9). In $\langle j, \phi(p')j\rangle$ the term $l = 0$ is $|u_k(p')|^2|V(p')^*j|^2/\Delta(p') = |p'|^{-2}|j|^2(1 + O(|p'|))$ by the first two facts. For the terms with $l \neq 0$ we use the proof of Lemma 7.3: $\Delta(p'+l) \geq 4$, $|V^*j| \leq \frac{8}{5}|j|$, and by (7.7) with $|\partial^1_\mu(p')| \leq |p'_\mu|$ the modulus of $u_k(p'+l)$ is at most $5\sqrt{2}\,\eta|p'|$ for the 24 classes which are not central and have $q = p'$, and at most $\prod_\mu \min(1, \pi|p'_\mu|/(2|p'_\mu + 2\pi n_\mu|))$ for the classes with $q = p' + 2\pi n$, $n \neq 0$, in which at least one factor is at most $|p'|/(2(2|n_\mu| - 1))$ and the others with $n_\mu \neq 0$ are at most $\frac{\pi}{2}(2|n_\mu| - 1)^{-1}$. The squares add up to $O(|p'|^2)$, as in that proof. Hence $\langle j, \phi(p')j\rangle \leq |p'|^{-2}|j|^2(1 + O(|p'|))$, and (30.8) gives, for every admissible family,

$$\sum_l \langle a_l, \mathcal{D}(p'+l)a_l\rangle \geq |p'|^2(1 - O(|p'|)) \sup_j \frac{|\langle j, b\rangle|^2}{|j|^2}\ , \tag{30.9}$$

the supremum over the vectors $j \neq 0$ orthogonal to $\partial^1(p') = ip'(1 + O(|p'|))$. This supremum is the squared norm of the projection of b to the orthogonal complement of $\partial^1(p')$, which is $|b|^2 - |p'\cdot b|^2/|p'|^2 + O(|p'|)|b|^2$. *q.e.d.*

The symbol $|p'|^2|b|^2 - |p'\cdot b|^2$ is the second order of $\sum_{\mathsf{a}} T^{\mathsf{a}}$ with equal coefficients 1, so that [9, (4.41)] holds for the tilted run; the reason is that the action of the fine lattice is the sum over all the planes of an orthonormal basis, whichever basis it is, and that the averages map constant fields to constant fields.

We can now say what the two statements mean for [9]. In [9, (4.34)] the function Π enters through three expressions: its quadratic form; its second moments, which multiply the terms of the third and the fourth order in the field [9, (4.43), (4.44)]; and its first moments, of which Bałaban says that the corresponding expression "should vanish" by the Euclidean symmetries [9, p. 290]. By Proposition 30.1 and its proof the first moments vanish under $\mathfrak{G}_0$ as well, the second moments are those of $\sum_{\mathsf{a}} \beta^{\mathsf{a}} T^{\mathsf{a}}$, and the quadratic form is the sum over the three classes of β^{a} times the sum of $(\partial\delta B)_{\kappa\lambda}(\partial B)_{\kappa\lambda}$ over the planes of the class, in the place of the first term of [9, (5.43)], plus terms with three derivatives, to which the bound [9, (5.44)] and the argument of [9, p. 298] apply as they stand. The term [9, (4.38)] of the effective action, $\beta_j(g_{j-1})$ times the action, contributes the same three expressions for the kernel of Δ_j [9, (4.39)–(4.45)], by Lemma 30.2 with one coefficient for all the planes, and since it enters [9, (1.6)] with the opposite sign, the definition of β_j as the coefficient of [9, (5.43)] makes the two cancel [9, pp. 292, 298]. With three coefficients β^{a}_j the cancellation is complete for one class of planes. For the other two there remain the expressions of [9, (4.42), (4.44)], restricted to the planes of the class, with the differences of the coefficients; they are of the kind which Bałaban describes as those "for which the sum over j has not a uniform

bound" [9, p. 292]. Since each coefficient is a second moment of Π, a bound on the difference of the vacuum polarisations of two runs, of which one has the symmetries of its lattice, would bound the differences of the three coefficients of the other.

Averages which are covariant. Proposition 30.1 assumes the covariance under $\mathfrak{G}_0$. In [9] the covariance of the effective actions under the symmetries of the lattice is obtained by a choice of the averages: the blocks are centred at the points of the coarse lattice, and the contours of a block are replaced by a family which the symmetries permute [9, (0.3), (0.4), pp. 252–253]. The averages of Definition 6.1 are those of the earlier papers and are not of this kind, since a block $y + [0, a)^4$ is not mapped to a block by a reflection. We record that the turned lattice admits the choice of [9]. Let $\mathfrak{G}$ be the group of the maps $x \mapsto rx + t$ with $r \in \mathfrak{G}_0$ and $t \in a\mathbb{Z}^4$, for a scale a; they map T_η and $a\mathbb{Z}^4$ to themselves.

Lemma 30.3 (Centred Blocks)**.** *Let a be a scale for which a/η is odd, and for $y \in a\mathbb{Z}^4$ let $B_a^\circ(y) \coloneqq \{x \in T_\eta \,|\, |x_\mu - y_\mu| < \frac{1}{2}a,\ \mu = 0, \dots, 3\}$.*

(i) The sets $B_a^\circ(y)$, $y \in a\mathbb{Z}^4$, are disjoint, their union is T_η, and each has $(a/\eta)^4$ points.

(ii) An element $x \mapsto rx + t$ of $\mathfrak{G}$ maps $B_a^\circ(y)$ onto $B_a^\circ(ry + t)$; it maps the path from y to x which changes the coordinates in the basis $\eta e'_\nu$ in a given order of the directions to the path from $ry + t$ to $rx + t$ of the image order; and it maps the staircase $\sigma_\mu^a(x)$ to the staircase $\sigma_{\mu'}^a(rx + t)$ if $re_\mu = e_{\mu'}$, and to the reverse of the staircase $\sigma_{\mu'}^a(rx + t - ae_{\mu'})$ if $re_\mu = -e_{\mu'}$.

(iii) The averages $Q_a'^\circ$ and Q_a° defined by (6.2) and (6.3) with the blocks B_a° in the place of B_a commute with the elements of $\mathfrak{G}$, acting on the functions by $\lambda \mapsto \lambda \circ g^{-1}$ and on the fields by $(gA)(gb) \coloneqq A(b)$, and they satisfy Lemma 6.4(i).

Proof. (i) No point x of T_η has $|x_\mu - y_\mu| = \frac{1}{2}a$: for $\mu \in \{0, 1\}$ the difference is a multiple of η, and for $\mu \in \{2, 3\}$ a multiple of $\frac{1}{5}\eta$, by (6.1), while $a/(2\eta)$ and $5a/(2\eta)$ are not integers. Hence every point of T_η lies in the open cube of the side a around exactly one point of $a\mathbb{Z}^4$, so that each block contains one point of every class of T_η modulo $a\mathbb{Z}^4$, and the number of the classes is the quotient $(a/\eta)^4$ of the volumes of the cells of the two lattices. (ii) An element of $\mathfrak{G}_0$ is a signed permutation of the coordinates $0, 1$ together with a rotation of the plane of the coordinates $2, 3$ by a multiple of a right angle (Lemma 5.1); it maps the cube $(-\frac{1}{2}a, \frac{1}{2}a)^4$ to itself, and since it preserves $\rho\mathbb{Z}^4$ it permutes the eight vectors $\pm e'_\nu$. This gives the statements on the blocks and on the paths. For the staircases it suffices to consider generators of $\mathfrak{G}_0$. For the signed permutations of the coordinates $0, 1$ the statement concerns straight paths. The rotation $z \mapsto iz$ maps e_2 to e_3 and e_3 to $-e_2$, and e'_2 to $-e'_3$ and e'_3 to e'_2; hence it maps the steps $e'_3, e'_2, e'_3, e'_2, e'_3, e'_2, e'_3$ of σ_2^a to the steps $e'_2, -e'_3, e'_2, -e'_3, e'_2, -e'_3, e'_2$ of σ_3^a, and the steps of σ_3^a to $-e'_3, -e'_2, -e'_3, -e'_2, -e'_3, -e'_2, -e'_3$, which are those of σ_2^a run through backwards, the pattern of σ_2^a being the same in both directions. (iii) follows from (ii), with $(Q_a^\circ A)(b^{-1}) = -(Q_a^\circ A)(b)$ for the case of a reversed bond, and the proof of Lemma 6.4(i) uses only that the translation by ae_μ maps the block of y onto that of $y + ae_\mu$. *q.e.d.*

The lemma concerns the definitions only. The momentum representation, the cells, and the contours of Sections 7 to 9 are written for the blocks B_a, with explicit constants, and they have not been redone for the centred blocks; nor has the family of the paths for all the orders of the directions been put in the place of the path $\pi_y(x)$ in Section 22 and in the sections which use it.

A remark on the printed formulae: the second term of [9, (5.16), (5.37)] is printed as $\overline{\partial_\mu(p)}\partial_\nu(p)$, and the corresponding factor of [9, (5.33), (5.35), (5.36)] as $(z_\mu^{-1}-1)(z_\nu-1)$; [9, (4.41)] has the same form. The tensor which satisfies [9, (5.15), (5.19)], which the substitution of [9, (5.30)] into [9, (5.26)] gives, and which is the transform [9, (5.11)] of the first term of [9, (5.43)], has the second term $\partial_\mu(p)\overline{\partial_\nu(p)}$, as above. The two expressions differ by terms of the third order, so that the printed formulae hold with another remainder and [9, (5.42)] is not affected; but with the printed term the remainder Π' does not satisfy [9, (5.15)], contrary to the sentence which follows [9, (5.37)].

31 The Three Couplings of the Tilted Run

Section 30 ended with a difficulty: the vacuum polarisation of a step of the tilted run has three coefficients, the action has one, and two quadratic terms of the dimension of the action are left in the effective action. We now ask what a renormalisation with three coefficients looks like and what has to be proved about them. The first question is whether there are terms which can absorb the three coefficients separately, and there are: the sums of the plaquette terms of the turned lattice over the three classes of planes, evaluated at the background field, have the three forms T^{a} as their second order (Lemma 31.1). With them Bałaban's expression for the effective action becomes one with a coupling and two further numbers, the accumulated differences of the coefficients, and the inductive description requires that these numbers stay bounded; it does not require that the differences be summable. The second question is how the two numbers evolve. Their equation has three terms, of which the first is a source which does not depend on them, the second is their own effect on the coefficients of the next step, and the third collects the higher orders, and Lemma 31.3 says which properties of the three give a bound. The two lemmata and the identities between them are proved here. The bound on the two numbers is not, and the last paragraphs of the section say what is missing for each of the three terms.

The actions of the three classes. A plaquette of T_η lies in a plane spanned by two of the vectors e'_κ, e'_λ, and we assign it to the class of the plane $\{\kappa,\lambda\}$ in the sense of Section 30: the class of $\{0,1\}$, the class of $\{2,3\}$, and the mixed class. An element of $\mathfrak{G}_0$ maps the span of e_0, e_1 and the span of e'_2, e'_3 to themselves and permutes the vectors $\pm e'_\nu$ (Lemma 5.1), so that it maps the plaquettes of each class to plaquettes of the same class. For a configuration U on T_η and a field A on the bonds of T_η we put

$$\mathcal{A}^{\mathsf{a}}(U) \coloneqq \sum_{p\in\mathsf{a}} (1 - \operatorname{Re}\operatorname{tr} U(\partial p)) \;, \qquad \|\partial A\|^2_{\mathsf{a}} \coloneqq \sum_{p\in\mathsf{a}} \eta^4 |(\partial A)(p)|^2 \;, \tag{31.1}$$

the sums over the plaquettes of the class a. Each $\mathcal{A}^{\mathsf{a}}$ is invariant under the gauge transformations and under the elements of the group $\mathfrak{G}$ of Section 30; the sum $\mathcal{A} \coloneqq \sum_{\mathsf{a}} \mathcal{A}^{\mathsf{a}}$ is the action of the lattice T_η, which is the function $A^\eta(U)$ of [9, (0.21), p. 256]; and $\sum_{\mathsf{a}} \|\partial A\|^2_{\mathsf{a}} = \|\partial A\|^2$. In the momentum representation of Lemma 30.2 the form $\|\partial A\|^2_{\mathsf{a}}$ has the symbol

$$\langle a, \mathcal{D}^{\mathsf{a}}(p) a\rangle \coloneqq \sum_{\{\kappa,\lambda\}\in\mathsf{a}} |\partial'_\kappa(p) a_\lambda - \partial'_\lambda(p) a_\kappa|^2 \;, \tag{31.2}$$

where $a = (a_\nu) \in \mathbb{C}^4$ are the components of a plane wave in the directions e'_ν, so that $\sum_{\mathsf{a}} \mathcal{D}^{\mathsf{a}} = \mathcal{D}$. For $p \in \mathbb{R}^4$ and $b \in \mathbb{C}^4$ we write

$$\mathsf{t}^{\mathsf{a}}(p,b) \coloneqq \sum_{\{\kappa,\lambda\}\in\mathsf{a}} |p_\kappa b_\lambda - p_\lambda b_\kappa|^2 \;, \tag{31.3}$$

with the components of p and b in the basis e_μ. By the formula for $T^{\mathsf{a}}(p)$ which follows (30.2), $\sum_{\mu,\nu} \bar{b}_\mu T^{\mathsf{a}}_{\mu\nu}(p) b_\nu = \sum_{\{\kappa,\lambda\}\in\mathsf{a}} |\partial_\kappa(p) b_\lambda - \partial_\lambda(p) b_\kappa|^2$, which is $\mathsf{t}^{\mathsf{a}}(p,b) + O(|p|^3)|b|^2$ because $\partial_\kappa(p) = ip_\kappa + O(|p|^2)$.

Lemma 31.1 (The Second Order of the Three Actions)**.** *Let $32\eta \le \frac{1}{2}$, $p' \in [-\pi,\pi]^4$, and $b \in \mathbb{C}^4$. As in Lemma 30.2, consider the families $a = (a_l)$ of vectors of $\mathbb{C}^4$ with $\sum_l u_k(p'+l)V(p'+l)a_l = b$, and on them the functions $\mathfrak{N}(a) := \sum_l \langle a_l, \mathcal{D}(p'+l)a_l\rangle$ and $\mathfrak{N}^{\mathsf{a}}(a) := \sum_l \langle a_l, \mathcal{D}^{\mathsf{a}}(p'+l)a_l\rangle$.*

(i) The minimum of $\mathfrak{N}$ over these families is attained, and $\mathfrak{N}^{\mathsf{a}}$ has the same value at all the families at which it is attained. We denote this value by $\langle b, \Delta^{\mathsf{a}}_k(p')b\rangle$. Then $\sum_{\mathsf{a}} \langle b, \Delta^{\mathsf{a}}_k(p')b\rangle = \langle b, \Delta_k(p')b\rangle$.

(ii) There is a number C, which does not depend on k, L, p', and b, such that

$$|\langle b, \Delta^{\mathsf{a}}_k(p')b\rangle - \mathsf{t}^{\mathsf{a}}(p',b)| \le C|p'|^{5/2}|b|^2 \,. \tag{31.4}$$

Proof. The families form a space of finite dimension, since l runs through η^{-4} classes, and $\mathfrak{N}$ and $\mathfrak{N}^{\mathsf{a}}$ are the values on the diagonal of Hermitian forms $\mathfrak{N}(a,a')$ and $\mathfrak{N}^{\mathsf{a}}(a,a')$ which are positive semidefinite, with $\mathfrak{N}^{\mathsf{a}}(a) \le \mathfrak{N}(a)$, because $\mathcal{D}^{\mathsf{a}}$ consists of a part of the squares which make up $\mathcal{D}$. We call a family *admissible* if it satisfies the condition with the given b; the admissible families form an affine space, which is not empty by the proof of Lemma 30.2, and its differences form the space $\mathfrak{F}_0$ of the families which satisfy the condition with $b = 0$.

(i) Let $\mathfrak{K}$ be the space of the $n \in \mathfrak{F}_0$ with $\mathfrak{N}(n) = 0$. Schwarz's inequality $|\mathfrak{N}(a,n)|^2 \le \mathfrak{N}(a)\mathfrak{N}(n)$ gives $\mathfrak{N}(a+n) = \mathfrak{N}(a)$ for $n \in \mathfrak{K}$. On a complement of $\mathfrak{K}$ in $\mathfrak{F}_0$ the form $\mathfrak{N}$ is positive definite, so that on the translate of this complement by an admissible family the function $\mathfrak{N}$ is a polynomial of the second degree with a positive definite leading part; it attains its minimum there, and this is its minimum over all admissible families. Let m be a family at which the minimum is attained. For $d \in \mathfrak{F}_0$ and a complex number s, $\mathfrak{N}(m+sd) \ge \mathfrak{N}(m)$, which is possible for all s only if $\mathfrak{N}(m,d) = 0$. Hence, for every admissible family a,

$$\mathfrak{N}(a) = \mathfrak{N}(m) + \mathfrak{N}(a-m) \,. \tag{31.5}$$

If m' is a second family at which the minimum is attained, (31.5) gives $\mathfrak{N}(m'-m) = 0$, hence $\mathfrak{N}^{\mathsf{a}}(m'-m) = 0$, and Schwarz's inequality for $\mathfrak{N}^{\mathsf{a}}$ gives $\mathfrak{N}^{\mathsf{a}}(m') = \mathfrak{N}^{\mathsf{a}}(m)$. The sum over the classes is $\mathfrak{N}(m)$, the minimum which defines $\langle b, \Delta_k(p')b\rangle$ in Lemma 30.2.

(ii) We write $O(t)$ as in the proof of Lemma 30.2, and we use the family a° of the upper bound in that proof: $a^\circ_0 := (u_k(p')V(p'))^{-1}b$ and $a^\circ_l := 0$ for $l \ne 0$. By that proof and by (30.7), both $\mathfrak{N}(a^\circ)$ and $\mathfrak{N}(m)$ differ from $|p'|^2|b|^2 - |p'\cdot b|^2$ by $O(|p'|^3)|b|^2$, so that (31.5) with $a = a^\circ$ gives

$$\mathfrak{N}(a^\circ - m) = \mathfrak{N}(a^\circ) - \mathfrak{N}(m) = O(|p'|^3)|b|^2 \,, \qquad \mathfrak{N}(a^\circ) = O(|p'|^2)|b|^2 \,. \tag{31.6}$$

Since $m = a^\circ + (m - a^\circ)$, the Hermitian form $\mathfrak{N}^{\mathsf{a}}$ satisfies $\mathfrak{N}^{\mathsf{a}}(m) - \mathfrak{N}^{\mathsf{a}}(a^\circ) = 2\,\mathrm{Re}\,\mathfrak{N}^{\mathsf{a}}(a^\circ, m - a^\circ) + \mathfrak{N}^{\mathsf{a}}(m-a^\circ)$, and by Schwarz's inequality, by $\mathfrak{N}^{\mathsf{a}} \le \mathfrak{N}$, and by (31.6),

$$|\mathfrak{N}^{\mathsf{a}}(m) - \mathfrak{N}^{\mathsf{a}}(a^\circ)| \le 2\,\mathfrak{N}(a^\circ)^{1/2}\,\mathfrak{N}(a^\circ - m)^{1/2} + \mathfrak{N}(a^\circ - m) = O(|p'|^{5/2})|b|^2 \,. \tag{31.7}$$

It remains to compute $\mathfrak{N}^{\mathsf{a}}(a^\circ) = \langle a^\circ_0, \mathcal{D}^{\mathsf{a}}(p')a^\circ_0\rangle$. Let $\mathbf{a}^\circ_0 := \sum_\nu (a^\circ_0)_\nu e'_\nu$. By the first two facts in the proof of Lemma 30.2, $\mathbf{a}^\circ_0 = b + O(|p'|)|b|$ and $\partial'_\nu(p') = i(p'\cdot e'_\nu)(1 + O(\eta|p'|))$. For vectors $v, v' \in \mathbb{R}^4$ let $F(v,v') := (p'\cdot v)(b\cdot v') - (p'\cdot v')(b\cdot v)$, with the bilinear product of $\mathbb{C}^4$, so that

$F(e_\kappa, e_\lambda) = p'_\kappa b_\lambda - p'_\lambda b_\kappa$. Replacing $\partial'_\nu(p')$ by $i(p' \cdot e'_\nu)$ and $(a^\circ_0)_\nu = \mathbf{a}^\circ_0 \cdot e'_\nu$ by $b \cdot e'_\nu$ in (31.2) changes each of the differences by $O(|p'|^2)|b|$, and each of them is $O(|p'|)|b|$; hence

$$\mathfrak{N}^{\mathsf{a}}(a^\circ) = \sum_{\{\kappa,\lambda\}\in\mathsf{a}} |F(e'_\kappa, e'_\lambda)|^2 + O(|p'|^3)|b|^2 \,. \tag{31.8}$$

For the class of $\{0,1\}$ the sum is $|F(e_0, e_1)|^2$, because $e'_0 = e_0$ and $e'_1 = e_1$. For the class of $\{2,3\}$ the function F is bilinear and antisymmetric, so that $F(e'_2, e'_3)$ is $F(e_2, e_3)$ times the determinant of the matrix which expresses e'_2, e'_3 by e_2, e_3, and this determinant is $\frac{3}{5} \cdot (-\frac{3}{5}) - \frac{4}{5} \cdot \frac{4}{5} = -1$ by (2.2). For the mixed class and each $a \in \{0,1\}$, the map $v \mapsto F(e_a, v)$ is a linear function on the plane of e_2 and e_3, and the sum of the squared moduli of its values on an orthonormal basis of the plane does not depend on the basis; hence $\sum_{j\in\{2,3\}} |F(e_a, e'_j)|^2 = \sum_{j\in\{2,3\}} |F(e_a, e_j)|^2$. In all three cases the sum in (31.8) is $\mathsf{t}^{\mathsf{a}}(p', b)$, and with (31.7) this is (31.4). *q.e.d.*

The value $\langle b, \Delta^{\mathsf{a}}_k(p')b\rangle$ is a Hermitian quadratic form in b, since a family at which the minimum is attained can be chosen as a linear function of b, in the translate of a fixed complement of $\mathfrak{K}$; we write $\Delta^{\mathsf{a}}_k(p')$ for its matrix as well. It is the momentum representation of the form $B \mapsto \|\partial H_k B\|^2_{\mathsf{a}}$ on the fields of the unit lattice, because $H_k B$ is a field at which the minimum of $\|\partial A\|^2$ under the condition $Q_k A = B$ is attained (Lemma 17.1(i)). This form is the second order of the function $B \mapsto \mathcal{A}^{\mathsf{a}}(U_k(\exp iB))$ at $B = 0$ in the same way in which $\langle B, \Delta_k B\rangle$ is that of the action: the computation [9, (4.39), (4.40), p. 291] is made plaquette by plaquette, and the sum over the plaquettes is restricted to the class.

Corollary 31.2 (The Polarisation after the Subtraction of the Three Actions)**.** *Let* $32\eta \le \frac{1}{2}$*, and let* Π *satisfy the hypotheses of Proposition 30.1, with the bound* E*, the rate* δ_1*, and the coefficients* β^{a}*. There is a number* C*, which depends on* δ_1 *and not on* k*,* L*, and* Π*, such that for every* $p \in [-\pi, \pi]^4$ *the norm of the matrix* $\Pi(p) - \sum_{\mathsf{a}} \beta^{\mathsf{a}} \Delta^{\mathsf{a}}_k(p)$ *is at most* $CE|p|^{5/2}$*.*

Proof. By (30.3), $\Pi - \sum_{\mathsf{a}} \beta^{\mathsf{a}} \Delta^{\mathsf{a}}_k = \Pi' + \sum_{\mathsf{a}} \beta^{\mathsf{a}}(T^{\mathsf{a}} - \Delta^{\mathsf{a}}_k)$. Each entry of $\Pi'(p)$ is a sum of products of three of the functions $\partial_\kappa(p)$, $\overline{\partial_\kappa(p)}$, which are at most $|p_\kappa|$ in modulus, with coefficients bounded by a constant times E at real p, so that $\|\Pi'(p)\| \le CE|p|^3$. The matrices $T^{\mathsf{a}}(p)$ and $\Delta^{\mathsf{a}}_k(p)$ are Hermitian, the norm of a Hermitian matrix is the supremum of the modulus of its quadratic form on the unit vectors, and by the remark which precedes Lemma 31.1 and by (31.4) the quadratic forms of the two differ by $O(|p|^{5/2})|b|^2$. Finally $|\beta^{\mathsf{a}}| \le E \sum_x |x|^2 e^{-\delta_1|x|_1}$ by the definition (30.3). *q.e.d.*

Thus the subtraction of $\sum_{\mathsf{a}} \beta^{\mathsf{a}} \mathcal{A}^{\mathsf{a}}$ removes the terms of the orders zero, one, and two of the vacuum polarisation at the momentum zero, as the subtraction of $\beta\mathcal{A}$ does when the three coefficients agree. The exponent $\frac{5}{2}$ is what the comparison of the two minimum problems gives without further information; on a torus the kernel of the form $\|\partial H_k B\|^2_{\mathsf{a}}$ decays exponentially, by Corollary 17.2, and for a smooth symbol the bound (31.4) implies the same bound with the exponent 3. We do not use this.

The effective action with three couplings. Bałaban writes the effective action after k steps as

$$A_k(U_k) = -\frac{1}{g_k^2}\,\mathcal{A}(U_k) + \sum_{j=0}^{k-1} \Big\{ -\beta_{j+1}(g_j)\,\mathcal{A}(U_k) + [\mathbf{E}^{(j+1)}(g_j, U_k) - \mathbf{E}^{(j+1)}(g_j, 1)] \Big\} \tag{31.9}$$

[9, (1.6), p. 261]. Here $g^{-2}_{j+1} = g^{-2}_j - \beta_{j+1}(g_j)$ [9, (0.20), p. 256], so that the terms with $\mathcal{A}$ add up to $-g_0^{-2}\mathcal{A}(U_k)$, and β_{j+1} is the coefficient [9, (1.22), p. 264] of the vacuum polarisation

of $\mathbf{E}^{(j+1)}$. The braces are what the induction estimates: in each of them the term with $\mathcal{A}$ cancels the part of the second order of $\mathbf{E}^{(j+1)}$ which has the dimension of the action, together with the parts of the third and the fourth order which the gauge invariance ties to it [9, (4.38)–(4.45), pp. 291–292, and p. 298]. For the tilted run let β^{a}_{j+1} be the three coefficients of the vacuum polarisation of $\mathbf{E}^{(j+1)}$ (Proposition 30.1), and put

$$\bar\beta_{j+1} \coloneqq \tfrac{1}{6}(\beta^{01}_{j+1} + \beta^{23}_{j+1} + 4\beta^{\mathrm{mix}}_{j+1}) \;, \qquad \frac{1}{g^2_{j+1}} \coloneqq \frac{1}{g^2_j} - \bar\beta_{j+1} \;, \qquad \zeta^{\mathsf{a}}_k \coloneqq -\sum_{j=0}^{k-1} (\beta^{\mathsf{a}}_{j+1} - \bar\beta_{j+1}) \;, \tag{31.10}$$

the mean of the coefficients over the six planes, the coupling which it defines, and the accumulated deviations of the three coefficients from the mean. Since $\sum_{\mathsf{a}} \mathcal{A}^{\mathsf{a}} = \mathcal{A}$, we have, for every configuration,

$$\begin{aligned} &-\frac{1}{g_0^2}\,\mathcal{A}(U_k) + \sum_{j=0}^{k-1} [\mathbf{E}^{(j+1)}(U_k) - \mathbf{E}^{(j+1)}(1)] \\ &\quad = -\frac{1}{g_k^2}\,\mathcal{A}(U_k) - \sum_{\mathsf{a}} \zeta^{\mathsf{a}}_k\, \mathcal{A}^{\mathsf{a}}(U_k) + \sum_{j=0}^{k-1} \Big\{ -\sum_{\mathsf{a}} \beta^{\mathsf{a}}_{j+1}\, \mathcal{A}^{\mathsf{a}}(U_k) + [\mathbf{E}^{(j+1)}(U_k) - \mathbf{E}^{(j+1)}(1)] \Big\} \;. \end{aligned} \tag{31.11}$$

Indeed the terms with the actions on the right are $-(g_0^{-2} - \sum_j \bar\beta_{j+1})\mathcal{A} + \sum_j \sum_{\mathsf{a}} (\beta^{\mathsf{a}}_{j+1} - \bar\beta_{j+1})\mathcal{A}^{\mathsf{a}} - \sum_j \sum_{\mathsf{a}} \beta^{\mathsf{a}}_{j+1} \mathcal{A}^{\mathsf{a}}$, and the terms with $\bar\beta_{j+1}$ cancel. The three numbers ζ^{a}_k satisfy $\zeta^{01}_k + \zeta^{23}_k + 4\zeta^{\mathrm{mix}}_k = 0$, so that two of them are independent, and

$$\zeta^{\mathsf{a}}_0 = 0 \;, \qquad \zeta^{\mathsf{a}}_{k+1} = \zeta^{\mathsf{a}}_k - (\beta^{\mathsf{a}}_{k+1} - \bar\beta_{k+1}) \;. \tag{31.12}$$

The identity (31.11) is the form which (31.9) takes for the tilted run. Its braces are those of Bałaban with three terms in the place of one, and by Corollary 31.2 the vacuum polarisation of each of them has no term of the orders zero, one, and two at the momentum zero; the identities [9, (4.39)–(4.45)] hold for each class separately, by Lemma 31.1 in the place of [9, (4.41)], so that the cancellation which was complete for one class of planes in Section 30 is now complete for all three. The price is the term $-\sum_{\mathsf{a}} \zeta^{\mathsf{a}}_k \mathcal{A}^{\mathsf{a}}(U_k)$. It is not small in the sense in which the braces are: it has the dimension of the action and does not decrease from one step to the next. It is, however, a single explicit term. In the fluctuation integral of the next step it enters as the term [9, (4.38)] does, with ζ^{a}_k in the place of β_j and $\mathcal{A}^{\mathsf{a}}$ in the place of the action, and it has no partner which cancels its part of the second order in the fluctuation field. This part is $\sum_{\mathsf{a}} \zeta^{\mathsf{a}}_k$ times the form of the class a, next to g_k^{-2} times the sum of the three forms, so that it changes the quadratic form of the fluctuation integral by at most the fraction

$$g_k^2 \max_{\mathsf{a}} |\zeta^{\mathsf{a}}_k| \tag{31.13}$$

of the form itself. A quadratic term of the exponent which is a small multiple of the Gaussian form is a perturbation of the kind which the paper absorbs into the Gaussian integral (Paper, Lemma 11.7). What the inductive description needs from the three coefficients is therefore that (31.13) stays below a fixed small number at all the steps, uniformly in the number of the steps. Since the couplings are bounded, a bound on the numbers ζ^{a}_k which is uniform in k and in the number of the steps suffices. We have not carried out the induction of [9] and [10] with the additional term; what is proved in this paragraph is the identity (31.11) and the statement on the braces.

The equation of the two numbers. By (31.12) the numbers ζ^{a}_k are the partial sums of the deviations $\beta^{\mathsf{a}}_j - \bar\beta_j$, and the question is what bounds these partial sums. A bound on

each deviation by a constant is of no use, and neither is the bound by a constant times g_{j-1}^2 which holds for the part of a coefficient beyond its value at the coupling zero (Paper, Proposition 4.3), since the sum of the squared couplings over the steps grows like the logarithm of their number. The deviations at the step $k+1$ depend on the numbers ζ_k^{a} themselves, because the term $\sum_{\mathsf{a}} \zeta_k^{\mathsf{a}} \mathcal{A}^{\mathsf{a}}$ takes part in the fluctuation integral of that step, and its contribution to the coefficients is of the first order in the fraction (31.13). One is thus led to write the deviation at the step $k+1$ as the sum of its value at the coupling zero and at $\zeta_k = 0$, of a term which is linear in $g_k^2 \zeta_k$, and of a remainder. The next lemma treats a recursion of this form for one number. It shows that everything depends on the sign of the linear term.

Lemma 31.3 (A Damped Recursion)**.** *Let $n \geq 1$, and for $1 \leq j \leq n$ let $\vartheta_j \in [0,1]$ and let f_j and ϱ_j be real numbers. Let $z_0 := 0$ and $z_j := (1-\vartheta_j) z_{j-1} - f_j - \varrho_j$ for $1 \leq j \leq n$, and let $S \geq 0$ be such that $|\sum_{j=m}^{m'} f_j| \leq S$ for all $1 \leq m \leq m' \leq n$.*

(i) For every $k \leq n$, $|z_k| \leq S + \sum_{j \leq k} |\varrho_j|$.

(ii) If $|\varrho_j| \leq \nu_j |z_{j-1}| + \tau_j$ with numbers $\nu_j, \tau_j \geq 0$, then $|z_k| \leq (S + \sum_{j \leq k} \tau_j) \prod_{j \leq k} (1+\nu_j)$ for every $k \leq n$.

Proof. (i) Fix k and let $\varpi_j := \prod_{i=j+1}^{k} (1-\vartheta_i)$ for $1 \leq j \leq k$, the empty product being 1. By induction on k the recursion gives $z_k = -\sum_{j=1}^{k} \varpi_j (f_j + \varrho_j)$. The weights satisfy $0 \leq \varpi_1 \leq \varpi_2 \leq \cdots \leq \varpi_k = 1$, because every factor lies in $[0,1]$. Let $T_j := \sum_{i=j}^{k} f_i$ and $T_{k+1} := 0$, so that $f_j = T_j - T_{j+1}$ and $|T_j| \leq S$. Summation by parts gives

$$\sum_{j=1}^{k} \varpi_j f_j = \varpi_1 T_1 + \sum_{j=2}^{k} (\varpi_j - \varpi_{j-1}) T_j \,, \tag{31.14}$$

and since the weights are nonnegative and increase, the right side is at most $S(\varpi_1 + \varpi_k - \varpi_1) = S$ in modulus. The terms with ϱ_j are bounded by $|\varrho_j|$, because $\varpi_j \leq 1$.

(ii) Fix k, let $c := S + \sum_{j \leq k} \tau_j$, and for $i \leq k$ let $v_i := c + \sum_{j \leq i} \nu_j |z_{j-1}|$. By (i) and the hypothesis, $|z_i| \leq v_i$ for $i \leq k$. Hence $v_i = v_{i-1} + \nu_i |z_{i-1}| \leq (1+\nu_i) v_{i-1}$, and $v_0 = c$, so that $|z_k| \leq v_k \leq c \prod_{j \leq k} (1+\nu_j)$. *q.e.d.*

The lemma uses the sources f_j only through their partial sums, which is what the coefficients at the coupling zero offer (see below), and it uses the sign of the numbers ϑ_j: for negative ϑ_j the weights ϖ_j exceed 1 and the bound of (i) fails, the solution of the recursion with one source being that source multiplied by the product of the factors $1-\vartheta_i$ of the later steps. If the linear term is known only through a bound on its modulus, it has to be counted in ϱ_j, with ν_j a constant times g_{j-1}^2, and the product in (ii) grows like a power of the number of the steps; such a bound does not keep (31.13) small. If it damps at a rate proportional to the squared coupling, on the other hand, a remainder which is bounded by a constant times the squared coupling, and by nothing better, is harmless: its contribution from the step j reaches the step k with a weight which is a power of the ratio of the two couplings, and these weights make the sum of the squared couplings converge.

The next lemma makes this precise, in the form which the problem has. There are two numbers, and the linear term is a matrix; what the coefficients at the coupling zero offer for it is, as for the sources, a statement on sums over the steps, a positive multiple of the logarithm of the ratio of the scales plus a part whose partial sums are bounded; and the inverse squared couplings need not decrease at every step. What the paper proves for them

is a statement on sums as well: by Paper, Theorem 4.4 and Proposition 4.3(i), the decrease over the steps $j < i \le k$ is between $b(k-j) - C_W$ and $b'(k-j) + C_W$, with the numbers $b = 2c \ln L - \delta'$, $b' = 2c \ln L + \delta' \le \frac{3}{2} b$, and C_W of Paper, Theorem 4.1. We say that numbers $x_0, \dots, x_n$ satisfy the *window* with the constants $0 < b \le b' \le \frac{3}{2} b$ and $C_W \ge 0$ if

$$b(k-j) - C_W \le x_j - x_k \le b'(k-j) + C_W \tag{31.15}$$

for all $0 \le j \le k \le n$. The second part of the lemma is what the equality (1.1) requires. We use the maximum norm on $\mathbb{R}^d$ and the corresponding norm of matrices.

Lemma 31.4 (A Recursion Which Is Damped on Sums)**.** *Let $d, n \ge 1$, and let $x_0, \dots, x_n$ be numbers which satisfy the window* (31.15) *with the constants b, b', and C_W, and $x_j \ge x_\star > 0$ for every j. Let $\mathbf{D}$ be a diagonal $d \times d$ matrix with entries in $[D_{\min}, D_{\max}]$, $D_{\min} > 0$, and $\epsilon := D_{\min}/b'$. For $1 \le j \le n$ let δ_j be real $d \times d$ matrices and $s_j \in \mathbb{R}^d$, and let*

$$z_0 := 0 \,, \qquad z_j := z_{j-1} - \frac{(\mathbf{D} + \delta_j)\, z_{j-1}}{x_{j-1}} - s_j \,. \tag{31.16}$$

Suppose that $\|\delta_j\| \le \bar\delta$ and $\|\sum_{j=m}^{m'} \delta_j\| \le S_\delta$ for all $1 \le m \le m' \le n$, and put $A_0 := 9/(4D_{\min})$, $c_1 := 3(D_{\max} + \bar\delta + b' + C_W)$, and $x_\circ := \max(5C_W, 2(b' + C_W), 2D_{\max})$.

(a) Let $s_j = f_j + \varrho_j$ with $|f_j| \le \bar f$, $|\sum_{j=m}^{m'} f_j| \le S$ for all $1 \le m \le m' \le n$, and $|\varrho_j| \le C_\varrho / x_{j-1}$. If $x_\star \ge \max(x_\circ, 2S_\delta A_0 c_1)$, then for every $k \le n$

$$|z_k| \le 2\Big[S + A_0 (C_\varrho + S_\delta \bar f) + \frac{C_\varrho}{4b'} \Big] \,. \tag{31.17}$$

(b) Let $|s_j| \le C_s \theta^j$ with $0 < \theta < 1$. If $x_\star \ge \max(x_\circ, S_\delta/\theta, 18 \cdot 4^\epsilon S_\delta c_1 / b')$, then for every $k \le n$

$$|z_k| \le 4C_s \sum_{j=1}^{k} \Big(\frac{x_k + C_W + b'}{x_k + C_W + b'(k-j+1)} \Big)^\epsilon \theta^j \le \frac{4C_s}{1-\theta} \Big[\Big(\frac{x_k + C_W + b'}{x_k + \frac{1}{2} b' k} \Big)^\epsilon + \theta^{k/2} \Big] \,. \tag{31.18}$$

Proof. Fix $k \le n$, and for $m \ge 0$ let $y_m := x_k + C_W + b'm$ and $\omega_m := (y_1 / y_{m+1})^\epsilon$. We first collect what the window gives, for $x_\star \ge x_\circ$. For $j \le k$,

$$\tfrac{2}{3}\, y_{k-j} \le x_j \le y_{k-j} \,, \tag{31.19}$$

the upper bound by (31.15), and the lower one because $x_k - C_W \ge \frac{2}{3}(x_k + C_W)$ for $x_k \ge 5C_W$ and $b \ge \frac{2}{3} b'$. Two consecutive numbers differ by at most $b' + C_W \le \frac{1}{2} x_\star$, so that their quotient lies between $\frac{2}{3}$ and $\frac{3}{2}$, and $|1/x_{j-1} - 1/x_{j-2}| \le \frac{3}{2}(b' + C_W)/x_{j-1}^2$. Let ϖ_j, $1 \le j \le k$, be the diagonal matrix $\prod_{i=j+1}^{k} (1 - \mathbf{D}/x_{i-1})$. Since $x_{i-1} \ge 2D_{\max}$, its entries are positive and increase with j, and $\varpi_j - \varpi_{j-1} = \varpi_j \mathbf{D} / x_{j-1}$. Each entry is at most $\exp(-D_{\min} \sum_{i=j+1}^{k} 1/x_{i-1})$, and by (31.19) the sum is at least $\sum_{m=1}^{k-j} 1/y_m \ge (1/b') \ln(y_{k-j+1}/y_1)$, by comparison with the integral; hence

$$\|\varpi_j\| \le \omega_{k-j} \,, \qquad \sum_{j=1}^{k} \frac{\omega_{k-j}}{x_{j-1}} \le \frac{3}{2}\, y_1^\epsilon \sum_{m \ge 1} y_m^{-1-\epsilon} \le \frac{3}{2} \Big(\frac{1}{y_1} + \frac{1}{b'\epsilon} \Big) \le A_0 \,, \tag{31.20}$$

the second statement by $x_{j-1} \ge \frac{2}{3} y_{k-j+1}$, the comparison of the sum with its first term plus an integral, $b'\epsilon = D_{\min}$, and $y_1 \ge 2D_{\min}$. By induction on k the recursion gives

$$z_k = -\sum_{j=1}^{k} \varpi_j \, s_j - \sum_{j=1}^{k} \varpi_j \, \delta_j \, w_j \;, \qquad w_j \coloneqq \frac{z_{j-1}}{x_{j-1}} \;. \tag{31.21}$$

We shall also use that $\sum_{i \le m < k} x_m^{-2} \le \frac{9}{4} \sum_{r \ge 1} y_r^{-2} \le \frac{9}{4}(y_1^{-2} + (b'y_1)^{-1}) \le \frac{9}{2}(b'x_\star)^{-1}$, by (31.19) and $y_1 \ge b'$.

The second sum. Let $\Delta_j \coloneqq \sum_{i=j}^{k} \delta_i$ and $\Delta_{k+1} \coloneqq 0$, so that $\delta_j = \Delta_j - \Delta_{j+1}$ and $\|\Delta_j\| \le S_\delta$. Summation by parts, with $w_1 = 0$, gives

$$\sum_{j=1}^{k} \varpi_j \, \delta_j \, w_j = \sum_{j=2}^{k} [(\varpi_j - \varpi_{j-1}) \Delta_j w_j + \varpi_{j-1} \Delta_j (w_j - w_{j-1})] \;. \tag{31.22}$$

Here $\|\varpi_j - \varpi_{j-1}\| \le D_{\max} \omega_{k-j} / x_{j-1}$ and $\|\varpi_{j-1}\| \le \omega_{k-j}$, and

$$w_j - w_{j-1} = \frac{z_{j-1} - z_{j-2}}{x_{j-1}} + z_{j-2} \Big(\frac{1}{x_{j-1}} - \frac{1}{x_{j-2}} \Big) \;, \qquad |z_{j-1} - z_{j-2}| \le \frac{(D_{\max} + \bar\delta)|z_{j-2}|}{x_{j-2}} + |s_{j-1}| \;, \tag{31.23}$$

with $1/x_{j-2} \le \frac{3}{2}/x_{j-1}$ and the bound on $|1/x_{j-1} - 1/x_{j-2}|$ above. Hence the modulus of (31.22) is at most

$$S_\delta \sum_{j=2}^{k} \omega_{k-j} \Big[\frac{c_1 \max(|z_{j-1}|, |z_{j-2}|)}{x_{j-1}^2} + \frac{|s_{j-1}|}{x_{j-1}} \Big] \;, \tag{31.24}$$

since $D_{\max} + \frac{3}{2}(D_{\max} + \bar\delta) + \frac{3}{2}(b' + C_W) \le c_1$.

(a) Let $Z \coloneqq \max_{j<k} |z_j|$. In the first sum of (31.21) the terms with f_j add up to at most S in modulus, by the summation by parts of the proof of Lemma 31.3(i) applied to each component, the weights being increasing and at most 1; and the terms with ϱ_j add up to at most $C_\varrho A_0$, by (31.20). In (31.24), $|s_{j-1}| \le \bar f + \frac{3}{2} C_\varrho / x_\star$, and by (31.20) the expression is at most $S_\delta A_0 [\bar f + (c_1 Z + \frac{3}{2} C_\varrho)/x_\star]$. By the hypothesis on $x_\star$, $S_\delta A_0 c_1 / x_\star \le \frac{1}{2}$, and $\frac{3}{2} S_\delta A_0 / x_\star \le 3/(4c_1) \le 1/(4b')$, so that $|z_k| \le B + \frac{1}{2} Z$ with B the bracket of (31.17). By induction on k, $|z_k| \le 2B$.

(b) Let $E_k \coloneqq \sum_{j \le k} \omega_{k-j} \theta^j$, the sum in (31.18), and let E_i, for $i < k$, be the same sum with i in the place of k, in the numbers y as well; suppose that $|z_i| \le 4 C_s E_i$ for $i < k$. The first sum of (31.21) is at most $C_s E_k$, by (31.20). In (31.24) the terms with s_{j-1} give at most $S_\delta C_s \theta^{-1} x_\star^{-1} E_k$. For the others we compare the weights of the step $j-1$ with those of the step k: by (31.19) and $C_W + b' \le \frac{1}{2} x_\star$, the number $x_{j-1} + C_W + b'$ is at most $\frac{3}{2} y_{k-j+1}$ and $x_{j-1} + C_W + b'(j-i)$ is at least $\frac{2}{3} y_{k-i+1}$, so that the weight of the term i in E_{j-1} is at most $(\frac{9}{4})^\epsilon (y_{k-j+1}/y_{k-i+1})^\epsilon$, and its product with ω_{k-j} is at most $(\frac{9}{4})^\epsilon \omega_{k-i}$. For E_{j-2} the same holds with the further factor $(y_{k-j+2}/y_{k-j+1})^\epsilon \le (\frac{3}{2})^\epsilon$. Hence $\omega_{k-j} \max(E_{j-1}, E_{j-2}) \le 4^\epsilon \sum_{i<j} \omega_{k-i} \theta^i$, and exchanging the order of the summations and using the bound on the sums of x_m^{-2}, the terms with z_{j-1} and z_{j-2} give at most $4C_s \cdot 4^\epsilon S_\delta c_1 \cdot \frac{9}{2} (b' x_\star)^{-1} E_k$. By the hypothesis on $x_\star$ the two contributions are at most $C_s E_k$ each, and $|z_k| \le 3 C_s E_k$. This proves the first inequality of (31.18) by induction. For the second we split the sum at $j = k/2$: for $j \le k/2$ we have $y_{k-j+1} \ge x_k + \frac{1}{2} b' k$ and $\sum \theta^j \le (1-\theta)^{-1}$, and the terms with $j > k/2$, in which the weight is at most 1, add up to at most $\theta^{k/2} (1-\theta)^{-1}$. *q.e.d.*

In (a) nothing is assumed of the single matrices δ_j beyond a bound, and nothing of the remainder beyond its order; the bound does not depend on n. In (b) the right side tends to

zero as $k \to \infty$ with x_k bounded, at the rate $k^{-\epsilon}$: a source of the size θ^j at an early step has decayed by the factor $(x_k/x_j)^\epsilon$ when it reaches the step k, and a late one is small by itself.

The sources: the coefficients at the coupling zero. At the coupling zero the term of a step is the logarithm of a Gaussian integral, and for its own run the paper shows that the sum of the coefficients b_j of the steps $m < j \le n$ differs from $2c(n-m)\ln L$ by at most a constant, where c is the coefficient of the logarithm of the continuum theory (Paper, Theorem 4.4). The proof, in the section of the Supplement on the coefficients at zero coupling, rests on an identity of linear algebra: the sum of the logarithms of the Gaussian integrals of k steps is the difference $F_k - F_0$ of a functional of the level k and of a functional of the bare lattice, up to terms of the current. The two sides are compared at one momentum $p_0 \neq 0$ of the lattice of the level k, where the left side is $\Phi_k \Delta_k(p_0)$ plus a bounded term, with $\Phi_k := \sum_{j \le k} b_j$, and the right side is $c \ln(1/\eta^2) \Delta_k(p_0)$ plus a bounded term, with $\eta = L^{-k}$. For the tilted run the left side has three coefficients $\Phi^{\mathsf{a}}_k := \sum_{j \le k} b^{\mathsf{a}}_j$, and one momentum does not separate them. Three momenta do, by Lemma 31.1.

Proposition 31.5 (Three Coefficients from Three Momenta)**.** *Let* $32\eta \le \frac{1}{2}$*, let* C_0 *be the number* C *of* (31.4)*, let* $0 < s \le \min(\pi, (6C_0)^{-2})$*, and let*

$$(p_1, b_1) := (se_0, e_1) \,, \qquad (p_2, b_2) := (se_2, e_3) \,, \qquad (p_3, b_3) := (se_0, e_2) \,. \tag{31.25}$$

If three real numbers x^{a}*, one for each class, and a number* $y \ge 0$ *satisfy* $|\sum_{\mathsf{a}} x^{\mathsf{a}} \langle b_i, \Delta^{\mathsf{a}}_k(p_i) b_i \rangle| \le y$ *for* $i = 1, 2, 3$*, then* $|x^{\mathsf{a}}| \le 2y/s^2$ *for every class.*

Proof. We number the classes so that the first is that of $\{0, 1\}$, the second that of $\{2, 3\}$, and the third the mixed class. By (31.3), $\mathsf{t}^{\mathsf{a}}(p_i, b_i)$ is s^2 if a is the i-th class and 0 otherwise: for (se_0, e_1) the only difference $p_\kappa b_\lambda - p_\lambda b_\kappa$ which does not vanish is that of the plane $\{0, 1\}$, for (se_2, e_3) that of the plane $\{2, 3\}$, and for (se_0, e_2) that of the plane $\{0, 2\}$. By (31.4) the matrix $M_{i\mathsf{a}} := \langle b_i, \Delta^{\mathsf{a}}_k(p_i) b_i \rangle$ is therefore $s^2(\mathbf{1} + N)$ with $|N_{i\mathsf{a}}| \le C_0 s^{1/2}$, so that the sums of the moduli of the entries of the rows of N are at most $3C_0 s^{1/2} \le \frac{1}{2}$. For the maximum $|x|_\infty$ of the three numbers $|x^{\mathsf{a}}|$ this gives $y \ge s^2(|x|_\infty - \frac{1}{2}|x|_\infty)$. *q.e.d.*

Suppose now that for the tilted run the two sides of the identity are known at the three momenta (31.25), in the form

$$\sum_{\mathsf{a}} \Phi^{\mathsf{a}}_k \, \langle b_i, \Delta^{\mathsf{a}}_k(p_i) b_i \rangle = c \ln \frac{1}{\eta^2} \, \langle b_i, \Delta_k(p_i) b_i \rangle + O(1) \tag{31.26}$$

with a bound on the last term which does not depend on k. Since $\Delta_k = \sum_{\mathsf{a}} \Delta^{\mathsf{a}}_k$ and $\ln(1/\eta^2) = 2k \ln L$, Proposition 31.5 with $x^{\mathsf{a}} = \Phi^{\mathsf{a}}_k - 2ck \ln L$ gives a constant C_W with $|\Phi^{\mathsf{a}}_k - 2ck \ln L| \le C_W$ for all k and all three classes. The differences $\Phi^{\mathsf{a}}_{m'} - \Phi^{\mathsf{a}}_{m-1}$ of the three classes then differ from one number by at most $2C_W$ each, and so does their mean, so that the partial sums of the deviations $b^{\mathsf{a}}_j - \bar{b}_j$ over $m \le j \le m'$ are at most $4C_W$ in modulus. This is the hypothesis on the sources in Lemma 31.3, with $S = 4C_W$; the deviations at the single steps need not be small, and we know no reason for which they should be summable in modulus without an examination of the dependence of the Gaussian integrals on the ratio of the two spacings.

The statement (31.26) is not proved for the tilted run. It consists of four statements, which stand as follows. *The identity.* The decomposition of the Gaussian integrals of k steps (Supplement, Proposition "The Exact Decomposition" and its corollary for one step) is proved for linear maps D, h, Q on spaces of finite dimension which satisfy the relations listed there, of which the one that concerns the averages is $QD = \bar{D}Q'$; it involves no lattice, and it

holds for the tilted run as soon as its averages satisfy this relation at the configurations of the background, which is the covariance of the averages. *The left side.* The form $\sum_{\mathsf{a}} \Phi_k^{\mathsf{a}} \Delta_k^{\mathsf{a}}$ plus a bounded term is obtained in the Supplement, for one coefficient, from the representation of the term of a step as β_j times the action plus terms which decrease by a power of the ratio of the scales from one step to the next [9, (0.28)–(0.30), p. 258]. For the tilted run this representation is the result of Sections 3 to 5 of [9] with the three subtractions of (31.11); Corollary 31.2 is the part of it in which the symmetries enter, and the rest has not been written out (Section 34). *The functional of the level k and the terms of the current.* The Supplement bounds them through the propagators of [5] near the trivial configuration and their derivatives with respect to the background. It uses the symmetries of the lattice at one place, to show that a matrix $T(p)$ which annihilates the vector $(e^{ip_\nu} - 1)_\nu$ and is covariant vanishes to the second order at $p = 0$ (Supplement, Lemma "Second-Order Vanishing"); this holds under $\mathfrak{G}_0$, since $T(0) = 0$ follows from the first condition alone, by differentiation at $p = 0$, and an entry of the gradient at $p = 0$ has three indices, among which the index 0, the index 1, or the indices of $\{2, 3\}$ together occur an odd number of times, so that the reflection of x_0, that of x_1, or the map $z \mapsto -z$ changes its sign. The bounds on the propagators themselves are those of Section 27, with the degree of detail stated in Section 34, and we have not gone through this part of the Supplement line by line. *The bare functional.* Here the turned lattice enters in a way which can be settled, and we do so.

Lemma 31.6 (The Curl of the Minimiser at the Aliases)**.** *Let* $32\eta \le \frac{1}{2}$, $p' \in [-\pi, \pi]^4 \setminus \{0\}$, *and* $b \in \mathbb{C}^4$, *and let* $m = (m_l)$ *be a family at which the minimum of Lemma 31.1 is attained. For a class* l *let* $\hat{q}_l$ *be the representative of* $p' + l$ *modulo* $2\pi\eta^{-1}\rho\mathbb{Z}^4$ *with* $|\eta\hat{q}_l \cdot e'_\nu| \le \pi$ *for every* ν. *There is a number* C, *which does not depend on* k, L, p', *and* b, *such that for every* l

$$\langle m_l, \mathcal{D}(p' + l) m_l \rangle \le C\, |u_k(p' + l)|^2\, \frac{|p'|^4}{|\hat{q}_l|^2}\, |b|^2 \,, \tag{31.27}$$

and

$$\sum_l (1 + |\ln|\hat{q}_l|^2|) \langle m_l, \mathcal{D}(p' + l) m_l \rangle \le C(1 + |\ln|p'||)|p'|^2 |b|^2 \,. \tag{31.28}$$

Proof. We construct one family at which the minimum is attained. Let $\partial^1 := \partial^1(p')$ and $\phi := \phi(p')$, the matrix (7.9), which is Hermitian and satisfies $\gamma_0 \Delta_0(p')^{-1} \le \phi \le \gamma_1 \Delta_0(p')^{-1}$ by Lemma 7.3. Let P_ϕ be the orthogonal projection onto the vector $\phi^{-1/2}\partial^1$, and

$$j := \phi^{-1/2}(1 - \mathsf{P}_\phi)\phi^{-1/2} b \,, \qquad \varsigma := \frac{\langle \partial^1, \phi^{-1} b \rangle}{\langle \partial^1, \phi^{-1} \partial^1 \rangle} \,, \tag{31.29}$$

so that $\phi j + \varsigma \partial^1 = b$, the vector j is orthogonal to ∂^1, and $|j| \le \|\phi^{-1}\| |b| \le \gamma_0^{-1} \Delta_0(p')|b| \le \gamma_0^{-1}|p'|^2|b|$. The symbol $\Delta(q)$ vanishes only if $\eta q \cdot e'_\nu \in 2\pi\mathbb{Z}$ for every ν, that is, for $q \in 2\pi\eta^{-1}\rho\mathbb{Z}^4$. This group is contained in $2\pi\mathbb{Z}^4$ (Lemma 7.1(iv)), and so is the set of the l, while the only point of $[-\pi, \pi]^4$ in $2\pi\mathbb{Z}^4$ is 0; hence $p' + l$ does not lie in it, and $\Delta(p' + l) > 0$ for every class. We put $c_l := \overline{u_k(p' + l)}\, V(p' + l)^* j$ and

$$a_l := \Delta(p' + l)^{-1} c_l \quad \text{for } l \ne 0 \,, \qquad a_0 := \Delta(p')^{-1} c_0 + \varsigma\, u_k(p')^{-1} \partial'(p') \,. \tag{31.30}$$

By the definition (7.9) of ϕ and by Lemma 7.1(iii), $\sum_l u_k(p' + l) V(p' + l) a_l = \phi j + \varsigma \partial^1 = b$, so that the family is admissible. As in the proof of Lemma 30.2, c_l is orthogonal to $\partial'(p' + l)$, and $\mathcal{D}(p) = \Delta(p)(1 - \mathsf{P}(p))$ with the projection $\mathsf{P}(p)$ onto $\partial'(p)$; hence $\mathcal{D}(p' + l) a_l = c_l$ for every l. For a family d of the space $\mathfrak{F}_0$ this gives $\mathfrak{N}(a, d) = \sum_l \langle c_l, d_l \rangle = \langle j, \sum_l u_k(p' + l) V(p' + l) d_l \rangle = 0$, so that $\mathfrak{N}(a + d) = \mathfrak{N}(a) + \mathfrak{N}(d) \ge \mathfrak{N}(a)$: the minimum is attained at a. Moreover

$\langle a_l, \mathcal{D}(p'+l)a_l\rangle = |c_l|^2/\Delta(p'+l)$. By Lemma 7.2(ii), $|V^*j| \le \frac{8}{5}|j|$, and $\Delta(p'+l) = \Delta(\hat q_l) \ge (4/\pi^2)|\hat q_l|^2$, as in the proof of Lemma 7.3; with the bound on j this is (31.27) for the family a. For a second family m at which the minimum is attained, $\mathfrak{N}(m-a) = 0$ by (31.5), hence $\langle (m-a)_l, \mathcal{D}(p'+l)(m-a)_l\rangle = 0$ for every l, and Schwarz's inequality for the form $\mathcal{D}(p'+l)$ gives $\langle m_l, \mathcal{D}(p'+l)m_l\rangle = \langle a_l, \mathcal{D}(p'+l)a_l\rangle$.

For (31.28) we use (31.27). For $l = 0$ we have $\hat q_0 = p'$ and $|u_k| \le 1$, and the term is at most $C(1+2|\ln|p'||)|p'|^2|b|^2$. For $l \ne 0$ the difference $\hat q_l - p'$ is a nonzero element of $2\pi\mathbb{Z}^4$, so that $|\hat q_l| \ge \pi$ and $(1+\ln|\hat q_l|^2)|\hat q_l|^{-2} \le 1$, the function $t \mapsto (1+\ln t)/t$ being decreasing for $t \ge 1$ and less than 1 at $t = \pi^2$. The terms with $l \ne 0$ are therefore at most $C|p'|^4|b|^2\sum_l |u_k(p'+l)|^2$, and the sum is at most $25(1+\pi^2/4)^4$ by the proof of Lemma 7.3; finally $|p'|^4 \le 4\pi^2|p'|^2$. *q.e.d.*

The lemma takes the place of the bound on the aliases which the Supplement derives from [2, (1.63), p. 28] (Supplement, Lemma "Alias Summability"), and it is used as follows. The functional F_0 is built from the bare lattice alone, and for the lattice $\mathbb{Z}^4$ its part of the second order at the momentum ξ is the form of the curl with the matrix $c\ln(1/|\xi|^2) + R_0(\xi)$ and a bounded function R_0 (Supplement, Proposition "The Bare Polarisation"). The turned lattice is the image of $\mathbb{Z}^4$ under the isometry ρ, so that the same holds for it with the same c and with the components in the basis e'_ν. Seen from the level k, the functional is evaluated at the background, whose part of the first order in the field B is the minimiser; at the plane wave $B = b\,e_{p'}$ its part of the second order is therefore the sum over the classes l of this form at the momentum $\xi = \eta\hat q_l$ of the bare lattice, applied to m_l. Since the form of the curl of m_l is $\langle m_l, \mathcal{D}(p'+l)m_l\rangle$, and $\ln(1/|\eta\hat q_l|^2) = \ln(1/\eta^2) - \ln|\hat q_l|^2$, this sum is

$$c\ln\frac{1}{\eta^2}\,\langle b, \Delta_k(p')b\rangle + \mathcal{R}\,, \qquad |\mathcal{R}| \le (c + \sup\|R_0\|)\,C(1+|\ln|p'||)|p'|^2|b|^2\,, \tag{31.31}$$

by Lemma 31.1(i) and (31.28). The bare functional thus contributes to the right side of (31.26) what it contributes in the run of the paper: the term with the logarithm, which is the same for the three classes, and a term which is bounded at every fixed momentum $p' \ne 0$ uniformly in k. The derivation uses, besides the lemma and the proposition of the Supplement, that the part of the first order of the bare functional at the trivial configuration vanishes and that the background is $\exp(i\eta H_k B)$ to the first order, which for the tilted run are statements of Section 29.

The linear term. We have no theorem on the effect of the numbers $\zeta_k^{\mathsf a}$ on the coefficients of the step $k+1$, and what follows is a heuristic which says what such a theorem would have to contain. To the first order in the coupling the term $\sum_{\mathsf a}\zeta_k^{\mathsf a}\mathcal{A}^{\mathsf a}$ changes the Gaussian integral of the step to that of an action in which the three classes carry the weights $\gamma^{\mathsf a} = 1 + g_k^2\zeta_k^{\mathsf a}$. For fixed weights the identity of linear algebra holds as before, since it holds for every form in the place of the Hessian of the action, and one expects the functional of the bare lattice to have a logarithm with three coefficients $c^{\mathsf a}(\gamma)$ which are smooth in the weights, with $c^{\mathsf a}(1,1,1) = c$. The linear term of the recursion of $\zeta_k^{\mathsf a}$ would then be $2\ln L$ times g_k^2 times the derivative of $c^{\mathsf a} - \bar c$ at equal weights, applied to ζ_k, up to terms without a logarithm. Two directions of the weights need no computation. Equal weights multiply the form by a number, which does not change the logarithm of a Gaussian integral beyond a constant, so that the derivative vanishes in this direction; this is the direction which (31.10) removes. The weights $(\lambda^{-2}, \lambda^2, 1)$ of the three classes are those of the isotropic action in coordinates in which x_0 and x_1 are multiplied by λ, for which $c^{\mathsf a}(\gamma) = c\,\gamma^{\mathsf a}$; in this direction the derivative is c, the number ϑ_j of Lemma 31.3 is $2c\ln L\,g_{j-1}^2$, which is positive and for small couplings less than 1, and the factor $1-\vartheta_j$ is the ratio g_{j-1}^2/g_j^2 to the first order: the weights themselves stay constant. The third direction, $(1,1,-\frac12)$, is that of the form $F_{01}^2 + F_{23}^2 - \frac12\sum_{a,j}F_{aj}^2$, which no change of

coordinates produces. Its derivative is a number of the continuum theory, and we compute it.

In the continuum the logarithm is that of the bubble of the background-field method, as in the Supplement's lemma on the coefficient of the logarithm, and the first order in the weights is the trace of the Hessian of the form $\frac{1}{4}\sum_{\mu,\nu} r_{\mu\nu}|F_{\mu\nu}|^2$ against the propagator of the isotropic theory in the Feynman gauge, to the second order in the background. For a background which is a plane wave of the momentum ξ and the amplitude β, every term contains two factors $\operatorname{ad}\beta_\mu$ and $\operatorname{ad}\bar\beta_\nu$, whose trace is a multiple of $\langle\bar\beta_\mu,\beta_\nu\rangle$; we write commuting variables a_μ and $\bar a_\nu$ in their place. The terms with one propagator do not depend on ξ and carry no logarithm. The others are built from three matrices. For $\ell,\ell'\in\mathbb{R}^4$, $q\coloneqq\ell'-\ell$, a symmetric matrix $r=(r_{\mu\nu})$ with $r_{\mu\mu}=0$, and $a\in\mathbb{C}^4$ let

$$\begin{aligned} v(\ell',\ell;a)_{\nu\mu} &\coloneqq \delta_{\nu\mu}\,(\ell+\ell')\cdot a + 2(q_\nu a_\mu - q_\mu a_\nu)\;,\\ k_0(\ell)_{\nu\mu} &\coloneqq \delta_{\nu\mu}\sum_\kappa r_{\kappa\nu}\ell_\kappa^2 - r_{\nu\mu}\ell_\nu\ell_\mu\;,\\ k_1(\ell',\ell;a)_{\nu\mu} &\coloneqq \delta_{\nu\mu}\sum_\kappa r_{\kappa\nu}(\ell+\ell')_\kappa a_\kappa - r_{\nu\mu}(\ell'_\mu a_\nu+\ell_\nu a_\mu) + r_{\nu\mu}(q_\nu a_\mu - q_\mu a_\nu)\;, \end{aligned} \tag{31.32}$$

the vertex of the first order of the operator $-D^2-2i\operatorname{ad}F$, the symbol of the Hessian of the form at the trivial background, and its vertex of the first order. With $a^+\coloneqq a$ and $a^-\coloneqq\bar a$, the integrand of the first order in r is

$$I_r(\ell,\xi)\coloneqq\frac{1}{2}\sum_\pm\left[\frac{\operatorname{tr}(k_0(\ell)\,v(\ell,\ell\pm\xi;a^\mp)\,v(\ell\pm\xi,\ell;a^\pm))}{|\ell|^4|\ell\pm\xi|^2}-\frac{\operatorname{tr}(k_1(\ell,\ell\pm\xi;a^\mp)\,v(\ell\pm\xi,\ell;a^\pm))}{|\ell|^2|\ell\pm\xi|^2}\right], \tag{31.33}$$

and the integrand of the isotropic theory itself is $I_1(\ell,\xi)\coloneqq[-\frac{1}{2}\operatorname{tr}(v(\ell,\ell+\xi;\bar a)\,v(\ell+\xi,\ell;a))+((2\ell+\xi)\cdot\bar a)((2\ell+\xi)\cdot a)]|\ell|^{-2}|\ell+\xi|^{-2}$, the bubble of the vector field with the factor $-\frac{1}{4}$ and that of the ghost with the factor $\frac{1}{2}$, each counted for the two orders of the modes. For a function $I(\ell,\xi)$ we write $I_{(2)}$ for the part of the second order in ξ of its Taylor expansion at $\xi=0$, which is homogeneous of the degree -4 in ℓ for the two functions above, and $\langle\cdot\rangle$ for the mean over the unit sphere in ℓ; the coefficient of the logarithm is this mean, up to a factor common to all terms (Supplement, Lemma "The Infrared Lemma", for the integrals with two propagators).

Lemma 31.7 (The Logarithm for Unequal Weights)**.** *Let $\xi\in\mathbb{R}^4$, $a,\bar a\in\mathbb{C}^4$, and $\Xi_{\mu\nu}\coloneqq(\xi_\mu a_\nu-\xi_\nu a_\mu)(\xi_\mu\bar a_\nu-\xi_\nu\bar a_\mu)$. For every symmetric r with vanishing diagonal,*

$$\begin{aligned} \langle I_{1,(2)}\rangle &= -\frac{11}{3}\sum_{\mu<\nu}\Xi_{\mu\nu}\;,\qquad \langle I_{r,(2)}\rangle = -\sum_{\mu<\nu}(\mathsf{M}r)_{\mu\nu}\,\Xi_{\mu\nu}\;,\\ (\mathsf{M}r)_{\mu\nu} &\coloneqq \frac{23}{6}\,r_{\mu\nu}-\sum_{\{\kappa,\lambda\}}r_{\kappa\lambda}+\frac{1}{6}\,r_{\mu'\nu'}\;, \end{aligned} \tag{31.34}$$

where the sum runs over the four planes $\{\kappa,\lambda\}$ which have exactly one index in common with $\{\mu,\nu\}$ and $\{\mu',\nu'\}$ is the plane of the other two indices.

Proof. Both sides are polynomials in ξ, a, $\bar a$, and r. On the unit sphere $|\ell\pm t\xi|^{-2}=1\mp 2t\,\ell\cdot\xi+t^2(4(\ell\cdot\xi)^2-|\xi|^2)+O(t^3)$, so that the part of the second order of each term of (31.33) is the coefficient of t^2 in the product of this polynomial with the numerator at ξ replaced by $t\xi$, a polynomial in ℓ of the degree at most 6. Its mean is obtained from the moments $\langle\ell_\mu\ell_\nu\rangle=\frac{1}{4}\delta_{\mu\nu}$, $\langle\ell_\mu^4\rangle=\frac{1}{8}$, $\langle\ell_\mu^2\ell_\nu^2\rangle=\frac{1}{24}$, $\langle\ell_\mu^6\rangle=\frac{5}{64}$, $\langle\ell_\mu^4\ell_\nu^2\rangle=\frac{1}{64}$, and $\langle\ell_\mu^2\ell_\nu^2\ell_\kappa^2\rangle=\frac{1}{192}$

for different indices, the moments with an odd exponent being zero. We have carried out the expansion and the comparison of the two sides of (31.34) with a computer algebra system, for general ξ, a, $\bar a$, and r; the difference of the two sides vanishes identically. For $\xi = e_0$ and $a = \bar a = e_1$, where $\Xi_{01} = 1$ and the other $\Xi_{\mu\nu}$ vanish, the first term of (31.33) contributes $\frac{11}{6} r_{01} + r_{02} + r_{03} + \frac{2}{3}(r_{12} + r_{13}) - \frac{1}{6} r_{23}$ to the mean and the second $-\frac{17}{3} r_{01} + \frac{1}{3}(r_{12} + r_{13})$, and in I_1 the bubble of the vector field has the mean $\frac{20}{3}$ and that of the ghost $-\frac{1}{3}$, which are the numbers $4 \cdot (-\frac{1}{3}) + 8$ and $-\frac{1}{3}$ of the Supplement's lemma on the coefficient of the logarithm. *q.e.d.*

The lemma contains two checks of the integrand (31.33), which we have derived from the Hessian of the form and not from a source. For $r_{\mu\nu} = 1$ for all pairs, $(\mathsf{M} r)_{\mu\nu} = \frac{23}{6} - 4 + \frac{1}{6} = 0$, as it must be for equal weights. For $r_{\mu\nu} = s_\mu + s_\nu$ with $\sum_\mu s_\mu = 0$, the first order of the weights which a change of the scales of the coordinates produces, $(\mathsf{M} r)_{\mu\nu} = \frac{23}{6}(s_\mu + s_\nu) - 2(s_\mu + s_\nu) - 2(s_{\mu'} + s_{\nu'}) + \frac{1}{6}(s_{\mu'} + s_{\nu'}) = \frac{11}{3}(s_\mu + s_\nu)$, that is, $\mathsf{M} r = \frac{11}{3} r$, which is the statement $c^{\mathsf{a}}(\gamma) = c\,\gamma^{\mathsf{a}}$ above. Since the isotropic integrand has the mean $-\frac{11}{3}\sum \Xi_{\mu\nu}$, the coefficients of the logarithm for the weights $1 + r_{\mu\nu}$ are $c(1 + \frac{3}{11}(\mathsf{M} r)_{\mu\nu})$ to the first order. For weights which are constant on the three classes this reads

$$\begin{aligned} c^{01}(\gamma)/c &= 1 + \tfrac{23}{22}\, r^{01} - \tfrac{12}{11}\, r^{\mathrm{mix}} + \tfrac{1}{22}\, r^{23} \,, \\ c^{23}(\gamma)/c &= 1 + \tfrac{23}{22}\, r^{23} - \tfrac{12}{11}\, r^{\mathrm{mix}} + \tfrac{1}{22}\, r^{01} \,, \\ c^{\mathrm{mix}}(\gamma)/c &= 1 + \tfrac{6}{11}\, r^{\mathrm{mix}} - \tfrac{3}{11}\, (r^{01} + r^{23}) \,, \end{aligned} \tag{31.35}$$

up to terms of the second order in $r = \gamma - 1$. The matrix of the derivatives has the eigenvectors $(1,1,1)$, $(-1,1,0)$, and $(1,1,-\frac{1}{2})$ with the eigenvalues 0, 1, and $\frac{18}{11}$. The number of the third direction is thus $m = \frac{18}{11} c$. It is positive: in the continuum the recursion is damped in both directions, with $\vartheta_j = 2c \ln L\, g_{j-1}^2$ in the one and $\vartheta_j = \frac{36}{11} c \ln L\, g_{j-1}^2$ in the other, and these are of the form which Lemma 31.4 requires. What remains is the passage from the continuum to the tilted run, and its first step can be made: to the first order in the weights, the functional of the bare lattice has the logarithm of Lemma 31.7.

The logarithm on the lattice. The turned lattice with weights on its three classes of planes is the image under ρ of the lattice $\mathbb{Z}^4$ with weights on its coordinate planes, so that we may work on $\mathbb{Z}^4$, in the setting and the notation of the Supplement's subsection on the polarisation of the bare lattice: $s(\ell) = \sum_\mu 4 \sin^2(\ell_\mu/2)$, the functional $B_{\mathrm{F}}(U) = -\frac{1}{2} \log\det(\mathbf{H} + D_U D_U^{\mathsf{T}}) + \log\det(D_U^{\mathsf{T}} D_U)$ of the Feynman gauge with the Hessian $\mathbf{H}$ of the action, and the polarisation in the direction of a background b. For a symmetric r with vanishing diagonal let $\mathcal{A}_r(U)$ be the sum over the pairs $\mu < \nu$ of $r_{\mu\nu}$ times the sum of $1 - \mathrm{Re}\,\mathrm{tr}\, U(\partial p)$ over the plaquettes of the plane $\{\mu, \nu\}$, and

$$\Phi_r(U) \coloneqq -\tfrac{1}{2}\, \mathrm{Tr}(\mathrm{Hess}\, \mathcal{A}_r(U)\, (\mathbf{H} + D_U D_U^{\mathsf{T}})^{-1}) \,, \tag{31.36}$$

the derivative at $\tau = 0$ of the functional B_{F} of the action $\mathcal{A} + \tau \mathcal{A}_r$. Its part of the second order in the background contains an integral with three propagators, to which the Supplement's infrared lemma does not apply as it stands. Two of the three propagators have the same momentum, however, and together with the symbol between them they form a bounded function of the loop momentum alone; the lemma holds with such a function as a weight.

Lemma 31.8 (The Infrared Lemma with a Weight)**.** *Let $N(\ell, \xi)$ be a trigonometric polynomial in $(\ell/2, \xi/2)$ with $N(-\ell, -\xi) = N(\ell, \xi)$ and $N(0,0) = 0$, let N_2 be its part of the second degree at 0, and let $f_2(\ell, \xi)$ be the part of the second order in ξ of the Taylor expansion at $\xi = 0$ of $N_2(\ell, \xi) |\ell|^{-2} |\ell + \xi|^{-2}$, as in the Supplement's Lemma "The Infrared*

Lemma". Let w *be a bounded measurable function on* $[-\pi,\pi]^4$ *with* $\mathsf{w}(-\ell)=\mathsf{w}(\ell)$ *and with* $|\mathsf{w}(\ell)-\mathsf{w}_\mathsf{h}(\ell)|\le C_\mathsf{w}|\ell|^2$ *for* $|\ell|\le\frac{\pi}{2}$*, where* w_h *is bounded and homogeneous of the degree* 0*. Let* $\Sigma_\mathsf{w}(\xi):=(2\pi)^{-4}\int_{S^3}\mathsf{w}_\mathsf{h}(\omega)f_2(\omega,\xi)\,\mathrm{d}\sigma(\omega)$*. Then, as* $\xi\to 0$*,*

$$\int_{[-\pi,\pi]^4}\frac{\mathsf{w}(\ell)\,N(\ell,\xi)}{s(\ell)\,s(\ell+\xi)}\,\frac{\mathrm{d}^4\ell}{(2\pi)^4}=\text{const}+\Sigma_\mathsf{w}(\xi)\ln\frac{1}{|\xi|}+O(|\xi|^2)\,, \tag{31.37}$$

the constant being the value of the integral at $\xi=0$*.*

Proof. We follow the proof of the Supplement's lemma, which is the case $\mathsf{w}=1$, with its notation: $G:=N/(s(\ell)s(\ell+\xi))$, $\Delta^2G:=G(\ell,\xi)+G(\ell,-\xi)-2G(\ell,0)$, and $f:=N_2|\ell|^{-2}|\ell+\xi|^{-2}$. The weight does not depend on ξ, so that the second difference of the integrand is $\mathsf{w}\,\Delta^2G$, and since w, N, and s are even, the substitution $\ell\mapsto-\ell$ shows that the integral is even in ξ; its difference from the value at $\xi=0$ is therefore $\frac12\int\mathsf{w}\,\Delta^2G$. That proof bounds $|\Delta^2G|$ by $C|\xi|^2$ in the region $|\ell|\ge\frac{\pi}{2}$; it bounds $|G(\ell,\eta)|$ for $\eta\in\{\xi,-\xi,0\}$ by $C(|\ell|^{-2}+|\ell+\eta|^{-2})$ in the ball $|\ell|<2|\xi|$; and in the annulus $2|\xi|<|\ell|<\frac{\pi}{2}$ it bounds $|\Delta^2(G-f)|$ by $C|\xi|^2|\ell|^{-3}$ and $|\Delta^2f-2f_2|$ by $C|\xi|^4|\ell|^{-6}$. Each of these bounds is on the modulus of the integrand, and after multiplication by the bounded function w its integral over the region is $O(|\xi|^2)$, as there. There remains $\int\mathsf{w}\,f_2$ over the annulus. Since $f_2(\cdot,\xi)$ is homogeneous of the degree -4 and quadratic in ξ, $|f_2|\le C|\xi|^2|\ell|^{-4}$, so that the part with $\mathsf{w}-\mathsf{w}_\mathsf{h}$ is at most $CC_\mathsf{w}|\xi|^2\int_{|\ell|<\pi/2}|\ell|^{-2}\,\mathrm{d}^4\ell=O(|\xi|^2)$; and $\mathsf{w}_\mathsf{h}f_2$ is homogeneous of the degree -4, so that in polar coordinates its integral over the annulus is $\ln(\pi/(4|\xi|))\int_{S^3}\mathsf{w}_\mathsf{h}f_2\,\mathrm{d}\sigma$, which is $(2\pi)^4\Sigma_\mathsf{w}(\xi)\ln(1/|\xi|)$ plus a quadratic form in ξ. *q.e.d.*

Proposition 31.9 (The Bare Functional to the First Order in the Weights)**.** *Let* r *be symmetric with vanishing diagonal, let* $\hat b_\mu(\xi)$ *be a mode of the background, and* $\hat f_{\mu\nu}:=\xi_\mu\hat b_\nu-\xi_\nu\hat b_\mu$*. In the normalisation in which the part of the second order of* B_F *in this mode has the logarithmic part* $c\ln(1/|\xi|^2)\cdot\frac12\sum_{\mu<\nu}\langle\overline{\hat f_{\mu\nu}},\hat f_{\mu\nu}\rangle$ *(Supplement, Lemma "The Coefficient of the Logarithm"), the part of the second order of* Φ_r *in this mode is*

$$\text{const}+\frac{3}{11}\,c\ln\frac{1}{|\xi|^2}\cdot\frac12\sum_{\mu<\nu}(\mathsf{M}r)_{\mu\nu}\,\langle\overline{\hat f_{\mu\nu}},\hat f_{\mu\nu}\rangle+O(|\xi|^2)|\hat b|^2 \tag{31.38}$$

as $\xi\to0$*, with the matrix* M *of* (31.34) *and a constant which is a quadratic form in* $\hat b$ *and does not depend on* ξ*.*

Proof. The expansion. We use the chart $\tilde X\mapsto e^{i(b+\tilde X)}$ and the operators $\tilde M=\tilde{\mathbf{H}}+\check D\check D^\mathrm{T}$, T, and E of the Supplement's paragraph on the chart of B_F, for which $\mathbf{H}+D_UD_U^\mathrm{T}=T^{-\mathrm{T}}(\tilde M+E)T^{-1}$ with $E=O(b^2)$. In the same way $\operatorname{Hess}\mathcal{A}_r(U)=T^{-\mathrm{T}}\tilde KT^{-1}+\mathbf{Gr}_r$, where $\tilde K$ is the Hessian of $\tilde X\mapsto\mathcal{A}_r(e^{i(b+\tilde X)})$ at $\tilde X=0$ and $\mathbf{Gr}_r$ is the pairing of the gradient of this function, which is $O(b)$, with the second derivative of the change of the chart, which is $O(b)$ and contains no difference. Hence $-2\Phi_r=\operatorname{Tr}(\tilde K(\tilde M+E)^{-1})+\operatorname{Tr}(\mathbf{Gr}_r\,T(\tilde M+E)^{-1}T^\mathrm{T})$. Let $\tilde K=K^0+K^1+K^2+\dots$ and $\tilde M=s+V^1+V^2+\dots$ be the expansions in b, with s the multiplication by $s(\ell)$ on every component. The part of the second order of $-2\Phi_r$ is

$$\begin{aligned}&\operatorname{Tr}(K^0s^{-1}V^1s^{-1}V^1s^{-1})-\operatorname{Tr}(K^1s^{-1}V^1s^{-1})\\&\quad-\operatorname{Tr}(K^0s^{-1}(V^2+E_2)s^{-1})+\operatorname{Tr}(K^2s^{-1})+\operatorname{Tr}(\mathbf{Gr}_{r,2}\,s^{-1})\,.\end{aligned} \tag{31.39}$$

The symbols. As in the Supplement's Lemma "The Naive Continuum Limit", all the operators are finite sums of shifts with coefficients which are polynomials in the values of b

nearby, and with the phases at the sites and at the midpoints of the bonds their symbols are trigonometric polynomials in $(\ell/2, \xi/2)$; the point inversion of that lemma leaves $\mathcal{A}_r$ invariant, so that the symbols of K^1 and V^1 are odd and that of K^0 is even. The symbol of K^0 is that of the form $\frac{1}{2}\sum_{\mu,\nu} r_{\mu\nu}|(d\tilde{X})_{\mu\nu}|^2$, which is the matrix k_0 of (31.32) with $\check{v}_\kappa(\ell) = 2i\sin(\ell_\kappa/2)$ in the place of $i\ell_\kappa$; it is even, it differs from $k_0(\ell)$ by $O(|\ell|^4)$, and its entries are at most $2\max|r_{\mu\nu}|\, s(\ell)$ in modulus. The part of $\mathcal{A}_r(e^{i\Xi})$ of the order two in $\tilde{X}$ and one in b is that of $\sum_p r_p \frac{i}{2}\operatorname{tr}((d\Xi)_p[\Xi,\Xi]_p)$, by the expansion of the plaquette in the proof of that lemma, with the weight of the plane at each plaquette; as there, the linear part of its symbol is that of the continuum form $\frac{1}{4}\sum_{\mu,\nu} r_{\mu\nu}\int \operatorname{tr} F(\Xi)^2_{\mu\nu}$ at the same order, which is the matrix k_1 of (31.32): the first two terms of k_1 come from $\frac{1}{2}\sum r_{\mu\nu}|D_\mu\tilde{X}_\nu - D_\nu\tilde{X}_\mu|^2$ and the third from $\sum r_{\mu\nu}\langle F_{\mu\nu}(b), i[\tilde{X}_\mu, \tilde{X}_\nu]\rangle$. The linear part of the symbol of V^1 is v, by that lemma.

The terms without a logarithm. In the last three traces of (31.39) the operator between the propagators has the momentum transfer zero, so that the integrand is the fibre trace of a trigonometric polynomial in (ℓ, ξ) divided by $s(\ell)$, in the third trace after multiplication by the bounded function $k_0^{\text{latt}}(\ell)/s(\ell)$. Since $1/s$ is integrable in four dimensions, these integrals are C^∞ in ξ, and they are even in ξ by the substitution $\ell \mapsto -\ell$; they contribute a constant and $O(|\xi|^2)$.

The two bubbles. The second trace is a sum over the two orders of the modes of integrals $\int N(\ell,\xi)s(\ell)^{-1}s(\ell+\xi)^{-1}$ whose numerators are entries of the product of the symbols of K^1 and V^1: trigonometric polynomials, even, and vanishing at 0, with the part of the second degree given by the product of the linear parts k_1 and v. The Supplement's infrared lemma applies. In the first trace two propagators have the momentum ℓ, and its integrand is the sum over the entries of $\mathsf{w}(\ell) := k_0^{\text{latt}}(\ell)/s(\ell)$ times entries of the product of two symbols of V^1 divided by $s(\ell)s(\ell+\xi)$. The function w is bounded and even, and it differs from $\mathsf{w}_\mathsf{h}(\ell) := k_0(\ell)/|\ell|^2$ by $O(|\ell|^2)$, because $k_0^{\text{latt}} = k_0 + O(|\ell|^4)$ and $s = |\ell|^2 + O(|\ell|^4)$; Lemma 31.8 applies. In both cases the coefficient of $\ln(1/|\xi|)$ is $(2\pi)^{-4}$ times the integral over the sphere of the part of the second order in ξ of the continuum expression, and for the sum of the two traces, with their signs and the factor $\frac{1}{2}$, this expression is the function I_r of (31.33), the variables a, $\bar{a}$ standing for $\operatorname{ad}\hat{b}(\xi)$ and $\operatorname{ad}\hat{b}(-\xi)$.

The coefficient. By Lemma 31.7 the mean of the part of the second order of I_r is $-\sum_{\mu<\nu}(\mathsf{M}r)_{\mu\nu}\Xi_{\mu\nu}$, and that of the isotropic integrand I_1 is $-\frac{11}{3}\sum_{\mu<\nu}\Xi_{\mu\nu}$. The functional B_F is minus the functional whose integrand is I_1, and Φ_r is minus the one whose integrand is I_r, with the same traces over the Lie algebra and the same factors; the logarithmic part of Φ_r is therefore that of B_F with $\frac{3}{11}(\mathsf{M}r)_{\mu\nu}$ in the place of 1 in each plane. *q.e.d.*

The proposition is the statement on the derivative of the coefficients $c^\mathsf{a}(\gamma)$ at equal weights which the heuristic used, for the functional of the Feynman gauge; for weights which are constant on the three classes the numbers $\frac{3}{11}(\mathsf{M}r)_{\mu\nu}$ are those of (31.35). It has the same form as the Supplement's statement on B_F, with a diagonal matrix on the planes in the place of the number c, and the argument which led to (31.31) then gives, at the level k and at a plane wave $b\,e_{p'}$, the term $\frac{3}{11}c\ln(1/\eta^2)\sum_\mathsf{a}(\mathsf{M}r)^\mathsf{a}\langle b, \Delta_k^\mathsf{a}(p')b\rangle$ plus a term which is bounded at fixed $p' \neq 0$, by Lemma 31.1(i) and Lemma 31.6; this is where the forms of the three classes enter with different coefficients. We have not written out the two steps which this requires besides the proposition, the passage from B_F to the functional F_0 of the decomposition and the representation of the polarisation through the curl with a bounded remainder (Supplement, Proposition "The Bare Polarisation", Steps 1 to 5), for Φ_r; the symmetries which the second of them uses are the reflections of the coordinates, which preserve the weights. For the linear term of the recursion one needs in addition a bound, uniform in k, on the derivative with

respect to the weights of the functional of the level k, of the kind of the Supplement's bound on that functional itself. With it the comparison of the two sides of the decomposition at the three momenta (31.25), which is linear in the functionals and applies to their derivatives, shows that the matrices M_j of the linear term, defined by the first order $g_{j-1}^2 M_j \zeta_{j-1}$ of the deviations at the step j in the two numbers, satisfy: in the basis of the directions $(-1, 1, 0)$ and $(1, 1, -\frac{1}{2})$,

$$M_j = \mathbf{D} + \delta_j \ , \qquad \mathbf{D} \coloneqq 2c \ln L \begin{pmatrix} 1 & 0 \\ 0 & \frac{18}{11} \end{pmatrix} \ , \tag{31.40}$$

with matrices δ_j which are bounded and whose sums over any range of steps are bounded by a constant. Nothing is said of the single δ_j, and nothing is needed: this is the hypothesis of Lemma 31.4.

The remainder. The parts of the deviations which are neither sources at the coupling zero nor linear in the numbers ζ_k^{a} are the terms of the second order in the weights at the coupling zero, which are at most a constant times $(g_k^2 \zeta_k)^2$, and the parts of the second order in the coupling. For the run of the paper the part of a coefficient beyond its value at the coupling zero is bounded by a constant times g_k^2 (Paper, Proposition 4.3), and the same argument gives this bound for the three coefficients of the tilted run. By Lemma 31.4(a) nothing more is needed of the remainder: the sum of the squared couplings over the steps, which grows like the logarithm of their number, is made convergent by the damping. In particular the bound on the two numbers requires no comparison of the tilted run with the run of the paper and no information on the coefficients beyond the order zero in the coupling, apart from the linear term.

The assembly. Let $\zeta_k \in \mathbb{R}^2$ be the components of the numbers ζ_k^{a} in the two directions of (31.40), and $x_k \coloneqq g_k^{-2}$. By the three preceding paragraphs the recursion (31.12) is the recursion (31.16), with the matrices of (31.40) and with $s_j = f_j + \varrho_j$, where f_j are the deviations of the coefficients at the coupling zero and at equal weights and ϱ_j is the remainder. The hypotheses of Lemma 31.4(a) are: the bound on the partial sums of the f_j, which is (31.26) with Proposition 31.5; the bound on the partial sums of the δ_j, which is (31.40); the bound C_ϱ / x_{j-1} on the remainder, in which the term of the second order in the weights is admitted by an induction on the step, since it is at most a constant times $|\zeta_{j-1}|^2 / x_{j-1}^2$; and the window (31.15) for the numbers x_j, with a lower bound $x_\star$ which is large when the coupling at the stopping scale is small. For the run of the paper the window is the sum of Paper, Theorem 4.4, and of Paper, Proposition 4.3(i); the paper does not show, and the lemma does not need, that the coupling increases at every step. For the coupling of the mean coefficient $\bar{\beta}_j$ of the tilted run it is the mean over the six planes of the bounds for the three classes, that is, (31.26) again and the counterpart of that proposition. The conclusion is a bound on $|\zeta_k|$ by a constant which does not depend on the step and on the number of the steps, so that the fraction (31.13) is at most a constant times g_k^2. This is what the inductive description requires.

In this argument the following is proved in this section: that the three coefficients can be subtracted (Lemma 31.1, Corollary 31.2, and the identity (31.11)); that three momenta separate them (Proposition 31.5); that the bare lattice contributes the same logarithm to the three classes (Lemma 31.6); the matrix $\mathbf{D}$, that is, the logarithm for unequal weights in the continuum and for the functional of the bare lattice (Lemma 31.7, Lemma 31.8, Proposition 31.9); and the recursion (Lemmata 31.3 and 31.4). These are the steps which have no counterpart in the run of the paper. The remaining steps have one, and we have not written them out for the tilted run: the representation of the term of a step as the sum of the three subtractions and of terms which decrease with the scale, which is that of [9, Sects. 3–5] with

Corollary 31.2 at the place where the symmetries enter; the bounds on the functional of the highest level and on the terms of the current, and on their derivatives with respect to the weights, which are those of the Supplement for the propagators of Sections 27 and 32, the action with weights being a small perturbation of the action in every estimate; the passage from the functional of the Feynman gauge to that of the decomposition for Φ_r; the bound on the remainder; and the induction of [9] and [10] with the term $-\sum_{\mathsf{a}} \zeta_k^{\mathsf{a}} \mathcal{A}^{\mathsf{a}}(U_k)$, which enters the fluctuation integral as the terms [9, (4.38)] do and changes its quadratic form by the fraction (31.13). They are not proved, and the bound on the two numbers is therefore not a theorem. We have looked at each of them for a place at which the three classes of planes or the turned lattice could enter in a way which the steps proved here do not cover, and we have found none; the reasons are the following, and they are reasons, not proofs.

The representation of a step. Section 3 of [9] expands the terms in the fluctuation field and involves the lattice through the earlier papers only. Its Section 4 derives the identities which express the terms of the third and the fourth order through the vacuum polarisation and its moments from the gauge invariance, which each of the functions $\mathcal{A}^{\mathsf{a}}(U_k(\cdot))$ has as well. The expressions [9, (4.42)–(4.44)] are linear in the second moments, and the second moments of Π are those of $\sum_{\mathsf{a}} \beta^{\mathsf{a}} T^{\mathsf{a}}$, which are those of $\sum_{\mathsf{a}} \beta^{\mathsf{a}} \Delta_j^{\mathsf{a}}$ by Lemma 31.1 and the decay of the kernel; so the three subtractions cancel them class by class, and what is left has three derivatives and decreases with the scale by the argument of [9, p. 298], which uses no symmetry.

The functional of the highest level and the weights. The derivative of a minimum under a constraint with respect to a parameter of the form is the derivative of the form at the minimiser. For the form of the level k with the weights $1 + \tau r$ this gives, at the trivial background,

$$\frac{\mathrm{d}}{\mathrm{d}\tau}\Big|_{\tau=0} \min\{\|\partial A\|^2 + \tau \textstyle\sum_{\mathsf{a}} r^{\mathsf{a}} \|\partial A\|_{\mathsf{a}}^2 \,|\, Q_k A = B\} = \sum_{\mathsf{a}} r^{\mathsf{a}} \|\partial H_k B\|_{\mathsf{a}}^2 \,, \tag{31.41}$$

the sum of the forms of the three classes of Lemma 31.1, and the same with the operators at a background. The derivative of the functional of the level k is therefore one half of the trace of the inverse of the form Δ_k against the form $\sum_{\mathsf{a}} r^{\mathsf{a}} \Delta_k^{\mathsf{a}}$, with the minimiser and the propagators of the action without weights: no propagator of an action with weights occurs to the first order. The second form is bounded by $\max|r^{\mathsf{a}}|$ times the first, its derivatives with respect to the background are those of the minimiser and of a local form, and the integrals over the momenta which the Supplement bounds for the logarithm of the determinant of Δ_k have, for this trace, one more propagator of Δ_k at the same momentum and the bounded symbol of the second form next to it; the singularities are those of the Supplement's proof. The same remark shows that the term $\sum_{\mathsf{a}} \zeta_k^{\mathsf{a}} \mathcal{A}^{\mathsf{a}}(U_k)$ enters the quadratic form of the step k through (31.41) and through a term with the gradient of $\mathcal{A}^{\mathsf{a}}$ at the background, which is of the kind of the Supplement's terms of the current. For the terms of the current and for the sums over the aliases with a first power, which the Supplement takes from [2, (1.63)], the proof of Lemma 31.6 gives $|u_k(p'+l)||p'|^2/|\hat{q}_l|$ for the curl of the minimiser at an alias; for the central classes $\hat{q}_l$ is the vector q of (7.7) and the sum is that of the Supplement, and for the others $|\hat{q}_l|$ is of the order of η^{-1} and (7.7) supplies a further factor $\eta|q|$.

The passage to the functional of the decomposition. The identity which relates the two functionals is algebraic and holds for every gauge invariant action; its terms are built from the gradient of the action through a map which does not depend on the action. For $\mathcal{A}_r$ the gradient at a plane wave is the symbol k_0 applied to the amplitude, of the second order in the momentum, as for the action itself, and the cancellation on which the Supplement's estimate rests, a total derivative in the loop momentum, comes from the gauge covariance alone.

The remainder and the induction. The bound of the paper on the part of a coefficient beyond the coupling zero is Cauchy's estimate for a second moment of a family which is analytic in the coupling; it applies to the second moment of each class. In the induction the additional term is analytic and small on the spaces of regular configurations, where a plaquette term is at most the square of the regularity constant and its coefficient is bounded; its part of the first order in the fluctuation field does not vanish, since the background is the minimiser of the action without weights, and it is the pairing of the fluctuation with the gradient of $\mathcal{A}^{\mathsf{a}}$ at the background, a term of the kind which [9, (1.5)] contains already; its part of the second order is the perturbation of the quadratic form by the fraction (31.13); and the term itself is carried to the next step unchanged, as the action is. In the regions of large fields the action with the weights $1 + g_k^2 \zeta_k^{\mathsf{a}}$ is bounded below by a multiple close to 1 of the action, which is what the bounds there use.

The equality. A bound on ζ_k leaves the fraction (31.13) of the order of the squared coupling at the stopping scale, where it is not small in the number of the steps, and the equality (1.1) requires that the two numbers tend to zero there. Lemma 31.4(b) gives this if the sources decrease geometrically in the step, and this is what one expects from the comparison of the tilted run with the run of the paper at their shared scales: the run of the paper has equal coefficients, so that each deviation of the tilted run is at most twice the largest difference between a coefficient of the tilted run and the coefficient of the run of the paper at the same scale, and for two runs of the paper this difference is bounded by the norms of the differences of their terms, which decrease geometrically in the scale (Paper, Section 9.1, Lemma 11.6). The linear term is not a difference of this kind, since it is produced by the two numbers themselves, and it is kept on the left of the recursion, as in (31.16). With sources of the size $C_s \theta^j$, the two numbers after k steps are at most a constant times $((x_k + C_W + b')/(x_k + \frac{1}{2} b' k))^{\epsilon} + \theta^{k/2}$, which tends to zero as $k \to \infty$ when x_k stays bounded, as it does at the stopping scale. The rate is a power of $1/K$, that is, of the bare coupling, and not of the lattice spacing: the first step of the tilted run produces a deviation which does not depend on K, and it is the damping over the K steps, and nothing else, which removes it. For the equality of two limits any rate suffices. The comparison of the two runs itself, with a fine lattice which is turned against the other, is the transfer of Paper, Sections 9.1 to 11, and we have not made it; Section 33 takes up what it requires besides this transfer.

32 The Symmetric Averages: A Sketch

Sections 30 and 31 assume that the terms of the effective action are covariant under the group $\mathfrak{G}$, and this requires the averages of [9]: blocks centred at the points of the coarse lattice and families of contours which the symmetries permute. Sections 6 to 29 are written for the averages of the earlier papers. This section sketches the transfer, and it is a sketch in the literal sense: it says how the definitions and each group of statements change, with the reason in a sentence, it verifies one elementary fact about the cell, and it does not rewrite a proof or follow a constant. Nothing in it has the status of the statements of the earlier sections.

The cell. Let a/η be an odd multiple of 5. The centred block $B_a^{\circ}(y)$ of Lemma 30.3 is the union of the centred cells $B_{5\eta}^{\circ}(z)$ over the $z \in y + 5\eta\mathbb{Z}^4$ with $|z_\mu - y_\mu| < \frac{1}{2} a$, because $a/(5\eta)$ is odd, and the cell of z consists of the points $z + \eta(n_0 e_0 + n_1 e_1 + c)$ with $|n_0|, |n_1| \leq 2$ and c in the set C° of the points of $\omega\mathbb{Z}[i]$ with real and imaginary parts of modulus less than $\frac{5}{2}$. This set is simpler than the set C of Lemma 8.1:

$$\mathsf{C}^{\circ} = \{ m e_2' + n e_3' \,|\, m, n \in \mathbb{Z},\ |m| + |n| \leq 3 \} \,. \tag{32.1}$$

Indeed, by (2.2) the components of $me'_2 + ne'_3$ are $\frac{1}{5}(3m+4n)$ and $\frac{1}{5}(4m-3n)$, of modulus at most $\frac{4}{5}(|m|+|n|) \le \frac{12}{5} < \frac{5}{2}$, so that the $1+4+8+12=25$ points on the right lie in C°, which has 25 points by Lemma 30.3(i) for $a = 5\eta$. Three things follow. The set C° is mapped to itself by $z \mapsto iz$ and by $z \mapsto -z$. A path from the centre of a cell to one of its points which changes the two coordinates in the basis e'_2, e'_3 monotonically, in either order, stays in the cell. And the staircase of the scale 5η from z to $z + 5\eta e_2$, whose partial sums have the coordinates $(0,1),(1,1),(1,2),(2,2),(2,3),(3,3),(3,4)$, stays in the union of the cells of z and of $z + 5\eta e_2 = z + \eta(3e'_2 + 4e'_3)$, the first three points having $|m| + |n| \le 3$ and the others $|m-3| + |n-4| \le 3$; by the rotation the same holds for the direction e_3. The connectors and the contours of Definition 8.2 can therefore be replaced by the staircases between the centres of the cells, in a given order of the directions e_μ, followed by a monotone path in the cell, in a given order of the directions e'_ν; these contours lie in the block, and the elements of $\mathfrak{G}$ permute the family of the contours for all the orders, by Lemma 30.3(ii).

The momentum representation. With the centred blocks the function u_k of (7.4) becomes

$$u_k^\circ(p) = \prod_{\mu\in\{0,1\}} \frac{\sin(p_\mu/2)}{\eta^{-1}\sin(\eta p_\mu/2)} \prod_{\mu\in\{2,3\}} \frac{\sin(p_\mu/2)}{(5\eta)^{-1}\sin(5\eta p_\mu/2)}\, \kappa^\circ(\eta p_2, \eta p_3)\ , \tag{32.2}$$

with $\kappa^\circ(\theta) := \frac{1}{25}\sum_{c\in\mathsf{C}^\circ} e^{i\theta\cdot c}$, a function which is real and invariant under the elements of $\mathfrak{G}_0$ acting on p. The first four factors have the moduli of the functions D in (7.4). The function κ° has the properties of κ in Lemma 7.2(i), whose proof uses of the set C only that it contains one point of every class of $\omega\mathbb{Z}[i]$ modulo $5\mathbb{Z}[i]$ and a bound on the moduli of its points, which is smaller for C°. The staircases are those of Definition 6.1, so that the matrix V of (7.5) does not change. We therefore expect Lemmata 7.1, 7.2, and 7.3, and with them the statements of Sections 12 to 16 and Lemmata 30.2, 31.1, and 31.6, to hold for the centred averages with the same proofs and with constants which are not larger; we have not gone through them with the new function.

The bounds in position space. The bounds of Section 9 count the points of a block whose staircase passes through a given bond, and such a count is made along one staircase and does not depend on the shape of the block. The Poincaré inequality of Lemma 8.3 is proved by joining the points of a block by contours inside it, and with the contours just described the same argument applies; the constant has to be computed anew.

The gauge fixing. In [9] the contour variable of a point x of a block is replaced by a mean over the contours from the centre to x for all the orders of the directions, and the axial gauge condition by an exponential function of these means [9, (0.4), (0.11)–(0.16), pp. 252–255]. For the linear theory this means the following. For a field A let $\lambda(x)$ be the mean, over the family of the contours from y to x, of the sums of A along them. Then $A - \partial\lambda$ has vanishing means, and for a field with vanishing means and a bond b from x to x' in the block, $A(b)$ is the mean over the pairs of contours of the sum of A over the closed path which consists of a contour to x, the bond, and the reversed contour to x'. A closed path in the block is the boundary of a surface of plaquettes (Lemma 22.3) whose number is at most a constant times $(a/\eta)^2$, so that $A(b)$ is bounded by the supremum of the curl as in Lemma 22.2, with another constant. This replaces the axial gauge of Section 22 in the statements which use it, and it gives the lower bound for the quadratic form with the gauge-fixing form of [9, (2.5)] by the argument of the second point of Section 34.

The averages of the configurations. For the fields with values in the group, Bałaban states that the results of [4] hold for the symmetric averages with proofs which are "in most cases unchanged" [9, p. 254], and we accept this for his lattice. For the tilted run the statements of Section 26 were derived from bounds on the plaquettes along contours and in the strips of

the staircases (Lemma 26.1); these bounds hold for each contour of the family, and a mean of elements of the group which are close to each other inherits the bounds of its terms.

The later sections. In Sections 25 to 29 the lattice enters through the statements above and through counts of plaquettes between contours, such as the numbers of Lemma 28.1. With the centred blocks the coordinates of $x - y$ in the turned basis are less than $\frac{1}{2}a$ in the directions $0, 1$ and less than $\frac{7}{10}a$ in the directions $2, 3$, in modulus, so that the contours are shorter than those of Section 22 and the counts are smaller; the constants B_1 and the thresholds of Sections 28 and 29 would have to be chosen again.

A proof along these lines would have to redo, with their constants, Lemma 9.1, Lemma 8.3, Lemma 22.2 with Lemma 22.4 and Proposition 22.6, Lemma 28.1, and Lemma 29.1, and to check the remaining statements against the new definitions. We see no step at which the turned lattice enters in a new way.

33 The Equality: What It Requires beyond the Bound

We now suppose that the tilted run has been constructed, with the bound on the two numbers of Section 31, and ask what the equality (1.1) requires in addition. The method is the comparison of two runs at their shared scales, by which the paper proves that its limit exists (Paper, Sections 9.1 to 11), and most of it is a transfer of that comparison to a pair of runs of which one has a turned fine lattice. Three points are not, and they are the subject of this section: the bare coupling of the tilted run cannot be chosen, so that its coupling at the stopping scale has to be shown to agree with that of the paper in the limit; this agreement rests, at the coupling zero, on a property of the bare lattice which the Supplement does not provide, the continuity of the bounded part of its polarisation at the momentum zero; and the normalisations of the two runs are related through a single loop. The first and the third are short. For the second we prove the infrared lemma to the next order.

The three runs. Let $\bar{g}$ be the coupling of the paper at the unit scale, and let *run A* and *run B* be the constructions of the paper for K and for $K+1$, on the lattices Λ_K and Λ_{K+1} and on the torus $\mathbb{T}_\Gamma$, with the bare couplings $g_0(K, \bar{g})$ and $g_0(K+1, \bar{g})$ for which the couplings after K and after $K+1$ steps are equal to $\bar{g}$ (Paper, Theorem 4.1). For an ensemble at the level q the two runs are stopped after $K' = K - j_q$ and $K'+1$ steps, at a lattice which they share, with a number j_q which does not depend on K (Paper, Section 9.1). By Proposition 4.2, for every g_0,

$$\mathbb{E}_{\Gamma,g_0}^{\Lambda_{K+1}}[W_{\mathcal{C}_d}] = \mathbb{E}_{\Gamma,g_0}^{\Lambda'_{K+1}}[W_{\mathcal{C}_{\rho d}}] \,, \tag{33.1}$$

with the same bare coupling on the two sides. The left side at $g_0 = g_0(K+1, \bar{g})$ is the object of run B for the ensemble $\mathcal{C}_d$. The tilted run which represents the right side has therefore to start from the bare coupling of run B. Its coupling after its $K'+1$ steps, at the stopping lattice which it shares with run A, is then not at our disposal: it is whatever the flow of the mean coefficients $\bar{\beta}'_j$ of the tilted run produces. Let x'_k and x_k^{B} be the inverse squared couplings of the tilted run and of run B after k steps, so that $x'_0 = x_0^{\mathrm{B}}$ and

$$x'_k - x_k^{\mathrm{B}} = -\sum_{j=1}^{k} (\bar{\beta}'_j - \beta_j^{\mathrm{B}}) \,. \tag{33.2}$$

In the comparison of run A with run B the difference of the couplings vanishes at the unit scale by the choice of the bare couplings, and the differences at the shared scales, that of the stopping lattice among them, are sums over the later steps, which the paper bounds by the

norms of the differences of the terms (Paper, Section 11.5). For the tilted run the sum (33.2) runs over the earlier steps, the first of which contributes a number which does not depend on K. The effective actions of the tilted run and of run A at the stopping scale differ by the action times $x'_{K'+1} - x^{\mathrm{A}}_{K'}$, which is the sum of (33.2) at $k = K' + 1$ and of the difference of run B and run A, and the first of the two has to be shown to tend to zero. Two blocking schemes for one lattice theory define two couplings, and in general these differ by a finite amount at the first order; the two schemes at hand differ in the finest lattice only, and one expects the amount to be zero. The next lemma says in which form this has to be shown.

Lemma 33.1 (The Couplings of Two Schemes)**.** *Let $0 < \theta < 1$, let $x_0, \dots, x_n$ satisfy the window* (31.15) *with $x_j > C_W$ for every j, and let $t_j = \tau_j + h_j$, $1 \le j \le n$, be real numbers with $|\sum_{j \le k} \tau_j| \le C_\tau \theta^k$ for every k and $|h_j| \le C_h \theta^j / x_{j-1}$. Then for every $k \le n$*

$$\Big|\sum_{j=1}^{k} t_j\Big| \le C_\tau \theta^k + \frac{C_h}{1-\theta}\Big[\frac{1}{x_k - C_W + \frac{1}{2}bk} + \frac{\theta^{k/2}}{x_k - C_W}\Big] . \tag{33.3}$$

Proof. By (31.15), $x_{j-1} \ge x_k - C_W + b(k - j + 1)$ for $j \le k$. For $j \le \frac{1}{2}k + 1$ this is at least $x_k - C_W + \frac{1}{2}bk$, and the sum of θ^j over these j is at most $(1-\theta)^{-1}$. For the other j it is at least $x_k - C_W$, and the sum of θ^j is at most $\theta^{k/2}(1-\theta)^{-1}$. *q.e.d.*

The window holds for the inverse squared couplings x^{B}_j of run B, as was said before Lemma 31.4. With $x_k = x^{\mathrm{B}}_k$ and $k = K' + 1$ the right side tends to zero as $K \to \infty$: the first term and the last geometrically, and the middle one like $1/K$, since $x^{\mathrm{B}}_{K'+1}$ lies between $\bar g^{-2} + bj_q - C_W$ and $\bar g^{-2} + b' j_q + C_W$ for every K, by Paper, (4.1). The two hypotheses are of different kinds. The numbers τ_j are the differences of the coefficients of the two runs at the coupling zero, $\bar b'_j - b_j$. They are not small at the first steps, and the hypothesis is on their sums: these are the differences of the accumulated coefficients, which the decomposition expresses through functionals of the level k and of the bare lattice for each of the two runs. The numbers h_j are the differences of the parts of the coefficients beyond the coupling zero; they carry the squared coupling of their step, and nothing is asked of their sums.

The coupling zero. By Proposition 31.5, the difference $\bar\Phi'_k - \Phi_k$ of the accumulated coefficients is bounded by the differences, between the tilted run and run B, of the four parts of the identity (31.26) at three momenta. Three of these are differences of objects of the level k for two fine lattices of the same spacing $\eta = L^{-k}$: the forms Δ_k and the minimisers, the functional of the level k, and the terms which decrease with the scale. The fine lattice of run A is contained in both, and one expects each difference to be bounded through run A by the estimates of the paper for two fine lattices of which one contains the other, hence by a power of η; we count this with the transfer. The fourth is the bare functional, seen from the level k. By (31.31) it is $c \ln(1/\eta^2)\Delta_k$, where the two runs differ through the forms only, plus the sum over the classes l of the form of the curl of m_l with the matrix $R_0(\eta\hat q_l) - c \ln|\hat q_l|^2$. In the classes with $|\hat q_l|$ of the order of η^{-1} the two lattices differ, and these are small by the factor $|u_k|^2$ of (31.27); in the others the momentum $\eta\hat q_l$ of the bare lattice tends to zero with η, the functions u_k and V of the two runs differ by $O(\eta|\hat q_l|)$, and the matrices are $R_0(\eta\hat q_l)$ for run B and $\rho R_0(\eta\rho\hat q_l)\rho$ for the tilted run. The difference of the two tends to zero if and only if R_0 has a limit at the momentum zero which commutes with ρ. The Supplement shows that R_0 is bounded, which is what the paper needs, and a bounded function of this kind could have a limit along every ray which depends on the ray. It has not: the bounded part is a quadratic form up to terms of the third order.

Lemma 33.2 (The Infrared Lemma to the Next Order)**.** *Let N, N_2, f_2, and s be as in Lemma 31.8, and $\Sigma(\xi) \coloneqq (2\pi)^{-4} \int_{S^3} f_2(\omega,\xi)\,\mathrm{d}\sigma(\omega)$. There is a quadratic form Q on $\mathbb{R}^4$ such that, as $\xi \to 0$,*

$$\int_{[-\pi,\pi]^4} \frac{N(\ell,\xi)}{s(\ell)\,s(\ell+\xi)}\,\frac{\mathrm{d}^4\ell}{(2\pi)^4} = \text{const} + \Sigma(\xi)\ln\frac{1}{|\xi|} + Q(\xi) + O(|\xi|^3)\,. \tag{33.4}$$

Proof. With the notation of the proof of Lemma 31.8, the left side minus its value at $\xi = 0$ is $\frac12\int \Delta^2 G\,\mathrm{d}^4\ell/(2\pi)^4$, and on the ball $|\ell| < \frac{\pi}{2}$ we have $G = f + F_r$ with $F_r \coloneqq N_r|\ell|^{-2}|\ell+\xi|^{-2} + N[h_s(\ell)|\ell+\xi|^{-2} + h_s(\ell+\xi)|\ell|^{-2} + h_s(\ell)h_s(\ell+\xi)]$, where $N_r \coloneqq N - N_2$ and $h_s(p) \coloneqq 1/s(p) - 1/|p|^2$, as in the Supplement's proof. We treat the complement of the ball, the function F_r, and the function f in turn. Let $0 < |\xi| \le \frac{\pi}{16}$.

The complement of the ball. For $|\ell| \ge \frac{\pi}{2}$ the function $G(\ell,\cdot)$ is smooth on $|\zeta| \le |\xi|$ with derivatives bounded uniformly in ℓ, since $s(\ell+\zeta)$ is bounded below there. The terms of the odd orders cancel in a symmetric second difference, so that $\Delta^2 G = \langle \xi, \partial_\xi^2 G(\ell,0)\xi\rangle + O(|\xi|^4)$, and the integral over this region is a quadratic form plus $O(|\xi|^4)$.

The function F_r. We say that a function $F(\ell,\zeta)$, defined for $0 < |\ell| \le \frac{\pi}{2}$ and $|\zeta| \le \frac12|\ell|$, has the order m if it is smooth in ζ and $|\partial_\zeta^n F| \le C_n|\ell|^{m-n}$ for every n. Sums and products of such functions have the orders given by the maximum and by the sum. The polynomial N has the order 2 and N_r the order 3, because they vanish at 0 to the orders 2 and 3 and $|(\ell,\zeta)| \le \frac32|\ell|$; the function $|\ell+\zeta|^{-2}$ has the order -2, being homogeneous and smooth where $|\ell+\zeta| \ge \frac12|\ell|$; and $h_s(\ell+\zeta) = (|p|^2 - s(p))/(s(p)|p|^2)$ at $p = \ell+\zeta$ has the order 0, the numerator being analytic and $O(|p|^4)$ and the denominator at least a constant times $|p|^4$. Hence F_r has the order -1. For $|\ell| > 2|\xi|$ this gives $\Delta^2 F_r = \langle \xi, \partial_\zeta^2 F_r(\ell,0)\xi\rangle + O(|\xi|^4|\ell|^{-5})$ with $|\partial_\zeta^2 F_r(\ell,0)| \le C|\ell|^{-3}$, which is integrable at $\ell = 0$. The integral of $\Delta^2 F_r$ over $2|\xi| < |\ell| < \frac{\pi}{2}$ is therefore the quadratic form $\langle \xi, \int_{|\ell|<\pi/2} \partial_\zeta^2 F_r(\ell,0)\,\mathrm{d}^4\ell\,\xi\rangle$, minus its part from $|\ell| < 2|\xi|$, which is at most $C|\xi|^2\int_{|\ell|<2|\xi|}|\ell|^{-3} = O(|\xi|^3)$, plus an error of at most $C|\xi|^4\int_{|\ell|>2|\xi|}|\ell|^{-5} = O(|\xi|^3)$. In the ball $|\ell| < 2|\xi|$ we bound F_r itself, at $\eta \in \{\xi, -\xi, 0\}$. For $\eta = \pm\xi$ we have $|N_r| \le C|\xi|^3$ and $|N| \le C|\xi|^2$, and h_s is bounded, so that $|F_r(\ell,\eta)| \le C|\xi|^3|\ell|^{-2}|\ell+\eta|^{-2} + C|\xi|^2(|\ell|^{-2} + |\ell+\eta|^{-2} + 1)$; the integral of $|\ell|^{-2}|\ell+\eta|^{-2}$ over the ball is a number which does not depend on ξ, by the substitution $\ell = |\xi|\lambda$, and those of $|\ell|^{-2}$ and $|\ell+\eta|^{-2}$ are $O(|\xi|^2)$. For $\eta = 0$ we have $|N_r(\ell,0)| \le C|\ell|^3$ and $|N(\ell,0)| \le C|\ell|^2$, so that $|F_r(\ell,0)| \le C|\ell|^{-1} + C$. In all three cases the integral over the ball is $O(|\xi|^3)$.

The function f. Let $J(\xi) \coloneqq \frac12\int_{|\ell|<\pi/2}\Delta^2 f\,\mathrm{d}^4\ell$. The polynomial N_2 is a combination of the monomials $\ell_\alpha\ell_\beta$, $\ell_\alpha\xi_\beta$, and $\xi_\alpha\xi_\beta$, and J is the same combination of the integrals $T_{\alpha\beta}(\xi) \coloneqq \frac12\int_{|\ell|<\pi/2}\Delta^2\varphi_{\alpha\beta}\,\mathrm{d}^4\ell$, where $\varphi_{\alpha\beta}(\ell,\xi)$ is the monomial divided by $|\ell|^2|\ell+\xi|^2$, the second difference being taken of the whole function of ξ. The integrals converge, since $\varphi_{\alpha\beta}$ is at most $C(|\ell|+|\xi|)^2|\ell|^{-2}|\ell\pm\xi|^{-2}$ and vanishes at $\xi = 0$ unless the monomial is $\ell_\alpha\ell_\beta$. The domain is a ball, and $\varphi_{\alpha\beta}(O\ell, O\xi) = \sum_{\alpha',\beta'} O_{\alpha\alpha'}O_{\beta\beta'}\varphi_{\alpha'\beta'}(\ell,\xi)$ for every rotation O, so that $T(O\xi) = OT(\xi)O^{\mathrm{T}}$ for each of the three kinds of monomials. A function of one vector with this property is of the form $T_{\alpha\beta}(\xi) = a_0(|\xi|)\delta_{\alpha\beta} + a_1(|\xi|)\xi_\alpha\xi_\beta$, and it suffices to find the functions a_0 and a_1 along one ray. Each $\varphi_{\alpha\beta}$ is homogeneous of the degree -2 in (ℓ,ξ), so that the substitution $\ell = |\xi|\lambda$ gives $T(\xi) = \frac12|\xi|^2\int_{|\lambda|<R}\Delta^2\varphi(\lambda,\hat\xi)\,\mathrm{d}^4\lambda$ with $\hat\xi = \xi/|\xi|$ and $R = \pi/(2|\xi|)$. For $|\lambda| \ge 2$ we have $|\Delta^2\varphi - 2\varphi_2| \le C|\lambda|^{-6}$, where $\varphi_2(\lambda,\hat\xi)$ is the part of the second order of the Taylor expansion, homogeneous of the degree -4 in λ; this is the bound of the Supplement's proof for f, which holds for every N_2. Hence the integral over $|\lambda| < R$ is a matrix $C_0(\hat\xi)$, the integral of $\Delta^2\varphi$ over $|\lambda| < 2$ plus that of $\Delta^2\varphi - 2\varphi_2$ over $|\lambda| > 2$, plus $2\ln(R/2)\int_{S^3}\varphi_2(\omega,\hat\xi)\,\mathrm{d}\sigma$, minus the integral of $\Delta^2\varphi - 2\varphi_2$ over $|\lambda| > R$, which

is $O(R^{-2}) = O(|\xi|^2)$. Along a ray, therefore, $T(t\hat{\xi}) = t^2(A + A' \ln(1/t)) + O(t^4)$ with matrices A, A', so that $a_0(t) = t^2(\alpha_0 + \alpha_0' \ln(1/t)) + O(t^4)$ and $a_1(t) = \alpha_1 + \alpha_1' \ln(1/t) + O(t^2)$ with numbers α_i, α_i'. Thus each $T_{\alpha\beta}(\xi)$ is a quadratic form, plus $\ln(1/|\xi|)$ times a quadratic form, plus $O(|\xi|^4)$, and so is J.

Adding the three parts, we obtain (33.4) with some quadratic form in the place of Σ; by the Supplement's lemma, whose remainder is $O(|\xi|^2)$, this form is Σ. *q.e.d.*

The lemma applies to every term with two propagators of the polarisation Ψ_{F} of the functional B_{F}, and the terms with one propagator are even and smooth, hence quadratic forms up to $O(|\xi|^4)$. Together with the Supplement's Lemma "The Coefficient of the Logarithm" and the vanishing of the affine part shown there, this gives $\Psi_{\mathrm{F}}(\xi) = c \ln(1/|\xi|^2)(|\xi|^2 \delta_{\mu\nu} - \xi_\mu \xi_\nu) + \mathcal{Q}_{\mu\nu}(\xi) + O(|\xi|^3)$ with a matrix $\mathcal{Q}$ of quadratic forms. The matrix $\mathcal{Q}(\xi)$ is symmetric; it is covariant under the signed permutations of the coordinates, as Ψ_{F} is; and it annihilates the vector ξ, by the terms of the third order of the identity $\Psi_{\mathrm{F}}(\xi)\check{v}(\xi) = 0$ with $\check{v}_\mu(\xi) = 2i \sin(\xi_\mu/2)$, in which the logarithmic term contributes $O(|\xi|^5 \ln(1/|\xi|))$. Its coefficients therefore form a tensor with the properties (30.4) and (30.5) for all the isometries of $\mathbb{Z}^4$, and by the proof of Proposition 30.1 such a tensor is a multiple of that of $|\xi|^2 \delta_{\mu\nu} - \xi_\mu \xi_\nu$:

$$\Psi_{\mathrm{F}}(\xi) = \Big(c \ln \frac{1}{|\xi|^2} + r_{\mathrm{F}} \Big) (|\xi|^2 \delta_{\mu\nu} - \xi_\mu \xi_\nu) + O(|\xi|^3) \tag{33.5}$$

with a number r_{F}. The form on the right is invariant under all the rotations, in particular under ρ, so that the functional of the Feynman gauge of the turned lattice has the same expansion with the same two numbers. For the functional F_0 of the decomposition the Supplement bounds the difference of the two polarisations by $O(|\xi|^2)$, by integrals of the same kind, a boundary term over a sphere of the radius $2|\xi|$ in the loop momentum included; we expect the same refinement for them, a quadratic form plus a term of higher order, and then the same conclusion with another number, by the same use of the covariance. With this, the matrix R_0 has the limit $r_0 \cdot 1$ at the momentum zero for both lattices, and the difference of the two bare functionals seen from the level k tends to zero with η.

The parts beyond the coupling zero. The hypothesis on the numbers h_j of Lemma 33.1 is a statement which the paper does not contain, for its own runs either, and which has to be added to the comparison. The paper bounds the difference of the whole coefficients of two runs at a shared scale by the norms of the differences of their terms, which decrease geometrically in the scale (Paper, Lemma 11.6); and for one run it bounds the part of a coefficient beyond the coupling zero by a constant times g_k^2, by replacing the coupling in the exponent of the fluctuation integral by a complex number z at fixed thresholds: the integral is then an even analytic function of z which vanishes at $z = 0$ up to a Gaussian tail (Paper, Proposition 4.3(ii) and its proof). What is needed is the two together: the difference of the fluctuation integrals of two runs, as a function of z, is even, analytic, and bounded by the norms of the differences on a disc whose radius does not depend on the step, so that its value at $z = g_k$ differs from its value at $z = 0$ by at most a constant times g_k^2 times that bound. We see no obstacle to this, since the bounds of the comparison are bounds on analytic functions on the same domains; but it is not in the paper, and we have not proved it.

The normalisations. The renormalised expectation of run B divides by the product of the normalisations $N^{\mathrm{B}}(C_i)$ of the single loops, which are products of the step factors of run B, and the tilted run produces its own factors $N'(C_i)$. Let S^{B} and S' denote the expectations at the stopping scale which the two runs obtain after their factors are taken out, so that $\mathbb{E}^{\Lambda_{K+1}}[W_{\mathcal{C}}] = \prod_i N^{\mathrm{B}}(C_i)\, S^{\mathrm{B}}(\mathcal{C})$ and $\mathbb{E}^{\Lambda'_{K+1}}[W_{\mathcal{C}}] = \prod_i N'(C_i)\, S'(\mathcal{C})$. A single loop C in the

plane $\{z = 0\}$ is fixed by ρ, so that (33.1) for it reads $N^{\mathrm{B}}(C)S^{\mathrm{B}}(C) = N'(C)S'(C)$. For the ensemble of the loop and its copy, (33.1) then gives

$$S^{\mathrm{B}}(\mathcal{C}_d) = S'(\mathcal{C}_{\rho d}) \Big(\frac{S^{\mathrm{B}}(C)}{S'(C)} \Big)^2 , \tag{33.6}$$

an identity at every K in which no normalisation occurs; it is the invariance under ρ of the quotient $\mathbb{E}[W_{\mathcal{C}}]/\prod_i \mathbb{E}[W_{\chi_i}(C_i)]$, which the paper singles out as the quantity in which the normalisation cancels (Paper, the remark which follows (1.12)). The left side tends to $W(\mathcal{C}_d)$ by the theorem of the paper. If the comparison of the tilted run with run A gives $S'(\mathcal{C}) - S^{\mathrm{A}}(\mathcal{C}) \to 0$ for the ensembles of paths of both lattices, then $S'(\mathcal{C}_{\rho d}) \to W(\mathcal{C}_{\rho d})$ and $S'(C) \to W(C)$, while $S^{\mathrm{B}}(C) \to W(C)$, and (33.6) gives (1.1) whenever $W(C) \neq 0$. The paper proves this for the loops which are not too long for the coupling: if $\bar{g} \leq \bar{g}_1$ and the loop satisfies the condition (12.19) on its length, then $|W_K(C)| \geq \frac{1}{2}$ for all large K and $|W(C)| \geq \frac{1}{2}$ (Paper, Proposition 12.5), and the bound on the length grows without limit as $\bar{g} \to 0$. For these loops the argument applies. The step factors themselves are exponentials of a term for the length of the segments of each orientation and of a term for each corner, with coefficients which do not depend on the loop (Paper, Proposition 7.13), so that the loop and its translates have the same factors in each run.

The comparison. What remains is the statement $S' - S^{\mathrm{A}} \to 0$, the estimate of the paper across two runs with the tilted run in the place of run B. Its inputs at the stopping scale are the three quantities discussed: the difference of the couplings, which tends to zero by Lemma 33.1; the two numbers ζ, which tend to zero by Lemma 31.4(b); and the differences of the terms of the two runs, for which the recursion of the paper has to be carried out with two further sources, the fractions $g_k^2|\zeta_k|$ and $g_k^2|x'_{k+1} - x_k^{\mathrm{A}}|$, next to the differences of the propagators, of the minimisers, and of the large-field terms of two fine lattices, proved for a fine lattice which is turned. The three recursions are coupled, as those of the paper are, and they have to be closed together in one induction over the scale. We have not done this, and we see nothing in it which is not of the kind of the paper's argument. The rates are worth recording, since they are not those of the paper: the differences of the terms decrease geometrically in the scale, but the two numbers and the difference of the couplings at the stopping scale decrease only like powers of $1/K$, the first by the damping and the second through the couplings of the first steps. The equality would thus be obtained with an error which is a power of the bare coupling, and the difference between the two orientations of the lattice at a finite spacing is not shown to be small in the spacing.

34 What Is Not Covered

The equality (1.1) is not proved. By Proposition 4.2 it is equivalent to a comparison of the construction of the paper on Λ_K, on the torus $\mathbb{T}_\Gamma$, with the tilted run. The seven papers of Bałaban on the propagators, the averaging operations, and the background fields, [2, 1, 3, 4, 5, 6, 7], have been gone through for the tilted run in Sections 10 to 29. We first say with what degree of detail, and then what has not been examined.

Statements proved here in full. The lemmata and propositions of Sections 6 to 23, of Section 25, and of Section 30; Lemma 24.2, Lemma 24.4, and Lemma 24.6; Lemma 26.1 and Proposition 26.2; Lemma 27.1; Lemma 28.1; and Lemma 29.1. None of them has had an independent review.

Statements for which the changes are named and the proofs are not written out. In Section 24 the bounds [3, (2.136)–(2.140), (2.149), (2.151)] are derived by the scheme of Sec-

tion 20 with the inputs named there. In Section 26 this holds for Propositions 3 to 10 of [4]; in Section 27 for the perturbation in the background field and for the expansions over walks of [5, Sections B and C], and for the unit lattice propagator of the first step at the boundary of its domain; in Sections 28 and 29 for everything except the two lemmata named above. In each case the text names the inputs in which the lattice occurs and where they are proved, and the constants which change; a reader who wants a proof has to follow Bałaban's with these substitutions.

Statements which are sketched in Bałaban's text. Two of them are proved here: the properties of the operators with Dirichlet conditions [3, p. 228], at the background field 1 (Proposition 25.1), and the smallness of the difference of the gauge-fixing projections [3, (2.134)] (Lemma 24.2). Two are accepted as his: the construction of the configuration V_0 at the boundary between two regions [7, p. 279], and the bound [7, (189)] on the second derivative of the interaction, for which he gives no computation.

In [1]: the statements for an external field $A \neq 0$ and for a box with Neumann conditions, for which the proof by reflections is not available. They are not used in the theory without matter fields (Section 11).

The papers on the renormalisation transformations. Of [9, 11, 10, 12, 13, 8] we have read the first, and the other five have not been gone through. Four points of [9] concern the tilted run.

First, [9] replaces the averages of [4] by averages which are invariant under the symmetries of the lattice. The lattices consist of the centres of cubes, and the blocks are centred at the points of the coarse lattice, L being odd [9, (0.1), (0.3)]; the contour $\Gamma_{y,x}$ is replaced by the family of the contours for all the orders of the directions, and the contour variable by a mean over this family [9, (0.4), (0.11), (0.12)]; and the axial gauge is fixed by exponential functions of these means in the place of conditions [9, (0.14)–(0.16)]. Bałaban states that the results of [4] hold for these averages, the proofs being "in most cases unchanged; in others only minor and obvious modifications are needed" [9, p. 254]. Sections 6 to 29 follow the earlier papers: the blocks are the sets $y + [0, a)^4$ of Definition 6.1, there is one path $\pi_y(x)$ and one staircase for each point, and the axial gauge is a condition. For the tilted run Lemma 30.3 shows that the symmetric definitions exist: the centred blocks, the family of the paths for all the orders of the directions, and the staircases are mapped to themselves by the symmetries of the run, and the linear averages over the centred blocks are covariant. The statements of Sections 6 to 29 have not been transferred to these definitions; the transfer requires the momentum representation, the cells, and the contours for the centred blocks, with their constants, and the axial gauge with the family of paths. Section 32 sketches it and names the statements whose proofs would have to be redone.

Secondly, with the exponential gauge fixing the quadratic form of the fluctuation integral contains the gauge-fixing form of [9, (2.5)] in the place of the axial condition [9, (2.11), p. 267]. For the first step of the tilted run, and for a gauge-fixing form which consists of the squares of the sums of the field over the paths $\pi_y(x)$, the lower bound follows from the bound for the fields which are axial, Proposition 22.6(ii): the map $\mathfrak{a}_\Lambda$ changes a field A by the gradient of a function whose norm is at most a constant times η times the root of this form, and it preserves the curl and the averages, so that $\|A\|^2$ is at most a constant times $\eta^2 \|\partial A\|^2$ plus a constant times η^2 times the form. For the means over the families of paths this belongs to the transfer just named.

Thirdly, the linearisation of the average in the fluctuation integral uses the operator h supported on the bonds $b_0(c)$ of the fine lattice which lie on the bonds c of the coarse lattice

[9, p. 267]. On the turned lattice a bond c of the directions e_2, e_3 contains no bond, and the right inverse of the average built from the fields f_c in the proof of Lemma 24.4 takes the place of h.

Fourthly, the invariance of the effective actions under the symmetries of the lattice, which [9, (2.17), (2.18), p. 269] derives from that of the averages and of the gauge fixing, is used for the vacuum polarisation; this is the subject of Section 30 and of the next item. The rest of [9] involves the fine lattice through the earlier papers only: its Section 3 expands the terms of the effective action in the fluctuation field with the functions $U_j(\square_0, \cdot)$ and $\mathbf{H}_j$ of [7], the averages, and the cubes of the coarse lattices [9, (3.3)–(3.54)], and its Section 4 derives from the gauge invariance the identities which reduce the terms of the third and the fourth order to the vacuum polarisation and its first and second moments [9, (4.9)–(4.34)]. We have read these two sections and have not written out their statements for the tilted run.

The coupling. Bałaban's inductive description of the effective actions assumes that the terms of the j-th step are invariant under the Euclidean transformations of the lattice $T^{(j+1)}$ [9, p. 263], and it uses this in [9, (1.21), (1.22)] and in Section 5 of that paper to define the coefficient β_{j+1} by one pair $\mu \neq \nu$. In the tilted run the transformations which map every lattice of the run to itself are the translations of $T^{(j+1)}$ and the elements of $\mathfrak{G}_0$. By Proposition 30.1 the vacuum polarisation then has three coefficients, one for each class of planes, whereas the term $g_k^{-2} A(U_k(V))$ of the effective action has one, by Lemma 30.2. Section 31 gives the form which the effective action takes when the three coefficients are subtracted with the actions of the three classes of planes of the turned lattice: besides the coupling it contains two numbers, the accumulated deviations of the three coefficients from their mean, which multiply terms of the dimension of the action, and the inductive description requires a bound on them which is uniform in the step and in the number of the steps. This is the first point at which the tilted run requires an estimate which is not a modification of one of Bałaban's. Section 31 proves the parts of it which have no counterpart in the run of the paper: the subtraction of the three coefficients, their separation by three momenta, the contribution of the bare lattice, the logarithm for unequal weights of the three classes, by which the two numbers are damped at the rates c and $\frac{18}{11}c$, and the recursion which turns bounds on sums over the steps into a bound on the two numbers. The estimate itself is not a theorem, because the parts which do have a counterpart are not written out for the tilted run; the paragraph on the assembly in Section 31 lists them. They are Sections 3 to 5 of [9] with the three subtractions, the bounds of the Supplement on the functional of the highest level and on the terms of the current with their derivatives in the weights, the bound of the paper on the part of a coefficient beyond the coupling zero, and the induction of [9] and of [10, Theorem 1, p. 262] with the additional term of (31.11); the choice of the bare coupling (Paper, Theorem 4.1) is made for the coupling of the mean coefficient. A bound on the two numbers is what the construction of the tilted run needs. The equality (1.1) needs more, namely that the two numbers tend to zero at the stopping scale as the number of the steps grows, since otherwise the effective actions of the two runs differ there by a term of the dimension of the action; by Lemma 31.4(b) this follows from the damping once the sources decrease geometrically in the scale, which is a statement of the comparison of the two runs.

The paper. Its construction on the torus $\mathbb{T}_\Gamma$, whose periods are not multiples of the vectors of the basis; the first step with a loop ensemble (Paper, Theorem 5.4) for the tilted run; and the comparison of two runs, in which the identities of Paper, Lemma 9.2, use of the passage between the two fine lattices only a linear map with a right inverse, while the bounds on the defect (Paper, Proposition 9.3), on the minimisers, and on the large-field terms are proved for two fine lattices of which one is the lattice of the cubes of the other. The normalisation of the tilted run is a product of its own step factors; since ρ fixes a loop in the plane $\{z = 0\}$,

the comparison for a single loop relates it to that of the paper whenever the limit for this loop is not zero. One point concerns the run of the paper itself. The linear isometries which preserve both $\mathbb{Z}^4$ and the group of periods Γ are the elements of $\mathfrak{G}_0$: a signed permutation of the coordinates which preserves Γ cannot exchange e_0 or e_1 with e_2 or e_3, because Te_0 and Te_1 are periods and Te_2 and Te_3 are not, and in the plane it preserves $(2+i)\mathbb{Z}[i]$, which the maps $z \mapsto i^k z$ do and the maps $z \mapsto i^k \bar{z}$ do not, since $(2-i)\mathbb{Z}[i] \neq (2+i)\mathbb{Z}[i]$. On the torus $\mathbb{T}_\Gamma$ the run of the paper has therefore no more symmetries than the tilted run, and in particular not the reflections of x_2 and of x_3 alone, which are among the symmetries of Paper, Corollary 1.6, on its own torus. The coefficients of the flow are not affected: the paper defines them on the whole lattice $\mathbb{Z}^4$, for the regular part which a step generates there (Paper, (4.5) and (4.6)), where all the symmetries of the lattice are present, and the same definition gives the three coefficients of the tilted run with the group $\mathfrak{G}_0$. What is affected is the statements of the paper on the torus which use the reflections of single coordinates, the symmetries and the reflection positivity of the limits among them (Paper, Corollaries 1.6 and 1.7): on $\mathbb{T}_\Gamma$ they hold for the reflections of x_0 and of x_1 and for $z \mapsto -z$ only. The construction of the step factors uses the values of the insertions at the trivial configuration, which the Supplement shows to be real and invariant under reflections; we have not checked which reflections that proof uses.

References

[1] Tadeusz Bałaban, *Regularity and Decay of Lattice Green's Functions*, Comm. Math. Phys. **89** (1983), 571–597.

[2] ———, *Propagators and Renormalization Transformations for Lattice Gauge Theories. I*, Comm. Math. Phys. **95** (1984), 17–40.

[3] ———, *Propagators and Renormalization Transformations for Lattice Gauge Theories. II*, Comm. Math. Phys. **96** (1984), 223–250.

[4] ———, *Averaging Operations for Lattice Gauge Theories*, Comm. Math. Phys. **98** (1985), 17–51.

[5] ———, *Propagators for Lattice Gauge Theories in a Background Field*, Comm. Math. Phys. **99** (1985), 389–434.

[6] ———, *Spaces of Regular Gauge Field Configurations on a Lattice and Gauge Fixing Conditions*, Comm. Math. Phys. **99** (1985), 75–102.

[7] ———, *The Variational Problem and Background Fields in Renormalization Group Method for Lattice Gauge Theories*, Comm. Math. Phys. **102** (1985), 277–309.

[8] ———, *Ultraviolet Stability of Three-Dimensional Lattice Pure Gauge Field Theories*, Comm. Math. Phys. **102** (1985), 255–275.

[9] ———, *Renormalization Group Approach to Lattice Gauge Field Theories. I. Generation of Effective Actions in a Small Field Approximation and a Coupling Constant Renormalization in Four Dimensions*, Comm. Math. Phys. **109** (1987), 249–301.

[10] ———, *Convergent Renormalization Expansions for Lattice Gauge Theories*, Comm. Math. Phys. **119** (1988), 243–285.

[11] ———, *Renormalization Group Approach to Lattice Gauge Field Theories. II. Cluster Expansions*, Comm. Math. Phys. **116** (1988), 1–22.

[12] ———, *Large Field Renormalization. I. The Basic Step of the* **R** *Operation*, Comm. Math. Phys. **122** (1989), 175–202.

[13] ———, *Large Field Renormalization. II. Localization, Exponentiation, and Bounds for the* **R** *Operation*, Comm. Math. Phys. **122** (1989), 355–392.